Comb Interpolation and Sampling Frontiers

Additive and geometric combs in the Fabius–Rvachev system

Consolidated frontier volume for Vladimir Reshetnikov's `ProveIt` project

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Abstract

This volume unifies two sampling geometries for the Fabius–Rvachev system. Its additive dyadic comb yields exact generating functions for each row, Bell–Prouhet compression, terminating Euler–Maclaurin formulae, phase-sensitive Rvachev cubature, and sharp global interpolation instability with stable local alternatives. On the geometric comb $cq^{\mathbb{N}_0}$ it develops Gaussian-Pascal and Jackson–Newton calculus, exact residual and stability formulae, Mellin and regular-variation filters, geometric-knot splines, inverse-dyadic Fabius transforms, and finite Rvachev synthesis. The common algebra explains why Gaussian binomials, Bernoulli and Bell polynomials, Thue–Morse cancellation, Richardson extrapolation, Lambert W, and the Rvachev sinc product repeatedly meet in the same problem.

The text consolidates four live manuscripts and, through the additive-dyadic manuscript, nine earlier source packages. Repeated geometric-comb derivations are stated once in their strongest form. Independent extensions are retained with their own proofs. Every theorem, proposition, lemma, and corollary in this volume has an explicit human-readable proof; every unproved mathematical claim is visibly labelled as a conjecture, problem, or research question.

Proof and evidence terminology

Lean-proved means that the theorem concordance names an exact compiled declaration in the repository. **Human-proved frontier result** means that a complete manuscript proof is present here but no exact Lean declaration has been validated. **Conjecture** and **problem** mean that a genuine proof obligation remains. Numerical tables, symbolic checks, and plots are reproducibility evidence, never proof premises. A module-level connection is not reported as a Lean proof unless an exact declaration has been checked. When compiled declarations cover only proper subclaims of a compound manuscript result, the concordance keeps that result at manuscript status and records the narrower Lean support without upgrading the whole row.

Editorial scope and reading map

The geometric-comb part comes first because its affine and multiplicative algebra supplies the reusable interpolation language. The strongest source is used for the finite spine: geometric Newton products, Jackson divided differences, Gaussian-Pascal basis changes, Lagrange cardinals, complete homogeneous residuals, and the endpoint/global stability dichotomy. Two independent reports then contribute only what is genuinely additional: confluent Puiseux–log filters, Mellin symbols, regular summability, superflat divergence, geometric-uniform laws, positive B-spline remainders, and the reciprocal sinc-product filter. The later parts return to the additive dyadic comb for exact quadrature and endpoint-flat interpolation.

The provenance appendix records every source at an immutable Git revision. The theorem concordance gives a one-to-one disposition for every source result environment. The source-disposition and companion-payload ledgers similarly account for every pre-retirement file. Git history remains the byte-for-byte archive; the live tree contains only the canonical publication and the unique evidence needed to reproduce it.

Normalizations

Throughout, $0 < q < 1$ unless a result explicitly says otherwise. A geometric comb has nodes cq^j; an additive dyadic comb has mesh 2^{-m}. The bounded Fabius distribution function is denoted F, and up denotes Rvachev's compactly supported even up-function. The shared notation catalogue imported by the master source governs q-Pochhammer symbols, Gaussian coefficients, expectations, valuations, asymptotic notation, and named number systems. Local symbols are introduced once at first use.

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Part I
Algebra of the geometric comb

1 Scope, provenance, and normalization

1.1 Relation to the canonical source and neighboring reports

The starting point is the forward-theory chapter on negative upper indices and geometric Newton interpolation in the canonical source `q_pochhammer_q_binomial_monograph.tex`. That chapter already records the basic divided-difference formula on $1, q, \dots, q^n$. The present report expands that seed in four directions:

1. it develops the full change-of-basis and residual algebra, rather than treating geometric interpolation as an isolated application;
2. it separates local convergence near the accumulation point from global stability on the convex hull and proves sharp asymptotics for both;
3. it specializes the transforms to the inverse-dyadic Fabius data and obtains exact boundary-residue and endpoint-jet estimates;
4. it composes the geometric interpolant with the existing Rvachev polynomial-synthesis operator, thereby connecting q-Pochhammer coordinates to finite dictionaries of compactly supported atomic functions.

The uniform dyadic-comb report is used only for comparison: its nodes $j/2^m$ fill $[0, 1]$ but suffer the familiar global high-degree instability. The geometric comb 2^{-j} has a different pathology: it samples one boundary on all binary scales, but it leaves a fixed gap $(1/2, 1)$. The Rvachev–Lagrange loop report supplies the exact inverse-moment Appell synthesis of polynomials by translates of up. Results imported from those reports are identified when used; the report-specific derivations carry manuscript proofs here.

Results developed here means beyond the explicitly inspected source set at the pinned repository snapshot; it is neither an exhaustive repository nonduplication certificate nor a claim of unrestricted world priority. Classical background on q-series and q-calculus is available in [36, 40, 41, 39]; the q-Newton viewpoint is particularly explicit in [42]. Barycentric interpolation and its numerical evaluation are treated in [22, 23].

1.2 Manuscript proof versus Lean proof

In this document, "proved" means proved in the displayed mathematical argument. It does not mean that the statement has a corresponding Lean declaration. The formal API was audited at the source snapshot `e5175d5eb78d66f4e31db3bc506541b9bae12c57`; the relevant files and declarations listed below were unchanged when rechecked at intake head `ffcdbdac47246bd553f272bd1ceac441f8dd4f0c`. The current crosswalk also audits the later compiled module `FabiusFunction.LagrangeRvachevSynthesis`; that current formal API is separate from the immutable manuscript source pin.

The repository already formalizes substantial finite foundations:

1. finite q-Pochhammer and Gaussian-binomial algebra, including `finite_qBinomial_theorem` and `gaussianBinomial_reciprocity`;
2. complete-homogeneous specialization on a geometric alphabet via `completeHomogeneousEvalOn_geometric_range`;

3. geometric evaluation-at-zero weights, all rational power residuals, and their exact finite ℓ^1 norm via `geometricLagrangeWeight_eq_qBinomial`, `geometricLagrangeQMoment_eq_residual_qBinomial`, and `sum_abs_geometricLagrangeWeight_eq_qPochhammer_ratio`;
4. the inverse-dyadic Fabius moment identity `halfMoment_eq_fabius_formula`;
5. Rvachev moment/Appell deconvolution and global and finite polynomial synthesis, in particular `integral_eval_rvachevDeconvolvedPolynomial_add_mul_rvachev` and `normalized_sum_Ioo_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`.
6. generic finite Lagrange–Rvachev synthesis for arbitrary node families: `normalized_sum_Ioo_lagrangeRvachevDecoder_mul_shifted_rvachevUp`, `normalized_sum_Ioo_lagrangeRvachevDecoder_eval_node`, `lagrangeRvachevAtomCoefficient_eq_deconvolved_interpolate`, `sum_Ioo_lagrangeRvachevAtomCoefficient_mul_shifted_rvachevUp`, and `sum_lagrangeRvachevDecoder_eq_one`.

No one-to-one Lean declaration was found for this report's Jackson divided-difference theorem, Gaussian Newton basis pair, arbitrary-evaluation complete-homogeneous interpolation remainder, analytic infinite Newton convergence theorem, boundary-layer or global Lebesgue asymptotics, Newton–Jackson quadrature, Fabius boundary-residue and fixed-jet bounds, gap-growth bound, the closed geometric Gaussian–Appell decoder formula and prefactor, or the Matrix-level right-inverse/stochastic-encoder package. The new scalar Lagrange–Rvachev API covers the generic synthesis, coefficient, nodewise-biorthogonality, and decoder-row-sum layers only. Some remaining statements follow on paper from formalized finite ingredients, but that is not the same as a checked Lean theorem. The conjectures and research problems have no proof status in either layer.

1.3 The affine geometric comb

Definition 1.1 (Affine geometric comb)**.** Let $\alpha, \beta, q \in \mathbb{C}$ with $\beta \neq 0$. The degree-n affine geometric comb is

$$\mathcal{G}_n(\alpha, \beta; q) := \{\alpha + \beta q^j : 0 \le j \le n\}.$$

When $|q| < 1$, its infinite continuation accumulates at α. The multiplicative normalization is $\alpha = 0$, $\beta = c$; the reflected boundary normalization used for up is $\alpha = 1$, $\beta = -1$.

Finite interpolation requires distinct nodes, equivalently $q^d \neq 1$ for $1 \le d \le n$. Most algebraic formulae remain valid over an arbitrary field under that hypothesis. The convergence and Lebesgue theory will assume

$$0 < q < 1, \qquad c > 0, \qquad x_j = cq^j. \tag{1.1}$$

The affine case is recovered by $z = (x - \alpha)/\beta$.

Proposition 1.2 (Affine covariance)**.** *Let P interpolate f on $\mathcal{G}_n(\alpha, \beta; q)$ and set $g(z) = f(\alpha + \beta z)$. If Q interpolates g on $1, q, \dots, q^n$, then*

$$P(x) = Q\left(\frac{x - \alpha}{\beta}\right).$$

The Lagrange cardinals transform by the same substitution, and the Lebesgue function is invariant under real affine changes that map the interpolation interval bijectively to itself.

Proof. Both sides are polynomials of degree at most n and agree at all $n + 1$ nodes. The cardinal and Lebesgue assertions follow immediately. ☐

Thus c is retained in algebraic formulae to expose homogeneity, but it can be set to 1 in all conditioning arguments.

1.4 Notation for q-analogues

For $k \in \mathbb{N}_0$,

$$(a;q)_k := \prod_{r=0}^{k-1}(1-aq^r), \qquad (a;q)_0 := 1, \tag{1.2}$$

$$[k]_q := \frac{1-q^k}{1-q}, \qquad [k]_q! := \prod_{r=1}^{k}[r]_q = \frac{(q;q)_k}{(1-q)^k}, \tag{1.3}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{(q;q)_n}{(q;q)_k(q;q)_{n-k}} \qquad (0 \le k \le n). \tag{1.4}$$

The Gaussian coefficient is a polynomial in q; quotient notation is used only where the displayed denominators are nonzero. Reciprocity reads

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q. \tag{1.5}$$

The complete homogeneous symmetric polynomial is characterized by

$$\prod_{j=0}^{m} \frac{1}{1-y_j t} = \sum_{r \ge 0} h_r(y_0, \dots, y_m) t^r. \tag{1.6}$$

Its geometric principal specialization is

$$h_r(c, cq, \dots, cq^m) = c^r \begin{bmatrix} m+r \\ m \end{bmatrix}_q. \tag{1.7}$$

This simple identity is the source of almost every Gaussian coefficient in the residual formulae below.

2 Newton interpolation as q-Taylor calculus

2.1 The q-Pochhammer Newton basis

Definition 2.1 (Geometric Newton polynomial)**.** For $k \ge 0$, define

$$N_k(x;c,q) := \prod_{r=0}^{k-1}(x - cq^r), \qquad N_0 := 1. \tag{2.1}$$

The factor locations are exactly the first k nodes of the comb. Pulling out $-cq^r$ from each factor gives the fundamental identification

$$\boxed{N_k(x;c,q) = (-c)^k q^{\binom{k}{2}} (x/c; q^{-1})_k.} \tag{2.2}$$

Thus the finite q-Pochhammer symbol is not merely analogous to a Newton basis: on a geometric grid it *is* the Newton basis, up to its monic normalization.

Let

$$(D_q f)(x) := \frac{f(x) - f(qx)}{(1-q)x}, \qquad x \ne 0, \tag{2.3}$$

with the removable value at 0 used when it exists. The usual q-Leibniz rule gives

$$D_q N_k(x;c,q) = [k]_q N_{k-1}(x;c,q). \tag{2.4}$$

A direct proof follows by writing $N_k = (x-c)_q^k$ in the standard q-falling-power notation. The next theorem supplies the exact data-side operator and fixes every power of q.

Theorem 2.2 (Iterated Jackson difference and geometric divided difference). *Let $E_q f(x) = f(qx)$. For every $k \geq 0$,*

$$D_q^k = \frac{x^{-k}}{(1-q)^k}(E_q; q^{-1})_k, \tag{2.5}$$

$$D_q^k f(x) = \frac{x^{-k}}{(1-q)^k} \sum_{j=0}^{k} (-1)^j q^{-jk+\binom{j+1}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q f(q^j x). \tag{2.6}$$

Consequently, for distinct nodes $c, cq, \ldots, cq^k$,

$$\boxed{[f; c, cq, \ldots, cq^k] = \frac{D_q^k f(c)}{[k]_q!} = c^{-k} \sum_{j=0}^{k} \frac{(-1)^j q^{-jk+\binom{j+1}{2}}}{(q;q)_j (q;q)_{k-j}} f(cq^j).} \tag{2.7}$$

Proof. As operators,

$$D_q = \frac{x^{-1}(1-E_q)}{1-q}.$$

The commutation relation $E_q x^{-1} = q^{-1} x^{-1} E_q$ implies inductively

$$D_q^k = \frac{x^{-k}}{(1-q)^k} \prod_{r=0}^{k-1} (1 - q^{-r} E_q),$$

which is (2.5). The finite q-binomial theorem at base q^{-1} gives

$$(E_q; q^{-1})_k = \sum_{j=0}^{k} (-1)^j q^{-\binom{j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} E_q^j.$$

Using reciprocity (1.5), the exponent becomes $-\binom{j}{2} - j(k-j) = -jk + \binom{j+1}{2}$, proving (2.6).

It remains to identify the normalization. Both functionals $[f; c, cq, \ldots, cq^k]$ and $D_q^k f(c)/[k]_q!$ vanish on polynomials of degree less than k. Applied to x^k, the divided difference is 1, while $D_q^k x^k = [k]_q!$. Equality follows on polynomials of degree at most k; the explicit divided-difference formula, being a fixed linear combination of point evaluations, then proves the equality for arbitrary data on the nodes. Substituting $x = c$ in (2.6) and using $(1-q)^k [k]_q! = (q;q)_k$ gives the last expression. □

Corollary 2.3 (q-Taylor form of the interpolant). *The polynomial of degree at most n interpolating f at $c, cq, \ldots, cq^n$ is*

$$\boxed{(\mathcal{I}_n f)(x) = \sum_{k=0}^{n} \frac{D_q^k f(c)}{[k]_q!} N_k(x; c, q).} \tag{2.8}$$

Proof. This is the ordinary Newton interpolation formula with Theorem 2.2 substituted for each divided difference. □

Remark 2.4 (Two meanings of "q-Taylor"). Formula (2.8) is simultaneously a polynomial q-Taylor formula and a finite interpolation formula. No limit process is involved. Infinite q-Taylor series require convergence hypotheses; those are developed in Section 6.

2.2 Triangular evaluation and the geometric Vandermonde determinant

At a node $x = cq^j$, the basis evaluation is triangular:

$$N_k(cq^j; c, q) = \begin{cases} (-c)^k q^{\binom{k}{2}} \dfrac{(q;q)_j}{(q;q)_{j-k}}, & k \le j, \\ 0, & k > j. \end{cases} \tag{2.9}$$

This gives an $O(n^2)$ forward-substitution algorithm for the coefficients, but it also exposes the determinant exactly.

Theorem 2.5 (Geometric Vandermonde determinant)**.** *Let $V_n = (x_j^m)_{0 \le j,m \le n}$ with $x_j = cq^j$, and put $N = n(n+1)/2$. Then*

$$\boxed{\det V_n = (-1)^N c^N q^{\binom{n+1}{3}} \prod_{k=1}^{n} (q;q)_k.} \tag{2.10}$$

Proof. The Vandermonde product is

$$\det V_n = \prod_{0 \le r < s \le n} (cq^s - cq^r).$$

For fixed s,

$$\prod_{r=0}^{s-1} (q^s - q^r) = (-1)^s q^{\binom{s}{2}} (q;q)_s.$$

Multiplying over $s = 1, \dots, n$ gives the sign $(-1)^{\sum s} = (-1)^N$, the power $q^{\sum_s \binom{s}{2}} = q^{\binom{n+1}{3}}$, and the stated product. There are N pairwise differences, giving c^N. $\square$

The factor $q^{\binom{n+1}{3}}$ already signals severe monomial-Vandermonde conditioning. Newton coordinates remove the linear solve, but they do not remove the intrinsic amplification of the interpolation operator. That amplification is measured exactly in Section 7.

2.3 Online extension of the comb

Adding the new node cq^{n+1} does not change any earlier Newton coefficient. Only

$$\gamma_{n+1} = \frac{f(cq^{n+1}) - \sum_{k=0}^{n} \gamma_k N_k(cq^{n+1})}{N_{n+1}(cq^{n+1})} \tag{2.11}$$

is appended. Since the denominator is $(-c)^{n+1} q^{\binom{n+1}{2}} (q;q)_{n+1}$, the update is explicit and exact in any coefficient field containing c and q. This nesting is valuable for Fabius data: every additional sample resolves one more dyadic boundary scale without invalidating the existing expansion.

3 The Gaussian Pascal change of coordinates

3.1 Monomials versus geometric falling powers

Theorem 3.1 (Gaussian Pascal basis pair)**.** *For every $m, k \in \mathbb{N}_0$,*

$$x^m = \sum_{k=0}^{m} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x; c, q), \tag{3.1}$$

$$N_k(x; c, q) = \sum_{m=0}^{k} (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q x^m. \tag{3.2}$$

The two triangular coefficient arrays are mutually inverse.

Proof. The second identity is the finite q-binomial theorem applied to

$$N_k(x; c, q) = x^k (c/x; q)_k.$$

For the first, apply the divided-difference functional to x^m. A standard identity for monomial divided differences gives

$$[x^m; c, cq, \dots, cq^k] = h_{m-k}(c, cq, \dots, cq^k) = c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q,$$

where the value is zero for $k > m$. Substitution in the Newton interpolation formula proves (3.1). Since both transformations are unit triangular, either identity also proves that they are inverses. □

Example 3.2. The first rows are

$$\begin{aligned} x &= c + N_1, \\ x^2 &= c^2 + c(1+q)N_1 + N_2, \\ x^3 &= c^3 + c^2(1+q+q^2)N_1 + c(1+q+q^2)N_2 + N_3. \end{aligned}$$

At $q \to 1$, these become the familiar binomial transforms between powers and falling factorials after the logarithmic scaling described in Section 3.5.

3.2 Coefficient transforms and Rogers–Szegő polynomials

Let

$$f(x) = \sum_{m=0}^{M} a_m x^m = \sum_{k=0}^{M} \gamma_k N_k(x; c, q). \tag{3.3}$$

Then Theorem 3.1 gives

$$\gamma_k = \sum_{m=k}^{M} a_m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q, \tag{3.4}$$

$$a_m = \sum_{k=m}^{M} \gamma_k (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q. \tag{3.5}$$

These are finite q-binomial inversions with a grading by the scale c. The diagonal entries are one, so the determinant of the basis-change matrix is one even though individual off-diagonal entries can be large.

Define the Rogers–Szegő polynomial

$$H_m(z; q) := \sum_{k=0}^{m} \begin{bmatrix} m \\ k \end{bmatrix}_q z^k. \tag{3.6}$$

Then the ordinary generating polynomial of the Newton coefficients satisfies

$$\sum_{k=0}^{M} \gamma_k z^k = \sum_{m=0}^{M} a_m c^m H_m(z/c; q). \tag{3.7}$$

Thus geometric interpolation naturally converts monomial coefficient data through a Rogers–Szegő transform. This is one route from interpolation to the orthogonal-polynomial side of q-analysis.

3.3 A matrix factorization of geometric sampling

Let $A = (a_0, \ldots, a_n)^T$, $\Gamma = (\gamma_0, \ldots, \gamma_n)^T$, and $Y = (f(c), f(cq), \ldots, f(cq^n))^T$. Introduce

$$T_{k,m} = \begin{cases} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q, & k \le m, \\ 0, & k > m, \end{cases}$$

$$L_{j,k} = \begin{cases} (-c)^k q^{\binom{k}{2}} \dfrac{(q;q)_j}{(q;q)_{j-k}}, & k \le j, \\ 0, & k > j. \end{cases}$$

Then

$$\Gamma = TA, \qquad Y = L\Gamma = LTA. \tag{3.8}$$

The geometric Vandermonde matrix is therefore factored into a unit upper Gaussian-Pascal matrix and an explicitly invertible lower Newton-evaluation matrix. Formula (2.7) is the closed form for L^{-1}.

This factorization is useful in three distinct senses.

1. It separates algebraic basis conversion from evaluation geometry.
2. It gives exact arithmetic algorithms over rational or algebraic q.
3. It identifies where instability resides: not in a mysterious dense solve, but in the growth of the explicit triangular factors and the interpolation projector they define.

3.4 Combinatorics at reciprocal integer base

When $q = b^{-1}$ with an integer $b \ge 2$, reciprocity rewrites many coefficients with Gaussian polynomials at base b:

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = b^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_b. \tag{3.9}$$

For prime-power b, $\begin{bmatrix} n \\ k \end{bmatrix}_b$ counts k-dimensional subspaces of $\mathbb{F}_b^n$. In particular, the half-base interpolation transforms for the Fabius function contain literal counts of binary subspaces, multiplied by the powers of two dictated by reciprocity. This is a more rigid connection than the generic statement that "Gaussian coefficients occur": the entire monomial/Newton conversion matrix is a rescaled incidence-count array for finite vector spaces.

3.5 The classical limit in logarithmic coordinates

As $q \to 1^-$ the nodes cq^j collide at c, so a direct limit in the x-coordinate is confluent. The correct resolved coordinate is

$$x = cq^\xi. \tag{3.10}$$

For fixed k and ξ,

$$\lim_{q \to 1^-} \frac{(-1)^k N_k(cq^\xi; c, q)}{c^k (1-q)^k} = \prod_{r=0}^{k-1} (\xi - r) = \xi^{\underline{k}}. \tag{3.11}$$

Indeed, $(q^\xi - q^r)/(1-q) \to r - \xi$. Hence geometric Newton interpolation is a multiplicative deformation of ordinary Newton interpolation on the integer grid in the logarithmic coordinate. Simultaneously, $\begin{bmatrix} m \\ k \end{bmatrix}_q \to \binom{m}{k}$, so (3.1) tends to the ordinary binomial/falling-factorial basis transform after the same rescaling.

4 Lagrange cardinals and geometric filters

4.1 Closed barycentric weights

Let

$$\omega_{n+1}(x) := \prod_{r=0}^{n}(x - cq^r) = N_{n+1}(x; c, q). \tag{4.1}$$

The leading-form barycentric weight is $\lambda_{j,n} = 1/\omega'_{n+1}(cq^j)$. Splitting factors below and above the jth node gives the following exact expression.

Theorem 4.1 (Geometric barycentric weights and cardinals). *For $0 \le j \le n$,*

$$\lambda_{j,n} = c^{-n}\frac{(-1)^j q^{\binom{j+1}{2}-nj}}{(q;q)_j(q;q)_{n-j}}, \tag{4.2}$$

$$\ell_{j,n}(x) = c^{-n}\frac{(-1)^j q^{\binom{j+1}{2}-nj}}{(q;q)_j(q;q)_{n-j}}\frac{\omega_{n+1}(x)}{x - cq^j}. \tag{4.3}$$

Consequently,

$$(\mathcal{I}_n f)(x) = \omega_{n+1}(x)\sum_{j=0}^{n}\frac{\lambda_{j,n} f(cq^j)}{x - cq^j}, \tag{4.4}$$

with the exact-node value supplied by continuity.

Proof. For $r < j$, $cq^j - cq^r = -cq^r(1 - q^{j-r})$; for $r > j$, $cq^j - cq^r = cq^j(1 - q^{r-j})$. Therefore

$$\begin{aligned}\prod_{r\neq j}(cq^j - cq^r) &= (-1)^j c^n q^{\binom{j}{2}+j(n-j)}(q;q)_j(q;q)_{n-j}\\ &= (-1)^j c^n q^{nj-\binom{j+1}{2}}(q;q)_j(q;q)_{n-j}.\end{aligned}$$

Inversion proves (4.2); the other formulas are the standard cardinal identities. □

The quotient form often used for numerical evaluation is

$$(\mathcal{I}_n f)(x) = \frac{\displaystyle\sum_{j=0}^{n}\frac{\lambda_{j,n}}{x - cq^j} f(cq^j)}{\displaystyle\sum_{j=0}^{n}\frac{\lambda_{j,n}}{x - cq^j}}. \tag{4.5}$$

In exact arithmetic (4.4) and (4.5) agree. In floating point, evaluation far from or outside the node cluster can make their stability behavior differ; see [23, 45]. The experiments bundled with this report use the product form for Lebesgue functions because it contains no alternating cancellation after absolute values are taken.

4.2 Cardinals in Newton coordinates

Proposition 4.2 (Triangular Newton expansion of a cardinal). *For each j,*

$$\boxed{\ell_{j,n}(x) = \sum_{k=j}^{n} c^{-k}\frac{(-1)^j q^{-jk+\binom{j+1}{2}}}{(q;q)_j(q;q)_{k-j}} N_k(x; c, q).} \tag{4.6}$$

Proof. Apply (2.7) to the data sequence that is one at node j and zero elsewhere. The kth divided difference vanishes for $k < j$ and equals the displayed coefficient for $k \ge j$. □

This formula makes the sample-to-Newton map manifestly lower triangular in the sample index. It is useful when the comb is extended online: the old cardinals change, but the data-to-Newton coefficient of order k still depends only on the first $k + 1$ samples.

4.3 Evaluation at the accumulation point: q-Richardson weights

Although 0 is not among the finite nodes, evaluating the interpolant there produces a classical geometric extrapolation filter.

Proposition 4.3 (Accumulation-point weights)**.** *Define* $\mu_{j,n} := \ell_{j,n}(0)$. *Then*

$$\boxed{\mu_{j,n} = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q;q)_j (q;q)_{n-j}}.} \tag{4.7}$$

They satisfy

$$\sum_{j=0}^{n} \mu_{j,n} q^{jm} = \delta_{m0}, \qquad 0 \le m \le n. \tag{4.8}$$

Thus $\sum_j \mu_{j,n} f(cq^j)$ *removes the first* n *powers in a local expansion of* f *at zero.*

Proof. Set $x = 0$ in (4.3) and simplify the powers of c and q. Equation (4.8) is polynomial reproduction applied to $1, x, \ldots, x^n$ and evaluated at zero. □

Theorem 4.4 (Exact Richardson tail)**.** *Let* $f(z) = \sum_{m \ge 0} a_m z^m$ *be a formal power series. Then*

$$\sum_{j=0}^{n} \mu_{j,n} f(cq^j z) = a_0 + (-1)^n c^{n+1} q^{\binom{n+1}{2}} \sum_{r \ge 0} a_{n+1+r} c^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q z^{n+1+r}. \tag{4.9}$$

Whenever both sides converge analytically, the same identity holds as an analytic equality.

Proof. By (4.8), only powers $m \ge n+1$ survive. The first surviving moment is

$$\sum_j \mu_{j,n} (cq^j)^{n+1} = (-1)^n c^{n+1} q^{\binom{n+1}{2}},$$

which is the negative of the nodal polynomial at zero. The universal moment residual theorem proved in Theorem 5.1, or a direct application of (1.7), gives the factor $c^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ for every higher moment. □

4.4 Generating function of the Richardson triangle

For arbitrary sample data $y_0, y_1, \ldots$, set

$$r_n := \sum_{j=0}^{n} \mu_{j,n} y_j. \tag{4.10}$$

Euler's expansion of an infinite q-Pochhammer product gives an exact transform identity.

Theorem 4.5 (Euler factorization of geometric extrapolation)**.** *As a formal power series in* z,

$$\boxed{\sum_{n \ge 0} r_n z^n = (qz;q)_\infty \sum_{j \ge 0} \frac{y_j}{(q;q)_j} z^j.} \tag{4.11}$$

Proof. Put $m = n - j$ in (4.7). The double sum factors as

$$\begin{aligned} \sum_{n \ge 0} r_n z^n &= \left(\sum_{m \ge 0} \frac{(-1)^m q^{\binom{m+1}{2}}}{(q;q)_m} z^m \right) \left(\sum_{j \ge 0} \frac{y_j}{(q;q)_j} z^j \right) \\ &= (qz;q)_\infty \sum_{j \ge 0} \frac{y_j}{(q;q)_j} z^j, \end{aligned}$$

where Euler's identity $(a;q)_\infty = \sum_{m \ge 0} (-1)^m q^{\binom{m}{2}} a^m / (q;q)_m$ was used with $a = qz$. □

This factorization will become a q-Borel-type transform of the Fabius moment sequence in Section 10.1.

Part II

Exact residuals, convergence, and stability

5 Complete-homogeneous residual calculus

5.1 A universal moment theorem

The Gaussian residual is best understood as a specialization of a statement that contains no q-notation.

Theorem 5.1 (Universal moment residual)**.** *Let $x_0, \dots, x_n$ and $w_0, \dots, w_n$ lie in a commutative ring, and put $M_m = \sum_{j=0}^{n} w_j x_j^m$. Suppose*

$$M_0 = \cdots = M_{n-1} = 0, \qquad M_n = K. \tag{5.1}$$

Then, for every $r \geq 0$,

$$M_{n+r} = K h_r(x_0, \dots, x_n). \tag{5.2}$$

Proof. In the formal power-series ring,

$$M(t) := \sum_{m \geq 0} M_m t^m = \sum_{j=0}^{n} \frac{w_j}{1 - x_j t}.$$

Let $E(t) = \prod_{j=0}^{n}(1 - x_j t)$. The product $E(t)M(t)$ is a polynomial of degree at most n. Its coefficient of t^m depends only on $M_0, \dots, M_m$. The hypotheses therefore make every coefficient below n vanish and make the coefficient at n equal K. Hence $E(t)M(t) = Kt^n$. Divide by $E(t)$ and use (1.6). □

The theorem explains why complete homogeneous functions, rather than an ad hoc sequence of correction terms, govern all higher residual moments. On a geometric alphabet they become Gaussian coefficients by (1.7).

5.2 Exact monomial interpolation error

Theorem 5.2 (All-degree monomial residual)**.** *Let $\mathcal{I}_n$ interpolate on $x_j = cq^j$, $0 \leq j \leq n$. For every $r \geq 0$,*

$$\boxed{x^{n+1+r} - \mathcal{I}_n(x^{n+1+r}) = N_{n+1}(x; c, q) h_r(x, c, cq, \dots, cq^n).} \tag{5.3}$$

Moreover,

$$\boxed{h_r(x, c, cq, \dots, cq^n) = \sum_{s=0}^{r} x^s c^{r-s} \begin{bmatrix} n + r - s \\ n \end{bmatrix}_q.} \tag{5.4}$$

Proof. The difference on the left is monic of degree $n + 1 + r$ and vanishes at every node, so it is divisible by $N_{n+1}(x)$. More precisely, the Newton remainder identity for the monomial t^{n+1+r} is

$$t^{n+1+r} - \mathcal{I}_n(t^{n+1+r}) = N_{n+1}(t)[x_0, \dots, x_n, t] z^{n+1+r}.$$

A monomial divided difference equals the complete homogeneous function of the nodes, giving (5.3). Alternatively, the quotient is symmetric and its generating function is immediate.

Indeed,

$$\begin{aligned}\sum_{r\ge 0} h_r(x,c,cq,\dots,cq^n)t^r &= \frac{1}{1-xt}\frac{1}{(ct;q)_{n+1}}\\ &= \left(\sum_{s\ge 0} x^s t^s\right)\left(\sum_{u\ge 0} c^u \begin{bmatrix} n+u \\ n\end{bmatrix}_q t^u\right).\end{aligned}$$

Extracting the coefficient of t^r proves (5.4). □

Example 5.3. The first three errors are

$$\begin{aligned} x^{n+1} - \mathcal{I}_n x^{n+1} &= N_{n+1}(x), \\ x^{n+2} - \mathcal{I}_n x^{n+2} &= N_{n+1}(x)\,(x + c[n+1]_q)\,, \\ x^{n+3} - \mathcal{I}_n x^{n+3} &= N_{n+1}(x)\left(x^2 + c[n+1]_q x + c^2\begin{bmatrix} n+2 \\ n\end{bmatrix}_q\right). \end{aligned}$$

The coefficients are positive for $0 < q < 1$, which is useful for one-sided error bounds when the Taylor coefficients of the target have a fixed sign.

5.3 Residual of a power series

Corollary 5.4 (Exact analytic residual kernel)**.** *Suppose $f(x) = \sum_{m\ge 0} a_m x^m$ and the rearrangements below are absolutely convergent. Then*

$$f(x) - \mathcal{I}_n f(x) = N_{n+1}(x;c,q)\sum_{r\ge 0} a_{n+1+r} h_r(x,c,cq,\dots,cq^n) \tag{5.5}$$

$$= N_{n+1}(x;c,q)\sum_{r\ge 0} a_{n+1+r}\sum_{s=0}^{r} x^s c^{r-s}\begin{bmatrix} n+r-s \\ n\end{bmatrix}_q. \tag{5.6}$$

The same identity is valid coefficientwise for a formal power series.

Proof. Apply Theorem 5.2 term by term. In the formal setting every coefficient receives only finitely many contributions; in the analytic setting absolute convergence justifies the interchange. □

Formula (5.6) is a geometric-grid counterpart of the Peano or Cauchy interpolation remainder, but it is exact and contains only the Taylor coefficients and Gaussian polynomials. It has three useful specializations:

1. $x = 0$ gives the Richardson tail (4.9);
2. $a_m = 0$ for $m > M$ makes the sum finite and exact over any coefficient ring;
3. coefficientwise positivity permits sharp majorants without unknown intermediate points.

5.4 Divided differences of analytic functions

The same algebra gives an explicit transform from Taylor coefficients to geometric divided differences.

Proposition 5.5 (Taylor-to-Newton transform)**.** *If $f(x) = \sum_{m\ge 0} a_m x^m$ converges in a disk containing the nodes, then*

$$\gamma_k := [f; c, cq, \dots, cq^k] = \sum_{m\ge k} a_m c^{m-k}\begin{bmatrix} m \\ k\end{bmatrix}_q. \tag{5.7}$$

Proof. Apply linearity to the monomial divided difference used in the proof of Theorem 3.1. Absolute convergence follows, for example, from the estimates in Theorem 6.1. □

This formula is often more useful than the nodal expression (2.7): the former exposes analytic growth, whereas the latter exposes exact arithmetic and sample dependence.

6 Infinite Newton series at an accumulating comb

6.1 Why an accumulation point changes the problem

For a generic infinite node sequence, matching all nodes need not determine an analytic function. Here the nodes cq^j accumulate at 0. If both the target and the Newton-series sum are analytic near 0, equality on the comb forces equality by the identity theorem. The main task is therefore to prove locally uniform convergence of the series

$$\sum_{k\geq 0} \gamma_k N_k(x; c, q). \tag{6.1}$$

Theorem 6.1 (Analytic convergence across the accumulation point). *Let $0 < q < 1$, $0 < c < R$, and suppose f is analytic in $|z| < R_0$ for some $R_0 > R$. Put*

$$M_R := \max_{|z|=R} |f(z)|.$$

For every $0 < r < R$, the geometric Newton coefficients satisfy

$$|\gamma_k| \leq \frac{M_R R^{-k}}{(q;q)_k(1-c/R)}, \tag{6.2}$$

and the infinite Newton series converges absolutely and uniformly on $|x| \leq r$ to $f(x)$. Quantitatively,

$$\boxed{\sup_{|x|\leq r} |f(x) - \mathcal{I}_n f(x)| \leq \frac{M_R(-c/r;q)_\infty}{(1-c/R)(q;q)_\infty} \frac{(r/R)^{n+1}}{1-r/R}.} \tag{6.3}$$

For $r = 0$, convergence follows directly from the endpoint formula.

Proof. Cauchy's estimate gives $|a_m| \leq M_R R^{-m}$. By (5.7),

$$|\gamma_k| \leq \sum_{m\geq k} M_R R^{-m} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q.$$

For $0 < q < 1$,

$$\begin{bmatrix} m \\ k \end{bmatrix}_q = \frac{(q^{m-k+1};q)_k}{(q;q)_k} \leq \frac{1}{(q;q)_k}.$$

Summing the geometric majorant proves (6.2).

If $|x| \leq r$, then

$$|N_k(x;c,q)| \leq \prod_{j=0}^{k-1}(r + cq^j) = r^k(-c/r;q)_k \leq r^k(-c/r;q)_\infty.$$

Since $(q;q)_k \geq (q;q)_\infty > 0$, the tail of (6.1) is bounded by the right side of (6.3). Hence the series converges locally uniformly to an analytic function g. For each fixed node cq^j, every partial sum with $n \geq j$ equals $f(cq^j)$; therefore $g(cq^j) = f(cq^j)$ for all j. The difference $g - f$ is analytic and has zeros accumulating at 0, so it vanishes identically. The same tail estimate gives the displayed error bound. ☐

Remark 6.2 (A convergence theorem, not a stability theorem). The bound is local in a disk $|x| < R$ on which f is analytic. It does not contradict the global Lebesgue growth proved later. Analytic data generated from one exact function can undergo cancellations that arbitrary perturbed sample vectors do not. Moreover, the Fabius function is flat and nonanalytic at 0, so Theorem 6.1 deliberately does not cover the most interesting dyadic specialization; that case needs its own arithmetic estimates.

6.2 Complex bases and spiralling combs

The finite algebra survives for complex q whenever the nodes are distinct. For $|q| < 1$, the same proof gives a complex convergence theorem after replacing every occurrence of q in a majorant by $|q|$ and assuming that the disk containing the comb lies inside the analytic domain. The nodes then approach 0 along a logarithmic spiral. Lebesgue constants cease to be orderable real quantities on an interval, but the norm of the evaluation functional at a chosen point remains meaningful. The spiral case is a natural setting for analytic continuation from exponentially separated samples and is listed as a research direction in Section 13.

6.3 Meromorphic targets

If f is meromorphic near the closed disk and its poles are known, one may subtract their principal parts before applying the Newton series. More systematically, a rational Newton–Padé construction can preserve the geometric sampling while moving the approximation space from polynomials to rational functions. The global stability theorem below strongly suggests that such rationalization is not optional when accurate reconstruction is required across the persistent gap.

7 A sharp stability dichotomy

7.1 Lebesgue functions

For real nodes $c, cq, \ldots, cq^n$, define

$$\mathscr{L}_n(x) := \sum_{j=0}^{n} |\ell_{j,n}(x)|, \qquad \mathscr{L}_n^* := \max_{0\le x\le c} \mathscr{L}_n(x). \tag{7.1}$$

The operator norm of interpolation from nodal ℓ^∞ data to the value at x is exactly $\mathscr{L}_n(x)$; the global norm on $C[0, c]$ is $\mathscr{L}_n^*$. By affine covariance, both quantities are independent of c after x is scaled by c.

7.2 Bounded extrapolation to the accumulation point

Theorem 7.1 (Exact accumulation-point Lebesgue constant). *For $0 < q < 1$,*

$$\boxed{\mathscr{L}_n(0) = \frac{(-q;q)_n}{(q;q)_n}.} \tag{7.2}$$

In particular,

$$\mathscr{L}_n(0) \longrightarrow \mathscr{L}_\infty(0) := \frac{(-q;q)_\infty}{(q;q)_\infty} < \infty. \tag{7.3}$$

Proof. By (4.7), with $r = n - j$,

$$\begin{aligned}\mathscr{L}_n(0) &= \sum_{j=0}^{n} \frac{q^{\binom{n-j+1}{2}}}{(q;q)_j (q;q)_{n-j}} \\ &= \frac{1}{(q;q)_n} \sum_{r=0}^{n} q^{\binom{r+1}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q.\end{aligned}$$

The finite q-binomial theorem with parameter q gives

$$\sum_{r=0}^{n} q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q q^r = (-q;q)_n.$$

This proves the finite formula. Both infinite products converge to positive limits for $0 < q < 1$. □

Thus geometric Richardson extrapolation to the point where the grid accumulates has a uniformly bounded worst-case amplification. For the half-base,

$$\mathscr{L}_\infty(0) = \frac{(-1/2;1/2)_\infty}{(1/2;1/2)_\infty} = 8.2559879357782500655\dots. \tag{7.4}$$

The number is not tiny, but it is independent of degree.

7.3 Boundary-layer profile near the isolated outer node

Set $c = 1$ and write $M_n = \binom{n}{2}$. The cardinal belonging to the smallest node q^n has, in the top gap $q < x < 1$, the exact form

$$|\ell_{n,n}(x)| = q^{-M_n}(1-x)x^{n-1}\frac{(q/x;q)_{n-1}}{(q;q)_n}. \tag{7.5}$$

Although this cardinal is attached to the node closest to zero, it becomes enormous near the opposite endpoint. The other small-node cardinals form a universal multiplicative cloud around it.

Theorem 7.2 (Scaled Lebesgue boundary profile)**.** *Fix* $0 < q < 1$. *Uniformly for* α *in compact subsets of* $(0,\infty)$,

$$\boxed{nq^{M_n}\mathscr{L}_n(1-\alpha/n) \longrightarrow 2(-q;q)_\infty\,\alpha\mathrm{e}^{-\alpha}.} \tag{7.6}$$

Proof. First, (7.5) gives

$$nq^{M_n}|\ell_{n,n}(1-\alpha/n)| = \alpha(1-\alpha/n)^{n-1}\frac{(q/(1-\alpha/n);q)_{n-1}}{(q;q)_n}. \tag{7.7}$$

The first two factors tend uniformly to $\alpha\mathrm{e}^{-\alpha}$. For the product ratio, take logarithms. On a compact α-set and for large n, the mean-value theorem gives

$$|\log(1-q^r/(1-\alpha/n)) - \log(1-q^r)| \le \frac{Cq^r}{n}.$$

after increasing C to handle the finitely many small r. Summation over r is $O(n^{-1})$, and the extra factor $(1-q^n)^{-1}$ tends to one. Hence the product ratio in (7.7) tends uniformly to one.

Now compare every cardinal to the last one. Put $j = n - r$. The barycentric weights in (4.2) imply

$$\frac{|\ell_{n-r,n}(x)|}{|\ell_{n,n}(x)|} = q^{\binom{r}{2}}\begin{bmatrix} n \\ r \end{bmatrix}_q \frac{|x-q^n|}{|x-q^{n-r}|}. \tag{7.8}$$

For $x = 1-\alpha/n$ and fixed r, the right side tends to $q^{\binom{r}{2}}/(q;q)_r$. For $0 \le r \le n-1$, the denominator $x - q^{n-r}$ is uniformly bounded below because $q^{n-r} \le q$, and

$$\begin{bmatrix} n \\ r \end{bmatrix}_q \le \frac{1}{(q;q)_r} \le \frac{1}{(q;q)_\infty}.$$

The summable majorant $Cq^{\binom{r}{2}}$ permits dominated convergence. The remaining term $r = n$, corresponding to the outer node 1, is $O(nq^{M_n})$ relative to the last cardinal and therefore vanishes. Hence

$$\begin{aligned}\frac{\mathscr{L}_n(1-\alpha/n)}{|\ell_{n,n}(1-\alpha/n)|} &\longrightarrow \sum_{r=0}^{\infty}\frac{q^{\binom{r}{2}}}{(q;q)_r}\\ &= (-1;q)_\infty = 2(-q;q)_\infty,\end{aligned}$$

by Euler's identity. Combining this with (7.7) proves the theorem. □

7.4 The global asymptotic

Theorem 7.3 (Sharp global Lebesgue constant)**.** *Let* $0 < q < 1$*, and let* x_n^* *be any maximizer of* $\mathscr{L}_n$ *on* $[0, c]$*. Then*

$$\boxed{\mathscr{L}_n^* \sim \frac{2(-q;q)_\infty}{\mathrm{e}} \frac{q^{-\binom{n}{2}}}{n},} \tag{7.9}$$

$$\boxed{\frac{x_n^*}{c} = 1 - \frac{1}{n} + o(n^{-1}).} \tag{7.10}$$

Proof. By affine covariance take $c = 1$. The lower bound follows from Theorem 7.2 at $\alpha = 1$.

We first show that every maximizing sequence approaches 1. Fix $\varepsilon > 0$. With $j = n - r$, the explicit cardinal formula gives

$$q^{M_n} |\ell_{n-r,n}(x)| = q^{\binom{r}{2}} \frac{\prod_{s \neq n-r} |x - q^s|}{(q;q)_{n-r}(q;q)_r}. \tag{7.11}$$

For $0 \le x \le 1 - \varepsilon$, choose R with $q^R \le \varepsilon/2$. All but at most one of the factors with $s \ge R$ are at most $1 - \varepsilon/2$. Since each denominator is at least $(q;q)_\infty$, summing (7.11) over r gives

$$q^{M_n} \mathscr{L}_n(x) \le C_{q,\varepsilon} (1 - \varepsilon/2)^{n-R-1}.$$

After multiplication by n, this tends to zero, whereas the lower bound at $1 - 1/n$ tends to the positive constant $2(-q;q)_\infty/\mathrm{e}$. Thus $x_n^* \to 1$.

Put $\delta_n = 1 - x_n^*$. On $x \ge x_0 := (1+q)/2$, the outer-node cardinal satisfies

$$|\ell_{0,n}(x)| = \prod_{s=1}^{n} \frac{x - q^s}{1 - q^s} \le 1.$$

For the remaining cardinals, (7.8) and the summability of $q^{\binom{r}{2}}$ give

$$\mathscr{L}_n(x) \le 1 + C_q |\ell_{n,n}(x)|.$$

Moreover, every factor of $(q/x;q)_{n-1}$ is no larger than the corresponding factor of $(q;q)_{n-1}$, so

$$|\ell_{n,n}(x)| \le \frac{q^{-M_n}}{1-q} (1-x) x^{n-1}. \tag{7.12}$$

It follows that

$$n q^{M_n} \mathscr{L}_n(x_n^*) \le n q^{M_n} + C'_q (n\delta_n)(1 - \delta_n)^{n-1}.$$

If $n\delta_n \to 0$ or $n\delta_n \to \infty$, the right side tends to zero, contradicting the positive lower bound. Hence every subsequence has a further subsequence with $n\delta_n \to \alpha \in (0, \infty)$. The locally uniform profile theorem then gives

$$\lim n q^{M_n} \mathscr{L}_n(x_n^*) = 2(-q;q)_\infty \alpha \mathrm{e}^{-\alpha} \le \frac{2(-q;q)_\infty}{\mathrm{e}}.$$

The lower bound at $\alpha = 1$ forces equality. Since $\alpha \mathrm{e}^{-\alpha}$ has a unique maximum at $\alpha = 1$, every subsequential limit is 1. This proves both assertions. □

For $q = 1/2$ the theorem becomes

$$\mathscr{L}_n^* \sim 1.7542191571673478084 \ldots \frac{2^{\binom{n}{2}}}{n}. \tag{7.13}$$

The coexistence of (7.4) and (7.13) is the central stability fact of this report.

7.5 Interpretation: stable at the accumulation end, unstable in the gap

The geometry has two radically different regimes.

1. Near 0, every new node is closer to the evaluation point, and the alternating Richardson weights have bounded total variation.
2. Near c, only one node lies at the outer endpoint while the remaining nodes retreat geometrically toward zero. The last few small-node cardinals form the Euler cloud $2(-q;q)_\infty$ and reach their maximum in a layer of width c/n.

The instability is stronger than ordinary exponential growth: its logarithm is quadratic in n. No choice between Newton, Lagrange, and barycentric representations can change this operator norm. Those representations can only determine whether a numerical algorithm adds avoidable roundoff error to the unavoidable sensitivity of the interpolation problem.

8 Jackson integration and confluent geometric data

The same basis that triangularizes sampling on $cq^{\mathbb{N}_0}$ also triangularizes Jackson integration. This gives a useful dual picture: geometric Newton coefficients are simultaneously interpolation coordinates and exact q-quadrature coordinates.

8.1 The Newton–Jackson integral formula

Recall the Jackson integral on $[0, c]$,

$$\int_0^c f(x)\ \mathrm{d}_q x := (1-q)c\sum_{j=0}^{\infty} q^j f(cq^j), \qquad 0<q<1. \tag{8.1}$$

The definition makes the geometric comb, including its multiplicative weights, into the native quadrature grid of q-calculus [41, 39].

Theorem 8.1 (Jackson moments of the geometric Newton basis)**.** *For every $k \ge 0$,*

$$\boxed{\int_0^c N_k(x;c,q)\ \mathrm{d}_q x = (-1)^k c^{k+1}\,\frac{q^{\binom{k+1}{2}}}{[k+1]_q}.} \tag{8.2}$$

Consequently, if $p = \sum_{k=0}^{d} \gamma_k N_k$ is a polynomial, then

$$\boxed{\int_0^c p(x)\ \mathrm{d}_q x = \sum_{k=0}^{d} \gamma_k (-1)^k c^{k+1}\,\frac{q^{\binom{k+1}{2}}}{[k+1]_q}.} \tag{8.3}$$

Proof. At a comb point $x = cq^j$ the factor $N_k(cq^j)$ vanishes for $j<k$. For $j \ge k$,

$$N_k(cq^j) = (-1)^k c^k q^{\binom{k}{2}}\,\frac{(q;q)_j}{(q;q)_{j-k}}. \tag{8.4}$$

Therefore

$$\begin{aligned}\int_0^c N_k(x)\ \mathrm{d}_q x &= (-1)^k c^{k+1} q^{\binom{k}{2}}(1-q)\sum_{j=k}^{\infty} q^j\,\frac{(q;q)_j}{(q;q)_{j-k}}\\ &= (-1)^k c^{k+1} q^{\binom{k}{2}+k}(1-q)(q;q)_k\sum_{r=0}^{\infty} q^r\,\frac{(q^{k+1};q)_r}{(q;q)_r}.\end{aligned}$$

The infinite q-binomial theorem gives

$$\sum_{r\geq 0} q^r \frac{(q^{k+1};q)_r}{(q;q)_r} = \frac{(q^{k+2};q)_\infty}{(q;q)_\infty} = \frac{1}{(q;q)_{k+1}}.$$

Since $(1-q)/(1-q^{k+1}) = 1/[k+1]_q$, this proves (8.2). Linearity proves (8.3). □

Corollary 8.2 (An analytic Newton–Jackson series)**.** *Under the hypotheses of Theorem 6.1, if* $f(x) = \sum_{k\geq 0} \gamma_k N_k(x;c,q)$ *on a neighborhood of* $[0,c]$*, then*

$$\int_0^c f(x)\, \mathrm{d}_q x = \sum_{k=0}^{\infty} \gamma_k (-1)^k c^{k+1} \frac{q^{\binom{k+1}{2}}}{[k+1]_q}, \tag{8.5}$$

and the series converges absolutely.

Proof. Uniform convergence on $[0,c]$ permits termwise Jackson integration; the absolute estimate in Theorem 6.1, together with the extra factor $q^{\binom{k+1}{2}}$, gives absolute convergence directly. □

The quadratic exponent in (8.5) is important. Even when the interpolation coefficients grow moderately, Jackson integration strongly damps the high Newton levels. Interpolation and quadrature therefore have very different condition profiles on the same comb.

8.2 Repeated geometric nodes

Ordinary Hermite interpolation replaces some simple nodes by repeated nodes. If every point of the finite comb has multiplicity s, the nodal polynomial remains a single q-product:

$$\prod_{j=0}^{n} (x-cq^j)^s = \left((-c)^{n+1} q^{\binom{n+1}{2}} (x/c;q^{-1})_{n+1}\right)^s. \tag{8.6}$$

Thus the q-Pochhammer structure survives confluence at the level of the annihilating polynomial, although the individual cardinal functions acquire higher-order principal parts.

For multiplicities $s_j \geq 1$, put

$$\Omega(x) = \prod_{j=0}^{n} (x-cq^j)^{s_j}.$$

The Hermite basis attached to the derivative functional $p \mapsto p^{(r)}(cq^j)$, $0 \leq r < s_j$, can be obtained by taking the first s_j terms in the Taylor expansion at cq^j of

$$\frac{(x-cq^j)^{s_j}}{\Omega(x)}. \tag{8.7}$$

All denominators in these coefficients are products of powers of $q^j - q^r$ and hence reduce to q-Pochhammer symbols. This gives a systematic, division-explicit route to geometric Hermite formulae without forming a confluent Vandermonde matrix.

Remark 8.3. A fully expanded formula is naturally indexed by compositions of the derivative order among the factors $(x-cq^r)^{-s_r}$. Complete homogeneous symmetric functions again appear, now on a multiset in which $(cq^j - cq^r)^{-1}$ occurs s_r times. This is the confluent analogue of the complete-homogeneous residual calculus in Section 5.

8.3 Anchoring ordinary derivatives at the accumulation point

For the Fabius and Rvachev applications it is often more useful to prescribe ordinary derivatives at the accumulation point 0 than to repeat every comb node. The following factorization converts that mixed Hermite problem into one ordinary geometric interpolation problem.

Theorem 8.4 (Accumulation-point Hermite factorization)**.** *Let f have derivatives through order m at 0, and define*

$$T_m f(x) = \sum_{r=0}^{m} \frac{f^{(r)}(0)}{r!} x^r, \qquad g_m(x) = \frac{f(x) - T_m f(x)}{x^{m+1}}, \tag{8.8}$$

where $g_m(0)$ is interpreted by continuous extension when it exists. Let $\mathcal{I}_n^{c,q}$ denote interpolation at $c, cq, \ldots, cq^n$. Then

$$\boxed{H_{n,m}^{c,q} f(x) := T_m f(x) + x^{m+1} \mathcal{I}_n^{c,q} g_m(x)} \tag{8.9}$$

is the unique polynomial of degree at most $n + m + 1$ satisfying

$$(H_{n,m}^{c,q} f)^{(r)}(0) = f^{(r)}(0), \qquad 0 \le r \le m, \tag{8.10}$$

$$H_{n,m}^{c,q} f(cq^j) = f(cq^j), \qquad 0 \le j \le n. \tag{8.11}$$

Its exact error is

$$f(x) - H_{n,m}^{c,q} f(x) = x^{m+1}\big(g_m(x) - \mathcal{I}_n^{c,q} g_m(x)\big). \tag{8.12}$$

Proof. The difference between (8.9) and $T_m f$ is divisible by x^{m+1}, proving all jet conditions. At a nonzero comb node, interpolation of g_m gives

$$T_m f(cq^j) + (cq^j)^{m+1} g_m(cq^j) = f(cq^j).$$

If two polynomials of degree at most $n+m+1$ satisfy the same conditions, their difference is divisible by x^{m+1} and by $N_{n+1}(x; c, q)$. These relatively prime factors have total degree $n + m + 2$, so the difference is zero. Subtracting (8.9) from f gives (8.12). □

Corollary 8.5 (Analytic convergence with prescribed jets)**.** *If f is analytic in a centered disk $|z| < R_0$ with $R_0 > c$, then for every fixed m, $H_{n,m}^{c,q} f \to f$ locally uniformly in that disk as $n \to \infty$. Moreover, all derivatives through order m agree exactly for every n.*

Proof. Apply theorem 8.4 to $g(x) = (f(x) - T_m f(x))/x^{m+1}$, whose removable singularity at zero is analytic in the same disk. The analytic Newton convergence theorem gives $\mathcal{I}_n g \to g$ locally uniformly; multiplication by x^{m+1} and addition of $T_m f$ preserve that convergence. The derivative identities through order m hold for every n by the defining Hermite conditions. □

The factorization also applies to nonanalytic but flat functions. If $f^{(r)}(0) = 0$ for every r, then

$$H_{n,m}^{c,q} f(x) = x^{m+1} \mathcal{I}_n^{c,q} \left(\frac{f(x)}{x^{m+1}} \right). \tag{8.13}$$

It is tempting to let m grow with n. The dyadic-comb experiments in the neighboring equispaced setting show, however, that too many exact derivative constraints can reintroduce a different large-cardinal mode. For a geometric comb, the correct joint scaling of m and n is an open conditioning problem; see Problem 8.6.

Problem 8.6 (Optimal geometric Hermite anchoring)**.** *For fixed $0 < q < 1$ and a target with a flat germ at 0, determine the largest jet order $m = m(n)$ for which the operators $H_{n,m}^{c,q}$ remain stable on a specified compact subset, and determine whether a transition occurs near $m \asymp n$, $m \asymp n/\log n$, or another scale.*

Part III
The dyadic Fabius–Rvachev specialization

9 Fabius moments and the exact half-base Gaussian transform

Set $q = 1/2$ throughout this chapter. The geometric comb $1, q, q^2, \ldots$ is then the inverse-dyadic scale selected by the defining self-similarity of the Fabius function. The interpolation data are not arbitrary: after a quadratic normalization, they are the moment coefficients of a compactly supported probability distribution. This produces a second, independent triangular structure on top of the Gaussian Pascal structure.

9.1 Probability model, moments, and the preferred normalization

Let $U_1, U_2, \ldots$ be independent random variables uniformly distributed on $[0, 1]$, and put

$$X = \sum_{r=1}^{\infty} 2^{-r} U_r. \tag{9.1}$$

The random series takes values in $[0, 1]$. Its distribution function is the Fabius function F; this probability model is equivalent to the customary functional-differential definition [1, 37]. The symmetry $X \overset{d}{=} 1 - X$ gives

$$F(x) + F(1 - x) = 1, \qquad 0 \le x \le 1. \tag{9.2}$$

On $[-1, 1]$ the Rvachev function is related by

$$\mathrm{up}(t) = F(1 - |t|). \tag{9.3}$$

Define the moments and their exponential normalization by

$$d_n := \mathbb{E}X^n, \qquad a_n := \frac{d_n}{n!}. \tag{9.4}$$

The notation d_n is especially natural for the Rvachev half-moments, because

$$\boxed{d_n = n \int_0^1 t^{n-1}\, \mathrm{up}(t)\ \mathrm{d}t.} \tag{9.5}$$

Indeed, the layer-cake identity and (9.2) give

$$\mathbb{E}X^n = n \int_0^1 t^{n-1} \mathbb{P}(X > t)\ \mathrm{d}t = n \int_0^1 t^{n-1} F(1 - t)\ \mathrm{d}t.$$

Thus d_n is the preferred moment notation throughout the dyadic specialization.

The moment-generating function is the entire product

$$A(z) := \mathbb{E}\mathrm{e}^{zX} = \sum_{n=0}^{\infty} a_n z^n \tag{9.6}$$

$$= \prod_{r=1}^{\infty} \frac{\mathrm{e}^{z/2^r} - 1}{z/2^r}. \tag{9.7}$$

Separating the first factor gives

$$A(2z) = \frac{\mathrm{e}^z - 1}{z} A(z). \tag{9.8}$$

Proposition 9.1 (Exact moment recurrence). *The coefficients $a_n \in \mathbb{Q}$ are determined by $a_0 = 1$ and*

$$\boxed{(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!}, \qquad n \geq 1.} \tag{9.9}$$

In particular $0 < a_n \leq 1/n!$.

Proof. Compare coefficients of z^n in (9.8):

$$2^n a_n = \sum_{j=0}^{n} \frac{a_j}{(n-j+1)!}.$$

Moving the $j = n$ term proves the recurrence. Rationality follows by induction. Since $0 \leq X \leq 1$, one has $0 < d_n \leq 1$, hence the stated bound. □

9.2 Inverse-dyadic values are normalized moments

The Fabius differential equation is

$$F'(x) = 2F(2x), \qquad 0 \leq x \leq \frac{1}{2}. \tag{9.10}$$

Repeated differentiation on the shrinking interval gives

$$F^{(n)}(x) = 2^{\binom{n+1}{2}} F(2^n x), \qquad 0 \leq x \leq 2^{-n}. \tag{9.11}$$

All derivatives of F vanish at 0.

Theorem 9.2 (Moment formula for inverse-dyadic values). *For every $n \geq 0$,*

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} a_n = 2^{-\binom{n}{2}} \frac{d_n}{n!}.} \tag{9.12}$$

Proof. The cases $n = 0$ and $n = 1$ are immediate. For $n \geq 1$, Taylor's formula with integral remainder and the flat germ at zero give

$$\begin{aligned} F(2^{-n}) &= \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t)\ \mathrm{d}t \\ &= \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u)\ \mathrm{d}u. \end{aligned}$$

After $t = 1 - u$, the last integral is d_n/n by (9.5). The result follows. □

The first values are

$$F(1) = 1, \quad F(1/2) = \frac{1}{2}, \quad F(1/4) = \frac{5}{72}, \quad F(1/8) = \frac{1}{288}, \quad F(1/16) = \frac{143}{2073600}. \tag{9.13}$$

Their rapidly growing denominators are partly neutralized by the quadratic factor $2^{\binom{n}{2}}$, leaving the much more regular moment sequence a_n.

9.3 The exact Gaussian transform of Fabius data

Let ν_k be the coefficient of $N_k(x; 1, q)$ in the Newton interpolant to F at $1, q, \dots, q^k$:

$$\nu_k = [F; 1, q, \dots, q^k]. \tag{9.14}$$

Theorem 9.3 (Half-base Fabius–Gaussian transform)**.** *For* $q = 1/2$*,*

$$\boxed{\nu_k = \frac{1}{(q;q)_k} \sum_{j=0}^{k} (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} a_j = \frac{1}{(1/2;1/2)_k} \sum_{j=0}^{k} (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_2 a_j.} \tag{9.15}$$

Consequently the degree-n *interpolant is*

$$\boxed{(\mathcal{I}_n F)(x) = \sum_{k=0}^{n} \nu_k \prod_{r=0}^{k-1} (x - 2^{-r}).} \tag{9.16}$$

Every coefficient is rational.

Proof. Insert $F(q^j) = q^{\binom{j}{2}} a_j$ from Theorem 9.2 into the explicit divided difference (2.7). The exponent simplifies:

$$-jk + \binom{j+1}{2} + \binom{j}{2} = j(j-k).$$

Reciprocity of Gaussian coefficients gives

$$\begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} = q^{-j(k-j)} \begin{bmatrix} k \\ j \end{bmatrix}_q = q^{j(j-k)} \begin{bmatrix} k \\ j \end{bmatrix}_q .$$

Factoring $(q;q)_k$ proves (9.15). The Newton formula and rationality are immediate. □

This transform is a meeting point of three triangular arrays:

1. the moment recurrence (9.9);
2. the reciprocal-base Gaussian Pascal array $\begin{bmatrix} k \\ j \end{bmatrix}_2$;
3. the Newton evaluation array on $1, 1/2, 1/4, \dots$.

The resulting coefficients need not be small. Exact values begin

$$\nu_0 = 1, \qquad \nu_1 = 1, \qquad \nu_2 = -\frac{26}{27}, \qquad \nu_3 = -\frac{128}{27}, \tag{9.17}$$

$$\nu_4 = -\frac{4253248}{637875}, \qquad \nu_5 = \frac{189673472}{19774125}. \tag{9.18}$$

The signs and magnitudes already foreshadow the gap instability studied in Section 10.

9.4 A finite Gaussian-cardinal formula

The Lagrange form packages the same transform into one rational kernel.

Theorem 9.4 (Finite Gaussian formula for the Fabius interpolant)**.** *Let*

$$\omega_{n+1}(x) = \prod_{r=0}^{n} (x - q^r), \qquad q = \frac{1}{2}.$$

Then

$$\boxed{(\mathcal{I}_n F)(x) = \frac{\omega_{n+1}(x)}{(q;q)_n} \sum_{j=0}^{n} \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} a_j}{x - q^j}.} \tag{9.19}$$

At a node the right side is interpreted by cancellation. Formula (9.19) *is an identity in* $\mathbb{Q}[x]$*.*

Proof. The Lagrange weight at q^j is

$$\lambda_{j,n} = \frac{(-1)^j q^{-nj+\binom{j+1}{2}}}{(q;q)_j (q;q)_{n-j}}.$$

Multiplication by $F(q^j) = q^{\binom{j}{2}} a_j$ changes the exponent to $j(j-n)$. Since

$$\frac{q^{j(j-n)}}{(q;q)_j (q;q)_{n-j}} = \frac{1}{(q;q)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}},$$

the ordinary Lagrange formula gives the result. □

The numerator in (9.19) is a finite base-2 Gaussian transform of the regularized moments. The poles are only an encoding device: the zero of ω_{n+1} cancels each one exactly.

10 Boundary recovery and gap divergence

The two stability regimes of Section 7 become unusually vivid for the Fabius data. At the accumulation endpoint, interpolation recovers not only $F(0) = 0$ but every fixed zero derivative. In the persistent gap $(1/2, 1)$, the same exact rational polynomials grow at a nearly $\exp((\log 2)n^2/4)$ scale.

10.1 The boundary residue and its generating function

Define

$$R_n := (\mathcal{I}_n F)(0). \tag{10.1}$$

Using the endpoint cardinal weights from Proposition 4.3 gives the exact finite sum below.

Theorem 10.1 (Fabius boundary residue)**.** *For $q = 1/2$,*

$$\boxed{R_n = \sum_{j=0}^{n} \frac{(-1)^{n-j} q^{\binom{n-j+1}{2} + \binom{j}{2}}}{(q;q)_j (q;q)_{n-j}} a_j.} \tag{10.2}$$

Its ordinary generating function is

$$\boxed{\sum_{n=0}^{\infty} R_n z^n = (qz;q)_\infty \sum_{j=0}^{\infty} \frac{q^{\binom{j}{2}} a_j}{(q;q)_j} z^j.} \tag{10.3}$$

Both factors on the right are entire functions of z.

Proof. The endpoint cardinal is

$$\ell_{j,n}(0) = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q;q)_j (q;q)_{n-j}}.$$

Multiplication by $F(q^j) = q^{\binom{j}{2}} a_j$ proves (10.2). Euler's expansion

$$(qz;q)_\infty = \sum_{r=0}^{\infty} \frac{(-1)^r q^{\binom{r+1}{2}}}{(q;q)_r} z^r$$

shows that the coefficient of z^n in the product on the right of (10.3) is precisely (10.2). The first factor is a normally convergent infinite product. In the second, $a_j \le 1/j!$ and the factor $q^{\binom{j}{2}}$ gives more than enough decay for entire convergence. □

Equation (10.3) may be viewed as a q-Borel regularization of the Fabius moment series, followed by multiplication by the Euler zero product. It explains why the boundary residues decay far faster than any geometric sequence.

Theorem 10.2 (Superexponential boundary recovery). *For every $n \geq 0$,*

$$\boxed{|R_n| \leq \frac{\mathrm{e}}{(q;q)_\infty^2} q^{\lfloor n^2/4 \rfloor}, \qquad q = \frac{1}{2}.} \tag{10.4}$$

In particular $R_n \to F(0) = 0$ superexponentially.

Proof. Since $a_j \leq 1/j!$ and each finite $(q;q)_r$ is at least $(q;q)_\infty$,

$$|R_n| \leq \frac{1}{(q;q)_\infty^2} \sum_{j=0}^{n} \frac{q^{E_n(j)}}{j!}, \quad E_n(j) = \binom{n-j+1}{2} + \binom{j}{2}.$$

A direct completion of the square gives $E_n(j) \geq \lfloor n^2/4 \rfloor$. Finally $\sum_{j=0}^{n} 1/j! \leq \mathrm{e}$. □

10.2 Recovery of every fixed flat jet

The value estimate extends to arbitrary fixed derivative order. For a finite set Y, write $e_m(Y)$ for its elementary symmetric polynomial of degree m.

Theorem 10.3 (Exact derivative formula and fixed-jet bound). *Let $0 \leq m \leq n$. Then*

$$\boxed{(\mathcal{I}_n F)^{(m)}(0) = (-1)^m m! \sum_{j=0}^{n} \ell_{j,n}(0) q^{\binom{j}{2}} a_j e_m\left(\{q^{-r} : 0 \leq r \leq n,\ r \neq j\}\right).} \tag{10.5}$$

Moreover,

$$\boxed{\left|(\mathcal{I}_n F)^{(m)}(0)\right| \leq \frac{\mathrm{e}m!}{(q;q)_\infty^2} \binom{n}{m} q^{\lfloor n^2/4 \rfloor - mn + \binom{m}{2}}.} \tag{10.6}$$

For every fixed m,

$$(\mathcal{I}_n F)^{(m)}(0) \longrightarrow F^{(m)}(0) = 0. \tag{10.7}$$

Proof. Factor a cardinal at zero as

$$\ell_{j,n}(x) = \ell_{j,n}(0) \prod_{\substack{0 \leq r \leq n \\ r \neq j}} (1 - xq^{-r}).$$

The coefficient of x^m is $(-1)^m \ell_{j,n}(0) e_m(\{q^{-r} : r \neq j\})$, proving (10.5) after summing against the nodal values.

There are $\binom{n}{m}$ products in the elementary symmetric polynomial, and each is bounded by the product of the m largest members of $\{q^{-r} : 0 \leq r \leq n\}$:

$$e_m(\{q^{-r} : r \neq j\}) \leq \binom{n}{m} q^{-mn + \binom{m}{2}}. \tag{10.8}$$

Apply the proof of Theorem 10.2. For fixed m, the quadratic term $\lfloor n^2/4 \rfloor$ dominates the linear loss mn and the polynomial factor $\binom{n}{m}$, proving convergence. □

This result is stronger than value convergence but weaker than convergence of all jets simultaneously. The order m must remain fixed while $n \to \infty$. If m grows proportionally to n, the exponent in (10.6) changes sign, and a new asymptotic problem begins.

Corollary 10.4 (Boundary jets for the Rvachev function). *Let*

$$J_n(t) := (\mathcal{I}_n F)(1-t). \tag{10.9}$$

Then $J_n(1-2^{-j}) = \mathrm{up}(1-2^{-j})$ *for* $0 \le j \le n$*, and for every fixed* m*,*

$$J_n^{(m)}(1) \longrightarrow \mathrm{up}^{(m)}(1) = 0. \tag{10.10}$$

Proof. The interpolation identities follow from $\mathrm{up}(1-x) = F(x)$ on $0 \le x \le 1$. Differentiating $J_n(t) = (\mathcal{I}_n F)(1-t)$ gives $J_n^{(m)}(1) = (-1)^m (\mathcal{I}_n F)^{(m)}(0)$; now apply theorem 10.3 and the flatness of up at 1. □

10.3 A rigorous gap-growth upper bound

Fix $x \in (q,1) = (1/2,1)$. This point remains separated from all nodes except the isolated node 1. The finite Gaussian formula gives a substantially sharper data-specific estimate than the full Lebesgue constant.

Theorem 10.5 (Quadratic-exponential gap bound). *For fixed* $x \in (q,1)$*,*

$$|(\mathcal{I}_n F)(x)| \le C_{q,x}\, x^n q^{-\lfloor n^2/4 \rfloor}, \tag{10.11}$$

where one may take

$$C_{q,x} = \frac{1}{(q;q)_\infty^2}\left(1 + \frac{\mathrm{e}(1-x)}{x-q}\right). \tag{10.12}$$

A sharper logarithmic estimate is

$$\boxed{\log^+ |(\mathcal{I}_n F)(x)| \le \frac{\log(1/q)}{4} n^2 - \frac{n}{2}\log n + O_{q,x}\big(n + (\log n)^2\big)\,.} \tag{10.13}$$

Proof. For $x \in (q,1)$,

$$|\omega_{n+1}(x)| = (1-x)\prod_{r=1}^{n}(x - q^r) \le (1-x)x^n.$$

In (9.19), the $j=0$ denominator is $1-x$, whereas $x - q^j \ge x - q$ for $j \ge 1$. Also

$$\frac{1}{(q;q)_n}\begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} = \frac{q^{-j(n-j)}}{(q;q)_j (q;q)_{n-j}}, \qquad a_j \le \frac{1}{j!}.$$

Taking absolute values and using $j(n-j) \le \lfloor n^2/4 \rfloor$ proves (10.11).

For the sharper form, it suffices to estimate

$$S_n = \sum_{j=0}^{n} \frac{q^{-j(n-j)}}{j!}.$$

With $L = \log(1/q)$ and $t_j = \exp(Lj(n-j))/j!$,

$$\frac{t_{j+1}}{t_j} = \frac{\exp(L(n-2j-1))}{j+1}. \tag{10.14}$$

This ratio is strictly decreasing, so the sequence is unimodal. Its maximizer j_n satisfies

$$j_n = \frac{n}{2} - \frac{\log n}{2L} + O_q(1). \tag{10.15}$$

Stirling's formula at this index gives

$$\log t_{j_n} = \frac{L}{4}n^2 - \frac{n}{2}\log n + O_q\big(n + (\log n)^2\big)\,.$$

Since $S_n \le (n+1)t_{j_n}$, substitution into the preceding cardinal bound proves (10.13). □

The dominant data index is not exactly $n/2$: the factorial normalization of the Fabius moments shifts it left by approximately $(2\log 2)^{-1}\log n$. This is a genuine competition between the reciprocal Gaussian coefficient $2^{j(n-j)}$ and the moment decay $1/j!$.

10.4 A saddle-point conjecture for the actual interpolation error

The theorem above discards alternating signs. Exact rational experiments show that cancellation changes the lower-order terms but apparently not the two leading scales.

Conjecture 10.6 (Fabius gap asymptotic)**.** *For every fixed $x \in (1/2, 1)$ outside a possibly discrete exceptional set,*

$$\boxed{\log|(\mathcal{I}_n F)(x) - F(x)| = \frac{\log 2}{4}n^2 - \frac{n}{2}\log n + O_x(n).} \tag{10.16}$$

More precisely, the $O_x(n)$ term should contain an explicit linear coefficient and a bounded discrete-saddle modulation determined by the fractional part of

$$\frac{n}{2} - \frac{\log n}{2\log 2} + \sigma(x). \tag{10.17}$$

A proof will require a signed saddle analysis of the finite sum (9.19), with sufficiently uniform asymptotics for a_j. The moment product (9.7) suggests that Hayman-type coefficient methods or a two-dimensional saddle combining the moment coefficient and Gaussian kernel may be effective.

Remark 10.7 (Why endpoint flatness does not prevent the gap catastrophe)*.* The interpolation nodes reveal the flat germ with exceptional accuracy: Theorem 10.3 recovers every fixed derivative at zero. Yet those same nodes leave the interval $(1/2, 1)$ permanently unsampled. A polynomial forced through increasingly many points near zero must transport their tiny but arithmetically structured values across this fixed gap. The reciprocal Gaussian coefficients provide precisely the required, and highly unstable, transport.

11 The Rvachev synthesis loop in geometric coordinates

Interpolation produces polynomials; Rvachev theory reproduces polynomials by compactly supported atoms. Composing the two operations gives an exact finite factorization of geometric-comb interpolation through translates of up. Because the geometric cardinal coefficients are Gaussian and the polynomial reproduction coefficients are inverse-moment Appell values, the result is naturally a Gaussian–Appell transform.

11.1 Fourier zeros and inverse-moment Appell polynomials

Use the Fourier convention

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} x\xi}\,\mathrm{d}x.$$

For $u = \mathrm{up}$,

$$\widehat{u}_{2\pi}(\xi) = \prod_{r=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\left(\frac{\pi\xi}{2^r}\right), \qquad \mathrm{sinc}_{\mathrm{rad}}(z) = \frac{\sin z}{z}. \tag{11.1}$$

Hence, for every nonzero integer m,

$$\mathrm{ord}_{\xi=m}\,\widehat{u}_{2\pi}(\xi) = 1 + \nu_2(m). \tag{11.2}$$

This exact multiplicity is the dyadic Strang–Fix mechanism behind the reproduction theorem below [43, 44].

The moment-generating function of the centered random series $Z = \sum_{r\geq 1} 2^{-r} V_r$, with $V_r \sim \mathrm{Unif}[-1,1]$, is

$$\mathcal{M}_u(z) = \int_{-1}^{1} \mathrm{e}^{zx} u(x)\,\mathrm{d}x = \prod_{r=1}^{\infty} \frac{\sinh(z/2^r)}{z/2^r}. \tag{11.3}$$

Its cumulants are

$$\log \mathcal{M}_u(z) = \sum_{r=1}^{\infty} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \qquad \kappa_{2r} = \frac{B_{2r}}{2r(1-4^{-r})}. \tag{11.4}$$

Define rational inverse moments γ_r by

$$\frac{1}{\mathcal{M}_u(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \tag{11.5}$$

Thus $\gamma_{2r+1} = 0$, and

$$\gamma_0 = 1, \qquad \gamma_2 = -\frac{1}{9}, \qquad \gamma_4 = \frac{31}{675}, \qquad \gamma_6 = -\frac{2347}{59535}. \tag{11.6}$$

For a polynomial p, introduce the terminating operator

$$\boxed{\mathcal{A}p := \mathcal{M}_u(D)^{-1} p = \sum_{r=0}^{\lfloor \deg p/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}.} \tag{11.7}$$

It is triangular with unit diagonal and therefore an automorphism on every finite polynomial space.

11.2 Exact fixed-radius polynomial reproduction

Theorem 11.1 (Dyadic fixed-radius synthesis)**.** *Let p have degree at most n, and set*

$$m = 2^n, \qquad h = 2^{-n}, \qquad K_n = \{-2m+1, -2m+2, \ldots, 2m-1\}. \tag{11.8}$$

Then

$$\boxed{p(x) = h \sum_{k\in\mathbb{Z}} (\mathcal{A}p)(kh) u(x-kh), \qquad x \in \mathbb{R},} \tag{11.9}$$

and, on $[-1,1]$, compact support reduces this to

$$\boxed{p(x) = h \sum_{k\in K_n} (\mathcal{A}p)(kh) u(x-kh).} \tag{11.10}$$

Proof. Define the sampled synthesis operator

$$(S_h v)(x) = h \sum_{k\in\mathbb{Z}} v(kh) u(x-kh).$$

Apply Poisson summation to $t \mapsto \mathrm{e}^{zt} u(x-t)$. For z near zero,

$$S_h(\mathrm{e}^{z\,\cdot})(x) = \sum_{\ell\in\mathbb{Z}} \mathrm{e}^{(z-2\pi\mathrm{i}\ell/h)x} \mathcal{M}_u(-z + 2\pi\mathrm{i}\ell/h). \tag{11.11}$$

Because u is even, the $\ell = 0$ term is $\mathrm{e}^{zx}\mathcal{M}_u(z)$. If $h = 2^{-n}$ and $\ell \neq 0$, then $\ell/h = 2^n \ell$; by (11.2), the corresponding factor has a zero of order at least $n+1$ at $z = 0$. Therefore the Taylor coefficients through degree n in (11.11) coincide with those of $\mathrm{e}^{zx}\mathcal{M}_u(z)$. Equating the coefficient of $z^r/r!$ gives

$$S_h(t^r)(x) = \mathcal{M}_u(D) x^r, \qquad 0 \leq r \leq n. \tag{11.12}$$

By linearity, $S_h v = \mathcal{M}_u(D)v$ for every polynomial v of degree at most n. Taking $v = \mathcal{A}p$ proves (11.9).

For $x \in [-1,1]$, a term can be nonzero only when $|x - kh| \le 1$, hence $kh \in [-2,2]$. The two extreme indices $k = \pm 2m$ contribute $u(\pm 1) = 0$, leaving precisely K_n. $\square$

The exponential relation between degree and spacing is sharp on uniform integer-ratio lattices: the first nonzero dual-lattice frequency has zero multiplicity $n+1$ only when $1/h$ is divisible by 2^n. Thus the same 2-adic valuation that controls the Fourier product determines the cost of exact polynomial synthesis.

11.3 A q-Appell deformation of geometric falling powers

Apply $\mathcal{A}$ to the geometric Newton basis. Define

$$\mathscr{A}_k(y;c,q) := (\mathcal{A}N_k(\,\cdot\,;c,q))(y). \tag{11.13}$$

Proposition 11.2 (Elementary-symmetric formula). *For $k \ge 0$,*

$$\boxed{\mathscr{A}_k(y;c,q) = \sum_{r=0}^{\lfloor k/2 \rfloor} \gamma_{2r} e_{k-2r}(y-c, y-cq, \ldots, y-cq^{k-1}).} \tag{11.14}$$

Consequently

$$N_k(x;c,q) = h \sum_{s \in K_n} \mathscr{A}_k(sh;c,q)u(x - sh), \qquad k \le n, \tag{11.15}$$

for $x \in [-1,1]$ and $h = 2^{-n}$.

Proof. For a product of k linear factors,

$$\frac{1}{r!}\frac{\mathrm{d}^r}{\mathrm{d}y^r} \prod_{j=0}^{k-1} (y - cq^j) = e_{k-r}(y-c, y-cq, \ldots, y-cq^{k-1}).$$

Substitute $r = 2s$ into (11.7), and then apply Theorem 11.1. $\square$

The polynomials $\mathscr{A}_k$ simultaneously deform two classical sequences: setting all inverse moments except γ_0 to zero recovers the q-Pochhammer Newton basis, while letting $q \to 1$ in logarithmic coordinates recovers inverse-moment Appell transforms of ordinary falling factorials. Their mixed recurrences, zero geometry, and connection coefficients appear to be unexplored.

11.4 Geometric cardinals as finite sums of up translates

Assume now $0 < c \le 1$, so the comb nodes lie in $[-1,1]$. Let $L^{c,q}_{j,n} := \ell_{j,n}$ be the geometric cardinal from Theorem 4.1.

Theorem 11.3 (Gaussian–Appell cardinal decoder). *For $0 \le j \le n$ and $x \in [-1,1]$,*

$$\boxed{L^{c,q}_{j,n}(x) = h \sum_{k \in K_n} D^{(n;c,q)}_{k,j} u(x - kh),} \tag{11.16}$$

where

$$D^{(n;c,q)}_{k,j} := (\mathcal{A}L^{c,q}_{j,n})(kh) \tag{11.17}$$

$$= \lambda^{c,q}_{j,n} \sum_{r=0}^{\lfloor n/2 \rfloor} \gamma_{2r} e_{n-2r}(\{kh - cq^s : 0 \le s \le n,\ s \ne j\}), \tag{11.18}$$

with

$$\lambda_{j,n}^{c,q} = c^{-n} \frac{(-1)^j q^{-nj+\binom{j+1}{2}}}{(q;q)_j (q;q)_{n-j}}. \tag{11.19}$$

Thus every cardinal coefficient is an explicit product of a Gaussian barycentric prefactor and a finite Bernoulli–Appell symmetric polynomial.

Proof. Apply Theorem 11.1 to the degree-n polynomial $L_{j,n}^{c,q}$. Since

$$L_{j,n}^{c,q}(y) = \lambda_{j,n}^{c,q} \prod_{\substack{0 \le s \le n \\ s \ne j}} (y - cq^s),$$

the same derivative calculation as in Proposition 11.2 yields (11.18). The prefactor is the barycentric weight in (4.2). ☐

Formalization boundary. The generic cardinal-synthesis identity in (11.16), with the decoder defined by (11.17), is formalized for arbitrary finite node families by `normalized_sum_Ioo_lagrangeRvachevDecoder_mul_shifted_rvachevUp`. That declaration does not derive the geometric elementary-symmetric formula (11.18) or the Gaussian prefactor (11.19). Because this theorem asserts all three formulas, its concordance row remains a human-proved frontier result.

Corollary 11.4 (Interpolation through an atomic dictionary). *For arbitrary nodal data $y_0, \ldots, y_n$,*

$$\boxed{(\mathcal{I}_n^{c,q} y)(x) = \sum_{k \in K_n} C_k(y) u(x - kh),} \tag{11.20}$$

where

$$C_k(y) = h \sum_{j=0}^{n} D_{k,j}^{(n;c,q)} y_j. \tag{11.21}$$

The map from data to atom amplitudes is finite and linear.

Proof. Write the interpolant as $\sum_j y_j L_j^{(n;c,q)}$. Substitute the decoder expansion of each cardinal from theorem 11.3, interchange the two finite sums, and collect the coefficient of each atom $u(x - kh)$. The collected coefficient is exactly (11.21). ☐

Formalization boundary. The generic scalar coefficient factorization is `lagrangeRvachevAtomCoefficient_eq_deconvolved_interpolate`, and the complete finite interpolation loop is `sum_Ioo_lagrangeRvachevAtomCoefficient_mul_shifted_rvachevUp`. They hold for arbitrary finite node families under the stated mesh-degree and interval hypotheses, so the latter declaration is the exact compiled crosswalk for this corollary.

11.5 Collocation, a stochastic encoder, and a signed right inverse

Let B be the $(n+1) \times (4 \cdot 2^n - 1)$ collocation matrix and D the $(4 \cdot 2^n - 1) \times (n+1)$ decoder matrix,

$$B_{i,k} = u(cq^i - kh), \qquad D_{k,j} = D_{k,j}^{(n;c,q)}. \tag{11.22}$$

Indices k range over K_n.

Theorem 11.5 (Exact Gaussian–Appell biorthogonality). *The matrices satisfy*

$$\boxed{(hB)D = I_{n+1}.} \tag{11.23}$$

In components,

$$h \sum_{k \in K_n} u(cq^i - kh)(\mathcal{A} L_{j,n}^{c,q})(kh) = \delta_{ij}. \tag{11.24}$$

Moreover, hB is nonnegative and row-stochastic, while every row of D sums to 1.

Proof. Evaluate (11.16) at $x = cq^i$ to obtain (11.24). Constant reproduction in Theorem 11.1 gives $h\sum_k u(x-kh) = 1$, so every row of hB sums to one; nonnegativity follows from $u \geq 0$. Finally

$$\sum_{j=0}^{n} D_{k,j} = \mathcal{A}\left(\sum_{j=0}^{n} L_{j,n}^{c,q}\right)(kh) = (\mathcal{A}1)(kh) = 1.$$

□

Formalization boundary. The component identity is formalized by `normalized_sum_Ioo_lagrangeRvachevDecoder_eval_node`, and `sum_lagrangeRvachevDecoder_eq_one` proves that the unnormalized decoder row sums to one for a nonempty distinct node family. The module does not package these scalar sums as a typed Matrix identity, nor does it formalize the row-stochastic encoder assertion. The compound theorem therefore remains a human-proved frontier result in the concordance.

Thus a positive averaging map has an exact signed polynomial decoder. The right inverse cannot generally be nonnegative: reconstructing high-degree cardinals requires cancellation, and the size of that cancellation reflects the geometric Lebesgue growth.

11.6 Closing the loop for the Rvachev function

Take $c = 1$, $q = 1/2$, and interpolate u at the center-directed comb $1, 1/2, 1/4, \ldots$. Since

$$u(2^{-j}) = F(1-2^{-j}) = 1 - F(2^{-j}), \tag{11.25}$$

all nodal values are rational and the polynomial interpolant satisfies

$$\mathcal{I}_n u = 1 - \mathcal{I}_n F. \tag{11.26}$$

Combining this identity with Corollary 11.4 gives a finite self-representation.

Theorem 11.6 (Finite geometric Lagrange–Rvachev loop). *Put*

$$U_n(x) := h \sum_{k\in K_n} (\mathcal{A}(\mathcal{I}_n u))(kh) u(x-kh), \qquad h = 2^{-n}. \tag{11.27}$$

Then

$$\boxed{U_n(x) = (\mathcal{I}_n u)(x), \qquad -1 \leq x \leq 1.} \tag{11.28}$$

Hence, in every norm or seminorm restricted to $[-1, 1]$,

$$\|u - U_n\| = \|u - \mathcal{I}_n u\|. \tag{11.29}$$

The atomic substitution creates no additional truncation error at a fixed level.

Proof. Equation (11.28) is Theorem 11.1 applied to the polynomial $\mathcal{I}_n u$. The error identity follows by subtraction. □

For degree 2, the center-comb interpolant is

$$(\mathcal{I}_2 u)(x) = \frac{40}{27} - \frac{22}{9}x + \frac{26}{27}x^2. \tag{11.30}$$

Since $h = 1/4$, $K_2 = \{-7, \ldots, 7\}$, and $\mathcal{A}p = p - p''/18$ for a quadratic, one obtains the concrete loop

$$\boxed{\frac{40}{27} - \frac{22}{9}x + \frac{26}{27}x^2 = \sum_{k=-7}^{7}\left(\frac{13k^2}{864} - \frac{11k}{72} + \frac{167}{486}\right) u\left(x - \frac{k}{4}\right),} \tag{11.31}$$

valid for $-1 \le x \le 1$.

The loop also transfers instability exactly. At the accumulation point $x = 0$, Theorem 10.3 gives recovery of every fixed center jet of u. In the gap $(1/2, 1)$, equation (11.26) transfers the Fabius divergence to u. A compactly supported atomic representation is therefore a change of coordinates, not a cure for a badly conditioned interpolation projector.

11.7 The reflected boundary comb

The affine reflection $t = 1 - x$ sends the inverse-dyadic comb to

$$t_j = 1 - 2^{-j}, \qquad j \ge 0, \tag{11.32}$$

which accumulates at the support boundary $t = 1$. The polynomial $J_n(t) = (\mathcal{I}_n F)(1 - t)$ interpolates $u(t)$ at these nodes. Its Newton basis is

$$\prod_{r=0}^{k-1} \big((1 - t) - 2^{-r}\big), \tag{11.33}$$

and all formulas above follow by affine covariance. This is the natural comb for boundary-flat approximation, while 2^{-j} is the natural comb for center-flat approximation. The two are conjugate through the Fabius symmetry, but their geometric interpretation inside the support of u is different.

Part IV

Experiments, conjectures, and implementation

12 Exact and high-dynamic-range numerical experiments

The accompanying program `geometric_comb_experiments.py` was written to test formulas whose normalizations are easy to misstate and to explore asymptotic regimes beyond ordinary floating-point range. It separates exact rational calculations from logarithmic floating-point calculations.

12.1 Reproducibility protocol

The exact path uses Python's `fractions.Fraction` for:

1. the Fabius recurrence (9.9);
2. every value $F(2^{-j})$ used in the experiments;
3. direct Lagrange evaluation at $x = 0$ and $x = 3/4$;
4. Gaussian Pascal coefficients at integer base 2;
5. verification of the basis transform and the complete-homogeneous residual identity.

No tolerance enters these checks.

The Lebesgue function is positive but astronomically large. It is therefore computed from logarithms of the absolute cardinal factors, followed by a log-sum-exp reduction. A coarse scan isolates the outer boundary layer and a golden-section search refines the maximum. The analytic constants $(-q; q)_\infty/(q; q)_\infty$ and $2(-q; q)_\infty/\mathrm{e}$ are computed with `mpmath` at elevated precision.

Running the script from its directory creates the tabular outputs `data/*.csv`, the exact-check transcript `data/identity_checks.txt`, and the figure files `figures/*.pdf` and `figures/*.png`. The PDF figures are used directly in this report.

12.2 Exact algebraic checks

The script verifies, in exact arithmetic,

$$x^m = \sum_{k=0}^{m} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x;c,q), \qquad 0 \le m \le 9, \tag{12.1}$$

$$x^{n+1+r} - \mathcal{I}_n(x^{n+1+r}) = N_{n+1}(x;c,q)h_r(x,c,cq,\ldots,cq^n), \qquad 0 \le n \le 7, \quad 0 \le r \le 5. \tag{12.2}$$

The test values $q = 1/2$, $c = 3/2$, $x = 7/5$ in the first check and $q = 1/2$, $c = 4/3$, $x = 11/13$ in the second are deliberately nonsingular and nontrivial. All tested identities pass exactly.

The first thirteen Fabius Newton coefficients are also generated exactly. The first six agree with (9.17)– (9.18); the later fractions are retained in `data/identity_checks.txt` rather than typeset here.

12.3 Lebesgue growth and boundary-layer location

For $q = 1/2$ and $c = 1$, the two constants are

$$\frac{(-q;q)_\infty}{(q;q)_\infty} = 8.2559879357782500655441408494З\ldots, \tag{12.3}$$

$$\frac{2(-q;q)_\infty}{\mathrm{e}} = 1.7542191571673478083613883041 4\ldots. \tag{12.4}$$

Selected computed values are shown in Table 1. The optimizer searches the persistent top gap $(1/2, 1)$; the theorem, rather than this finite search, establishes that this peak is globally dominant for all sufficiently large degrees.

Table 1: Geometric-comb Lebesgue diagnostics for $q = 1/2$, $c = 1$. Here x_n^{top} is the numerically maximized point in $(1/2, 1)$; the last column is that top-gap maximum divided by the leading global asymptotic in (7.13).

n	$\mathscr{L}_n(0)$	x_n^{top}	$n(1 - x_n^{\mathrm{top}})$	$\mathscr{L}_n^{\mathrm{top}}/\mathscr{L}^*_{n,\mathrm{asymp}}$
4	7.285714	0.830861	0.676556	0.86792
8	8.191739	0.899264	0.805886	0.86854
12	8.251958	0.927713	0.867445	0.90841
16	8.255736	0.943752	0.899970	0.93089
20	8.255972	0.954011	0.919782	0.94464
24	8.255987	0.961122	0.933075	0.95384
30	8.255988	0.968453	0.946418	0.96306

The endpoint column rapidly approaches its finite limit. At the same time, $\log_{10} \mathscr{L}_n^{\mathrm{top}}$ reaches approximately 129.70 by $n = 30$. The ratio in the last column and the scaled top-gap location both approach one, in agreement with Theorem 7.3.

12.4 Exact Fabius interpolation in the gap

At $x = 3/4$ the target value is exact:

$$F(3/4) = 1 - F(1/4) = \frac{67}{72}. \tag{12.5}$$

Every interpolant value in the experiment is a rational number computed from the exact inverse-dyadic samples. The error initially fluctuates at a moderate scale and then enters a clear quadratic-logarithmic growth regime.

By $n = 50$, the same exact interpolant that predicts $F(0)$ to more than 215 decimal digits is wrong at $3/4$ by about 10^{154}. This is a particularly clean numerical manifestation of the stability dichotomy.

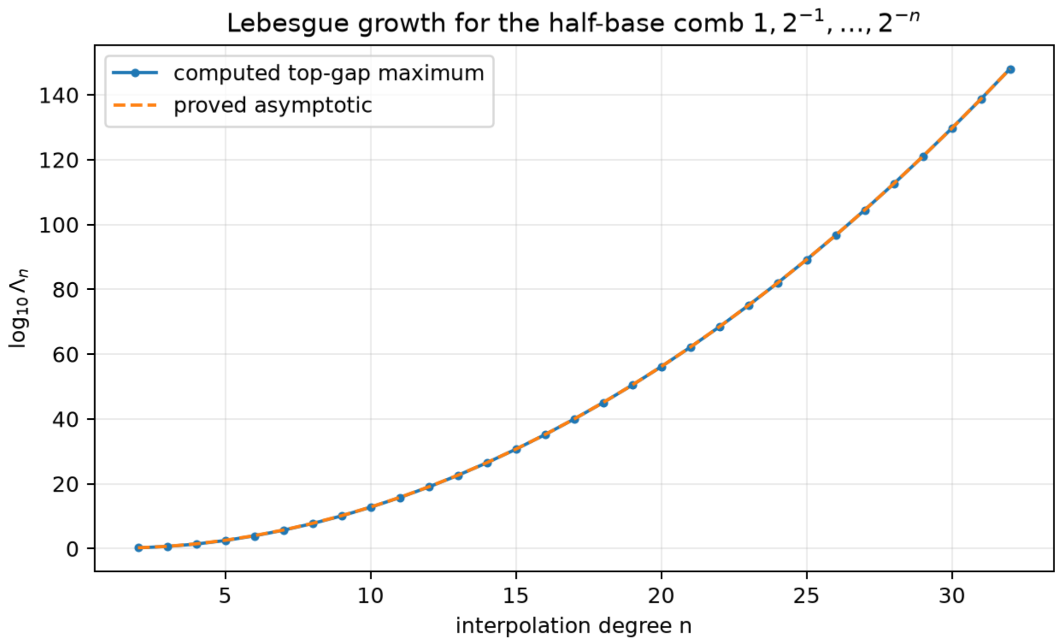

Figure 1: Top-gap maxima and the sharp global asymptotic $2(-q;q)_\infty q^{-\binom{n}{2}}/(\mathrm{e}n)$ for $q = 1/2$. Logarithmic arithmetic is essential: the ordinate already exceeds 10^{148} at $n = 32$.

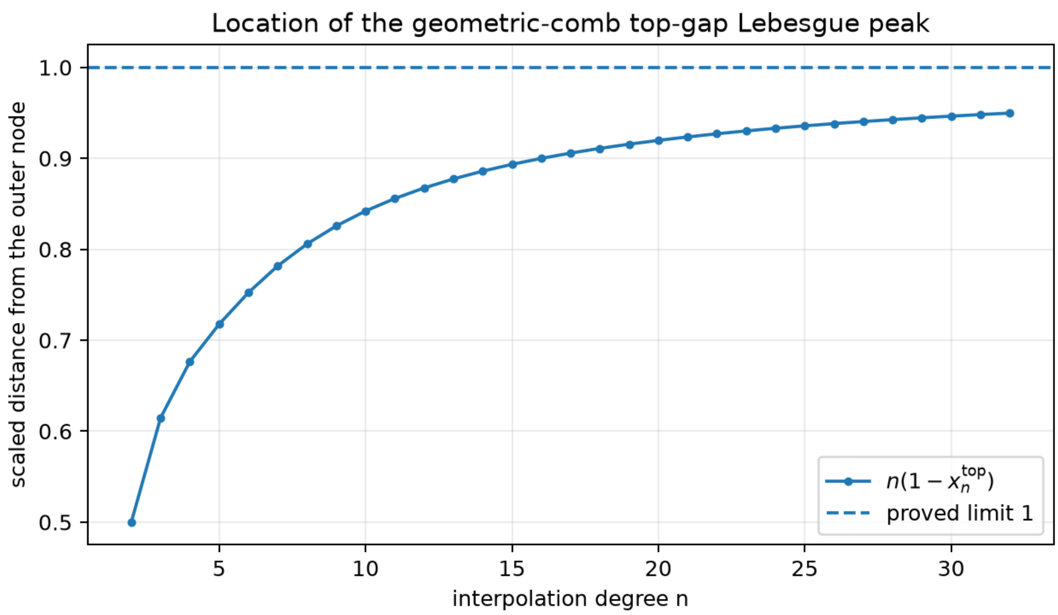

Figure 2: The scaled top-gap location $n(1 - x_n^{\mathrm{top}})$. The limiting value 1 is the maximizer of $\alpha \mathrm{e}^{-\alpha}$ in Theorem 7.2.

Table 2: Exact Fabius interpolation diagnostics. The third column is the number of correct decimal digits in the extrapolated boundary value $R_n = (\mathcal{I}_n F)(0)$, measured as $-\log_{10}|R_n|$.

n	$\log_{10}\lvert(\mathcal{I}_n F)(3/4) - 67/72\rvert$	$-\log_{10}\lvert R_n\rvert$
4	−0.8013	2.1595
8	0.8750	6.7622
12	3.7620	14.3862
16	10.5655	24.8392
20	19.0624	37.9904
24	29.6591	53.8998
30	49.7229	81.9068
40	94.5437	140.6070
50	153.7151	215.2423

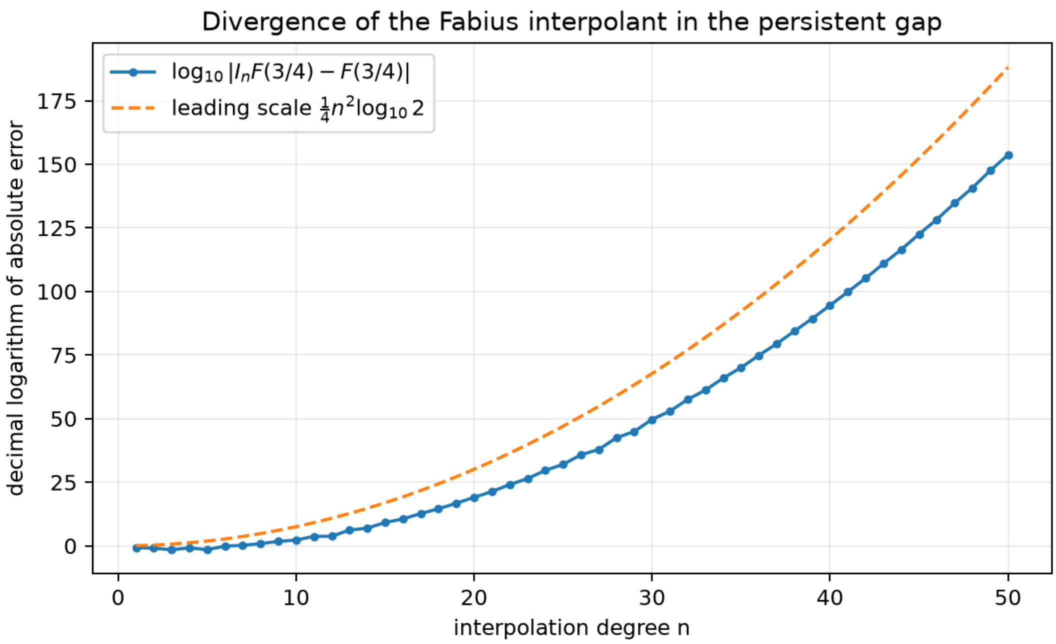

Figure 3: Exact-rational Fabius interpolation error at $x = 3/4$. The plotted quadratic comparison is the leading $n^2/4$ term suggested by Conjecture 10.6; the negative $(n/2)\log n$ correction is visible as a substantial downward displacement.

Numerical observation 12.1 (Sign oscillation)**.** The gap error does not have a simple alternating sign. Long runs of two consecutive signs occur, reflecting the motion of a discrete saddle across integer indices. This supports the modulation clause in Conjecture 10.6 and argues against fitting a single smooth prefactor from small n.

12.5 What the experiments do and do not establish

The exact checks strongly guard against transcription errors in powers of q, Gaussian reciprocity, and factorial normalizations. The numerical top-gap Lebesgue data are consistent with the proved global asymptotic but are not needed for its proof. The Fabius gap data, by contrast, are evidence for a conjecture: they show rapid divergence at one exact rational point but do not prove the proposed general saddle law. The script records raw CSV data so alternative asymptotic fits can be tested without repeating the expensive rational interpolation.

13 Further deductions, conjectures, and research directions

This chapter separates deductions that follow mechanically from the proved framework from genuinely open asymptotic and structural questions.

13.1 Second-order Lebesgue asymptotics

The proof of Theorem 7.2 can be expanded one order further. The factors $(q/x;q)_{n-1}$, the ratio cloud (7.8), and $(1-\alpha/n)^{n-1}$ each contribute an explicit $1/n$ correction. This suggests the following refinement.

Conjecture 13.1 (Second-order boundary layer)**.** *There exist explicit constants $A(q)$ and $B(q)$ such that*

$$\frac{x_n^*}{c} = 1 - \frac{1}{n} + \frac{A(q)}{n^2} + O_q(n^{-3}), \tag{13.1}$$

$$\mathscr{L}_n^* = \frac{2(-q;q)_\infty}{\mathrm{e}} \frac{q^{-\binom{n}{2}}}{n} \left(1 + \frac{B(q)}{n} + O_q(n^{-2})\right). \tag{13.2}$$

The coefficients should be expressible through Lambert series such as $\sum_{r\geq 1} q^r/(1-q^r)$ and weighted Euler-cloud moments.

A proof should follow from a uniform expansion of the ratios of the last $O(\sqrt{\log n})$ cardinals, with the remainder controlled by the $q^{\binom{r}{2}}$ tail. The result would turn the qualitative convergence in Table 1 into a high-accuracy formula.

13.2 A function-space criterion for infinite geometric Newton series

For analytic targets, Theorem 6.1 supplies a convenient sufficient condition. A more intrinsic question starts from a data sequence $y_j = f(cq^j)$ and asks whether the formal Newton series

$$\sum_{k\geq 0} \left(c^{-k} \sum_{j=0}^{k} \frac{(-1)^j q^{-jk+\binom{j+1}{2}}}{(q;q)_j (q;q)_{k-j}} y_j \right) N_k(x;c,q) \tag{13.3}$$

converges in a prescribed domain.

Problem 13.2 (Geometric Newton sequence spaces)**.** *Characterize weighted sequence spaces Y on the data $y = (y_j)$ and coefficient spaces G on the divided differences $\gamma = (\gamma_k)$ for which the triangular Gaussian transform is an isomorphism and the Newton synthesis map is bounded into $C(K)$, a Hardy space, or a Bergman space. Determine the exact dependence of the norm on the location of K relative to the isolated outer node.*

The Richardson generating factorization (4.11) indicates that Euler products provide the correct multipliers on the data side. A Wiener-algebra formulation may make inversion and perturbation estimates especially transparent.

13.3 Signed saddle analysis for Fabius data

The proof of Theorem 10.5 isolates the unsigned saddle at (10.15). To prove Conjecture 10.6, one needs asymptotics for the signed local packet

$$\sum_{|j-j_n|\le C\sqrt{\log n}} (-1)^j \frac{2^{j(n-j)}a_j}{x-2^{-j}} \tag{13.4}$$

with relative error smaller than the packet itself.

A promising route has three stages:

1. derive uniform saddle asymptotics for $a_j = [z^j]A(z)$ from the product (9.7);
2. insert them into (13.4) and apply a discrete Gaussian or theta transformation;
3. retain the fractional saddle displacement, which should produce a bounded periodic or antiperiodic modulation.

The functional equation (9.8) suggests that the moment saddle itself carries a logarithmically periodic correction. Its interaction with the integer Gaussian saddle may be the source of the sign blocks seen in the experiment.

Problem 13.3 (Exceptional gap points)**.** *Determine whether there are points $x \in (1/2, 1)$ at which the leading signed saddle cancels identically or along an infinite subsequence. If such points exist, classify them arithmetically and determine the next asymptotic scale.*

13.4 Boundary-residue entire functions

The generating function

$$\mathcal{R}(z) = (qz;q)_\infty \sum_{j\ge 0} \frac{q^{\binom{j}{2}}a_j}{(q;q)_j} z^j \tag{13.5}$$

encodes all endpoint extrapolation errors. The Euler factor has prescribed zeros at $z = q^{-r}$, while the second factor is an entire transform of the Fabius moments.

Problem 13.4 (Zero cancellation and coefficient asymptotics)**.** *Determine which Euler zeros survive in $\mathcal{R}$, obtain the order and indicator of $\mathcal{R}$, and use its zero distribution to derive a sharp asymptotic for R_n. In particular, decide whether*

$$\log|R_n| = -\frac{\log 2}{4}n^2 + O(n\log n) \tag{13.6}$$

has a full expansion with a parity-dependent linear term.

The exact coefficient formula makes this problem accessible both to complex-analytic saddle methods and to 2-adic valuation experiments.

13.5 Geometric Hermite stability

The anchored factorization (8.9) removes confluent linear algebra but not operator growth. For a flat target, its data are rescaled by $(cq^j)^{-(m+1)}$, changing the Gaussian saddle. A first model replaces a_j by $q^{\binom{j}{2}-j(m+1)}a_j$ in the Fabius formulas. This predicts a movable saddle and a phase transition when m is allowed to grow with n.

Conjecture 13.5 (Hermite saddle displacement)**.** *For $q = 1/2$ and Fabius data, the dominant index in the gap for the m-anchored interpolant is*

$$j_{n,m} = \frac{n}{2} + m + 1 - \frac{\log n}{2\log 2} + O(1), \tag{13.7}$$

until it collides with the endpoint $j = n$. The collision marks a change from an interior Gaussian saddle to an endpoint saddle and should determine the largest useful jet order.

This conjecture follows from the unsigned ratio heuristic; the signed problem may shift the practical optimum. It can be tested by extending the bundled exact code to Theorem 8.4.

13.6 The mixed q-Appell family

The polynomials $\mathscr{A}_k(y; c, q)$ in (11.14) deserve study independently of interpolation. They combine a Sheffer/Appell deconvolution with a q-falling-factorial basis.

Problem 13.6 (Structure of the q-Appell falling powers)**.** *Find:*

1. *raising and lowering operators for $\mathscr{A}_k$ involving D_q and the ordinary derivative D;*
2. *their connection coefficients with monomials, ordinary Appell polynomials, and N_k;*
3. *generating functions in k and continued-fraction representations;*
4. *zero-location and interlacing theorems for $0 < q < 1$;*
5. *asymptotics in the joint limits $k \to \infty$, $q \to 1$, and $k(1-q) \to \tau$.*

A useful starting identity is

$$\sum_{k\geq 0} \mathscr{A}_k(y; c, q)\frac{t^k}{(q;q)_k} = \mathcal{M}_u(D_y)^{-1} \sum_{k\geq 0} N_k(y; c, q)\frac{t^k}{(q;q)_k}, \tag{13.8}$$

where the second series can be rewritten by the infinite q-binomial theorem after normalization. Making the operator interchange analytic is itself a worthwhile problem.

13.7 Decoder variation, total positivity, and optimal gauges

The matrix hB in Theorem 11.5 is a positive row-stochastic sampling operator. Its right inverse D has row sum one but must use signs. The size of its row variation controls amplification from nodal perturbations to atom amplitudes.

Problem 13.7 (Minimum-variation atomic decoder)**.** *Among all right inverses of the collocation matrix hB, minimize a weighted ℓ^1 or total-variation norm subject to exact polynomial reproduction. Compare the canonical Appell decoder with the minimizer, and determine whether a symmetry-preserving minimizer has a closed q-binomial formula.*

The dense dictionary has an alias kernel inherited from the dyadic Fourier zeros. Adding an alias vector changes amplitudes without changing the synthesized polynomial. Convex optimization over this affine space may substantially improve numerical conditioning while preserving every exact identity.

13.8 Stable alternatives to global gap interpolation

The superexponential Lebesgue growth is a property of the all-node global projector, not of geometric data in general. Several alternatives retain the dyadic arithmetic:

1. piecewise low-degree Newton or Hermite interpolation on geometric annuli $[cq^{j+1}, cq^j]$;
2. rational barycentric interpolation with fixed local degree;
3. oversampled least squares in a logarithmic coordinate;
4. direct approximation in scaled translates of up rather than first passing through one global polynomial;
5. hybrid Taylor–Newton schemes that use the flat germ near zero and a separate approximation across the outer gap.

For the Fabius function, the functional equation can transport a local approximation between adjacent dyadic scales, suggesting multiresolution algorithms with exact consistency constraints.

13.9 Complex and arithmetic geometric combs

When q is complex with $0 < |q| < 1$, the comb spirals into the origin. The algebraic formulas remain valid, while the stability problem becomes potential-theoretic and angle-dependent. At a root of unity the nodes collide and ordinary interpolation becomes confluent; the denominator-free Gaussian recurrence is then the correct algebraic replacement.

Over finite fields, the condition that $q^0, \dots, q^n$ be distinct is exactly an order condition in the multiplicative group. The geometric Vandermonde determinant in Theorem 2.5 records its failure cyclotomically. This suggests a finite-field interpolation theory whose transition matrices are literal Gaussian coefficients counting subspaces.

Problem 13.8 (Cyclotomic confluent limit)**.** *Develop a division-free specialization of the cardinal and residual identities when q approaches a primitive root of unity. Identify the resulting Hermite blocks, their cyclotomic valuations, and their relation to q-Lucas congruences.*

13.10 Formalization program

The manuscript layer separates into finite algebra, analytic convergence, and asymptotics, making the remaining gap well suited to staged formalization in Lean. The bullets below distinguish reuse of existing declarations from genuinely missing statements; they should not be read as a claim that the whole report is already formalized.

1. **Finite products.** Reuse the existing finite q-Pochhammer API, then define N_k, prove (2.2), and derive the denominator product in Theorem 4.1 without division until the final field specialization.
2. **Gaussian Pascal equivalence.** Build on the existing Gaussian reciprocity and inversion API to formalize Theorem 3.1 and its inverse as mutually inverse triangular matrices over a commutative ring.
3. **Residual calculus.** Reuse the formal complete-homogeneous and geometric residual-moment layers to state the arbitrary-evaluation interpolation residual; derive the analytic residual only after importing convergence lemmas.
4. **Endpoint stability.** Reuse the formal rational finite ℓ^1 identity and extend its presentation as needed; the real infinite-product limit remains to be formalized.
5. **Boundary profile.** Isolate the elementary limits $(1 - \alpha/n)^n \to \mathrm{e}^{-\alpha}$, dominated convergence for the Euler cloud, and the tail bound from $q^{\binom{r}{2}}$.

6. **Fabius arithmetic.** Reuse the existing exact dyadic-value and moment development to prove Theorem 9.3, the residue generating function, and the fixed-jet bound.

7. **Atomic synthesis.** The generic scalar decoder synthesis, coefficient factorization, full interpolation loop, componentwise node biorthogonality, and unnormalized decoder row-sum law are formalized in `FabiusFunction.LagrangeRvachevSynthesis`. The remaining tasks are to derive the geometric elementary-symmetric decoder and Gaussian prefactor in (11.18)–(11.19) and to package the resulting scalar identities as the Matrix statement (11.23). Decoder optimization remains the open problem Problem 13.7.

The sharp global Lebesgue theorem is the most analytically demanding new formal target. Its proof was deliberately organized into local profile, away-from-boundary decay, and compactness of the scaled maximizer so these pieces can be formalized independently.

Part V

Analytic, probabilistic, and Mellin extensions

Editorial map of the independent extensions

The finite geometric-comb algebra is stated once in the preceding part. This part retains only the independent material not subsumed by that spine: confluent modal filters, regular variation, geometric-knot splines, geometric-uniform laws, Mellin symbols, superflat divergence, and the reciprocal Rvachev filter.

14 The Gaussian origin filter as a dilation polynomial

The preceding part derives the origin weights from geometric Lagrange interpolation. The following compact operator form is the common engine for the extensions in this part. For $0 < q < 1$ put

$$\lambda_{n,k}^{(q)} := \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q;q)_k (q;q)_{n-k}}, \qquad 0 \le k \le n, \tag{14.1}$$

$$(\mathcal{G}_{n,q} f)(A) := \sum_{k=0}^{n} \lambda_{n,k}^{(q)} f(Aq^k). \tag{14.2}$$

Theorem 14.1 (Dilation-polynomial factorization). *Let $(T_q f)(x) = f(qx)$. Then*

$$W_{n,q}(z) := \sum_{k=0}^{n} \lambda_{n,k}^{(q)} z^k = \prod_{j=1}^{n} \frac{z - q^j}{1 - q^j}, \tag{14.3}$$

$$\mathcal{G}_{n,q} = W_{n,q}(T_q) = \frac{1}{(q;q)_n} \prod_{j=1}^{n} (T_q - q^j I), \tag{14.4}$$

$$\mathcal{G}_{0,q} = I, \qquad \mathcal{G}_{n,q} = \frac{T_q - q^n I}{1 - q^n} \mathcal{G}_{n-1,q}. \tag{14.5}$$

Consequently the filter can be evaluated in place: starting with $y_k = f(Aq^k)$, at stage $j = 1, \dots, n$ replace

$$y_k \longleftarrow \frac{y_{k+1} - q^j y_k}{1 - q^j} \qquad (0 \le k \le n - j). \tag{14.6}$$

The final entry is $(\mathcal{G}_{n,q} f)(A)$.

Proof. The finite q-binomial theorem expands the product in (14.3) and gives exactly the coefficients (14.1). Substituting the commuting operator T_q for z proves the factorization; separating its last factor proves the recursion. Applied to the sample vector, that recursion is precisely the in-place update (14.6). □

Corollary 14.2 (Exact power multiplier and all integer moments). *For $A > 0$ and a chosen branch of A^α, q^α,*

$$(\mathcal{G}_{n,q}x^\alpha)(A) = A^\alpha W_{n,q}(q^\alpha) = A^\alpha q^{\alpha n}\frac{(q^{1-\alpha};q)_n}{(q;q)_n}. \tag{14.7}$$

In particular, for every $d \in \mathbb{N}_0$,

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)}(Aq^k)^d = \begin{cases} 1, & d = 0, \\ 0, & 1 \le d \le n, \\ (-1)^n A^d q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \ge n+1. \end{cases} \tag{14.8}$$

Proof. The function x^α is an eigenfunction of T_q with eigenvalue q^α, so (14.3) gives the first equality. Factoring q^α from every numerator factor gives the Pochhammer form. For integral d, its factors give the vanishing range; when $d \ge n+1$, reversing the factors and using the factorial definition of the Gaussian coefficient gives (14.8). □

Theorem 14.3 (Exact analytic tail and reciprocal-q majorant). *Let $f(z) = \sum_{m\ge 0} a_m z^m$. Formally, and analytically whenever the series is absolutely convergent,*

$$(\mathcal{G}_{n,q}f)(A) - f(0) = (-1)^n q^{\binom{n+1}{2}} \sum_{r\ge 0} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+1+r}A^{n+1+r}. \tag{14.9}$$

If $|a_m| \le CR^{-m}$ and $t = |A|/R < 1$, then

$$|(\mathcal{G}_{n,q}f)(A) - f(0)| \le Cq^{\binom{n+1}{2}}\frac{t^{n+1}}{(t;q)_{n+1}}. \tag{14.10}$$

This majorant is coefficientwise sharp: equality in the majorizing sum is attained when the relevant coefficients have the common phase and magnitudes CR^{-m}.

Proof. Apply (14.8) term by term and reindex $m = n + 1 + r$. Taking absolute values leaves the series

$$Cq^{\binom{n+1}{2}}t^{n+1}\sum_{r\ge 0}\begin{bmatrix} n+r \\ n \end{bmatrix}_q t^r.$$

The reciprocal finite q-binomial theorem identifies the last sum with $(t;q)_{n+1}^{-1}$. The asserted phase choice makes every triangle inequality an equality. □

15 Confluent modal extrapolation

Ordinary polynomial cancellation assumes an integer-power Taylor expansion. Endpoint asymptotics in differential equations, Mellin analysis, and inverse special functions often contain fractional powers and logarithms. The dilation-operator viewpoint gives the correct generalization immediately.

Definition 15.1 (Mode polynomial). Let $\alpha_1, \dots, \alpha_J \in \mathbb{C}$ have pairwise distinct spectral points q^{α_j}, none equal to 1, and let $m_1, \dots, m_J \in \mathbb{N}_0$, and put $N = \sum_j m_j$. Define

$$W_{\boldsymbol{\alpha},\boldsymbol{m}}(z) := \prod_{j=1}^{J}\left(\frac{z - q^{\alpha_j}}{1 - q^{\alpha_j}}\right)^{m_j} = \sum_{k=0}^{N} w_k z^k, \tag{15.1}$$

and

$$(\mathcal{R}_{\boldsymbol{\alpha},\boldsymbol{m}}f)(x) := \sum_{k=0}^{N} w_k f(q^k x).$$

Theorem 15.2 (Confluent annihilation)**.** *The filter* (15.1) *preserves constants and, for every j and $0 \le s < m_j$, annihilates*

$$x^{\alpha_j}(\log x)^s.$$

It is the unique filter of degree at most N with these properties.

Proof. For a general polynomial $W(z) = \sum_k w_k z^k$,

$$\begin{aligned}\sum_k w_k (q^k x)^\alpha (\log(q^k x))^s &= x^\alpha \sum_k w_k q^{k\alpha} (\log x + k \log q)^s \\ &= x^\alpha (\log x + \partial_\alpha)^s W(q^\alpha).\end{aligned}$$

A zero of multiplicity m_j at q^{α_j} makes the right side vanish for $s < m_j$. The normalization $W(1) = 1$ preserves constants. Conversely, the annihilation equations imply zeros with the stated multiplicities; degree and normalization then force (15.1). □

Corollary 15.3 (Exact general multiplier)**.** *For any $\alpha \in \mathbb{C}$ and $s \in \mathbb{N}_0$,*

$$\mathcal{R}_{\boldsymbol{\alpha},\boldsymbol{m}}[x^\alpha (\log x)^s] = x^\alpha (\log x + \partial_\alpha)^s W_{\boldsymbol{\alpha},\boldsymbol{m}}(q^\alpha). \tag{15.2}$$

Proof. Because $x^\alpha (\log x)^s = \partial_\alpha^s x^\alpha$ and the filter is a finite linear combination of dilations, differentiation in α commutes with the filter. Apply the pure-power multiplier and expand $\partial_\alpha^s (x^\alpha W_{\boldsymbol{\alpha},\boldsymbol{m}}(q^\alpha))$ by writing ∂_α on x^α as multiplication by $\log x$. □

Theorem 15.4 (Puiseux–log acceleration)**.** *Suppose, as $x \downarrow 0$,*

$$f(x) = a_0 + \sum_{j=1}^{J} \sum_{s=0}^{m_j - 1} a_{j,s} x^{\alpha_j} (\log x)^s + O\Big(x^\beta (1 + |\log x|^r)\Big), \tag{15.3}$$

where $\Re \alpha_j > 0$ and $\beta > 0$. Then

$$(\mathcal{R}_{\boldsymbol{\alpha},\boldsymbol{m}}f)(x) = a_0 + O\Big(x^\beta (1 + |\log x|^r)\Big). \tag{15.4}$$

Proof. The finite sum of displayed modes is annihilated exactly. For each fixed $k \le N$, replacing x by $q^k x$ preserves the stated remainder order, with a constant depending on q, k, β, r. Summing finitely many weighted remainders proves the assertion. □

Remark 15.5 (Relation to Richardson extrapolation). This is geometric Richardson extrapolation in its natural spectral form. The usual integer-power Gaussian filter corresponds to $\alpha_j = j$, $m_j = 1$. Repeated exponents handle logarithmic resonance without introducing derivative data at the nodes.

Problem 15.6 (Stability-optimal modal design)**.** *For a prescribed list of modes,* (15.1) *is algebraically forced at minimum degree. If extra degrees of freedom are allowed, determine the polynomial W with $W(1) = 1$ and the required confluent zeros that minimizes $\sum_k |w_k|$, or a weighted noise norm adapted to the expected precision of $f(q^k x)$. This is a finite convex optimization problem whose asymptotic structure appears unexplored in the Fabius/Rvachev setting.*

16 Stability, boundary layers, and regular variation

16.1 Exact total variation

Because the signs of (14.1) alternate, the total variation can be read from the row polynomial at -1.

Theorem 16.1 (Exact noise amplification). *For $0 < q < 1$,*

$$K_n(q) := \sum_{k=0}^{n} |\lambda_{n,k}^{(q)}| = \frac{(-q;q)_n}{(q;q)_n} = \prod_{j=1}^{n} \frac{1+q^j}{1-q^j}. \tag{16.1}$$

Consequently

$$K_n(q) \uparrow K_\infty(q) := \frac{(-q;q)_\infty}{(q;q)_\infty} < \infty. \tag{16.2}$$

If each sample has absolute error at most δ, the extrapolated absolute error is at most $K_n(q)\delta$.

Proof. Since $\operatorname{sgn} \lambda_{n,k} = (-1)^{n-k}$,

$$\sum_k |\lambda_{n,k}| = (-1)^n \sum_k \lambda_{n,k}(-1)^k = (-1)^n W_{n,q}(-1).$$

Substitute (14.3). Monotone convergence of the positive product gives (16.2). □

For reference,

$$K_\infty(1/4) \approx 1.9692603537, \qquad K_\infty(1/2) \approx 8.2559879358, \qquad K_\infty(3/4) \approx 803.0132718.$$

Thus the Fourier base $Q = 1/4$ associated with dyadic Rvachev scaling is much better conditioned than a direct dyadic $q = 1/2$ extrapolation.

16.2 The singular regime $q \uparrow 1$

Two limits must be distinguished. For fixed n,

$$K_n(q) \sim \frac{2^n}{n!(1-q)^n} \qquad (q \uparrow 1). \tag{16.3}$$

If the order tends to infinity first, the growth is nonperturbative.

Theorem 16.2 (Modular asymptotic). *Write $q = e^{-t}$ with $t \downarrow 0$. Then*

$$K_\infty(e^{-t}) \sim \frac{\sqrt{t}}{2\sqrt{\pi}} \exp\left(\frac{\pi^2}{4t}\right). \tag{16.4}$$

Proof. Use $(-q;q)_\infty = (q^2;q^2)_\infty/(q;q)_\infty$, so

$$K_\infty(q) = \frac{(q^2;q^2)_\infty}{(q;q)_\infty^2}.$$

The standard Dedekind-eta asymptotic

$$(e^{-t};e^{-t})_\infty \sim \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right)$$

applied at t and $2t$ gives (16.4); the elementary exponential corrections cancel. □

The plot in figure 5 shows the practical consequence: ratios close to one are unsuitable for high-order accumulation-point extrapolation unless precision grows very rapidly.

16.3 The reversed-row boundary layer

Put $r = n - k$. For fixed r,

$$\lambda_{n,n-r}^{(q)} \longrightarrow \beta_r^{(q)} := \frac{(-1)^r q^{\binom{r+1}{2}}}{(q;q)_\infty (q;q)_r}. \tag{16.5}$$

The limiting signed generating function is

$$\sum_{r=0}^{\infty} \beta_r^{(q)} z^r = \frac{(qz;q)_\infty}{(q;q)_\infty}. \tag{16.6}$$

In particular, its mass is 1 and its total variation is $K_\infty(q)$.

Theorem 16.3 (ℓ^1 boundary-layer convergence)**.** *Extend the reversed row by zero for $r > n$. Then*

$$\sum_{r=0}^{\infty} \left| \lambda_{n,n-r}^{(q)} - \beta_r^{(q)} \right| \longrightarrow 0. \tag{16.7}$$

Proof. Pointwise convergence follows from $(q;q)_{n-r} \to (q;q)_\infty$. After multiplication by $(-1)^r$, all terms are nonnegative. Their total masses are $K_n(q)$ and $K_\infty(q)$, respectively, and the former tends to the latter. Scheffe's lemma for counting measure gives convergence in ℓ^1. □

This theorem explains the stability mechanism. For large n, the effective weights sit a bounded number of steps behind the smallest node Aq^n; the large-node coefficients are superexponentially small.

16.4 Convergence for merely continuous data

Analyticity is not needed for consistency.

Theorem 16.4 (Continuous accumulation-point convergence)**.** *If f is continuous on $[0, A]$, then*

$$(\mathcal{G}_{n,q} f)(A) \longrightarrow f(0). \tag{16.8}$$

More quantitatively, if $\omega_f(\delta) = \sup_{0 \le x \le \delta} |f(x) - f(0)|$, then for $1 \le J \le n+1$,

$$\begin{aligned} |(\mathcal{G}_{n,q} f)(A) - f(0)| \le{}& K_\infty(q)\, \omega_f(Aq^J) \\ &+ \frac{2\|f\|_\infty J}{(q;q)_\infty^2} q^{\binom{n-J+2}{2}}. \end{aligned} \tag{16.9}$$

Proof. Split the weighted sum into $k \ge J$ and $k < J$. On the first part, every node is at most Aq^J, so the total is bounded by $K_n \omega_f(Aq^J)$. On the second part,

$$|\lambda_{n,k}^{(q)}| \le \frac{q^{\binom{n-k+1}{2}}}{(q;q)_\infty^2} \le \frac{q^{\binom{n-J+2}{2}}}{(q;q)_\infty^2}.$$

There are at most J such terms, and $|f(Aq^k) - f(0)| \le 2\|f\|_\infty$. First fix J and let $n \to \infty$, then let $J \to \infty$. □

16.5 Regularly varying endpoint germs

A function f is regularly varying at $0+$ with index $\alpha \in \mathbb{R}$ if $f(tx)/f(t) \to x^\alpha$ as $t \downarrow 0$ for every $x > 0$, with f eventually of one sign. Standard background is in [48].

Theorem 16.5 (Regular-variation transfer). *Let f be regularly varying at $0+$ with index α. Then, for fixed $A > 0$,*

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} \longrightarrow C_q(\alpha) := \frac{(q^{1-\alpha};q)_\infty}{(q;q)_\infty}. \tag{16.10}$$

For a positive integer $\alpha = m$, the limit is zero because the numerator product contains the factor $1 - q^0$.

Proof. Reverse the row:

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} = \sum_{r=0}^{n} \lambda^{(q)}_{n,n-r} \frac{f(Aq^{n-r})}{f(Aq^n)}.$$

For fixed r, regular variation gives a limit $q^{-\alpha r}$, while (16.5) gives the coefficient limit. Potter bounds dominate the ratio by $Cq^{-(\alpha+\varepsilon)r}$ away from a finite initial segment; the factor $q^{r(r+1)/2}$ in β_r makes the resulting series summable for every α. The finitely many terms corresponding to nodes bounded away from zero have superexponentially small coefficients. Dominated convergence and (16.6) now give

$$\sum_{r\ge 0} \beta^{(q)}_r q^{-\alpha r} = \frac{(q^{1-\alpha};q)_\infty}{(q;q)_\infty}.$$

□

For pure powers, (16.10) is already visible in the exact formula (14.7). The theorem says that slowly varying factors do not change the limiting multiplier.

17 Smooth remainders and geometric-knot B-splines

17.1 Hermite–Genocchi representation

The exact Taylor tail is best for analytic data. For finite smoothness, a divided-difference representation reveals positivity and sign.

Theorem 17.1 (Geometric Hermite–Genocchi remainder). *Let $f \in C^{n+1}[0, A]$ and let $(T_*, T_0, \dots, T_n)$ be uniformly distributed on the $(n+1)$-simplex, that is, Dirichlet with all parameters equal to 1. Put*

$$Y_{n,A,q} := A\sum_{k=0}^{n} q^k T_k. \tag{17.1}$$

Then

$$(\mathcal{G}_{n,q}f)(A) - f(0) = \frac{(-1)^n A^{n+1} q^{\binom{n+1}{2}}}{(n+1)!} \mathbb{E}\big[f^{(n+1)}(Y_{n,A,q})\big]. \tag{17.2}$$

Proof. Let $I_n f$ be the interpolating polynomial through $A, Aq, \dots, Aq^n$. The ordinary interpolation remainder at 0 gives

$$I_n f(0) - f(0) = (-1)^n A^{n+1} q^{\binom{n+1}{2}} f[0, A, Aq, \dots, Aq^n].$$

The Hermite–Genocchi formula for a divided difference of order $n + 1$ is

$$f[0, A, Aq, \dots, Aq^n] = \frac{1}{(n+1)!} \mathbb{E} f^{(n+1)}\left(A \sum_{k=0}^{n} q^k T_k\right).$$

Since $I_n f(0) = (\mathcal{G}_{n,q}f)(A)$, the result follows. □

Corollary 17.2 (Sharp derivative bound and sign)**.** *Under the hypotheses of the theorem,*

$$|(\mathcal{G}_{n,q}f)(A) - f(0)| \le \frac{A^{n+1}q^{\binom{n+1}{2}}}{(n+1)!}\|f^{(n+1)}\|_{L^\infty(0,A)}. \tag{17.3}$$

If $f^{(n+1)}$ has a fixed nonzero sign on $(0, A)$, then the remainder has the sign of $(-1)^n f^{(n+1)}$.

Proof. The Hermite–Genocchi formula expresses the remainder as $(-1)^n A^{n+1}q^{\binom{n+1}{2}}/(n+1)!$ times an average of $f^{(n+1)}$ over points in $[0, A]$. Taking absolute values gives the bound, while a fixed nonzero sign of the derivative fixes the sign of that average. □

17.2 Peano kernel and a positive spline density

For $n \ge 1$, define the Peano kernel

$$K_{n,A,q}(t) := \frac{1}{n!}\sum_{k=0}^{n} \lambda_{n,k}^{(q)}(Aq^k - t)_+^n, \qquad 0 \le t \le A. \tag{17.4}$$

Then

$$(\mathcal{G}_{n,q}f)(A) - f(0) = \int_0^A f^{(n+1)}(t)K_{n,A,q}(t)\,\mathrm{d}t.$$

Normalize it by

$$B_{n,A,q}(t) := \frac{(-1)^n(n+1)!}{A^{n+1}q^{\binom{n+1}{2}}}K_{n,A,q}(t). \tag{17.5}$$

Theorem 17.3 (Geometric-knot B-spline)**.** *The function $B_{n,A,q}$ is the probability density of $Y_{n,A,q}$ in* (17.1)*. In particular,*

(i) $B_{n,A,q} \ge 0$ *and is supported on* $[0, A]$*;*

(ii) $\int_0^A B_{n,A,q}(t)\,\mathrm{d}t = 1$*;*

(iii) it is the classical univariate B-spline associated with the knots $0, Aq^n, \dots, Aq, A$*, up to the displayed normalization;*

(iv) it is log-concave and hence unimodal on the interior of its support.

Proof. Comparing the Peano representation with (17.2) for arbitrary continuous $f^{(n+1)}$ identifies the normalized kernel with the law of $Y_{n,A,q}$. This is also the standard Curry–Schoenberg interpretation of a B-spline as the density of a Dirichlet average of its knots; see [50]. Positivity, support, and normalization follow immediately. The uniform density on a simplex is log-concave, and linear images preserve log-concavity by the Prekopa theorem, proving the last assertion. □

Remark 17.4 (Not the same as a "quantum B-spline"). The density above is a classical B-spline whose knots happen to be geometric. It should be distinguished from the quantum B-splines of Simeonov–Goldman [56], where continuity conditions are formulated using q-derivatives. The two theories meet through divided differences and Jackson calculus, but their definitions are different. The geometric-knot density is the one canonically forced by the interpolation remainder in this report.

17.3 Exact moments

Theorem 17.5 (Gaussian-binomial B-spline moments)**.** *For every* $r \in \mathbb{N}_0$*,*

$$\mathbb{E}Y_{n,A,q}^r = \frac{r!(n+1)!}{(n+r+1)!}A^r\begin{bmatrix} n+r \\ n \end{bmatrix}_q. \tag{17.6}$$

Equivalently,

$$\int_0^A t^r B_{n,A,q}(t)\,\mathrm{d}t = \frac{r!(n+1)!}{(n+r+1)!}A^r\begin{bmatrix} n+r \\ n \end{bmatrix}_q. \tag{17.7}$$

Proof. For a Dirichlet$(1,\dots,1)$ vector with $n+2$ components,

$$\mathbb{E}\left(\sum_i x_i T_i\right)^r = \frac{r!(n+1)!}{(n+r+1)!} h_r(x_0,\dots,x_{n+1}).$$

Take the knots $0, A, Aq, \dots, Aq^n$ and apply (1.7). $\square$

Thus the Gaussian coefficient is not only an interpolation residual: it is the moment sequence of the positive Peano-kernel distribution after a simple beta normalization.

17.4 A q-Pochhammer scaling limit

The geometric B-spline has a nontrivial boundary-scale limit. Let $E_*, E_0, E_1, \dots$ be independent exponential random variables of mean 1.

Theorem 17.6 (Scaling limit of the geometric B-spline)**.** *For fixed* $0<q<1$*,*

$$\frac{n+2}{A} Y_{n,A,q} \xrightarrow{d} Z_q := \sum_{k=0}^{\infty} q^k E_k. \tag{17.8}$$

The limit has moments and Laplace transform

$$\mathbb{E}Z_q^r = \frac{r!}{(q;q)_r}, \tag{17.9}$$

$$\mathbb{E}e^{-sZ_q} = \frac{1}{(-s;q)_\infty}, \qquad s \ge 0. \tag{17.10}$$

For $y>0$ *its density admits the absolutely convergent expansion*

$$p_q(y) = \frac{1}{(q;q)_\infty} \sum_{k=0}^{\infty} \frac{(-1)^k q^{\binom{k}{2}}}{(q;q)_k} e^{-q^{-k}y}. \tag{17.11}$$

Proof. Use the exponential representation of a uniform Dirichlet vector:

$$(T_*, T_0, \dots, T_n) \overset{d}{=} \frac{(E_*, E_0, \dots, E_n)}{E_* + E_0 + \cdots + E_n}.$$

The denominator divided by $n+2$ converges almost surely to 1, while $\sum_{k=0}^n q^k E_k$ converges almost surely and in every L^r to Z_q. Slutsky's theorem proves (17.8). Independence gives

$$\mathbb{E}e^{-sZ_q} = \prod_{k\ge 0} (1+sq^k)^{-1} = (-s;q)_\infty^{-1}.$$

Expanding $1/(t;q)_\infty = \sum_{r\ge 0} t^r/(q;q)_r$ around $t=0$ and comparing moment-generating coefficients gives (17.9).

For the density, first decompose the finite product $\prod_{j=0}^N (1+sq^j)^{-1}$ into simple fractions. Its coefficient at the pole $s=-q^{-k}$ is the finite version of

$$\frac{1}{(-s;q)_\infty} = \sum_{k\ge 0} \frac{(-1)^k q^{\binom{k+1}{2}}}{(q;q)_k (q;q)_\infty} \frac{1}{1+sq^k}.$$

For $s \ge 0$, the finite coefficients converge to the displayed coefficients and are dominated by a constant times $q^{\binom{k+1}{2}}/(q;q)_k$; dominated convergence therefore gives the infinite partial-fraction identity. Termwise Laplace inversion gives (17.11). For each $y>0$, the Gaussian factor in k together with $e^{-q^{-k}y}$ gives absolute convergence and justifies the inversion. $\square$

Remark 17.7. The limit satisfies the perpetuity equation

$$Z_q \overset{d}{=} E_0 + qZ'_q,$$

where Z'_q is an independent copy. The q-Pochhammer product is therefore both an interpolation object and the Laplace transform of the boundary-scale limit of the associated B-splines.

18 Geometric-uniform random sums and finite splines

18.1 The generic law

Let $0 < b < 1$ and let $U_0, U_1, \ldots$ be independent uniforms on $[0, 1]$. Define

$$X_b := (1-b)\sum_{j=0}^{\infty} b^j U_j. \tag{18.1}$$

The weights sum to 1, so $X_b \in [0, 1]$. Splitting off the first summand gives the distributional self-similarity

$$X_b \overset{d}{=} (1-b)U_0 + bX'_b, \tag{18.2}$$

where X'_b is an independent copy. If F_b is the distribution function, then

$$F_b(x) = \int_0^1 F_b\left(\frac{x-(1-b)u}{b}\right) \mathrm{d}u, \tag{18.3}$$

with the conventions $F_b = 0$ to the left of 0 and $F_b = 1$ to the right of 1. At $b = 1/2$, this is the Fabius distribution.

18.2 Finite generalized Irwin–Hall splines

Let

$$X_{b,N} := (1-b)\sum_{j=0}^{N-1} b^j U_j, \qquad a_j = (1-b)b^j.$$

This is a convolution of uniforms on intervals of geometrically decreasing width.

Theorem 18.1 (Finite box-spline formula)**.** *For $N \geq 1$, the density and distribution function of $X_{b,N}$ are*

$$f_{b,N}(x) = \frac{1}{(N-1)!\prod_{j=0}^{N-1} a_j} \sum_{\varepsilon\in\{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=0}^{N-1} \varepsilon_j a_j\right)_+^{N-1}, \tag{18.4}$$

$$F_{b,N}(x) = \frac{1}{N!\prod_{j=0}^{N-1} a_j} \sum_{\varepsilon\in\{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=0}^{N-1} \varepsilon_j a_j\right)_+^{N}, \tag{18.5}$$

where

$$\prod_{j=0}^{N-1} a_j = (1-b)^N b^{N(N-1)/2}. \tag{18.6}$$

Proof. The density of a uniform on $[0, a]$ is $a^{-1}(H(x) - H(x-a))$, where H is the Heaviside function. Convolving N such factors and expanding the product of translation differences gives the alternating subset sum. The N-fold convolution of H is $x_+^{N-1}/(N-1)!$. Integration gives the CDF, and (18.6) is immediate. □

For a general b, the knots are subset sums of a geometric progression. At the dyadic value they collapse to an arithmetic comb, and the coefficient sign is the Thue–Morse sign.

Corollary 18.2 (Dyadic Thue–Morse collapse). *For* $b = 1/2$,

$$F_{1/2,N}(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{r=0}^{2^N-1} (-1)^{\mathrm{wt}_2(r)} \left(x - \frac{r}{2^N}\right)_+^N, \tag{18.7}$$

where $\mathrm{wt}_2(r)$ *is the binary digit sum of* r.

Proof. Here $a_j = 2^{-(j+1)}$. Every subset sum is uniquely $r/2^N$, and the subset cardinality has the parity of $\mathrm{wt}_2(r)$. The product of widths is $2^{-N(N+1)/2}$. □

This identity explains why a geometric collection of widths produces an arithmetic dyadic knot comb: binary representation converts subset selection into position on the uniform grid, while the inclusion–exclusion sign becomes the Thue–Morse sequence.

18.3 Convergence to the infinite law

The omitted tail is bounded deterministically:

$$0 \le X_b - X_{b,N} \le b^N.$$

Hence

$$F_b(x) \le F_{b,N}(x) \le F_b(x + b^N). \tag{18.8}$$

Because F_b is continuous, $F_{b,N} \to F_b$ uniformly. Thus the Fabius function is a uniform limit of explicitly known compactly supported splines whose coefficients are a Thue–Morse finite difference.

18.4 Centered law, Bernoulli cumulants, and sinc products

Put

$$Z_b := 2X_b - 1 = 2(1-b) \sum_{j=0}^{\infty} b^j (U_j - 1/2). \tag{18.9}$$

Then Z_b is symmetric and supported on $[-1, 1]$. Under the Fourier convention

$$\widehat{u_b}_{\,2\pi}(z) = \mathbb{E}\, \mathrm{e}^{-2\pi \mathrm{i} z Z_b},$$

its characteristic function is

$$\widehat{u_b}_{\,2\pi}(z) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\big(2\pi(1-b) b^j z\big), \qquad \mathrm{sinc}_{\mathrm{rad}}\, w := \frac{\sin w}{w}. \tag{18.10}$$

At $b = 1/2$, this is

$$\widehat{\mathrm{up}}_{\,2\pi}(z) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(\pi z / 2^j), \tag{18.11}$$

which is the standard Rvachev product.

The centered uniform on $[-1/2, 1/2]$ has cumulants $\kappa_m = B_m/m$ for $m \ge 2$, where B_m are Bernoulli numbers and odd cumulants vanish. Independence and scaling give the following explicit q-denominators.

Proposition 18.3 (Cumulants of the geometric-uniform law). *For $m \ge 1$,*

$$\kappa_{2m}(Z_b) = \frac{[2(1-b)]^{2m} B_{2m}}{2m(1-b^{2m})}, \qquad \kappa_{2m+1}(Z_b) = 0. \tag{18.12}$$

Consequently every even moment is an explicit complete Bell polynomial in the numbers (18.12).

Proof. Cumulants add under independent sums and scale homogeneously. Therefore

$$\kappa_{2m}(Z_b) = [2(1-b)]^{2m} \frac{B_{2m}}{2m} \sum_{j\ge 0} b^{2mj},$$

and the geometric sum gives (18.12). □

The factors $1 - b^{2m}$ are the simplest q-analog denominators. In Fourier space the same even degree $2m$ is responsible for the squared interpolation base developed in section 20.

19 Fabius and Rvachev functions at the flat boundary

19.1 Definitions and endpoint differential scaling

Let F be the Fabius distribution function on $[0,1]$. It is the CDF of $X_{1/2}$ and satisfies

$$F(0) = 0, \qquad F(1) = 1, \qquad F(1-x) = 1 - F(x), \tag{19.1}$$

and, for $0 \le x \le 1/2$,

$$F'(x) = 2F(2x). \tag{19.2}$$

The compactly supported Rvachev function is related by

$$\mathrm{up}(x) = F(1 - |x|), \qquad |x| \le 1, \tag{19.3}$$

with $\mathrm{up}(x) = 0$ for $|x| \ge 1$. These normalizations agree with the primary Fabius/Rvachev literature [1, 4, 47] and with the inspected ProveIt exposition [54].

Repeated differentiation of (19.2) gives the exact endpoint scaling.

Lemma 19.1 (Endpoint derivatives). *For $m \ge 1$ and $0 \le x \le 2^{-m}$,*

$$F^{(m)}(x) = 2^{m(m+1)/2} F(2^m x). \tag{19.4}$$

Proof. The case $m = 1$ is (19.2). If the result holds for m, then on $x \le 2^{-(m+1)}$,

$$F^{(m+1)}(x) = 2^{m(m+1)/2} 2^m F'(2^m x) = 2^{m(m+1)/2+m+1} F(2^{m+1} x),$$

which has exponent $(m+1)(m+2)/2$. □

19.2 An exact boundary-comb identity

The smooth Peano remainder becomes self-similar when $f = F$.

Theorem 19.2 (Fabius boundary filter). *Let $n \ge 0$, $0 < A \le 2^{-(n+1)}$, and let $(T_*, T_0, \ldots, T_n)$ be uniform Dirichlet. Put*

$$S_n := \sum_{k=0}^{n} 2^{-k} T_k \in [0,1]. \tag{19.5}$$

Then

$$\sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} F(A2^{-k}) = \frac{(-1)^n (2A)^{n+1}}{(n+1)!} \mathbb{E} F(2^{n+1} A S_n). \tag{19.6}$$

In particular, the left side has strict sign $(-1)^n$ *and obeys*

$$0 < (-1)^n \sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} F(A2^{-k}) \le \frac{(2A)^{n+1}}{(n+1)!}. \tag{19.7}$$

Proof. Apply (17.2) with $q = 1/2$ and $f = F$. Since $F(0) = 0$,

$$\sum_k \lambda_{n,k}^{(1/2)} F(A2^{-k}) = \frac{(-1)^n A^{n+1} 2^{-n(n+1)/2}}{(n+1)!} \mathbb{E} F^{(n+1)}(AS_n).$$

The hypothesis on A allows (19.4) with $m = n+1$ throughout the support of AS_n. The powers of 2 reduce to 2^{n+1}, giving (19.6). The random argument lies in $[0,1]$, so $0 \le F \le 1$. It is positive almost surely because $S_n > 0$ almost surely, which gives strict sign and the bound. □

Corollary 19.3 (Exact dyadic form). *If* $M \ge n+1$, *then*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} F(2^{-(M+k)}) = \frac{(-1)^n 2^{-(M-1)(n+1)}}{(n+1)!} \mathbb{E} F(2^{n+1-M} S_n). \tag{19.8}$$

For $M = n+1$, *the ratio of the absolute value to the upper bound is exactly* $\mathbb{E}F(S_n)$.

Proof. Set $A = 2^{-M}$ in equation (19.6); the hypothesis $M \ge n+1$ is exactly the required boundary-range condition. Then $(2A)^{n+1} = 2^{-(M-1)(n+1)}$, proving the formula. At $M = n+1$ the argument of F is S_n, and division by the preceding theorem's upper bound leaves precisely $\mathbb{E}F(S_n)$. □

Through (19.3), the same result is an identity on the left boundary of up:

$$\sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} \operatorname{up}(-1 + A2^{-k}) = \frac{(-1)^n (2A)^{n+1}}{(n+1)!} \mathbb{E} F(2^{n+1} A S_n). \tag{19.9}$$

Thus a signed combination of values of a nowhere-analytic C^∞ function is represented by a positive expectation and has a completely determined sign.

19.3 The boundary B-spline and its q-exponential scale

The variable S_n is $Y_{n,1,1/2}$ from section 17. Therefore

$$(n+2)S_n \xrightarrow{d} Z_{1/2} = \sum_{k \ge 0} 2^{-k} E_k, \qquad \mathbb{E} e^{-sZ_{1/2}} = \frac{1}{(-s;1/2)_\infty}. \tag{19.10}$$

This identifies the random scale sampled by the normalized Fabius remainder. The typical size of S_n is $2/(n+2)$, but its limiting rescaled distribution is not degenerate; it is the q-Pochhammer perpetuity in (17.10).

19.4 Exact rational computation

All dyadic values of F are rational. A convenient triangular recurrence, also used in the linked monograph, is

$$V_0 = 1, \qquad V_m := F(2^{-m}) = \frac{2^{-\binom{m}{2}}}{2^m - 1} \sum_{k=0}^{m-1} \frac{2^{\binom{k}{2}}}{(m-k+1)!} V_k. \tag{19.11}$$

It begins

$$V_0 = 1, \quad V_1 = \frac{1}{2}, \quad V_2 = \frac{5}{72}, \quad V_3 = \frac{1}{288}, \quad V_4 = \frac{143}{2073600}.$$

Because both (19.11) and the dyadic weights $\lambda_{n,k}^{(1/2)}$ are rational, (19.8) can be checked with exact integer arithmetic. The companion Python program does so before switching to multiprecision arithmetic for larger plotted orders.

20 Fourier-scale interpolation for the Rvachev function

20.1 The base-squaring principle

Let Φ be an even entire function normalized by $\Phi(0) = 1$, and write

$$\Phi(z) = \sum_{m=0}^{\infty} c_m z^{2m}. \tag{20.1}$$

Fix a scale ratio $0 < c < 1$ and set

$$Q := c^2, \qquad H_z(t) := \Phi(z\sqrt{t}) = \sum_{m=0}^{\infty} c_m z^{2m} t^m. \tag{20.2}$$

Evenness makes H_z entire in t, independent of the choice of square root. Moreover

$$H_z(Q^k) = \Phi(c^k z). \tag{20.3}$$

Therefore geometric interpolation of Fourier scales $c^k z$ is interpolation on the comb $1, Q, Q^2, \ldots$, not on base c itself. In the dyadic Rvachev problem, $c = 1/2$ and the interpolation base is $Q = 1/4$.

20.2 Exact moment-flat filter

Let $\lambda_{n,k}^{(Q)}$ denote the origin weights with base Q.

Theorem 20.1 (Even-entire scale filter)**.** *For H_z as in (20.2),*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(Q)} \Phi(c^k z) = 1 + (-1)^n Q^{\binom{n+1}{2}} \sum_{r=0}^{\infty} \begin{bmatrix} n+r \\ n \end{bmatrix}_Q c_{n+1+r} z^{2n+2+2r}. \tag{20.4}$$

In particular, the residual has a zero of order at least $2n+2$ at $z = 0$.

Proof. Apply (14.9) to the entire function $H_z(t)$ at $A = 1$ and base Q. □

For a characteristic function under the convention $\Phi(z) = \mathbb{E}e^{-2\pi i z X}$ with a symmetric random variable X,

$$c_m = (-1)^m \frac{(2\pi)^{2m} \mu_{2m}}{(2m)!}, \qquad \mu_{2m} = \mathbb{E}X^{2m}. \tag{20.5}$$

Hence the first nonzero term of the filtered residual is always

$$-Q^{\binom{n+1}{2}} \frac{(2\pi z)^{2n+2} \mu_{2n+2}}{(2n+2)!}. \tag{20.6}$$

For real z near zero this is negative, regardless of the parity of n.

20.3 Application to up

For the Rvachev function,

$$\Phi(z) = \widehat{\mathrm{up}}_{2\pi}(z) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(\pi z/2^j).$$

Taking $c = 1/2$ and $Q = 1/4$ in (20.4) gives an exact finite multiscale identity. Its uniform sample-noise amplification is bounded by $K_\infty(1/4) \approx 1.96926$, which is notably mild.

More generally, the centered geometric-uniform density u_b in (18.10) has an even transform, and samples at scales $b^k z$ are governed by the interpolation base $Q = b^2$. The cumulants (18.12) determine the coefficients through Bell polynomials, so the filter simultaneously links Gaussian coefficients, Bernoulli numbers, Bell polynomials, and the infinite sinc product.

20.4 Entire Newton reconstruction of all scales

The finite filter recovers the constant term approximately. The infinite Newton series reconstructs the complete scale function.

Theorem 20.2 (Scale reconstruction from a geometric sample ray)**.** *Let Φ be even entire, fix $z \in \mathbb{C}$, and set H_z by (20.2). Then for every $t \in \mathbb{C}$,*

$$\Phi(z\sqrt{t}) = \sum_{j=0}^{\infty} \frac{D_Q^j H_z(1)}{[j]_Q!} \prod_{r=0}^{j-1} (t - Q^r), \tag{20.7}$$

locally uniformly in t. Each coefficient is a finite linear combination of $\Phi(z), \Phi(cz), \dots, \Phi(c^j z)$:

$$\frac{D_Q^j H_z(1)}{[j]_Q!} = \sum_{k=0}^{j} \frac{(-1)^k Q^{-kj+\binom{k+1}{2}}}{(Q;Q)_k (Q;Q)_{j-k}} \Phi(c^k z). \tag{20.8}$$

At $t = 0$ this gives the exact infinite sum rule

$$1 = \sum_{j=0}^{\infty} \frac{(-1)^j Q^{\binom{j}{2}}}{[j]_Q!} D_Q^j H_z(1). \tag{20.9}$$

Proof. Apply the locally uniform Newton series (6.1), with coefficients given by (2.7), to the entire function H_z at $A = 1$ and base Q. Evaluating that series at $t = 0$ gives the final identity. □

This is a strong determinacy statement: at any fixed frequency z, the multiscale values on one geometric ray determine the transform at every complex scale parameter t. For the Rvachev product, this turns the self-similar sample sequence into a complete Newton coordinate system.

21 Algorithms and numerical implementation

The formulas admit several computational realizations. They should not be used interchangeably without regard to conditioning.

21.1 Barycentric interpolation away from zero

For an arbitrary target x, use the second barycentric formula (4.5). The raw weights $b_{n,k}^{A,q} = 1/d_{n,k}^{A,q}$ can span a large range, so in floating-point arithmetic they should be rescaled by a common factor. Only their ratios enter the formula. For large n, compute logarithms of absolute values from (4.2) and attach the sign $(-1)^k$.

At a node, return the corresponding datum directly rather than evaluating a removable $0/0$ expression.

21.2 Newton–Horner evaluation

Once the coefficients $c_k = D_q^k f(A)/[k]_q!$ are known, evaluate (2.8) in nested form:

$$\mathcal{I}_n f(x) = c_0 + (x - A)\big(c_1 + (x - Aq)(c_2 + \cdots + (x - Aq^{n-1})c_n) \cdots\big). \tag{21.1}$$

This uses $O(n)$ arithmetic per target. Coefficients can be generated from samples by a divided-difference table or by (2.7).

21.3 Origin extrapolation by factor recursion

For the special target 0, the recursion (14.6) is preferable to solving for an interpolating polynomial. In pseudocode:

Algorithm 21.1 (Factorized geometric-comb filter)**.** Given $y_k = f(Aq^k)$ for $0 \le k \le n$:

1. Set `work[k] = y[k]`.
2. For $j = 1, \dots, n$, replace
$$\texttt{work[k]} \leftarrow \frac{\texttt{work[k+1]} - q^j \texttt{work[k]}}{1 - q^j}, \qquad 0 \le k \le n - j.$$
3. Return `work[0]`.

The induced ℓ^∞ error factor of the jth stage is at most $(1 + q^j)/(1 - q^j)$; their product is exactly $K_n(q)$. Thus the recursive algorithm exposes the same stability constant as the explicit row.

21.4 Exact dyadic arithmetic

When $q = 1/2$ and all samples are rational, every weight is rational. For the Fabius sequence, (19.11) and the filter can therefore be implemented with arbitrary-precision integer fractions. This is valuable for separating true cancellation from roundoff. Numerators and denominators grow quickly, so the companion code uses exact fractions only for moderate order and switches to high-precision `mpmath` arithmetic for plots.

21.5 Precision budgeting

Suppose the desired absolute output accuracy is ε and each sample is evaluated with absolute accuracy δ. A sufficient condition is
$$\delta \le \frac{\varepsilon}{K_n(q)}. \tag{21.2}$$
For fixed $q < 1$, the right side remains comparable to ε as $n \to \infty$. Near $q = 1$, (16.4) shows that the required number of guard digits grows like
$$\frac{\pi^2}{4(-\log q)\log 10}$$
in the infinite-order regime. Relative precision is subtler: if the samples themselves are superflat, as for $F(2^{-m})$, a small absolute error may still be a large relative error. Exact rational or ball arithmetic is preferable in that case.

22 Numerical experiments

The file `geometric_comb_experiments.py` included in the archive reproduces all data and figures in this section. It uses 140 decimal digits for alternating sums and exact rational arithmetic for the first dyadic Fabius checks. The numerical experiments are confirmations of proved identities, not substitutes for proofs.

22.1 Moment cancellation and factor recursion

For $q = 1/2$ and orders $1 \le n \le 20$, the script verifies

$$\sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} 2^{-kd} = 0, \qquad 1 \le d \le n,$$

and compares the first residual with $(-1)^n 2^{-n(n+1)/2}$. The largest observed canceled moment is at the level of 10^{-140}, consistent with the working precision. The direct weighted sum and the triangular factor recursion agree to the same scale on the nonpolynomial test function e^x.

22.2 Noise amplification

The data file also compares $K_\infty(e^{-t})$ to the asymptotic (16.4). For the plotted near-one values the relative difference is already below the displayed precision, reflecting the exponentially small modular correction.

22.3 Regular-variation constants

For example,

$$\begin{aligned} C_{1/2}(0.7) &\approx 0.2493175488440439431, \\ C_{1/2}(1.3) &\approx -0.1538705960439442305, \\ C_{1/2}(2.4) &\approx 0.2913748265379615867. \end{aligned}$$

At positive integer exponents the limit is exactly zero, in agreement with polynomial cancellation.

22.4 Exact Fabius boundary remainders

For $A = 2^{-(n+1)}$, define

$$R_n := \sum_{k=0}^{n} \lambda_{n,k}^{(1/2)} F(2^{-(n+1+k)}), \qquad B_n := \frac{2^{-n(n+1)}}{(n+1)!}.$$

The theorem gives $(-1)^n R_n > 0$ and $|R_n|/B_n = \mathbb{E}F(S_n) \in (0,1)$. Exact fraction arithmetic produces the following decimal summary.

Table 3: Normalized exact dyadic Fabius boundary remainders.

n	sign of R_n	$\lvert R_n\rvert/B_n$
1	$-$	0.500000000000
2	$+$	0.392067901235
3	$-$	0.285827999336
4	$+$	0.204528650771
5	$-$	0.146778506870
6	$+$	0.106554660592
7	$-$	0.078481279976
8	$+$	0.058671832771
9	$-$	0.044492131277
10	$+$	0.034188031436

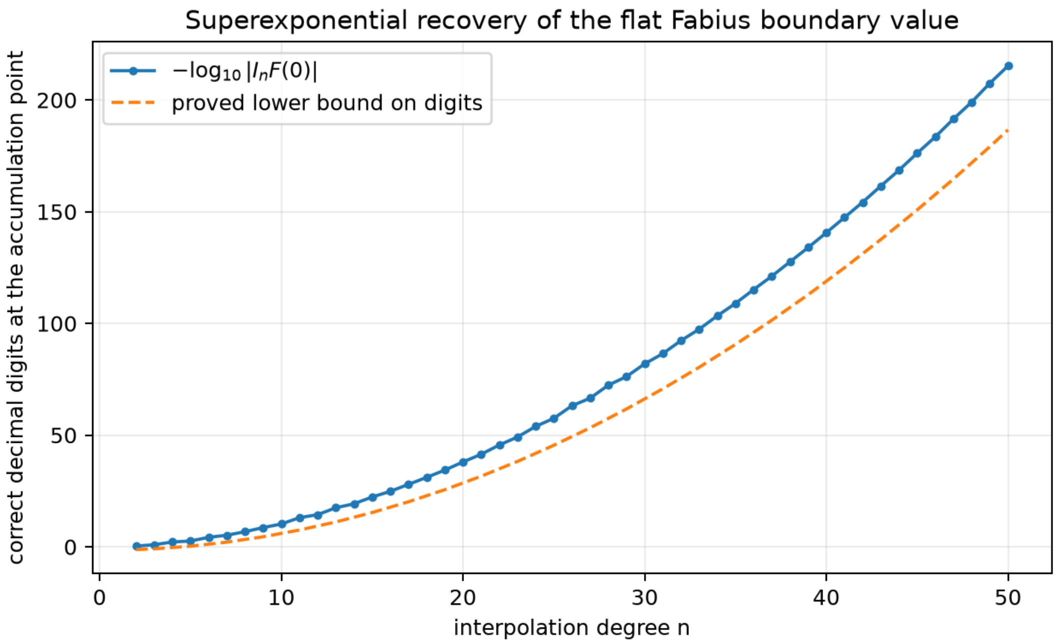

Figure 4: Decimal digits recovered at the accumulation point. The exact residue is compared with the rigorous quadratic-exponential estimate from Theorem 10.2.

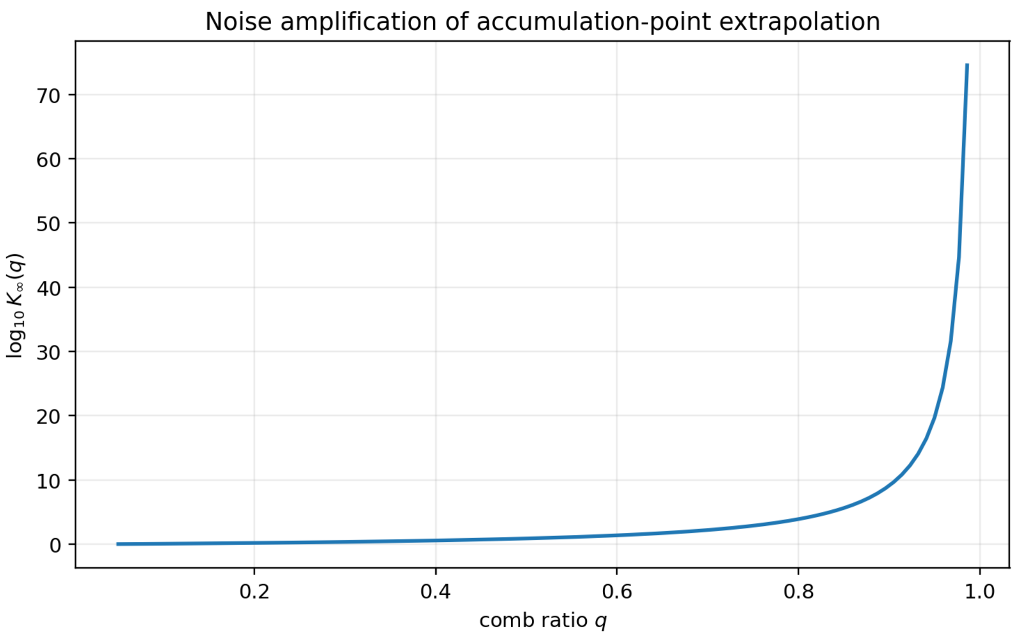

Figure 5: The infinite-order total variation $K_\infty(q) = (-q;q)_\infty/(q;q)_\infty$. The vertical axis is logarithmic in base 10. Stability is excellent for small and moderate q but deteriorates nonperturbatively as $q \uparrow 1$.

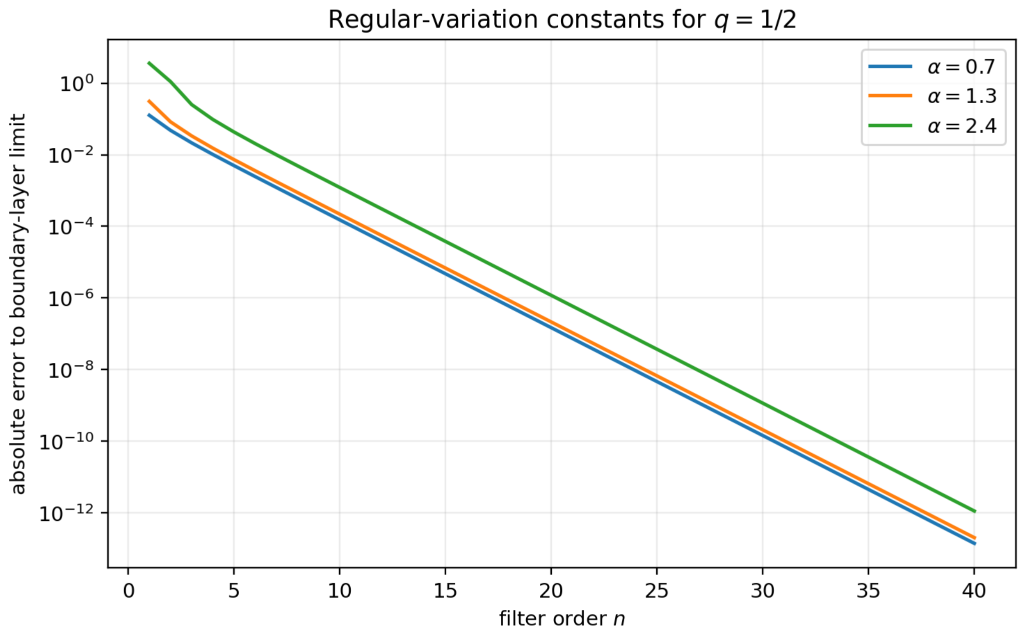

Figure 6: For pure powers x^α at $q = 1/2$, the normalized exact multiplier $(q^{1-\alpha}; q)_n/(q; q)_n$ converges geometrically to $C_q(\alpha)$. The three exponents illustrate positive and negative limiting constants.

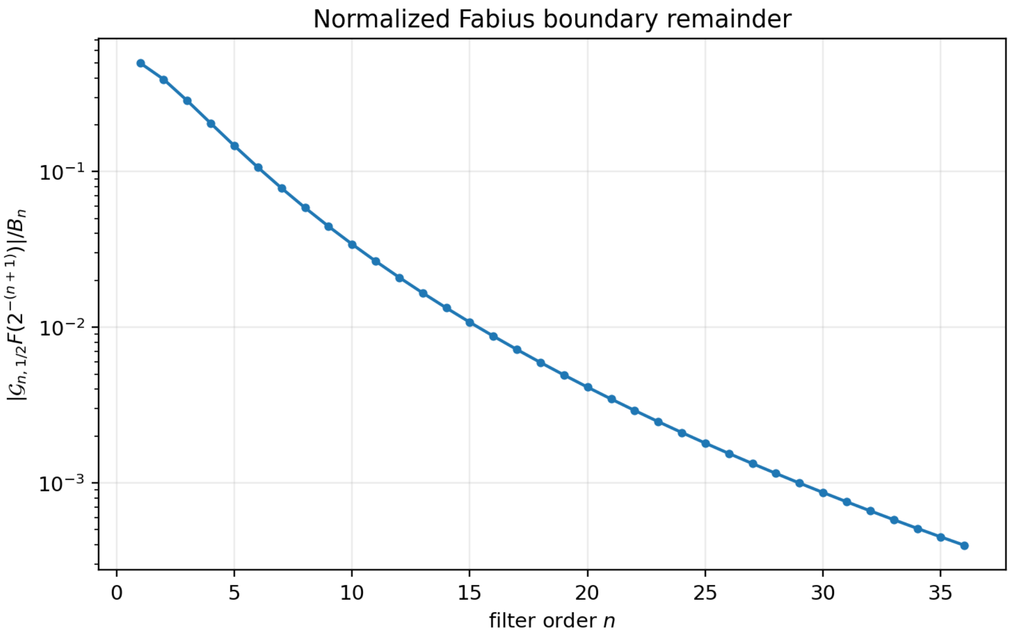

Figure 7: The normalized Fabius boundary remainder $|R_n|/B_n = \mathbb{E}F(S_n)$ through order 36. The decrease is much slower than the already extracted factorial-supergeometric factor B_n; its finer asymptotics are posed as a conjecture in section 23.

22.5 Reproducibility files

The script writes four CSV files:

File	Contents
`moment_checks.csv`	canceled moments and recursion comparison
`stability_growth.csv`	K_∞, modular approximation, ratio
`regular_variation.csv`	finite multipliers and limiting constants
`fabius_boundary.csv`	exact rational filters, bounds, sign checks

The archive includes these files and the three PNG figures, so the LaTeX source can be recompiled without rerunning Python.

23 Research frontiers and conjectures

23.1 A local limit theorem for geometric B-splines

equation (17.8) proves weak convergence. The explicit spline and limit densities suggest a substantially stronger statement.

Conjecture 23.1 (Local density limit)**.** *For every fixed* $0 < q < 1$ *and compact interval* $K \subset (0, \infty)$*,*

$$\sup_{y\in K}\left|\frac{A}{n+2}B_{n,A,q}\left(\frac{Ay}{n+2}\right)-p_q(y)\right|\longrightarrow 0, \tag{23.1}$$

where p_q *is given by* (17.11)*. The convergence should extend to all derivatives on compact subsets of* $(0, \infty)$ *after the natural scaling.*

A proof might combine the exponential/Dirichlet representation with Fourier or Laplace inversion and uniform control of the denominator $(E_* + \cdots + E_n)/(n + 2)$. Such a theorem would turn finite interpolation kernels into quantitative approximations of a canonical q-exponential density.

23.2 Fabius–Dirichlet small-value asymptotics

Let

$$\rho_n := \mathbb{E}F(S_n) = \frac{|R_n|}{B_n}.$$

The scaling limit $(n + 2)S_n \Rightarrow Z_{1/2}$ suggests that S_n is of order $1/n$, while the Fabius function has a logarithmic-square flatness scale at zero. Random fluctuations of $Z_{1/2}$ should affect lower-order terms but not the leading quadratic logarithm.

Conjecture 23.2 (Leading decay of the normalized Fabius remainder)**.**

$$\log\rho_n \sim -\frac{(\log n)^2}{2\log 2} \qquad (n\to\infty). \tag{23.2}$$

More precisely, the next terms should be obtainable by combining the sharp Lambert-W endpoint expansion of F developed elsewhere in the ProveIt project with the density (17.11) *and a Laplace saddle point.*

The experiment in figure 7 is consistent with a slowly converging logarithmic-square law, but orders below 40 are far too small to distinguish the lower-order $\log n \log\log n$ and $\log n$ corrections.

23.3 Second-order regular variation

equation (16.10) gives only the first boundary-layer limit. If $f(x) = x^\alpha L(x)$ and L belongs to a de Haan second-order class, one should be able to expand

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} = C_q(\alpha) + A_1(Aq^n)C_{q,1}(\alpha,\rho) + o(A_1(Aq^n)),$$

where A_1 is the auxiliary function and the correction constant is a q-Mellin moment of the signed boundary layer $\beta_r^{(q)}$. This would provide rigorous error estimators for nonanalytic asymptotic extrapolation.

Problem 23.3 (Second-order transfer theorem)**.** *Formulate minimal second-order regular-variation hypotheses and compute the correction constants explicitly from $(qz;q)_\infty/(q;q)_\infty$. Treat the resonant integer cases, where $C_q(\alpha) = 0$, separately; there the correction may become the leading term.*

23.4 The double-scaling transition $q \uparrow 1$, $n \to \infty$

The fixed-order behavior (16.3) and infinite-order behavior (16.4) do not commute. A natural transition parameter is $n(1-q)$ or nt when $q = e^{-t}$.

Problem 23.4 (Universal transition kernel)**.** *Determine asymptotics of the row polynomial, its total variation, and the Peano-kernel density when*

$$n \to \infty, \qquad t \downarrow 0, \qquad nt \to \tau \in (0,\infty).$$

One expects a continuum limit connecting geometric-comb extrapolation to confluent Hermite/Taylor extrapolation at A. The relevant asymptotics should involve dilogarithms and a saddle point for finite q-products.

23.5 Optimal filters under analytic priors

The Gaussian filter is algebraically unique when exactly n integer modes are canceled with $n+1$ samples. With more samples than constraints, stability and bias can be traded.

Problem 23.5 (Minimax geometric extrapolation)**.** *Given $M > n$, choose weights on $A, Aq, \ldots, Aq^M$ that preserve constants and annihilate $x, \ldots, x^n$, while minimizing either total variation or the worst error over an analytic coefficient-majorant class. Compare the optimizer with truncated, damped, and least-squares modifications of $W_{n,q}$.*

The problem is finite-dimensional and convex for total variation, but its large-M, n structure is likely governed by weighted extremal polynomials on the spectral set $\{q^m : m \geq 0\}$.

23.6 Inverse Fabius and nonlinear combs

The inverse Fabius function has endpoint behavior governed by asymptotic inversion and Lambert W, not a regular Taylor series. A promising strategy is to choose a nonlinear sample comb x_k so that the dominant inverse asymptotic modes become geometric in a transformed coordinate. Confluent modal filters from section 15 can then remove the leading Lambert-W transseries terms.

Problem 23.6 (Lambert-adapted Newton coordinates)**.** *Find a coordinate $u = u(x)$ in which the inverse Fabius germ admits an expansion in modes $e^{-\alpha u}u^s$, construct the corresponding geometric Newton series, and determine whether its interpolation coefficients have a positive kernel or a controlled boundary layer analogous to* (16.5)*.*

23.7 Multivariate geometric combs

Tensor products immediately give interpolation on $\{A_1 q_1^{k_1}\} \times \cdots \times \{A_d q_d^{k_d}\}$, with products of Gaussian rows. More interesting are diagonal or simplex combs with a shared scale index, which interact with multivariate self-similar distributions and box splines. The Dirichlet-average interpretation suggests that multivariate Peano kernels may again have positive probabilistic representations.

23.8 Formalization boundary and remaining roadmap

The live Lean corpus has moved beyond a roadmap for the elementary finite filter. The modules `FiniteQBinomialCore.lean`, `GeometricLagrangeQBinomial.lean`, `GeometricLagrangeQMoments.lean`, and `GeometricLagrangeCompleteHomogeneous.lean` contain named theorems for finite q-products, Gaussian weights, cancellation/residual formulas, the complete-homogeneous specialization, and the exact finite row norm. `LimitConditionNumber.lean` supplies the fixed-q infinite-product limit, while the geometric-uniform and Rvachev modules named in the opening status box cover related probability and Fourier infrastructure.

The report-specific obligations that remain are therefore narrower:

1. identify geometric divided differences with iterated Jackson derivatives under explicit field and differentiability hypotheses;
2. construct the infinite Newton series and prove its local uniform convergence;
3. formalize confluent spectral zeros and the Puiseux–log remainder bound;
4. prove the reversed-row ℓ^1 limit and the regular-variation transfer;
5. add Hermite–Genocchi/Dirichlet-average infrastructure, the geometric-knot B-spline moments, and the perpetuity scaling limit;
6. derive the Fabius boundary expectation and the entire scale-reconstruction theorem from the already formalized finite core.

Until those items exist as checked declarations, the correspondingly labelled results here remain manuscript theorems.

24 Complementary Mellin and superflat theory

24.1 A target on the same multiplicative line

A target $x = aq^\alpha$, where α need not be integral, preserves all q-structure.

Theorem 24.1 (Arbitrary geometric target). *Choose a branch of q^α. If q^α is not one of $1, q, \ldots, q^n$, then*

$$\ell_{n,j}(aq^\alpha) = (-1)^{n-j} q^{\binom{n-j+1}{2}} \frac{(q^{\alpha-n};q)_{n+1}}{(1-q^{\alpha-j})(q;q)_j (q;q)_{n-j}}. \tag{24.1}$$

At $\alpha = j$ the singularity is removable and the value is 1; at every other integral node it is 0. For $0 < q < 1$ and $\Re\alpha \to +\infty$, the weights tend to the zero-target row

$$\lambda_{n,j}^{(q)} = \ell_{n,j}(0) = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q;q)_j (q;q)_{n-j}}. \tag{24.2}$$

Proof. Use $\ell_{n,j}(x) = \Omega_{n+1}(x)/((x - aq^j)\Omega'_{n+1}(aq^j))$, (4.1) and (4.2). The product at the target is

$$\prod_{r=0}^{n} (q^\alpha - q^r) = (-1)^{n+1} q^{\binom{n+1}{2}} (q^{\alpha-n};q)_{n+1}.$$

Extracting $-q^j$ from $q^\alpha - q^j$ leaves exactly (24.1). The limiting assertions follow directly from the product. □

Remark 24.2 (q-gamma form). Since

$$(q^{\alpha-n};q)_{n+1} = (1-q)^{n+1}\frac{\Gamma_q(\alpha+1)}{\Gamma_q(\alpha-n)},$$

formula (24.1) is meromorphic in α and can be rewritten entirely with q-gamma functions and the q-number $[\alpha - j]_q$. This is the natural analytic continuation of the cardinal row, not merely a formal replacement of factorials by q-factorials.

25 Mellin symbols and exact residuals

25.1 Complex powers

The integer moment law in the source monograph is one specialization of a stronger identity. Since dilations act diagonally on every chosen branch of x^s, the full spectral response is available at once.

Theorem 25.1 (Exact Mellin symbol). *Let $s \in \mathbb{C}$, choose a branch of a^s and q^s, and set*

$$M_{n,q}(s) = \sum_{k=0}^{n} \lambda_{n,k}^{(q)} q^{ks}. \tag{25.1}$$

Then

$$M_{n,q}(s) = \frac{\prod_{j=1}^{n}(q^s - q^j)}{(q;q)_n} \tag{25.2}$$

$$= q^{ns}\frac{(q^{1-s};q)_n}{(q;q)_n} \tag{25.3}$$

$$= (-1)^n q^{\binom{n+1}{2}}\frac{(q^{s-n};q)_n}{(q;q)_n}. \tag{25.4}$$

Equivalently,

$$(\mathcal{G}_{n,q}x^s)(a) = a^s M_{n,q}(s). \tag{25.5}$$

The zeros at $s = 1, \dots, n$ are simple modulo the imaginary period $2\pi\mathrm{i}/\log q$.

Proof. Evaluate (14.3) at $z = q^s$. Factoring q^s from each factor gives (25.3); factoring $-q^j$ gives (25.4). □

Corollary 25.2 (Natural moments and the first residual). *For $d \in \mathbb{N}_0$,*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} q^{kd} = \begin{cases} 1, & d = 0, \\ 0, & 1 \le d \le n, \\ (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \ge n+1. \end{cases} \tag{25.6}$$

Proof. The first two cases follow from the zeros in (25.2). For $d \ge n+1$,

$$(q^{d-n};q)_n = \frac{(q;q)_{d-1}}{(q;q)_{d-n-1}},$$

and substitution into (25.4) gives the Gaussian coefficient. □

Remark 25.3 (Fractional endpoint laws). For fixed s with $\Re s > 0$ and $s \notin \mathbb{N}_0$,

$$M_{n,q}(s) \sim q^{ns}\frac{(q^{1-s};q)_\infty}{(q;q)_\infty} \qquad (n \to \infty). \tag{25.7}$$

Thus a fractional power x^s is not annihilated, but its extrapolation error has a completely explicit geometric rate. Integer powers are exceptional because the finite product terminates at an exact zero.

25.2 Logarithmic powers by spectral differentiation

Differentiate (25.5) with respect to s. This gives closed responses for singular terms common in endpoint asymptotics.

Proposition 25.4 (Powers times logarithms). *For every integer $r \geq 0$,*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} (aq^k)^s \big(\log a + k \log q\big)^r = \frac{\partial^r}{\partial s^r}\big(a^s M_{n,q}(s)\big). \tag{25.8}$$

In particular, if $1 \leq d \leq n$, then

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} (aq^k)^d \log(aq^k) = a^d M'_{n,q}(d), \tag{25.9}$$

where

$$M'_{n,q}(d) = (-1)^{d-1} (\log q) q^{d(2n-d+1)/2} \frac{(q;q)_{d-1}(q;q)_{n-d}}{(q;q)_n}. \tag{25.10}$$

Proof. Termwise differentiation is legitimate because the sum is finite. At $s = d$, the factor $q^s - q^d$ is the unique vanishing factor in (25.2). Differentiating that factor and splitting the remaining product below and above d gives

$$q^d \log q \prod_{\substack{1 \leq j \leq n \\ j \neq d}} (q^d - q^j) = (-1)^{d-1} (\log q) q^{d(2n-d+1)/2} (q;q)_{d-1}(q;q)_{n-d}.$$

Division by $(q;q)_n$ proves (25.10). □

Remark 25.5. A filter that is exact on polynomials need not suppress logarithmic corrections attached to those powers. Formula (25.10) quantifies the leakage exactly. Higher s-derivatives introduce finite sums of q-harmonic terms $(1 - q^j)^{-r}$ and can be organized with complete Bell polynomials.

25.3 Mellin-transform interpretation

For a function on $(0, \infty)$ with Mellin transform

$$(\mathcal{M}f)(z) = \int_0^\infty f(x) x^{z-1} \, \mathrm{d}x,$$

one has

$$\mathcal{M}[f(q^k x)](z) = q^{-kz} (\mathcal{M}f)(z).$$

Hence, on a common strip of convergence,

$$\mathcal{M}[\mathcal{G}_{n,q} f](z) = M_{n,q}(-z)(\mathcal{M}f)(z). \tag{25.11}$$

The geometric Lagrange filter is therefore a finite Mellin multiplier. Its spectral zeros are the same integer moments that the interpolation formula annihilates in physical space. This is the multiplicative analogue of an ordinary finite-difference stencil whose Fourier symbol has a prescribed zero at the origin.

25.4 The limits $q \to 1$ and $q \to 0$

Proposition 25.6 (Taylor limit)**.** *For fixed k, as $q \to 1$,*

$$D_q^k f(a) \longrightarrow f^{(k)}(a), \qquad [k]_q! \longrightarrow k!, \qquad \Phi_k(x) \longrightarrow (x-a)^k.$$

Thus (2.8) tends termwise to the ordinary Taylor polynomial about a.

Proof. The first statement follows by repeated use of the standard Jackson derivative limit on a sufficiently differentiable function; the other two are immediate from their finite products. □

At the opposite endpoint, $D_0 f(x) = (f(x) - f(0))/x$ is a tail-quotient operator. For $f(x) = \sum c_m x^m$,

$$\lim_{q\to 0} \frac{D_q^k f(a)}{[k]_q!} = \sum_{r\geq 0} c_{k+r} a^r, \qquad \lim_{q\to 0} \Phi_k(x) = \begin{cases} 1, & k = 0, \\ (x-a)x^{k-1}, & k \geq 1. \end{cases}$$

The limiting expansion telescopes. It is not Hermite interpolation at a multiple node: the geometric rates retain ordered information about how the nodes approach zero. A genuine confluent theory requires a further renormalization and is listed as Problem 29.3.

25.5 Neville, Richardson, and Romberg recurrences

Theorem 25.7 (Geometric Neville recurrence)**.** *For $i, m \geq 0$, let $P_{i,m}$ be the interpolant characterized by*

$$P_{i,m}(aq^{i+r}) = f(aq^{i+r}) \qquad (0 \leq r \leq m).$$

Then

$$P_{i,m}(x) = \frac{(x - aq^{i+m})P_{i,m-1}(x) - (x - aq^i)P_{i+1,m-1}(x)}{aq^i - aq^{i+m}}. \tag{25.12}$$

At $x = 0$, with $R_i^{(0)} = f(aq^i)$,

$$R_i^{(m)} = \frac{R_{i+1}^{(m-1)} - q^m R_i^{(m-1)}}{1 - q^m}, \qquad R_0^{(n)} = (\mathcal{G}_{n,q} f)(a). \tag{25.13}$$

Proof. The numerator in (25.12) has degree at most m and takes the required values at both endpoint nodes and all shared interior nodes; uniqueness proves the formula. Substitute $x = 0$ and cancel aq^i. □

Corollary 25.8 (Generalized Richardson cancellation)**.** *Suppose approximations at scales $h_i = h_0 r^i$ have an expansion*

$$T(h) = L + c_1 h^\rho + c_2 h^{2\rho} + \cdots.$$

Apply (25.13) with $q = r^\rho$ to the data $T(h_i)$. After n levels, the first n error powers are annihilated exactly. Romberg quadrature is the special case in which the trapezoidal error uses even powers and hence $q = r^2$.

Proof. The recurrence is the degree-n interpolation functional at the point 0 on nodes $1, q, \ldots, q^n$. It fixes the constant mode and annihilates q^{ki} for $1 \leq k \leq n$, equivalently the error modes $h^{k\rho}$. Linearity therefore removes the first n displayed error powers. For the trapezoidal expansion only even powers occur, so replacing r by r^2 gives the Romberg case. □

Remark 25.9. This recurrence is often numerically preferable to forming the closed weights. It also explains the historical term "deferred approach to the limit": each new level eliminates one Mellin eigenmode of the asymptotic error.

26 Convergence at an accumulation point and divergence away from it

26.1 The zero row is a regular summability method

Theorem 26.1 (Regularity at the accumulation point). *Let $0 < q < 1$ and let $y_k \to L$. Then*

$$\sum_{k=0}^{n} \lambda_{n,k}^{(q)} y_k \longrightarrow L. \tag{26.1}$$

Equivalently, if f is continuous at 0, the interpolating polynomial through $(aq^k, f(aq^k))$, $0 \le k \le n$, satisfies

$$(\mathcal{I}_n f)(0) \longrightarrow f(0). \tag{26.2}$$

Proof. The rows reproduce constants by (25.6); their ℓ^1 norms are bounded by $K(q)$ by Theorem 16.1; and for each fixed k,

$$|\lambda_{n,k}^{(q)}| = \frac{q^{\binom{n-k+1}{2}}}{(q;q)_k (q;q)_{n-k}} \longrightarrow 0.$$

These are precisely the Toeplitz regularity conditions. For completeness, subtract L, split the sum at a fixed K, make the tail small using the bounded row norm, and then let $n \to \infty$ in the finite head. □

26.2 The limiting profile viewed from the finest tooth

Reindex the row by $r = n - k$. For each fixed r,

$$\lambda_{n,n-r}^{(q)} \longrightarrow \mu_r^{(q)} := \frac{(-1)^r q^{\binom{r+1}{2}}}{(q;q)_\infty (q;q)_r}. \tag{26.3}$$

Euler's infinite theorem gives the generating function

$$\sum_{r=0}^{\infty} \mu_r^{(q)} z^r = \frac{(qz;q)_\infty}{(q;q)_\infty}, \qquad \sum_{r=0}^{\infty} \mu_r^{(q)} = 1. \tag{26.4}$$

Thus the mass does not remain attached to any fixed physical node; it travels with the finest scale aq^n and approaches a signed, absolutely summable profile in the reversed index. This is the mechanism behind Theorem 26.1.

26.3 Power, Puiseux, and logarithmic endpoint expansions

Suppose a germ has a convergent or asymptotic expansion

$$f(x) = L + \sum_r c_r x^{\alpha_r} (\log x)^{m_r}, \qquad \Re\alpha_r > 0.$$

Theorems 25.1 and 25.4 give the filter of each term exactly. In particular, a nonintegral leading exponent α produces the rate $q^{n\alpha}$ times an explicit q-product, while an integral polynomial jet is deleted once n reaches its degree. The filter is therefore an endpoint asymptotic microscope: its decay rate distinguishes fractional regularity from an analytic integer jet.

Proposition 26.2 (A simple Hölder estimate). *If $|f(x) - f(0)| \le C|x|^\alpha$ on the comb, with $\alpha > 0$, then*

$$|(\mathcal{G}_{n,q} f)(a) - f(0)| \le C|a|^\alpha \sum_{k=0}^{n} |\lambda_{n,k}^{(q)}| q^{k\alpha}. \tag{26.5}$$

For fixed q and α, the right side is $O(q^{n\alpha})$.

Proof. The first statement is the triangle inequality. Reindex $r = n - k$. Since $(q;q)_m \geq (q;q)_\infty$ for every m, one has

$$q^{-n\alpha} \sum_{k=0}^{n} |\lambda_{n,k}^{(q)}| q^{k\alpha} = \sum_{r=0}^{n} \frac{q^{\binom{r+1}{2} - r\alpha}}{(q;q)_{n-r}(q;q)_r}$$
$$\leq \frac{1}{(q;q)_\infty^2} \sum_{r=0}^{\infty} q^{\binom{r+1}{2} - r\alpha} < \infty.$$

The last series converges for every fixed α because its exponent is quadratic in r. This uniform bound proves the stated $O(q^{n\alpha})$ estimate. Dominated convergence also gives the sharper limit

$$q^{-n\alpha} \sum_{k=0}^{n} |\lambda_{n,k}^{(q)}| q^{k\alpha} \longrightarrow \sum_{r=0}^{\infty} |\mu_r^{(q)}| q^{-r\alpha}.$$

□

26.4 Superflat data force divergence off the comb

The favorable result at 0 has a nearly opposite counterpart everywhere else. It is useful to formulate it directly for data, without assuming that they initially came from a function.

Definition 26.3. A sequence $y_j \to L$ is *q-superflat at its limit* if

$$y_j - L = o(q^{mj}) \qquad (j \to \infty) \tag{26.6}$$

for every $m \in \mathbb{N}_0$.

Theorem 26.4 (Exact convergence set for nontrivial superflat data)**.** *Let $0 < q < 1$, $a \neq 0$, and let $y_j \to L$ be q-superflat. Assume $y_j \neq L$ for infinitely many j. Let*

$$P_n(x) = \sum_{k=0}^{n} A_k \Phi_k(x) \tag{26.7}$$

be the polynomial interpolating $P_n(aq^j) = y_j$ for $0 \leq j \leq n$. Then

$$\limsup_{k\to\infty} |A_k|^{1/k} = \infty. \tag{26.8}$$

Moreover:

1. *$P_n(0) \to L$;*
2. *for each $m \geq 0$, $P_n(aq^m) = y_m$ once $n \geq m$;*
3. *for every $x \in \mathbb{C} \setminus (\{0\} \cup \{aq^m : m \geq 0\})$, the increments $A_n\Phi_n(x)$ fail to tend to zero, and the sequence $P_n(x)$ diverges; indeed it is unbounded along a subsequence.*

Thus the convergence set is exactly the geometric comb together with its accumulation point.

Proof. Subtract L; this changes only A_0 and reduces to $L = 0$. Suppose, contrary to (26.8), that $|A_k| \leq CR^{-k}$ for some $R > 0$. For every $0 < r < R$, the estimate

$$\sup_{|z|\leq r} |\Phi_k(z)| \leq r^k \prod_{j=0}^{\infty} (1 + |a|q^j/r)$$

shows that $g(z) = \sum_{k\geq 0} A_k \Phi_k(z)$ is analytic on $|z| < R$. For every sufficiently large j, $|aq^j| < R$, and all terms with $k > j$ vanish at aq^j. Hence

$$g(aq^j) = P_j(aq^j) = y_j.$$

If the first nonzero Taylor coefficient of g at 0 had order m, then $g(aq^j)/(aq^j)^m$ would tend to a nonzero constant. This contradicts q-superflatness. Therefore every Taylor coefficient vanishes, so g is identically zero near 0, contradicting the assumption that infinitely many y_j are nonzero. This proves (26.8).

Assertion 1 is Theorem 26.1, and assertion 2 is interpolation. For a point x in assertion 3,

$$\Phi_n(x) = x^n \prod_{r=0}^{n-1} \left(1 - \frac{aq^r}{x}\right). \tag{26.9}$$

The infinite product tends to $(a/x;q)_\infty \neq 0$, because x is not a node. Hence $|\Phi_n(x)|^{1/n} \to |x| > 0$. Together with (26.8), this implies $\limsup |A_n \Phi_n(x)|^{1/n} = \infty$; the increments do not tend to zero. Finally, $|P_n - P_{n-1}| = |A_n \Phi_n|$ is unbounded along a subsequence, so at least one of each adjacent pair P_n, P_{n-1} is unbounded along a subsequence. □

Remark 26.5 (A precise geometric Runge phenomenon). This is not the classical Runge phenomenon caused by equispaced nodes on a fixed interval. The comb is not dense away from 0. Superflat nonzero data cannot be the trace of a nonzero analytic germ at the accumulation point; the interpolation coefficients must therefore leave every exponential class. The instability is an analytic-continuation obstruction encoded in a Vandermonde problem.

Remark 26.6 (Derivative-enhanced interpolation). Adding finitely many Hermite conditions at any fixed collection of nonzero teeth does not remove the obstruction: an exponentially bounded coefficient sequence would still define an analytic germ with a superflat nonzero trace. A successful regularization must either change the approximation space, use a number of derivative conditions growing with n, or explicitly build flatness into the basis. Rvachev atoms are natural candidates for the last option.

27 The geometric-uniform family

27.1 Probability law, support, and refinement

Let $U_0, U_1, \ldots$ be independent uniform random variables on $[0, 1]$ and set

$$X_q = (1-q) \sum_{j=0}^{\infty} q^j U_j, \qquad 0 < q < 1. \tag{27.1}$$

This is a geometric member of the infinite-convolution class studied by Jessen and Wintner [58]. The coefficients are positive and sum to 1, so $0 \leq X_q \leq 1$. Reflection of every U_j shows

$$X_q \stackrel{d}{=} 1 - X_q. \tag{27.2}$$

Let $F_q(x) = \mathbb{P}(X_q \leq x)$, extended by 0 to $x \leq 0$ and by 1 to $x \geq 1$. Removing the first coordinate gives the affine fixed-point law

$$X_q \stackrel{d}{=} (1-q)U + qX_q', \tag{27.3}$$

where X_q' is an independent copy. Consequently

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1-q)u}{q}\right) \mathrm{d}u \tag{27.4}$$

$$= \frac{q}{1-q} \int_{(x-(1-q))/q}^{x/q} F_q(t) \, \mathrm{d}t. \tag{27.5}$$

Differentiation gives

$$F_q'(x) = \frac{F_q(x/q) - F_q((x-(1-q))/q)}{1-q}. \tag{27.6}$$

Proposition 27.1 (Smooth compactly supported density)**.** *The centered random variable*

$$Z_q = 2X_q - 1 = (1-q)\sum_{j=0}^{\infty} q^j V_j, \qquad V_j \sim \mathrm{Unif}[-1,1], \tag{27.7}$$

has an even C^∞ density u_q supported on $[-1,1]$. With the Fourier convention $\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i}\xi x}\,\mathrm{d}x$,

$$\widehat{u_q}_{\,2\pi}(\xi) = \prod_{j=0}^{\infty} \mathrm{sinc}_\pi\left(2(1-q)q^j\xi\right), \qquad \mathrm{sinc}_\pi(t) = \frac{\sin(\pi t)}{\pi t}. \tag{27.8}$$

Moreover

$$u_q(z) = \frac{1}{2}F_q'\left(\frac{z+1}{2}\right). \tag{27.9}$$

Proof. The characteristic function factors by independence and gives (27.8). To see rapid decay, use $|\mathrm{sinc}_\pi(t)| \le \min(1, 1/(\pi|t|))$ on the first $N \asymp \log|\xi|/|\log q|$ factors. The logarithm of the resulting product is $-c_q(\log|\xi|)^2 + O_q(\log|\xi|)$, hence the product decreases faster than any inverse power. Fourier inversion then gives a C^∞ density. The support and evenness follow from (27.7); the density change of variables gives (27.9). □

Remark 27.2. The repository's formal development treats a still broader geometric-uniform law, including negative ratios and uniqueness of the affine distributional fixed point. The positive-ratio case above is the one naturally matched to a one-sided interpolation comb.

27.2 Moments and Bernoulli cumulants

Write

$$\mu_n(q) = \mathbb{E}X_q^n. \tag{27.10}$$

Theorem 27.3 (Moment recurrence)**.** *The moments satisfy $\mu_0(q) = 1$ and, for $n \ge 1$,*

$$(1-q^n)\mu_n(q) = \sum_{k=0}^{n-1} \binom{n}{k} q^k (1-q)^{n-k} \frac{\mu_k(q)}{n-k+1}. \tag{27.11}$$

At $q = 1/2$ this reduces to

$$(2^n - 1)\mu_n(1/2) = \sum_{k=0}^{n-1} \binom{n}{k} \frac{\mu_k(1/2)}{n-k+1}. \tag{27.12}$$

Proof. Raise (27.3) to the nth power, expand, and use $\mathbb{E}U^r = 1/(r+1)$. Move the $k = n$ term to the left. At $q = 1/2$ every factor $q^k(1-q)^{n-k}$ equals 2^{-n}, and multiplication by 2^n gives (27.12). □

The centered moment generating function is

$$\mathsf{M}_q(z) = \mathbb{E}\mathrm{e}^{zZ_q} = \prod_{j=0}^{\infty} \frac{\sinh((1-q)q^j z)}{(1-q)q^j z}. \tag{27.13}$$

Using

$$\log\frac{\sinh z}{z} = \sum_{m\ge 1} \frac{2^{2m-1}B_{2m}}{m(2m)!} z^{2m},$$

we obtain the even cumulants

$$\kappa_{2m}(Z_q) = \frac{2^{2m}B_{2m}}{2m}\frac{(1-q)^{2m}}{1-q^{2m}}, \qquad \kappa_{2m+1}(Z_q) = 0. \tag{27.14}$$

Thus the same geometric series that produces the interpolation denominators also produces the cumulant denominators. At $q = 1/2$,

$$\kappa_{2m}(Z_{1/2}) = \frac{B_{2m}}{2m(1-4^{-m})}. \tag{27.15}$$

Moments can be recovered from these cumulants by complete Bell polynomials, providing a second exact route parallel to (27.12).

27.3 The left boundary layer

For $0 \le x \le 1-q$, the second term in (27.6) vanishes, so

$$F_q'(x) = \frac{1}{1-q}F_q(x/q). \tag{27.16}$$

Iteration gives the geometric derivative tower.

Lemma 27.4 (Boundary derivative tower)**.** *If $m \ge 1$ and $0 \le x \le (1-q)q^{m-1}$, then*

$$F_q^{(m)}(x) = (1-q)^{-m}q^{-\binom{m}{2}}F_q(x/q^m). \tag{27.17}$$

Proof. Induct on m. The case $m = 1$ is (27.16). Differentiating the induction hypothesis contributes a factor q^{-1}, and the domain assumption ensures that the next use of (27.16) remains in the left boundary layer. □

27.4 Comb samples are normalized moments

The ratio $q = 1/2$ is geometrically critical: the first uniform block has length $1-q$, while the tail has length q, and these are equal exactly at $q = 1/2$. For every $q \le 1/2$, however, the tooth q^n still lies inside the nth iterated boundary layer, which is enough for the following formula.

Theorem 27.5 (Geometric comb–moment identity)**.** *For $0 < q \le 1/2$ and every $n \ge 0$,*

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!}\,\mu_n(q). \tag{27.18}$$

Proof. The case $n = 0$ is $F_q(1) = 1$. For $n \ge 1$, compact support and smoothness imply that F_q and all its derivatives vanish at 0. Since $q^n \le (1-q)q^{n-1}$ when $q \le 1/2$, Taylor's integral formula and Lemma 27.4 give

$$\begin{aligned} F_q(q^n) &= \frac{1}{(n-1)!}\int_0^{q^n}(q^n-t)^{n-1}F_q^{(n)}(t)\,\mathrm{d}t \\ &= \frac{(1-q)^{-n}q^{-\binom{n}{2}}}{(n-1)!}\int_0^{q^n}(q^n-t)^{n-1}F_q(t/q^n)\,\mathrm{d}t \\ &= \frac{q^{n^2-\binom{n}{2}}}{(1-q)^n(n-1)!}\int_0^1(1-u)^{n-1}F_q(u)\,\mathrm{d}u. \end{aligned}$$

By Fubini,

$$\int_0^1(1-u)^{n-1}F_q(u)\,\mathrm{d}u = \frac{1}{n}\mathbb{E}(1-X_q)^n.$$

Symmetry (27.2) changes the last moment to $\mu_n(q)$, and $n^2 - \binom{n}{2} = \binom{n+1}{2}$. □

Remark 27.6 (What fails for $q > 1/2$). The formula is not expected to survive unchanged. The point q^n lies beyond the pure left boundary layer, so the reflected term in (27.6) enters before the nth differentiation. The resulting correction should be indexed by affine words recording boundary crossings. Deriving a closed word-sum or automaton formula is Problem 29.2.

27.5 The classical Fabius and Rvachev normalizations

Put

$$F = F_{1/2}, \qquad X = X_{1/2}, \qquad u = u_{1/2}.$$

Then F is the bounded Fabius function. On $0 \le x \le 1/2$,

$$F'(x) = 2F(2x), \qquad F(1-x) = 1 - F(x).$$

The centered density u is Rvachev's up-function in the probability normalization. From (27.9),

$$u(z) = \begin{cases} F(z+1), & -1 \le z \le 0, \\ F(1-z), & 0 \le z \le 1, \\ 0, & |z| \ge 1. \end{cases} \tag{27.19}$$

Its Fourier transform is the canonical infinite sinc product

$$\widehat{u}_{2\pi}(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_\pi(\xi/2^j). \tag{27.20}$$

The construction and these identities agree with the classical treatments of Fabius [1] and Arias de Reyna [37], and with the normalization used in the repository's primary exposition [54].

Write $\mu_n := \mu_n(1/2)$ and introduce the Mersenne product

$$\mathfrak{M}_n := \prod_{r=1}^{n} (2^r - 1), \qquad \mathfrak{M}_0 := 1. \tag{27.21}$$

Proposition 27.7 (Mersenne form of the dyadic origin row). *For every $n \ge 0$,*

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{M}_n, \tag{27.22}$$

$$\lambda_{n,k}^{(1/2)} = (-1)^{n-k} \frac{2^{\binom{k+1}{2}}}{\mathfrak{M}_k \mathfrak{M}_{n-k}} \qquad (0 \le k \le n). \tag{27.23}$$

Proof. In the Pochhammer product, write $1 - 2^{-r} = 2^{-r}(2^r - 1)$ and multiply the powers 2^{-r} for $1 \le r \le n$. Substitution of that identity at k and $n-k$ into (14.1) cancels the remaining powers of two and yields the second formula. □

27.6 The endpoint extrapolant as a Mersenne–moment convolution

Let P_n interpolate F at $1, 1/2, \ldots, 2^{-n}$ and put $E_n = P_n(0)$.

Theorem 27.8 (Exact endpoint convolution). *For every $n \ge 0$,*

$$E_n = \sum_{k=0}^{n} (-1)^{n-k} \frac{2^k \mu_k}{k!\, \mathfrak{M}_k \mathfrak{M}_{n-k}}. \tag{27.24}$$

The first values are

$$1, \quad 0, \quad -\frac{13}{27}, \quad \frac{1}{9}, \quad \frac{4418}{637875}, \quad -\frac{3218}{1318275}, \ldots.$$

Despite the initially nonmonotone behavior, $E_n \to 0$.

Proof. Substitute (9.12) and (27.23) into $E_n = \sum_k \lambda_{n,k}^{(1/2)} F(2^{-k})$. The powers combine as $2^{\binom{k+1}{2}-\binom{k}{2}} = 2^k$. Convergence follows independently from Theorem 26.1. □

The same mechanism yields a quantitative bound throughout the geometric-uniform family.

Theorem 27.9 (Quadratic endpoint bound). *Let $0 < q \le 1/2$, and let $P_{n,q}$ interpolate F_q at $1, q, \dots, q^n$. Then*

$$|P_{n,q}(0)| \le \frac{\exp((1-q)^{-1})}{(q;q)_\infty^2} q^{\binom{n+1}{2} - \lfloor n^2/4 \rfloor}. \tag{27.25}$$

In particular the convergence is at least as fast as $q^{n^2/4+O(n)}$, and hence faster than every fixed geometric rate.

Proof. Use the exact sample formula (27.18), the bound $0 \le \mu_k(q) \le 1$, and (24.2):

$$|P_{n,q}(0)| \le \sum_{k=0}^{n} \frac{q^{\binom{n-k+1}{2}+\binom{k+1}{2}}}{(q;q)_k (q;q)_{n-k}} \frac{(1-q)^{-k}}{k!}.$$

The exponent is

$$\binom{n+1}{2} - k(n-k) \ge \binom{n+1}{2} - \lfloor n^2/4 \rfloor .$$

Also $(q;q)_k, (q;q)_{n-k} \ge (q;q)_\infty$. Extract the smallest power of q and majorize the remaining sum by $\sum_{k\ge 0}(1-q)^{-k}/k! = \exp((1-q)^{-1})$. □

27.7 Newton coefficients as a Gaussian moment transform

Write the Newton form

$$P_n(x) = \sum_{k=0}^{n} A_k \prod_{r=0}^{k-1} (x - 2^{-r}). \tag{27.26}$$

The coefficient A_k is independent of the final interpolation degree once the first $k+1$ nodes are present.

Theorem 27.10 (General geometric-uniform moment transform). *For $0 < q \le 1/2$, let $A_n(q)$ be the Newton coefficient for the data $F_q(1), F_q(q), \dots$. Then*

$$A_n(q) = \frac{1}{(q;q)_n} \sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} \left(\frac{q}{1-q}\right)^j \frac{\mu_j(q)}{j!}. \tag{27.27}$$

The transform is triangular and invertible. If $B_k = (q;q)_k A_k(q)$ and $Q = q^{-1}$, then

$$\frac{\mu_n(q)}{n!} = \left(\frac{1-q}{q}\right)^n \sum_{k=0}^{n} (-1)^k Q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_Q B_k. \tag{27.28}$$

Proof. Insert (27.18) into the explicit divided difference (2.7) with $c = 1$. The powers simplify by

$$q^{-jn+\binom{j+1}{2}+\binom{j+1}{2}} (1-q)^{-j} = q^{-j(n-j)} \left(\frac{q}{1-q}\right)^j,$$

and reciprocity $q^{-j(n-j)} \begin{bmatrix} n \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}}$ proves (27.27). For inversion, factor the transform into a diagonal matrix and the Gaussian Pascal matrix, and use the mutually inverse basis transforms in Theorem 3.1 with base Q; direct multiplication gives (27.28). □

Corollary 27.11 (Dyadic Gaussian and Mersenne forms). *For the Fabius function,*

$$A_n = \frac{1}{(1/2;1/2)_n} \sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 \frac{\mu_j}{j!} \tag{27.29}$$

$$= 2^{\binom{n+1}{2}} \sum_{j=0}^{n} \frac{(-1)^j \mu_j}{j!\,\mathfrak{M}_j \mathfrak{M}_{n-j}}. \tag{27.30}$$

Moreover

$$E_n - E_{n-1} = (-1)^n 2^{-\binom{n}{2}} A_n \qquad (n \geq 1). \tag{27.31}$$

Proof. Set $q = 1/2$ in (27.27). Then $q^{-1} = 2$ and $q/(1-q) = 1$. Use $\begin{bmatrix} n \\ j \end{bmatrix}_2 = \mathfrak{M}_n/(\mathfrak{M}_j \mathfrak{M}_{n-j})$ and (27.22) for the second form. Finally, the difference between two successive Newton partial sums at zero is $A_n \Phi_n(0) = (-1)^n 2^{-\binom{n}{2}} A_n$. □

27.8 Generating-function factorization

Define

$$B(z) = \sum_{r \geq 0} \frac{z^r}{\mathfrak{M}_r} = (-z/2;1/2)_\infty, \qquad H(z) = \sum_{r \geq 0} \frac{(-1)^r \mu_r}{r!\,\mathfrak{M}_r} z^r. \tag{27.32}$$

The equality for B is Euler's infinite q-binomial theorem. The two exact convolutions above are equivalent to

$$\sum_{n \geq 0} 2^{-\binom{n+1}{2}} A_n z^n = B(z) H(z), \tag{27.33}$$

$$\sum_{n \geq 0} (-1)^n E_n z^n = B(z) H(2z). \tag{27.34}$$

These identities exhibit the Fabius Newton coefficients as a q-Borel-type transform of the moment generating function multiplied by a universal Euler product. They may be useful for saddle-point analysis of A_n.

27.9 Exact divergence of the Fabius interpolants

The Fabius function is positive on $(0,1]$ and flat at 0: $F(x) = o(x^m)$ for every m. Therefore its dyadic samples satisfy the hypotheses of Theorem 26.4.

Corollary 27.12 (Exact convergence set for Fabius comb interpolation). *Let P_n interpolate F at $1, 1/2, \ldots, 2^{-n}$. Then*

$$P_n(x) \text{ converges as } n \to \infty \quad \Longleftrightarrow \quad x \in \{0, 1, 1/2, 1/4, \ldots\}. \tag{27.35}$$

At 0 the limit is 0 and obeys the quadratic bound (27.25); at a sampled point 2^{-m} the sequence is eventually equal to $F(2^{-m})$; at every other nonzero complex point it is unbounded along a subsequence.

Proof. Apply Theorem 26.4 with $a = 1$, $q = 1/2$, and $y_j = F(2^{-j})$. □

Corollary 27.13 (A rigorous upper scale for coefficient growth). *The Fabius Newton coefficients satisfy*

$$|A_n| \leq 2^{n^2/4 + O(n)}. \tag{27.36}$$

On the other hand, $\limsup |A_n|^{1/n} = \infty$.

Proof. Combine (27.31) with the bounds for E_n and E_{n-1} in Theorem 27.9. The lower qualitative statement is part of Theorem 26.4. □

Conjecture 27.14 (Quadratic coefficient exponent). *For the Fabius coefficients in* (27.26),

$$\lim_{n\to\infty} \frac{\log_2 |A_n|}{n^2} = \frac{1}{4}. \tag{27.37}$$

More generally, for $0 < q \le 1/2$,

$$\lim_{n\to\infty} \frac{\log |A_n(q)|}{n^2 \log(1/q)} = \frac{1}{4}. \tag{27.38}$$

A weaker and probably more robust version replaces the limit by a limsup. The Gaussian coefficient in (27.27) *is largest near* $j = n/2$, *where* $q^{-j(n-j)} = q^{-n^2/4+O(1)}$; *the open issue is to control alternating cancellation and the subquadratic moment factor. The order-*80 *companion computation recorded in* `assets/companion-evidence/geometric_comb_interpolation_report-3/NUMERICAL_SUMMARY.txt` *is consistent with* (27.37).

28 The Rvachev Fourier product through the same filter

28.1 Tail products along a geometric comb

Put

$$s_q(\xi) = \operatorname{sinc}_\pi(2(1-q)\xi), \qquad S_{q,k}(\xi) = \prod_{j=0}^{k-1} s_q(q^j\xi), \quad S_{q,0} = 1. \tag{28.1}$$

The infinite product (27.8) satisfies

$$\widehat{u}_{q\,2\pi}(q^k\xi) = \frac{\widehat{u}_{q\,2\pi}(\xi)}{S_{q,k}(\xi)}. \tag{28.2}$$

Thus values of $\widehat{u}_{q\,2\pi}$ on a geometric frequency comb are exactly the tails of its defining product.

Theorem 28.1 (Reciprocal sinc-product filter). *Let* $0 < q < 1$ *and suppose* $\widehat{u}_{q\,2\pi}(\xi) \neq 0$. *Define*

$$R_{n,q}(\xi) = \sum_{k=0}^{n} \frac{\lambda_{n,k}^{(q)}}{S_{q,k}(\xi)}. \tag{28.3}$$

Then

$$R_{n,q}(\xi) \longrightarrow \frac{1}{\widehat{u}_{q\,2\pi}(\xi)}. \tag{28.4}$$

More precisely,

$$\left| R_{n,q}(\xi) - \frac{1}{\widehat{u}_{q\,2\pi}(\xi)} \right| \le \frac{q^{\binom{n+1}{2}}}{|\widehat{u}_{q\,2\pi}(\xi)|(q;q)_n} e^{2\pi|\xi|} \frac{(2\pi|\xi|)^{n+1}}{(n+1)!}. \tag{28.5}$$

Proof. Apply the geometric filter to the entire function of t

$$h_\xi(t) = \widehat{u}_{q\,2\pi}(t\xi) = \mathbb{E}e^{-2\pi i t\xi Z_q}.$$

Since $h_\xi(0) = 1$, Theorem 26.1 already gives $\sum_k \lambda_{n,k} h_\xi(q^k) \to 1$. By (28.2), this sum is $\widehat{u}_{q\,2\pi}(\xi) R_{n,q}(\xi)$, proving the limit. For the quantitative bound, the Taylor coefficients of h_ξ satisfy $|c_m| \le (2\pi|\xi|)^m/m!$ because $|Z_q| \le 1$. Formula (14.9) with $A = 1$ and $\left[{n+r \atop n}\right]_q \le 1/(q;q)_n$ gives

$$|\widehat{u}_{q\,2\pi}(\xi) R_{n,q}(\xi) - 1| \le \frac{q^{\binom{n+1}{2}}}{(q;q)_n} \sum_{m=n+1}^{\infty} \frac{(2\pi|\xi|)^m}{m!}.$$

The exponential-tail bound $\sum_{m\ge n+1} t^m/m! \le e^t t^{n+1}/(n+1)!$ completes the proof. □

Remark 28.2 (Zeros and regularized reciprocals). The hypothesis excludes the union of zero lattices $2(1-q)q^j\xi \in \mathbb{Z}\setminus\{0\}$. Near a zero, the individual prefix reciprocals are badly conditioned. Multiplying by the known vanishing factor, or differentiating the filtered identity to reconstruct the finite part, should produce stable formulas for residues and zero multiplicities; see Problem 29.5.

28.2 Moment interpretation of the residual

Expanding the characteristic function gives

$$\widehat{u_{q\,2\pi}}(t\xi) = \sum_{m=0}^{\infty} \frac{(-2\pi\mathrm{i}\xi)^m}{m!} \mathbb{E} Z_q^m \, t^m. \tag{28.6}$$

Substitution into (14.9) expresses the error in (28.4) as a Gaussian transform of the centered moments. Odd moments vanish, so the actual leading term is the first even degree above n. The reciprocal product identity is therefore the Fourier counterpart of the Fabius moment transform: the same filter reads moments from the CDF comb and cancels moments in the characteristic-function comb.

28.3 The additive Thue–Morse companion

Let

$$X^{(N)} = \sum_{j=1}^{N} 2^{-j} U_j. \tag{28.7}$$

The CDF of a sum of independent uniforms with widths w_j is a truncated-power spline. For the dyadic widths, all subset sums form the uniform grid $m/2^N$ and their inclusion–exclusion signs become the Thue–Morse signs. Thus

$$\mathbb{P}(X^{(N)} \le x) = \frac{2^{N(N+1)/2}}{N!} \sum_{m=0}^{2^N-1} (-1)^{\mathrm{wt}_2(m)} \left(x - \frac{m}{2^N}\right)_+^N, \tag{28.8}$$

where $\mathrm{wt}_2(m)$ is the binary digit sum of m. Differentiating gives the density with power $N-1$ and factor $(N-1)!^{-1}$.

The Gaussian comb filter and the Thue–Morse spline are two different projections of the same hierarchy of scales:

- the Lagrange filter keeps the scale index j and uses only $n+1$ multiplicative nodes q^j; Gaussian coefficients encode its exact moment cancellations;
- the finite convolution expands all 2^N subsets of dyadic scales; subset sums become additive knots and parity produces Thue–Morse cancellation.

At $q = 1/2$, the products $(1-2^{-j})$, the Mersenne factors (2^j-1), Gaussian coefficients at bases $1/2$ and 2, and the binary subset lattice all coexist. This is why the Fabius/Rvachev system is not merely an example of the general geometric theory: it is the point at which multiplicative interpolation, additive splines, and binary arithmetic lock together exactly.

29 Research frontiers

The problems below are ordered from sharply formulated local questions to broader structural programs. None is needed for the proved results.

29.1 Asymptotics of the Fabius Newton coefficients

Conjecture 27.14 asks only for the leading quadratic exponent. A more informative target is a saddle-point expansion of (27.29) or (27.33). The Gaussian coefficient favors $j \approx n/2$, while $\mu_j/j!$ has its own lower-Lambert saddle and periodic correction. The interaction should produce a two-scale saddle problem.

Problem 29.1 (Full coefficient asymptotic). *Determine functions $c_2, c_1, c_{\log}$ and a bounded oscillatory term Ψ such that*

$$\log |A_n| = c_2 n^2 + c_1 n \log n + c_{\log} n + \Psi(\vartheta_n) + o(1), \tag{29.1}$$

with an explicit Lambert-type phase ϑ_n. Decide whether sign changes and near-cancellations are controlled by the same periodic function that occurs in the sharp Fabius endpoint asymptotic.

A first rigorous step would be matching upper and lower bounds $2^{n^2/4+o(n^2)}$. The upper bound is already in Corollary 27.13; a lower bound must prevent exponential-scale cancellation in the alternating central block.

29.2 Boundary-crossing formulas for $q > 1/2$

Problem 29.2 (The supercritical comb-moment law). *For $1/2 < q < 1$, derive a closed formula for $F_q(q^n)$ in terms of moments and finite affine words generated by*

$$x \mapsto x/q, \qquad x \mapsto (x - (1 - q))/q.$$

A satisfactory formula should specialize to (27.18) when no word crosses the reflected boundary, and should make the breakpoints in q explicit.

The number of admissible boundary words may change when polynomial relations among q-powers are crossed. This suggests a piecewise-algebraic structure related to beta-expansions, but with continuous uniform digits rather than Bernoulli digits.

29.3 Confluent limits and roots of unity

Problem 29.3 ($q \to 0$ confluent calculus). *Find a renormalized cardinal and Newton basis whose $q \to 0$ limit separates the infinitely many nodes collapsing at 0. Determine whether the correct limit is a triangular mixture of value, derivative, and tail-quotient functionals.*

Problem 29.4 (Cyclotomic geometric interpolation). *When q is a primitive mth root of unity, replace repeated nodes by a canonical Hermite data set and factor the resulting confluent Vandermonde matrix into cyclotomic analogues of P_q and $D_{a,q}$. The denominator-free operator $\prod_{j=1}^{n}(E_q - q^j)$ should survive as a periodic difference operator even when normalized Lagrange weights do not.*

These limits may connect the present theory to divided powers, quantum groups at roots of unity, and cyclic q-difference equations.

29.4 Regularization near Fourier zeros

Problem 29.5 (Filtered finite parts at sinc zeros). *Let ξ_0 be a zero of $\widehat{u_q}_{\,2\pi}$ of known multiplicity m. Construct from $R_{n,q}$ and its first m derivatives a sequence converging to the finite part of $1/\widehat{u_q}_{\,2\pi}$ at ξ_0, with a bound uniform in a shrinking neighborhood of the zero.*

Because every zero comes from explicit finite sinc factors, one can first divide out the known local factor and then apply the geometric filter to the nonvanishing tail. The main issue is numerical cancellation when several zero lattices overlap, as they do extensively at $q = 1/2$.

29.5 Exact Lebesgue functions below the finest node

For $x = aq^\alpha$ with real $\alpha > n$, the cardinal signs are $(-1)^{n-j}$, so the Lebesgue function is the explicit positive sum below. The formula extends continuously to $\alpha = n$, where the target is the finest node and the Lebesgue function equals 1.

$$\Lambda_n(aq^\alpha) = (q^{\alpha-n}; q)_{n+1} \sum_{j=0}^{n} \frac{q^{\binom{n-j+1}{2}}}{(1 - q^{\alpha-j})(q; q)_j (q; q)_{n-j}}. \tag{29.2}$$

At $\alpha = \infty$ it reduces to (16.1). It is natural to ask whether the finite sum has a single basic-hypergeometric closed form and whether Λ_n is monotone on $[0, aq^n]$.

Conjecture 29.6 (Maximum below the comb)**.** *For $0 < q < 1$ and $a > 0$,*

$$\max_{0 \le x \le aq^n} \Lambda_n(x) = \Lambda_n(0) = \frac{(-q;q)_n}{(q;q)_n}. \tag{29.3}$$

Low-degree symbolic and numerical checks support the claim, but a proof appears to require a variation-diminishing or total-positivity argument rather than termwise monotonicity.

29.6 Flatness-adapted approximation spaces

Polynomial interpolation fails off the comb for a structural reason. The basis must be changed, not merely evaluated more accurately. Classical geometric-knot spline constructions [59] provide one nearby approximation-theoretic model, while Rvachev atoms add exact C^∞ compact support and endpoint flatness.

Problem 29.7 (Rvachev-preconditioned comb interpolation)**.** *Construct finite-dimensional spaces generated by scale-dependent Rvachev atoms, for example*

$$u(2^j x - r), \qquad 0 \le j \le n,$$

that interpolate the Fabius dyadic data and incorporate endpoint flatness. Determine whether one can obtain convergence on $[0, 1]$ while retaining exact rational or algebraic coefficient formulas.

The Gaussian–Appell decoder (11.16) supplies an exact preconditioner for each finite polynomial interpolant, but it cannot remove the divergence because it reproduces the same polynomial. A genuinely new scheme must let the atom space grow in smoothness or scale, rather than merely re-expressing the space of polynomials of degree at most n.

Problem 29.8 (Growing Hermite data)**.** *At the tooth 2^{-j}, include derivatives through an order r_j that grows with j. Find the minimal growth of r_j for which the Hermite interpolants can converge to F on a nontrivial interval. Relate the threshold to the sharp derivative and endpoint asymptotics of F.*

29.7 Multivariate geometric meshes

Schoenberg's one-dimensional problem has a triangular-mesh extension due to Lee and Phillips [60]. The present operator and probability viewpoints suggest several further versions.

Problem 29.9 (Tensor and simplex filters)**.** *For nodes $(a_1 q_1^{j_1}, \ldots, a_d q_d^{j_d})$, develop tensor and total-degree Lagrange filters with explicit multivariate Mellin symbols. On a simplex-like geometric mesh, identify the correct multivariate Gaussian coefficients and complete homogeneous residuals. Apply them to joint geometric-uniform laws and tensor products of up-functions.*

The tensor case is immediate algebraically but may suffer severe anisotropic conditioning. The simplex case should involve q-multinomial coefficients and may admit a probabilistic interpretation through flags of coordinate subspaces.

29.8 Adaptive ratios and asymptotic optimization

The exact error factors make ratio selection an optimization problem. Small q gives powerful Gaussian suppression but spreads the first nodes; q near 1 resolves a local germ but causes the condition number (16.4) to explode.

Problem 29.10 (Optimal q under noise)**.** *Suppose the data $f(aq^k)$ have absolute or relative noise of known scale and f belongs to a specified analytic, Gevrey, or endpoint-asymptotic class. Minimize a rigorous sum of truncation and amplified-noise bounds over both q and n. Determine whether the optimum has a universal scaling law as the noise tends to zero.*

For the Fabius function, the superflat signal and the large off-comb coefficients make this a non-standard regularization problem; the optimum for evaluating $F(0)$ need not resemble the optimum for reconstructing $F(x)$ at $x > 0$.

29.9 Formal verification roadmap

The finite algebra is particularly well suited to Lean because most identities can be stated over a commutative ring before any division is introduced. A productive dependency order is:

1. prove the denominator-free nodal product and geometric Vandermonde factorization;
2. define the Gaussian Pascal matrix and prove the inverse basis pair of Theorem 3.1;
3. derive the barycentric denominator and general-target cardinal after adding field hypotheses;
4. define the dilation polynomial and prove the full polynomial identity (14.3);
5. obtain natural moment annihilation by evaluation, then extend to the analytic Mellin symbol in a complex-power layer;
6. formalize Jackson divided differences first for polynomials, then for functions with iterated q-derivatives;
7. prove the row-norm identity and Toeplitz regularity using summable reversed rows;
8. reuse the repository's geometric-uniform law to prove (27.18);
9. instantiate at $q = 1/2$ and connect to existing Fabius moments and Mersenne products;
10. formalize the superflat divergence theorem using a local power-series contradiction;
11. feed the existing sinc-product theorem through the analytic filter to obtain Theorem 28.1.

The distinction between denominator-free polynomial statements and normalized field statements should be maintained throughout. It will make the later root-of-unity and confluent developments substantially easier.

Part VI
Additive dyadic foundations

30 Common foundations

Everything in this volume rests on a small set of established facts about the Fabius–Rvachev system, all proved in the repository's primary exposition [34] and, where indicated, in its Lean development. This section fixes them in the unified notation; nothing here is claimed as new.

30.1 Normalization and the probability model

Let $U_1, U_2, \dots$ be independent random variables uniform on $[0, 1]$ and put

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j, \qquad X = 2Y - 1 = \sum_{\nu=1}^{\infty} 2^{-\nu} V_\nu, \tag{30.1}$$

where the $V_\nu = 2U_\nu - 1$ are uniform on $[-1,1]$. The Fabius function is the distribution function of Y,

$$F(x) = \mathbb{P}(Y \le x) \qquad (0 \le x \le 1), \tag{30.2}$$

and Rvachev's function up is the density of X: the unique even C^∞ function supported on $[-1,1]$ with $\int_{-1}^{1} \mathrm{up} = 1$. The two are folds of each other,

$$\mathrm{up}(x) = F(1-|x|) \quad (|x| \le 1), \qquad \text{so} \qquad \mathrm{up}(x) = 1 - F(x) \quad (0 \le x \le 1), \tag{30.3}$$

and the Fabius density is a rescaled copy of up:

$$\rho(x) := F'(x) = 2\,\mathrm{up}(2x-1). \tag{30.4}$$

The basic structural identities are the reflection symmetry and the delay-differential self-similarity

$$F(1-x) = 1 - F(x), \qquad F'(x) = 2F(2x) \quad (0 \le x \le \tfrac{1}{2}), \tag{30.5}$$

together with infinite flatness at the endpoints:

$$F^{(q)}(0) = F^{(q)}(1) = 0 \quad (q \ge 1), \qquad \mathrm{up}^{(q)}(\pm 1) = 0, \quad (\mathrm{up} - 1)^{(q)}(0) = 0 \quad (q \ge 0). \tag{30.6}$$

Equivalently $F(x) = \mathcal{O}(x^A)$ as $x \downarrow 0$ for every $A > 0$; this flatness is what makes negative powers harmless in Part I and what makes endpoint derivative data so tempting in Part II.

Iterating (30.5) through the signed extension F_{gl} of F (the solution of $F_{\mathrm{gl}}'(x) = 2F_{\mathrm{gl}}(2x)$ on all of $\mathbb{R}$) gives the derivative tower and the sharp norms

$$F_{\mathrm{gl}}^{(q)}(x) = 2^{\binom{q+1}{2}} F_{\mathrm{gl}}(2^q x), \qquad \left\| F^{(q)} \right\|_{\infty,[0,1]} = \left\| \mathrm{up}^{(q)} \right\|_{\infty,[-1,1]} = 2^{\binom{q+1}{2}} = 2^{q(q+1)/2}. \tag{30.7}$$

The tension between (30.6) and (30.7) — flat endpoints, superexponentially growing global derivative norms, nowhere analyticity — is the single most important analytic feature of the system for this volume: it disables every Taylor- or remainder-based convergence argument while enabling exact spectral and arithmetic ones.

30.2 The Fourier product and its integer zeros

With $\widehat{f}_{2\pi}(\xi) = \int f(x) \mathrm{e}^{-2\pi \mathrm{i} \xi x}\,\mathrm{d}x$ and $\operatorname{sinc}_\pi z = \sin(\pi z)/(\pi z)$,

$$\Phi(\xi) := \widehat{\mathrm{up}}_{2\pi}(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_\pi\left(\frac{\xi}{2^j}\right), \tag{30.8}$$

an even, real, rapidly decreasing entire function of exponential type. The characteristic function of the Fabius law is

$$\chi(\xi) := \mathbb{E}\,\mathrm{e}^{-2\pi \mathrm{i} \xi Y} = \widehat{\rho}_{2\pi}(\xi) = \mathrm{e}^{-\pi \mathrm{i} \xi}\,\Phi(\xi/2). \tag{30.9}$$

At a nonzero integer n, write $n = \sigma 2^v q$ with q odd, $\sigma = \pm 1$, $v = \nu_2(n)$. Exactly the factors $j = 0, \ldots, v$ of (30.8) vanish at n, each simply, so

$$\operatorname{ord}_{\xi = n} \Phi = \nu_2(n) + 1, \tag{30.10}$$

and the leading jet at the zero is explicit [34]: with $d = \nu_2(n) + 1$,

$$\Phi^{(d)}(n) = -\frac{d!}{n^d}\,\Phi\left(\frac{q}{2}\right). \tag{30.11}$$

Consequently, at the dual lattice of the level-m comb,

$$\operatorname{ord}_{\xi = M\ell} \chi = m + \nu_2(\ell) \qquad (\ell \ne 0), \tag{30.12}$$

a zero order that *grows with the level.* This linear growth of dual-lattice zero multiplicity is the engine of every exactness theorem in Part I; the half-integer values $\Phi(q/2)$, q odd, which are nonzero and satisfy the sign law of theorem 35.1, control every first failure.

30.3 Moments

Three interlocking moment families appear throughout.

Even Rvachev moments. Let

$$c_r := \int_{-1}^{1} x^{2r}\,\mathrm{up}(x)\,\mathrm{d}x = \mathbb{E}X^{2r}, \qquad c_0 = 1; \tag{30.13}$$

odd moments vanish by symmetry. Splitting off V_1 in (30.1) gives the exact rational recurrence

$$(2r+1)\big(2^{2r}-1\big)c_r = \sum_{j=0}^{r-1}\binom{2r+1}{2j}c_j, \tag{30.14}$$

with first values

$$c_1 = \tfrac{1}{9}, \quad c_2 = \tfrac{19}{675}, \quad c_3 = \tfrac{583}{59535}, \quad c_4 = \tfrac{132809}{32531625}, \quad c_5 = \tfrac{46840699}{24405225075}. \tag{30.15}$$

The moment generating function is the infinite product

$$\mathsf{M}(t) := \mathbb{E}\,\mathrm{e}^{tX} = \prod_{\nu=1}^{\infty}\frac{\sinh(t/2^{\nu})}{t/2^{\nu}}, \tag{30.16}$$

whose logarithmic expansion yields the even cumulants

$$\kappa_{2r} = \frac{2^{2r-1}B_{2r}}{r\,(2^{2r}-1)}, \qquad \mathbb{E}X^n = Y_n(0,\kappa_2,0,\kappa_4,\dots), \tag{30.17}$$

one of several places where complete Bell polynomials organize the dyadic structure.

Fabius-law moments. For $s \in \mathbb{C}$ put

$$d_s := \mathbb{E}Y^s = \int_0^1 x^s \rho(x)\,\mathrm{d}x. \tag{30.18}$$

Flatness of ρ at both endpoints makes d_s an *entire* function of s. For integers,

$$d_n = 2^{-n}\sum_{j=0}^{\lfloor n/2\rfloor}\binom{n}{2j}c_j, \qquad d_1 = \tfrac{1}{2}, \quad d_2 = \tfrac{5}{18}, \quad d_3 = \tfrac{1}{6}, \quad d_4 = \tfrac{143}{1350}, \quad d_5 = \tfrac{19}{270}, \tag{30.19}$$

and d_n also satisfies the self-contained recurrence

$$d_0 = 1, \qquad d_n = \frac{1}{2^n-1}\sum_{q=1}^{n}\binom{n}{q}\frac{d_{n-q}}{q+1} \quad (n \ge 1), \tag{30.20}$$

obtained by conditioning on U_1 in (30.1).

Fabius monomial integrals. For $p \in \mathbb{N}_0$, integration by parts gives

$$I_p := \int_0^1 x^p F(x)\,\mathrm{d}x = \frac{1-d_{p+1}}{p+1} \in \mathbb{Q}, \tag{30.21}$$

and the same formula defines the entire continuation

$$I(\alpha) := \int_0^1 x^{\alpha}F(x)\,\mathrm{d}x = \frac{1-d_{\alpha+1}}{\alpha+1} \qquad (\alpha \in \mathbb{C}), \tag{30.22}$$

the singularity at $\alpha = -1$ being removable with

$$I(-1) = -\left.\frac{\mathrm{d}}{\mathrm{d}s}d_s\right|_{s=0} = \mathbb{E}\big[-\log Y\big] = 0.7582400\dots. \tag{30.23}$$

All three families are tied together by one exponential generating function.

Theorem 30.1 (Moment EGF)**.** *With* M *as in* (30.16),

$$\sum_{p=0}^{\infty} I_p \frac{t^p}{p!} = \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t \mathsf{M}(t)}{\mathrm{e}^t - 1}, \tag{30.24}$$

and consequently

$$I_p = \frac{1}{p+1}\left[1 - \sum_{j=0}^{p+1}\binom{p+1}{j}\mathbb{E}X^j\, B_{p+1-j}(1)\right]. \tag{30.25}$$

Proof. Set $A(t) = \int_0^1 \mathrm{up}(x)\mathrm{e}^{tx}\,\mathrm{d}x$. Evenness of up gives $A(t) + A(-t) = \mathsf{M}(t)$, while the partition identity $\mathrm{up}(x) + \mathrm{up}(1-x) = 1$ on $[0,1]$ (a restatement of (30.3) and (30.5)) gives $A(t) + \mathrm{e}^t A(-t) = (\mathrm{e}^t - 1)/t$. Solving the linear system, $A(-t) = 1/t - \mathsf{M}(t)/(\mathrm{e}^t - 1)$. By (30.3), $\int_0^1 F(x)\mathrm{e}^{tx}\,\mathrm{d}x = \mathrm{e}^t A(-t)$, which is (30.24). Expanding $t\mathrm{e}^t/(\mathrm{e}^t-1) = \sum_n B_n(1)t^n/n!$ and $\mathsf{M}(t) = \sum_j \mathbb{E}X^j\, t^j/j!$ gives (30.25). □

The positive-half Rvachev moments follow from the same computation:

$$a_r := \int_0^1 x^r\,\mathrm{up}(x)\,\mathrm{d}x = \frac{1}{r+1} - I_r = \frac{d_{r+1}}{r+1}, \tag{30.26}$$

where the last form uses (30.21); equivalently the absolute Rvachev moments are

$$\int_{-1}^{1} |x|^{\alpha}\,\mathrm{up}(x)\,\mathrm{d}x = \frac{2d_{\alpha+1}}{\alpha+1} \qquad (\Re\alpha > -1), \tag{30.27}$$

an identity that continues meromorphically in section 38.3.

30.4 Exact dyadic values: one identity, three algorithms

Every sample used in this volume is an exact rational number. The repository's dyadic evaluator can be organized as a single convolution identity with three computationally different specializations; we state it with a self-contained probabilistic proof because both parts of this volume lean on the structure of the proof, not only on the statement.

Define the *Appell moment polynomial* and the *dyadic prefactor*

$$G_m(y) := \mathbb{E}\,(y-X)^m = \sum_{j=0}^{\lfloor m/2\rfloor}\binom{m}{2j}c_j\, y^{m-2j}, \qquad C_m := \frac{2^{-m(m+1)/2}}{m!}. \tag{30.28}$$

G_m is monic of degree m, even or odd according to the parity of m, with $G_m' = mP_{m-1}$ (an Appell sequence for the law of X).

Theorem 30.2 (Dyadic Appell–Thue–Morse convolution)**.** *For every* $m \in \mathbb{N}_0$ *and* $0 \le k \le M = 2^m$,

$$\boxed{F\!\left(\frac{k}{M}\right) = C_m \sum_{a=0}^{k-1}\varepsilon_a\, G_m\big(2(k-a)-1\big),} \tag{30.29}$$

the empty sum at $k=0$ *being zero.*

Proof. Split the series (30.1) for X after $n = m+1$ terms and rescale:

$$2^n X = \sum_{\nu=1}^{n} 2^{n-\nu}V_\nu + X',$$

where X' is an independent copy of X. The finite sum is an Irwin–Hall-type variable: a sum of independent uniforms on $[-2^{n-\nu}, 2^{n-\nu}]$, $\nu = 1, \dots, n$. Because the half-widths $1, 2, \dots, 2^{n-1}$ have subset sums exhausting $\{0, 1, \dots, 2^n - 1\}$ *without repetition*, inclusion–exclusion over the 2^n sign choices collapses to a Thue–Morse–signed spline: the density of the finite sum is

$$f_n(y) = \frac{2^{-n(n+1)/2}}{(n-1)!} \sum_{a=0}^{2^n-1} \varepsilon_a \left(y + 2^n - 1 - 2a\right)_+^{n-1}.$$

The self-similarity $2^{-n}\,\mathrm{up}(y/2^n) = \mathbb{E} f_n(y - X')$ of the scaled density, evaluated at the dyadic point $y = 2(k - M)$ and combined with $\mathrm{up}(k/M - 1) = F(k/M)$ from (30.3), turns each truncated power into the expectation $\mathbb{E}(2(k-a) - 1 - X')^m = G_m(2(k-a) - 1)$ precisely for $a \le k - 1$ (for larger a the truncated power vanishes because $|X'| \le 1$), and the constants assemble to C_m. □

Example 30.3. For $m = 2$: $P_2(y) = y^2 + \frac{1}{9}$, $C_2 = \frac{1}{16}$, and $F(1/4) = C_2\, P_2(1) = \frac{1}{16}\left(1 + \frac{1}{9}\right) = \frac{5}{72}$; for $m = 3$ the complete row is

$$\left(F(k/8)\right)_{k=0}^{8} = \left(0, \tfrac{1}{288}, \tfrac{5}{72}, \tfrac{73}{288}, \tfrac{1}{2}, \tfrac{215}{288}, \tfrac{67}{72}, \tfrac{287}{288}, 1\right).$$

Reflection $F(k/M) + F((M-k)/M) = 1$ is visible and is used as a standing self-check in every experiment of this volume.

Three specializations of (30.29) serve different purposes.

(a) Streaming the complete row. Expanding $G_m(2z+1) = 2^m \sum_j \binom{m}{j} d_j\, z^{m-j}$ (the binomial transform connecting (30.28) to the Fabius-law moments (30.19)) turns (30.29) into

$$F\left(\frac{a}{2^m}\right) = \frac{2^{-\binom{m}{2}}}{m!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{j=0}^{m} \binom{m}{j} d_j\, (a - 1 - h)^{m-j}. \tag{30.30}$$

Algorithm 30.4 (Streaming exact comb)**.** Maintain the signed prefix power sums $S_k(a) = \sum_{h=0}^{a-1} \varepsilon_h (a - 1 - h)^k$ for $0 \le k \le m$ via

$$S_0(a+1) = S_0(a) + \varepsilon_a, \qquad S_k(a+1) = \sum_{q=0}^{k} \binom{k}{q} S_q(a) \quad (k \ge 1),$$

and output $F(a/2^m) = 2^{-\binom{m}{2}} (m!)^{-1} \sum_j \binom{m}{j} d_j S_{m-j}(a)$. The complete level-$m$ row costs $\mathcal{O}(Mm^2)$ exact rational operations. An equivalent sparse form multiplies the coefficient array of $\sum_{d \ge 1} G_m(2d - 1) z^d$ successively by $(1 - z), (1 - z^2), \dots, (1 - z^{2^{m-1}})$; each factor is one backward difference, for $\mathcal{O}(mM)$ operations total.

(b) Single values by bit recursion. With $V_n := F(2^{-n})$, conditioning on the leading binary digit gives

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}} V_k}{(n-k+1)!}, \qquad F\left(\frac{a}{2^e}\right) = \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} \left(\frac{q}{2^e}\right)^j - F\left(\frac{q}{2^e}\right), \tag{30.31}$$

where $2^b \le a < 2^{b+1}$, $r = e - b$, $q = a - 2^b$. This computes one value in $\mathcal{O}(e^2)$ operations and underlies the classical arithmetic of Arias de Reyna and Haugland [6, 7].

(c) Symbolic convolution. Identity (30.29) itself, read as a coefficient convolution, is the input to the generating-function calculus of section 32.

All experiments behind this volume implement at least two of the three routes and compare them exactly; see section 41 and section 48.

Part VII
Additive dyadic comb sums and quadrature

31 The dyadic combs

31.1 Left, right, and trapezoidal Fabius combs

For $\alpha \in \mathbb{C}$ define the *punctured left Fabius comb*

$$L_{m,\alpha} := h \sum_{k=1}^{M-1} \left(\frac{k}{M}\right)^{\alpha} F\left(\frac{k}{M}\right), \tag{31.1}$$

with the real logarithm on $(0,1]$. For $\alpha = p \in \mathbb{N}_0$ this agrees with the ordinary left Riemann sum of $x^p F(x)$ including $k = 0$, since $F(0) = 0$; the puncture is the correct convention for negative and nonintegral powers, where flatness of F at 0 makes even that convention innocuous. At fixed m, $\alpha \mapsto L_{m,\alpha}$ is a finite exponential sum, hence *entire*.

The right and trapezoidal combs differ only through the endpoint $F(1) = 1$:

$$R_{m,\alpha} = L_{m,\alpha} + h, \qquad T_{m,p} = L_{m,p} + \frac{h}{2}. \tag{31.2}$$

The trapezoidal normalization is the natural one for Fourier aliasing (the periodic extension of $x^p F(x)$ has a unit jump at the integers, and its Fourier series converges to the midpoint value there); the right rule R produces the cleanest single-formula closed forms (corollary 34.5); the left rule L matches the generated tables of the absorbed sources. All statements below convert freely via (31.2).

31.2 Shifted and midpoint combs

For a phase $\theta \in \mathbb{R}$ define

$$L^{(\theta)}_{m,\alpha} := h \sum_{\substack{k\in\mathbb{Z} \\ 0<k+\theta<M}} \left(\frac{k+\theta}{M}\right)^{\alpha} F\left(\frac{k+\theta}{M}\right). \tag{31.3}$$

A dyadic phase is a residue-class extraction from a finer level: the midpoint comb, for instance, satisfies the exact two-scale identity

$$L^{(1/2)}_{m,p} = 2L_{m+1,p} - L_{m,p}, \tag{31.4}$$

because the level-$(m+1)$ nodes split into the level-m nodes and the midpoints. More generally:

Proposition 31.1 (Dyadic phase as a roots-of-unity filter). *Let $\theta = a/2^s$ with $0 \le a < 2^s$, put $D = m + s$ and $\omega = \exp(2\pi i/2^s)$, and let A_D be the row polynomial of theorem 32.1 at level D. Then, with the spectral Euler power (32.11),*

$$L^{(a/2^s)}_{m,\alpha} = \frac{2^{-D\alpha}}{M\,2^s} \sum_{r=0}^{2^s-1} \omega^{-ar}\,(\mathcal{D}^{\alpha} A_D)(\omega^r). \tag{31.5}$$

Proof. The character projector $2^{-s}\sum_r \omega^{r(k-a)}$ selects the indices $k \equiv a \pmod{2^s}$ from the level-D row, and $\mathcal{D}^{\alpha}$ implements the power weight coefficientwise. □

Every standard one-dimensional quadrature convention on the dyadic comb is therefore an exact linear functional of finitely many objects $(\mathcal{D}^{\alpha} A_D)(\zeta)$ at 2-power roots of unity ζ. This is the first indication that the *row generating function* is the right primitive object; we develop it next.

31.3 Rvachev combs

The corresponding two-sided objects, studied in sections 36 and 38, are the shifted monomial combs

$$Q_{m,p}(\theta) := h \sum_{k\in\mathbb{Z}} \big(h(k+\theta)\big)^p \,\mathrm{up}\big(h(k+\theta)\big), \tag{31.6}$$

finite sums because $\operatorname{supp}\mathrm{up} = [-1,1]$, and the punctured absolute combs

$$U_m(\alpha) := h \sum_{k\in\mathbb{Z}\setminus\{0\}} |kh|^\alpha \,\mathrm{up}(kh). \tag{31.7}$$

On $[0,1]$ the fold (30.3) links them to the Fabius combs; with the pure power comb $\Pi_{m,\alpha} := h\sum_{k=1}^{M-1}(k/M)^\alpha = M^{-\alpha-1}[\zeta(-\alpha)-\zeta(-\alpha,M)]$,

$$U_m(\alpha) = 2\big(\Pi_{m,\alpha} - L_{m,\alpha}\big) \qquad (\alpha\in\mathbb{C}), \tag{31.8}$$

so every Fabius-comb theorem has an immediate Rvachev shadow and conversely. This elementary bridge carries surprising weight: in section 36 it converts the exact Euler–Maclaurin collapse into exact quadrature statements, and in section 38 it transports the Hurwitz-zeta calculus to the two-sided combs.

32 The rational row generating function

32.1 Pole collapse

Encode a complete level-m sample row in the ordinary generating function

$$A_m(z) := \sum_{k=0}^{M-1} F(k/M)\, z^k. \tag{32.1}$$

Let $\mathcal{D} = z\,\mathrm{d}/\mathrm{d}z$ be the Euler operator and recall the finite Thue–Morse product $\Theta_m(z) = \prod_{j=0}^{m-1}(1-z^{2^j}) = \sum_{a=0}^{M-1}\varepsilon_a z^a$.

Theorem 32.1 (Complete sample-row generating function)**.** *Define*

$$\mathscr{F}_m(z) := C_m\,\Theta_m(z)\,G_m(2\mathcal{D}-1)\,\frac{z}{1-z}. \tag{32.2}$$

Then $\mathscr{F}_m$ is a rational function with a single simple pole at $z=1$, and

$$\mathscr{F}_m(z) = \sum_{k\ge 0} D_m^{\mathrm{cl}}(k) z^k, \qquad D_m^{\mathrm{cl}}(k) = \begin{cases} F(k/M), & 0\le k\le M,\\ 1, & k\ge M.\end{cases} \tag{32.3}$$

In particular

$$\boxed{A_m(z) = \mathscr{F}_m(z) - \frac{z^M}{1-z}} \tag{32.4}$$

is a polynomial of degree at most $M-1$.

Proof. The kernel $\sum_{r\ge1} G_m(2r-1)z^r = G_m(2\mathcal{D}-1)\,z/(1-z)$ has coefficient $G_m(2r-1)$ at z^r, so the coefficient of z^k in (32.2) is exactly the convolution (30.29); this proves the coefficient identity for $0\le k\le M$. For the pole structure, each application of $(2\mathcal{D}-1)$ to a rational function with denominator $(1-z)^{r+1}$ raises the denominator order by one, so $G_m(2\mathcal{D}-1)z/(1-z)$ has denominator $(1-z)^{m+1}$. On the other hand,

$$\Theta_m(z) = (1-z)^m \prod_{j=0}^{m-1} \bigl(1 + z + \cdots + z^{2^j - 1}\bigr), \tag{32.5}$$

so the finite Thue–Morse mask cancels exactly m of the $m+1$ poles, leaving one simple pole; the numerator degree count $(M-1)+(m+1)-m = M$ shows that the coefficients of $\mathscr{F}_m$ are constant from index M on. The coefficient at index M is $F(1)=1$, so the constant tail is 1, and subtracting the tail $z^M/(1-z)$ leaves the polynomial A_m. □

Remark 32.2 (What the pole collapse says). Prouhet cancellation is usually quoted as $\sum_a \varepsilon_a\, a^r = 0$ for $r < m$. Theorem 32.1 is its generating-function upgrade: the Thue–Morse mask annihilates the entire order-m polar part generated by the degree-m moment kernel, and the surviving simple pole encodes precisely the saturation $F \equiv 1$ past the right endpoint. The low levels are

$$\mathscr{F}_1(z) = \frac{z(z+1)}{2(1-z)}, \tag{32.6}$$

$$\mathscr{F}_2(z) = \frac{z(z+1)(z+5)(5z+1)}{72(1-z)}, \tag{32.7}$$

$$\mathscr{F}_3(z) = \frac{z(z+1)^3(z^2+1)(z^2+16z+1)}{288(1-z)}. \tag{32.8}$$

After removing the initial factor z, the numerators are reciprocal polynomials — an exact reflection of $F(k/M) + F((M-k)/M) = 1$, made precise by the functional equation

$$A_m(z) + z^M A_m(1/z) = \frac{z\,(1 - z^{M-1})}{1-z}, \tag{32.9}$$

proved in Appendix H. A factorization theory for these row numerators (cyclotomic multiplicities, the limiting law of the noncyclotomic zeros) is posed as problem 42.1.

32.2 Comb sums as Euler derivatives of the row

Corollary 32.3 (All powers from one polynomial). *For every $p \in \mathbb{N}_0$,*

$$L_{m,p} = M^{-p-1}\,(\mathcal{D}^p A_m)(1), \tag{32.10}$$

and with the spectral Euler power

$$\mathcal{D}^\alpha\Bigl(\sum_{k\ge 0} a_k z^k\Bigr) := \sum_{k\ge 1} k^\alpha a_k z^k \qquad (\alpha \in \mathbb{C}) \tag{32.11}$$

the same formula $L_{m,\alpha} = M^{-\alpha-1}(\mathcal{D}^\alpha A_m)(1)$ holds for every complex exponent.

Proof. By definition, the coefficient of z^k in A_m is $F(k/M)$. The Euler operator satisfies $\mathcal{D}^p z^k = k^p z^k$ for natural p, so evaluation at $z=1$ gives $M^{-p-1}\sum_k k^p F(k/M) = L_{m,p}$. The spectral definition of $\mathcal{D}^\alpha$ makes the identical coefficientwise calculation for every $\alpha \in \mathbb{C}$; the sum is finite, so no convergence interchange is involved. □

Together with proposition 31.1 this reduces *every* comb convention of section 31 to coefficient extractions from A_m — a picture completed by the closed forms of the next section, which eliminate the 2^m-term row itself.

33 Closed forms for nonnegative integer powers

33.1 The Bernoulli–Thue–Morse formula

Inserting the convolution (30.29) into (31.1) and summing the resulting power sums by Faulhaber's formula $\sum_{r=1}^{K} r^q = \big[B_{q+1}(K+1) - B_{q+1}(1)\big]/(q+1)$ gives an exact finite formula at every level and degree.

Theorem 33.1 (Exact closed form for every m, p)**.** *For $p \in \mathbb{N}_0$,*

$$L_{m,p} = C_m M^{-p-1} \sum_{a=0}^{M-1} \varepsilon_a \, H_{m,p}(a), \tag{33.1}$$

where

$$\begin{aligned} H_{m,p}(a) = &\sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{r=0}^{p} \binom{p}{r} a^{p-r} \\ &\times \sum_{s=0}^{m-2j} \binom{m-2j}{s} 2^s (-1)^{m-2j-s} \frac{B_{r+s+1}(M-a) - B_{r+s+1}(1)}{r+s+1}. \end{aligned} \tag{33.2}$$

Proof. By (30.29),

$$M^{p+1} C_m^{-1} L_{m,p} = \sum_{a=0}^{M-2} \varepsilon_a \sum_{r=1}^{M-1-a} (a+r)^p \, G_m(2r-1).$$

Expand $(a+r)^p$ binomially, expand $(2r-1)^d$ for each monomial of G_m, and apply Faulhaber with $K = M - 1 - a$. □

Formula (33.1) is manifestly rational and is one of the three independent exact routes used in the verification suite (section 41). It still contains a complete 2^m-term signed block; the next theorem shows that only a bounded number of its coefficients can ever matter.

33.2 Bell–Prouhet collapse of the signed block

Define the signed Thue–Morse power moments

$$T_{m,n} := \sum_{a=0}^{M-1} \varepsilon_a \, a^n, \tag{33.3}$$

whose exponential generating function is the finite product $\sum_n T_{m,n} t^n / n! = \Theta_m(\mathrm{e}^t) = \prod_{j=0}^{m-1} (1 - \mathrm{e}^{2^j t})$.

Theorem 33.2 (Bell formula for complete Thue–Morse power blocks)**.** *Let*

$$\lambda_{m,1} = \frac{M-1}{2}, \qquad \lambda_{m,2r} = \frac{B_{2r}}{2r} \, \frac{2^{2rm}-1}{2^{2r}-1} \quad (r \geq 1), \qquad \lambda_{m,2r+1} = 0 \quad (r \geq 1). \tag{33.4}$$

Then

$$\Theta_m(\mathrm{e}^t) = (-1)^m 2^{m(m-1)/2} \, t^m \exp\Big(\sum_{r \geq 1} \lambda_{m,r} \frac{t^r}{r!} \Big), \tag{33.5}$$

whence $T_{m,n} = 0$ for $n < m$ (Prouhet cancellation) and, for $s \geq 0$,

$$\boxed{T_{m,m+s} = (-1)^m 2^{m(m-1)/2} \, \frac{(m+s)!}{s!} \, Y_s(\lambda_{m,1}, \ldots, \lambda_{m,s}).} \tag{33.6}$$

Proof. Factor $1 - \mathrm{e}^x = -x\,\mathrm{e}^{x/2}\,\dfrac{\sinh(x/2)}{x/2}$ and use the classical expansion $\log\bigl[(\mathrm{e}^x - 1)/x\bigr] = x/2 + \sum_{r\geq 1} \frac{B_{2r}}{2r}\frac{x^{2r}}{(2r)!}$, summed over $x = 2^j t$, $0 \leq j < m$: the prefactors contribute $(-1)^m 2^{0+1+\cdots+(m-1)} t^m$, the linear terms sum to $(M-1)t/2$, and each even term is the finite geometric sum in (33.4). Comparing $\exp(\sum_r x_r t^r/r!) = \sum_s Y_s(x_1, \ldots, x_s)t^s/s!$ with the definition of $T_{m,n}$ gives (33.6). □

Corollary 33.3 (Only $p+2$ signed moments survive)**.** *The polynomial $H_{m,p}$ of (33.2) has degree at most $m+p+1$, so*

$$\boxed{L_{m,p} = C_m M^{-p-1} \sum_{s=0}^{p+1} \bigl[a^{m+s}\bigr] H_{m,p}(a)\, T_{m,m+s},} \tag{33.7}$$

with $T_{m,m+s}$ given by (33.6). The full 2^m-term Thue–Morse block collapses to at most $p+2$ Bell-polynomial terms, uniformly in m.

Proof. In (33.2) the Bernoulli polynomial has degree $r+s+1$ in a after the substitution $a \mapsto M - a$; multiplied by a^{p-r} the total degree is at most $p+s+1 \leq m+p+1$. Monomials of degree below m are annihilated by (33.6). □

A computational dichotomy

For fixed m and many degrees p, stream the row once (algorithm 30.4, $\mathcal{O}(mM)$ operations) and read off every $L_{m,p}$ from corollary 32.3. For fixed p and large or symbolic m, use the Bell collapse: its signed part has constant length $p+2$, and all dependence on the exponentially long block is compressed into the explicit jets $\lambda_{m,r}$ — a closed form in which m appears only through 2^m. The two routes, together with the direct sample sum, are compared exactly in section 41.

Remark 33.4 (Eulerian form of the kernel)*.* For symbolic work in z rather than t, the kernel of theorem 32.1 expands through negative-index polylogarithms,

$$\sum_{d\geq 1} G_m(2d-1)z^d = \sum_{j=0}^{\lfloor m/2\rfloor} \binom{m}{2j} c_j \sum_{i=0}^{m-2j} \binom{m-2j}{i} 2^i (-1)^{m-2j-i}\, \mathrm{Li}_{-i}(z), \tag{33.8}$$

each $\mathrm{Li}_{-i}(z) = z\,\mathcal{A}_i(z)/(1-z)^{i+1}$ rational with Eulerian numerator $\mathcal{A}_i$. This makes (32.2) effective in any computer algebra system without operator calculus.

34 The exact Euler–Maclaurin collapse

For a smooth integrand, the Euler–Maclaurin formula is an asymptotic expansion with a remainder that is small but almost never zero. For $x^p F(x)$ on the dyadic comb the situation is remarkably better: the expansion *terminates and becomes an exact identity* as soon as the level clears an explicit threshold linear in p. This section proves the collapse, identifies the sharp-looking threshold, and prepares the exact description of the first failure.

34.1 Trapezoidal aliasing through the characteristic function

Put $g_p(x) = x^p F(x)$ on $[0,1]$ and $\widehat{g_p}_{\mathrm{FS}}[n] = \int_0^1 g_p(x)\mathrm{e}^{-2\pi i n x}\,\mathrm{d}x$. The periodic extension of g_p has a unit jump at the integers, and its Fourier series converges there to the midpoint value, which is exactly the trapezoidal endpoint convention:

$$T_{m,p} = \sum_{\ell\in\mathbb{Z}} \widehat{g_p}_{\mathrm{FS}}[M\ell]\,, \tag{34.1}$$

the zero-frequency term being I_p. Every nonzero coefficient is a finite expression in the Fabius characteristic function (30.9).

Lemma 34.1 (Fourier coefficients of $x^pF(x)$). *For $n \in \mathbb{Z} \setminus \{0\}$,*

$$\widehat{g_p}_{\,\mathrm{FS}}[n] = \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi \mathrm{i} n)^{j+1}} \left(\frac{\chi^{(p-j)}(n)}{(-2\pi \mathrm{i})^{p-j}} - 1 \right). \tag{34.2}$$

In particular, for $p = 0$,

$$\widehat{g_0}_{\,2\pi}(\xi) = -\frac{\mathrm{e}^{-2\pi \mathrm{i}\xi}}{2\pi \mathrm{i}\xi} + \frac{\chi(\xi)}{2\pi \mathrm{i}\xi} \qquad (\xi \neq 0), \tag{34.3}$$

valid for all real ξ, not only integers.

Proof. Since $F(x) = \mathbb{P}(Y \le x)$, Fubini gives the layer representation $\widehat{g_p}_{\,\mathrm{FS}}[n] = \mathbb{E}\int_Y^1 x^p \mathrm{e}^{-2\pi \mathrm{i} n x}\,\mathrm{d}x$. Integrating by parts $p+1$ times, using $\mathrm{e}^{-2\pi \mathrm{i} n} = 1$ at the fixed endpoint,

$$\int_Y^1 x^p \mathrm{e}^{-2\pi \mathrm{i} n x}\,\mathrm{d}x = \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi \mathrm{i} n)^{j+1}} \Big(Y^{\,p-j}\mathrm{e}^{-2\pi \mathrm{i} n Y} - 1 \Big),$$

and $\mathbb{E}\big[Y^r \mathrm{e}^{-2\pi \mathrm{i} n Y}\big] = \chi^{(r)}(n)/(-2\pi \mathrm{i})^r$. For (34.3), one integration by parts of $\int_0^1 F\,\mathrm{e}^{-2\pi \mathrm{i}\xi x}\,\mathrm{d}x$ with $F' = \rho$, $\widehat{\rho}_{\,2\pi} = \chi$ suffices. □

Splitting (34.2) into its "-1" part (a pure boundary term) and its χ part (a spectral term) splits (34.1) accordingly. The boundary part sums by Euler's evaluation of $\zeta(2r)$ into the classical Bernoulli corrections; the spectral part is an exact, absolutely convergent alias remainder.

Theorem 34.2 (Euler–Maclaurin with exact alias remainder). *For every $m \ge 0$ and $p \in \mathbb{N}_0$,*

$$L_{m,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}}\, h^{2r} + \mathcal{R}_{m,p}, \tag{34.4}$$

where

$$\mathcal{R}_{m,p} = \sum_{\ell \in \mathbb{Z}\setminus\{0\}} \sum_{j=0}^{p} \frac{p^{\underline{j}}}{(2\pi \mathrm{i} M\ell)^{j+1}} \frac{\chi^{(p-j)}(M\ell)}{(-2\pi \mathrm{i})^{p-j}}. \tag{34.5}$$

Proof. Insert (34.2) into (34.1). In the boundary part only odd $j = 2r-1$ survives the symmetric ℓ-sum, and

$$-\frac{p^{\underline{2r-1}}}{(2\pi \mathrm{i} M)^{2r}} \sum_{\ell \neq 0} \ell^{-2r} = \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}} M^{-2r}$$

by $\zeta(2r) = (-1)^{r+1} B_{2r} (2\pi)^{2r}/(2(2r)!)$. Finally $T_{m,p} = L_{m,p} + h/2$. □

34.2 The collapse and its threshold

Define the parity threshold

$$\tau(0) := 0, \qquad \tau(p) := 2\lceil p/2 \rceil = \begin{cases} p, & p \text{ even}, \\ p+1, & p \text{ odd}, \end{cases} \quad p \ge 1; \tag{34.6}$$

equivalently, $m \ge \tau(p)$ if and only if $p \le 2\lfloor m/2 \rfloor$.

Theorem 34.3 (Exact Euler–Maclaurin collapse). *If $m \ge \tau(p)$, then $\mathcal{R}_{m,p} = 0$; that is,*

$$\boxed{L_{m,p} = I_p - \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}}\, h^{2r}.} \tag{34.7}$$

Equivalently, the level-m comb agrees exactly with its finite Euler–Maclaurin polynomial through degree m when m is even, and through degree $m-1$ when m is odd.

Proof. For $m = 0$, $p = 0$ the identity reads $L_{0,0} = 0 = I_0 - \frac{1}{2}$, which is the symmetry value $I_0 = \int_0^1 F = \frac{1}{2}$. Let $m \geq 1$. By (30.12), χ vanishes at $M\ell$ to order $m + \nu_2(\ell) \geq m$. If $p < m$, every derivative order $p - j \leq p$ appearing in (34.5) is below the zero order, so the remainder vanishes term by term.

It remains to treat $p = m$ for even m. Aliases with even ℓ still vanish ($\nu_2(\ell) \geq 1$), and at odd ℓ only the $j = 0$ term, carrying the derivative order exactly m, can survive. From $\chi(\xi) = \mathrm{e}^{-\pi \mathrm{i} \xi}\Phi(\xi/2)$ and the vanishing of all lower derivatives of $\Phi(\cdot/2)$ at $M\ell$, the Leibniz rule leaves a single term,

$$\chi^{(m)}(M\ell) = \mathrm{e}^{-\pi \mathrm{i} M\ell}\, 2^{-m}\, \Phi^{(m)}\left(\frac{M\ell}{2}\right), \qquad \mathrm{e}^{-\pi \mathrm{i} M\ell} = 1 \ \ (m \geq 1).$$

Since Φ is even, $\Phi^{(m)}$ is even for even m, so $\chi^{(m)}(M\ell)$ is then even in ℓ, whereas the accompanying factor $(2\pi \mathrm{i} M\ell)^{-1}$ is odd; the aliases ℓ and $-\ell$ cancel in pairs. For odd m no cancellation is forced, and theorem 35.1 shows the surviving sum is generically nonzero. This is precisely the range $m \geq \tau(p)$. □

Remark 34.4 (Why this is not ordinary polynomial exactness). The summand contains the non-polynomial factor F. The theorem does not assert that a fixed rule integrates polynomials against a fixed weight; it asserts that the *nonzero Fourier aliases of the particular integrand* are annihilated by the growing zeros of χ at the dual lattice. It is a spectral cancellation theorem for a family of ordinary Riemann sums — a Strang–Fix phenomenon [12, 13] in which the reproduced polynomial degree grows with the refinement level because each refinement contributes a fresh vanishing sinc factor.

34.3 One-formula closed form and low-degree examples

Because the finite Euler–Maclaurin polynomial is itself a Bernoulli polynomial in disguise, the collapse compresses to a single formula.

Corollary 34.5 (Exact right-comb formula)**.** *For $p \in \mathbb{N}_0$ and $m \geq \tau(p)$,*

$$\boxed{R_{m,p} = \frac{h^{p+1} B_{p+1}(M+1) - d_{p+1}}{p+1}.} \tag{34.8}$$

Proof. Expanding $B_{p+1}(M+1)$ around $M = 1/h$,

$$\frac{h^{p+1} B_{p+1}(M+1) - 1}{p+1} = \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} \underline{p^{2r-1}} h^{2r},$$

which combined with $I_p = (1 - d_{p+1})/(p+1)$ and $R_{m,p} = L_{m,p} + h$ is (34.7). □

The Bernoulli polynomial encodes the pure Faulhaber sum, and d_{p+1} carries all the Fabius arithmetic: the exact right comb is *the power sum minus one Fabius-law moment.* The first cases of (34.7), generated and exactly verified by the companion script of the absorbed sums package (`assets/companion-evidence/Fabius_Dyadic_Comb_Sums_Report_Package/`), are:

p	I_p	threshold $\tau(p)$	exact $L_{m,p}$ for $m \geq \tau(p)$
0	$\frac{1}{2}$	0	$\frac{1}{2} - \frac{h}{2}$
1	$\frac{13}{36}$	2	$\frac{13}{36} - \frac{h}{2} + \frac{1}{12}h^2$
2	$\frac{5}{18}$	2	$\frac{5}{18} - \frac{h}{2} + \frac{1}{6}h^2$
3	$\frac{1207}{5400}$	4	$\frac{1207}{5400} - \frac{h}{2} + \frac{1}{4}h^2 - \frac{1}{120}h^4$
4	$\frac{251}{1350}$	4	$\frac{251}{1350} - \frac{h}{2} + \frac{1}{3}h^2 - \frac{1}{30}h^4$
5	$\frac{22661}{142884}$	6	$\frac{22661}{142884} - \frac{h}{2} + \frac{5}{12}h^2 - \frac{1}{12}h^4 + \frac{1}{252}h^6$
6	$\frac{16427}{119070}$	6	$\frac{16427}{119070} - \frac{h}{2} + \frac{1}{2}h^2 - \frac{1}{6}h^4 + \frac{1}{42}h^6$
7	$\frac{63446897}{520506000}$	8	$\frac{63446897}{520506000} - \frac{h}{2} + \frac{7}{12}h^2 - \frac{7}{24}h^4 + \frac{1}{12}h^6 - \frac{1}{240}h^8$
8	$\frac{7096441}{65063250}$	8	$\frac{7096441}{65063250} - \frac{h}{2} + \frac{2}{3}h^2 - \frac{7}{15}h^4 + \frac{2}{9}h^6 - \frac{1}{30}h^8$

Table 4: Exact terminating Euler–Maclaurin identities. Each row is an identity of rational functions of $h = 2^{-m}$, valid for every level $m \geq \tau(p)$; below the threshold the defect of section 35 appears.

The complete exactness pattern through level 8 was verified in exact rational arithmetic (three independent routes; see section 41) and matches $\{(m,p) : m \geq \tau(p)\}$ perfectly — including the *first* failures, which is the subject of the next section.

35 The first defect and spectral Dirichlet values

Below the threshold $\tau(p)$ the alias remainder (34.5) is nonzero — but it is not an opaque error term. At the first failing degree it is a single, explicitly evaluable Dirichlet series over the half-integer values of the sinc product, and the exactness of everything else turns that series into a machine for producing closed forms.

35.1 The odd-level defect series

Theorem 35.1 (First odd-level defect). *Let $m \geq 1$ be odd. Then*

$$\boxed{\mathcal{R}_{m,m} = \frac{2\,(-1)^{(m+1)/2}\,m!}{(2\pi)^{m+1}\,2^{m(m+1)}}\,D(m+1), \qquad D(s) := \sum_{\substack{q\geq 1\\ q \text{ odd}}} \frac{\Phi(q/2)}{q^s},} \tag{35.1}$$

where D is entire in s by the rapid decay of Φ. Moreover the summands obey the Thue–Morse sign law

$$\operatorname{sgn}\Phi\left(\frac{2t+1}{2}\right) = \varepsilon_t \qquad (t \geq 0). \tag{35.2}$$

Proof. For $p = m$ odd, the proof of theorem 34.3 leaves in (34.5) only the $j = 0$ term at odd ℓ, with

$$\chi^{(m)}(M\ell) = 2^{-m}\,\Phi^{(m)}\big(2^{m-1}\ell\big) = -\,2^{-m}\,\frac{m!}{(2^{m-1}\ell)^m}\,\Phi\left(\frac{\ell}{2}\right)$$

by the leading-jet formula (30.11) at the order-m zero $2^{m-1}\ell$. Substituting into (34.5) and pairing ℓ with $-\ell$ (both terms are equal for odd m) gives (35.1) after collecting powers of 2. The sign law (35.2) follows by counting sign changes of the real even product Φ along $[0, q/2]$: each factor $\operatorname{sinc}_\pi(\xi/2^j)$ changes sign at the odd multiples of 2^j, and the total number of integer zeros (with multiplicity) in $(0, q/2)$ crossed by the product has the parity of the binary digit sum of $t = (q-1)/2$; alternatively, peel one factor as in (35.3) below and induct on t. □

Two structural identities frame D. Peeling the $j = 0$ factor of the product gives the *half-scale relation*

$$\Phi\left(\frac{q}{2}\right) = \operatorname{sinc}_\pi\left(\frac{q}{2}\right)\Phi\left(\frac{q}{4}\right) = \frac{2\,(-1)^{(q-1)/2}}{\pi q}\,\Phi\left(\frac{q}{4}\right) \qquad (q \text{ odd}), \tag{35.3}$$

which connects $D(s)$ to a twisted quarter-integer series and is the natural starting point for the functional-equation program of problem 35.7. And since $\Phi(q/2) \to 0$ rapidly as $q \to \infty$ while the $q = 1$ term is fixed,

$$D(s) = \Phi(\tfrac{1}{2}) + \Phi(\tfrac{3}{2})\,3^{-s} + \mathcal{O}(5^{-s}) \qquad (s \to \infty), \tag{35.4}$$

with $\Phi(\frac{1}{2}) = 0.5537712\ldots > 0$ and $\Phi(\frac{3}{2}) = -0.0462022\ldots < 0$: the values $D(s)$ increase to the limit $\Phi(1/2)$ from below.

35.2 Exact rational multiples of powers of π

The defect identity can be read in both directions. Read forward, it evaluates the first comb defect; read backward, it evaluates the Dirichlet series, because everything else in (34.4) is an explicitly computable rational number.

Theorem 35.2 (Spectral rationality). *For every $r \geq 1$,*

$$\boxed{\frac{D(2r)}{\pi^{2r}} \in \mathbb{Q},} \qquad \textit{effectively: } D(2r) = (-1)^r \frac{2^{(2r)^2}}{2\,(2r-1)!} \mathcal{R}_{2r-1,\,2r-1}\, \pi^{2r}, \tag{35.5}$$

where $\mathcal{R}_{2r-1,2r-1} = L_{2r-1,2r-1} - \big[$the finite Euler–Maclaurin polynomial of (34.7)$\big]$ is an exact rational number computable from the level-$(2r-1)$ comb.

Proof. Solve (35.1) for $D(m+1)$ with $m = 2r-1$; the prefactor is

$$\frac{(-1)^{(m+1)/2}(2\pi)^{m+1}2^{m(m+1)}}{2\,m!} = \frac{(-1)^r 2^{(m+1)^2}\pi^{m+1}}{2\,m!}.$$

The defect on the right of (35.1) is rational by theorem 33.1 and (30.21). □

Carrying this out with exact arithmetic gives the first four values:

$$\boxed{D(2) = \frac{\pi^2}{18}, \qquad D(4) = \frac{23\,\pi^4}{4050}, \qquad D(6) = \frac{1543\,\pi^6}{2679075}, \qquad D(8) = \frac{1196113\,\pi^8}{20494923750}.} \tag{35.6}$$

Numerically $0.548311\ldots$, $0.553187\ldots$, $0.553707\ldots$, $0.553764\ldots$, converging to $\Phi(1/2) = 0.553771\ldots$ as (35.4) predicts. The same route through levels 9 and 11 gives two further exact values,

$$D(10) = \frac{2045676559\,\pi^{10}}{345944065438125}, \qquad D(12) = \frac{1158575464019117\,\pi^{12}}{1933714893977851359375}, \tag{35.7}$$

numerically $0.5537704925659628\ldots$ and $0.5537711889156489\ldots$. The values $D(2)$ and $D(4)$ were derived in one absorbed source and re-derived here; $D(6)$ through $D(12)$ were computed for this consolidation by an independent implementation of (35.5) (bit-recursion evaluator (30.31), exact `Fraction` arithmetic) that first reproduced all eight first-defect values of table 5 and the complete exactness matrix through level 8, and each closed form was checked against a direct high-precision evaluation of the series to 15 digits.

m	first failing degree p	exact defect $\mathcal{R}_{m,p}$
1	1	$-1/144$
2	3	$-97/1382400$
3	3	$23/22118400$
4	5	$9671/11985978654720$
5	5	$-1543/767102633902080$
6	7	$-9903527/14650941422775671193600$
7	7	$1196113/37506410042305718255616000$
8	9	$21035141003/590082134533477495427314445451264000$

Table 5: Exact first defects through level eight (from the absorbed sums package, independently reproduced for this volume). At odd m the first failing degree is $p = m$ and the defect is (35.1); at even m it is $p = m + 1$.

Remark 35.3 (Even levels and the 2-adic layer structure). At even m the degree-$(m+1)$ defect mixes two contributions: the $j \in \{0, 1\}$ jets of the odd aliases (order-m zeros, now differentiated $m+1$ times) and the leading jet of the $\nu_2(\ell) = 1$ alias layer (order-$(m+1)$ zeros). Each piece is a Dirichlet series of the same half-integer type with explicitly computable local coefficients — the general jet calculus is proposition 36.5 — so the even-level defects in table 5 admit, in principle, the same closed-form treatment. Making that two-layer decomposition arithmetically explicit is part of conjecture 35.6.

35.3 Sharpness and positivity

The exactness matrix verified through level 8 shows the first defect occurring exactly at $p = \tau^{-1}$-boundary, and every entry of table 5 is nonzero. Of the two central problems stated here in the absorbed sources' equivalent forms, the second (spectral positivity) has since been settled in full by the first-harmonic bound of theorem 37.11, which also proves the odd-m half of the first; the even-m half of the sharp-threshold conjecture remains open.

Conjecture 35.4 (Even-level sharp threshold)**.** *For every even $m \geq 2$, the first degree beyond the proved exactness range is genuinely inexact: $\mathcal{R}_{m,m+1} \neq 0$. Equivalently, the two surviving alias-jet contributions described in remark 37.13 never cancel. This is verified exactly for even $m \leq 8$. The corresponding odd-level statement is already corollary 37.12, not part of this conjecture.*

Remark 35.5 (Spectral positivity — resolved). $D(2r) > 0$ holds for *every* $r \geq 1$, and equivalently $\operatorname{sgn} \mathcal{R}_{m,m} = (-1)^{(m+1)/2}$ for all odd m: although the summands alternate in sign by (35.2), the first harmonic dominates the whole tail uniformly in r — theorem 37.11 below proves $D(2r) \geq \Phi(\frac{1}{2}) - (\frac{\pi^2}{8} - 1) > \frac{4}{9} - \frac{1}{4} > 0$ by two elementary estimates. The exact values (35.6) and (35.7) remain the sharp quantitative data.

Conjecture 35.6 (Arithmetic closure of higher alias jets)**.** *After grouping by 2-adic valuation and Fourier parity, every parity-compatible jet series $\sum_{q\,\mathrm{odd}} \Phi^{(j)}(q/2)\, q^{-s}$-combination occurring in an integer comb defect $\mathcal{R}_{m,p}$ lies in $\mathbb{Q}\,\pi^{s-j}$. (Rationality of each* total *defect is a theorem, by theorem 33.1; the conjecture asserts that the valuation/jet decomposition is arithmetically diagonal, with no cross-series cancellation needed.)*

Problem 35.7 (Functional equations for D)**.** *Use the half-scale relation* (35.3) *to split odd residues modulo powers of two and search for a Mahler-type functional equation for the entire function $D(s)$. The exact values* (35.6) *provide unusually rigid interpolation data; a closed recurrence would in particular give a second, structural proof of the positivity theorem theorem 37.11.*

36 Shifted Rvachev quadrature and the alias calculus

The two-sided comb sums of up upgrade the Fabius collapse to a quadrature statement that is uniform in an *arbitrary real shift*, with a complete finite calculus for everything the exactness misses.

36.1 Exact polynomial quadrature for every shift

Recall the shifted monomial comb $Q_{m,p}(\theta)$ of (31.6). Shifted Poisson summation for $f_p(x) = x^p \operatorname{up}(x)$, together with $\widehat{f_p}_{2\pi}(\xi) = (-2\pi \mathrm{i})^{-p} \Phi^{(p)}(\xi)$, gives the exact absolutely convergent representation

$$Q_{m,p}(\theta) = \int_{-1}^{1} x^p \operatorname{up}(x)\,\mathrm{d}x + \Big(-\frac{1}{2\pi\mathrm{i}}\Big)^p \sum_{\ell \in \mathbb{Z}\setminus\{0\}} \mathrm{e}^{2\pi\mathrm{i}\ell\theta}\, \Phi^{(p)}(M\ell). \tag{36.1}$$

Theorem 36.1 (Shifted dyadic exactness)**.** *For every level $m \in \mathbb{N}_0$, every $\theta \in \mathbb{R}$, and every polynomial P with* $\deg P \leq m$*,*

$$\boxed{h \sum_{k\in\mathbb{Z}} P\big(h(k+\theta)\big)\, \operatorname{up}\big(h(k+\theta)\big) = \int_{-1}^{1} P(x) \operatorname{up}(x)\,\mathrm{d}x.} \tag{36.2}$$

Formal crosswalk. The full polynomial statement is kernel-verified as `Fabius.tsum_shifted_polynomial_eq_integral` in `PolynomialCombExactness.lean` (with the finite support of the comb samples as `Fabius.finite_support_comb`), by linearity from the monomial core, which is kernel-verified: kernel-verified: for every real shift θ and every $p \leq m$, `Fabius.tsu` `m_shifted_monomial_eq_integral_real` in `MonomialCombExactness.lean` proves $\sum_{k\in\mathbb{Z}}(\theta+k)^p \operatorname{up}\big(2^{-m}(\theta+k)\big) = \int_{\mathbb{R}} x^p \operatorname{up}(2^{-m}x)\,\mathrm{d}x$ (the display above is this identity multiplied by h^{p+1} and substituted, and extends to polynomials by linearity). The proof is the aliasing argument of this section made formal: `MonomialCombFourier.lean` packages $x^p \operatorname{up}(2^{-m}x)$ as a Schwartz map with $(-2\pi\mathrm{i})^p \widehat{f}_{p\,2\pi} = (\widehat{f}_{0\,2\pi})^{(p)}$, `MonomialCombDerivatives.lean` evaluates $(\widehat{f}_{0\,2\pi})^{(p)}$ through the entire transform via the real–complex derivative bridge, `MonomialCombAlias.lean` kills every nonzero alias by the integer-zero order theorem ($\operatorname{ord}_{2^m\ell}\Phi = v_2(2^m\ell)+1 > m$), and Poisson summation for Schwartz maps collapses the comb to the zero-frequency integral. The odd centered moments (corollary 36.2) vanish at every real scale with no level restriction by evenness alone (`Fabius.tsum_odd_pow_mul_rvachevUp` in `DyadicCombTrapezoid.lean`), and the trapezoidal $p=0$ exactness holds on every uniform grid (`Fabius.fabius_trapezoid_exact`).

Proof. By (30.10), $\operatorname{ord}_{\xi=M\ell}\Phi = m+\nu_2(\ell)+1 \ge m+1$ for $\ell\neq 0$, so every alias term in (36.1) vanishes for $p\le m$; linearity extends monomials to polynomials. □

Corollary 36.2. *For every m and θ, the samples form an exact partition of unity, $h\sum_k \operatorname{up}(h(k+\theta)) = 1$. For the centered comb ($\theta=0$),*

$$h\sum_k (kh)^{2r}\operatorname{up}(kh) = c_r \quad (2r\le m), \qquad h\sum_k (kh)^{2r+1}\operatorname{up}(kh) = 0 \quad (\textit{all } m), \tag{36.3}$$

and for $0<2r\le m$ the half-grid identity $h\sum_{k=0}^{M}(kh)^{2r}\operatorname{up}(kh) = c_r/2$ follows by evenness. Tensor products give exact cubature on dyadic lattices in any dimension for all coordinatewise degrees at most m.

Proof. Apply theorem 36.1 first to the constant polynomial and then to x^{2r} when $2r\le m$. On the centered comb, odd moments cancel termwise because up is even. The even half-grid sum is half of the centered full-grid sum: the two halves pair under $k\mapsto -k$ and the $k=0$ term vanishes for $r>0$. Finally, both the compactly supported density and a coordinatewise polynomial factor over Cartesian products, so repeated application of the one-dimensional identity proves the tensor statement. □

Remark 36.3 (Base-b extension). Only the alignment of the sinc zeros with the dual lattice is used. For an integer base $b\ge 2$, the compactly supported density up_b with $\widehat{\operatorname{up}_b}_{2\pi}(\xi) = \prod_{j\ge 0}\operatorname{sinc}_\pi(\xi/b^j)$ (an infinite convolution of uniform laws at geometrically decreasing scales) satisfies the same theorem on the comb $h\mathbb{Z}+\theta$, $h=b^{-m}$: at $b^m\ell$ the factors $j=0,\dots,m$ all vanish. The finer 2-adic valuation structure below, and the Thue–Morse arithmetic of Part II, are special to $b=2$.

36.2 The finite 2-adic alias hierarchy

For $p>m$ the alias sum survives, but only through finitely many 2-adic shells.

Proposition 36.4 (Valuation hierarchy). *For every $p\in\mathbb{N}_0$,*

$$Q_{m,p}(\theta) - \int_{-1}^{1} x^p \operatorname{up}(x)\,\mathrm{d}x = \Big(-\frac{1}{2\pi\mathrm{i}}\Big)^p \sum_{v=0}^{p-m-1} \sum_{\substack{q\in\mathbb{Z}\\ q \text{ odd}}} \mathrm{e}^{2\pi\mathrm{i}2^v q\theta}\,\Phi^{(p)}\big(2^{m+v}q\big), \tag{36.4}$$

the outer sum being empty for $p\le m$.

Proof. Write $\ell = 2^v q$ with q odd; the zero order at $M\ell$ is $m+v+1$, so $\Phi^{(p)}(M\ell)=0$ unless $v\le p-m-1$. □

Increasing the degree by one activates exactly one new shell. This finite stratification is what makes the defects *computable*: each shell is an odd-integer Dirichlet series built from a finite jet of the zero-stripped product, and the jets close under a Bell–Bernoulli calculus.

Proposition 36.5 (Bell–Bernoulli alias jets). *Let $n \neq 0$ be an integer, $d = \nu_2(n) + 1$, $n = \sigma 2^{d-1} q$ with q odd, $\sigma = \pm 1$. Factoring the d vanishing sinc terms of (30.8) at n gives the exact local identity*

$$\Phi(n+w) = w^d\, \Psi_n(w), \qquad \Psi_n(w) = -\frac{P_d^{\mathrm{sinc}}(w)}{(n+w)^d}\, \Phi\Big(\frac{q}{2} + \frac{\sigma w}{2^d}\Big), \qquad P_d^{\mathrm{sinc}}(w) := \prod_{j=0}^{d-1} \mathrm{sinc}_\pi\Big(\frac{w}{2^j}\Big), \tag{36.5}$$

so Ψ_n is smooth and nonvanishing near $w = 0$ with $\Psi_n(0) = -\Phi(q/2)/n^d$. Let $\lambda_s(n) = (\log \Psi_n)^{(s)}(0)$. Then for every $r \geq 0$

$$\Phi^{(d+r)}(n) = \frac{(d+r)!}{r!}\, \Psi_n(0)\, Y_r\big(\lambda_1(n), \ldots, \lambda_r(n)\big), \tag{36.6}$$

with

$$\lambda_s(n) = \big(\log P_d^{\mathrm{sinc}}\big)^{(s)}(0) + \frac{\sigma^s}{2^{ds}}\, (\log \Phi)^{(s)}\Big(\frac{q}{2}\Big) + (-1)^s\, \frac{d\,(s-1)!}{n^s}, \tag{36.7}$$

where the finite sinc block contributes nothing for odd s and, for $s = 2j$,

$$\big(\log P_d^{\mathrm{sinc}}\big)^{(2j)}(0) = -\frac{2^{2j-1}\,|B_{2j}|}{j}\, \pi^{2j}\, \frac{1 - 2^{-2jd}}{1 - 2^{-2j}}. \tag{36.8}$$

Proof. Equation (36.6) is the Leibniz expansion of $w^d \Psi_n(w)$ at $w = 0$ combined with the Bell formula $\Psi_n^{(r)}(0) = \Psi_n(0) Y_r(\lambda_1, \ldots, \lambda_r)$ for derivatives of $\exp(\log \Psi_n)$. Logarithmic differentiation of the three factors of Ψ_n gives (36.7); for the finite sinc block use $\log \mathrm{sinc}_\pi z = -\sum_{j \geq 1} \frac{2^{2j-1}|B_{2j}|}{j\,(2j)!} (\pi z)^{2j}$ and sum the geometric progression over the d scales. ☐

Together, (36.4) and (36.6) are a fully algorithmic exact description of $Q_{m,p}(\theta)$ for every natural degree: finitely many active shells, and every active derivative a finite Bell polynomial in explicit local data — Bernoulli logarithmic jets of the finite sinc block, logarithmic jets of Φ at half-integers, and inverse powers of n.

36.3 First failure and phase superconvergence

Theorem 36.6 (Degree-$(m+1)$ error). *For $p = m+1$,*

$$Q_{m,m+1}(\theta) - \int_{-1}^{1} x^{m+1}\, \mathrm{up}(x)\, \mathrm{d}x = -\Big(-\frac{1}{2\pi \mathrm{i}}\Big)^{m+1} \frac{(m+1)!}{2^{m(m+1)}} \sum_{\substack{\ell \in \mathbb{Z} \\ \ell \text{ odd}}} \frac{\Phi(|\ell|/2)}{\ell^{m+1}}\, \mathrm{e}^{2\pi \mathrm{i} \ell \theta}. \tag{36.9}$$

As a trigonometric series in θ this is not identically zero (its first Fourier coefficient is a nonzero multiple of $\Phi(1/2)$), so the uniform degree m of theorem 36.1 is sharp.

Proof. Only the shell $v = 0$ survives in (36.4), and the leading jet (30.11) at the order-$(m+1)$ zero $M\ell = 2^m \ell$ gives $\Phi^{(m+1)}(M\ell) = -(m+1)!\,(M\ell)^{-(m+1)} \Phi(\ell/2)$. ☐

Corollary 36.7 (Phase superconvergence). *At the first failing degree $p = m+1$:*

1. *if m is even, the centered and half-shifted combs are still exact: $Q_{m,m+1}(0) = Q_{m,m+1}(1/2) = 0$;*
2. *if m is odd, the quarter-shifted combs are still exact: $Q_{m,m+1}(1/4) = Q_{m,m+1}(3/4) = \int x^{m+1}$ up.*

Proof. For even m, $m+1$ is odd, and pairing $\ell \leftrightarrow -\ell$ in (36.9) produces $\sin(2\pi\ell\theta)$-type terms vanishing at $\theta \in \{0, \frac{1}{2}\}$ (the integral is also zero by symmetry). For odd m, the paired sum is proportional to $\cos(2\pi\ell\theta)$, which vanishes for every odd ℓ at $\theta \in \{\frac{1}{4}, \frac{3}{4}\}$. ☐

Exact rational experiments through $m = 4$ (companion script of the absorbed quadrature report, `assets/companion-evidence/fabius_dyadic_comb_report_final/`) illustrate both the failure and the special phases:

m	$p = m+1$	shift θ	$Q_{m,p}(\theta) - \int x^p\,\mathrm{up}$
1	2	0	$1/72$
1	2	$1/4$	0
2	3	0	0
2	3	$1/4$	$-4643/11059200$
3	4	0	$-23/5529600$
3	4	$1/4$	0
4	5	0	0
4	5	$1/4$	$4966069/383551316951040$

Table 6: First-failure errors at the two distinguished phases; every tested shift was exact for all degrees $p \le m$.

Remark 36.8 (Phase classification — resolved). Modulo 1, the parity-forced zeros of corollary 36.7 are the only superconvergent shifts at degree $m+1$ — not merely for every level of a parity, but *at each individual level*, with no accidental extra zeros: theorem 37.10 below proves that the first-failure series has the sign of its first harmonic wherever the forced trigonometric factor is nonzero, via the relative dominance lemma 37.8.

Remark 36.9 (Crosswalk: the first-failure series is a polynomial-synthesis defect). The trigonometric series (36.9) reappears, up to the explicit factor $(-1)^{m+1}2^{m(m+1)}$, as the Fourier residue of the canonical nonpolynomial defect of the local up-spline space in the companion volume *Exact Polynomial Synthesis from Rvachev Up-Atoms* (`representations/Up_Polynomial_Synthesis/`), whose defect–quadrature duality also converts the exact phase-error values of table 6 into the defect special values $\mathcal{E}_1(0) = \frac{1}{18}$, $\mathcal{E}_3(0) = -\frac{23}{1350}$ and, through the Dirichlet values (35.6), $\mathcal{E}_5(0) = \frac{1543}{119070}$, $\mathcal{E}_7(0) = -\frac{1196113}{65063250}$.

36.4 Absolute integer powers: the exact zeta correction

Write $U_{m,p} := U_m(p)$ for the punctured absolute comb (31.7) at integer degree. The Fabius collapse upgrades polynomial exactness to a closed form with a single correction term.

Theorem 36.10 (Absolute integer powers). *For $p \in \mathbb{N}_0$ and $m \ge \tau(p)$,*

$$\boxed{U_{m,p} = \frac{2d_{p+1}}{p+1} + 2\zeta(-p)\,h^{p+1}.} \tag{36.10}$$

In particular: for positive even p the comb equals the integral (30.27) exactly*; for odd p the sole error is the rational correction $2\zeta(-p)h^{p+1} = -B_{p+1}h^{p+1} \cdot \frac{2}{p+1}$; and for $p = 0$ the punctured comb is $1 - h$.*

Proof. Let $p \ge 1$. By evenness and the fold (30.3), with $a = M - k$,

$$U_{m,p} = 2h \sum_{k=1}^{M} (kh)^p F(1 - kh) = 2 \sum_{j=0}^{p} (-1)^j \binom{p}{j} R_{m,j}$$

(the substitution produces the left-rule sums, which differ from the $R_{m,j}$ by h each; the correction $2h \sum_j (-1)^j \binom{p}{j} = 0$ vanishes precisely because $p \ge 1$). The same alternating binomial combination of the integrals I_j equals the absolute moment $2d_{p+1}/(p+1)$. Insert the exact formula (34.8) for each $R_{m,j}$ (all thresholds $\tau(j) \le \tau(p)$ are met). The alternating difference annihilates every Euler–Maclaurin term of degree $< p$ in h-power count; the only surviving boundary term arises at odd p and equals $2\zeta(-p)h^{p+1}$ by $\zeta(-p) = -B_{p+1}/(p+1)$. The case $p = 0$ is the partition of unity minus the central sample $h\,\mathrm{up}(0) = h$. □

Once the threshold is reached, the first instances are exact identities:

$$U_{m,0} = 1 - h, \quad U_{m,1} = \frac{5}{18} - \frac{h^2}{6}, \quad U_{m,2} = \frac{1}{9}, \quad U_{m,3} = \frac{143}{2700} + \frac{h^4}{60}, \quad U_{m,4} = \frac{19}{675}. \tag{36.11}$$

Consistency bridge

The corpus's self-weighted shifted quadrature (samples of up used as weights) and the unweighted Fabius combs of section 34 are distinct constructions, but the elementary relation (31.8) makes them two faces of one mechanism: exact up-moments are *equivalent* to the cancellation between an ordinary Faulhaber sum and the exact Euler–Maclaurin polynomial of $L_{m,p}$. The zeta correction in (36.10) is precisely the odd-degree mismatch of the two boundary conventions, and its vanishing at even p is $\zeta(-2r) = 0$.

36.5 Finite arithmetic form

For exact rational work at fixed level, insert the convolution (30.29) directly: with $L = M - a - 1$,

$$U_{m,p} = \frac{2C_m}{M^{p+1}} \sum_{a=0}^{M-2} \varepsilon_a \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{u=0}^{p} (-1)^u \binom{p}{u} L^{p-u} \sum_{v=0}^{m-2j} \binom{m-2j}{v} 2^v \frac{B_{u+v+1}(L) - B_{u+v+1}}{u+v+1}, \tag{36.12}$$

an identity of rational numbers for every m, p. When $m \geq \tau(p)$ it collapses to (36.10), producing a family of nontrivial finite Thue–Morse–Bernoulli identities; below threshold it evaluates the defect layers of proposition 36.4 exactly.

37 Bernoulli periodization, composite meshes, and the exhaustion closure

The collapse of section 34 was stated for monomial weights on the dyadic endpoint comb. This section, an editorial merge of three independently written same-topic reports — the tenth-wave Euler–Maclaurin report, its eleventh-wave twin (*Exhaustion, Euler–Maclaurin, and Spectral Exactness for Fabius–Rvachev Monomial Integrals*), and the twelfth-wave *Bernoulli–Ruffa Phase–Resolution Calculus* (overlapping theorems stated once, each source's unique layers kept, shared constants verified identical) — extends the same spectral mechanism in several independent directions: general polynomial weights at *every real shift*; *arbitrary integer meshes*, classified by their 2-adic valuation; a telescoping "generalized exhaustion" hierarchy whose infinite tail closes into finitely many geometric terms past the threshold, with an exact derivative-free Richardson extraction; and the composite-mesh extension of the shifted Rvachev quadrature of section 36. As a by-product, the first-harmonic bound proved here settles the spectral-positivity conjecture of section 35 in full (theorem 37.11).

Throughout this section $M \geq 1$ is an *arbitrary* integer mesh (not necessarily a power of two) and $s = \nu_2(M)$; the dyadic case $M = 2^m$ recovers the earlier sections.

37.1 The Bernoulli corrector and the shifted rule

Let $B_k(x)$ denote the Bernoulli polynomials, $te^{xt}/(e^t - 1) = \sum_k B_k(x) t^k / k!$. For a polynomial P of degree d put

$$H_P(x) = \sum_{j=0}^{d} \frac{P^{(j)}(1)}{(j+1)!} B_{j+1}(x), \qquad R_P(x) = P(x)F(x) - H_P(x). \tag{37.1}$$

Lemma 37.1 (Endpoint-jump interpolation). *$H_P^{(k)}(1) - H_P^{(k)}(0) = P^{(k)}(1)$ for every $k \geq 0$; since F is flat at both endpoints, $(PF)^{(k)}(1) - (PF)^{(k)}(0) = P^{(k)}(1)$ as well, so every derivative of R_P matches at 0 and 1 and R_P extends to a C^∞ one-periodic function.*

Proof. $B_n' = nB_{n-1}$ and $B_n(1) = B_n(0)$ except $B_1(1) - B_1(0) = 1$; in the kth derivative of (37.1) only $j = k$ contributes an endpoint difference. ☐

Definition 37.2 (Corrected shifted Fabius rule). For $\theta \in \mathbb{R}$ and $x_a = (a+\theta)/M$,

$$\mathcal{A}_{M,\theta}[P] = \frac{1}{M}\sum_{a=0}^{M-1} P(x_a)F(x_a) - \sum_{j=0}^{d} \frac{P^{(j)}(1)\,B_{j+1}(\theta)}{(j+1)!\,M^{j+1}}. \tag{37.2}$$

By the Bernoulli multiplication identity $M^{-1}\sum_{a<M} B_k((a+\theta)/M) = M^{-k}B_k(\theta)$ and $\int_0^1 B_k = 0$ $(k \geq 1)$,

$$\mathcal{A}_{M,\theta}[P] = \frac{1}{M}\sum_{a=0}^{M-1} R_P(x_a), \qquad \int_0^1 R_P = \int_0^1 PF =: I_P, \tag{37.3}$$

so the corrected rule is exactly a finite sample average of a smooth periodic function, and the finite-grid alias identity gives the pure alias error

$$\begin{aligned} \mathcal{A}_{M,\theta}[P] - I_P &= \sum_{\ell\neq 0} \mathrm{e}^{2\pi \mathrm{i}\ell\theta}\,\widehat{R_P}_{\,\mathrm{FS}}[M\ell]\,, \\ \widehat{R_P}_{\,\mathrm{FS}}[q] &= \sum_{j=0}^{d} \frac{1}{(2\pi \mathrm{i}q)^{j+1}} \int_0^1 P^{(j)}(x)\,\rho(x)\,\mathrm{e}^{-2\pi \mathrm{i}qx}\,\mathrm{d}x \quad (q \neq 0), \end{aligned} \tag{37.4}$$

the coefficient formula by iterating $R_P' = R_{P'} + P\rho - P(1)$ (from the Appell relation $H_P' = P(1) + H_{P'}$) and periodic integration by parts. For $P = x^p$, $\theta = 0$, $M = 2^m$, formula (37.4) is exactly the χ-form coefficient (34.2), and (37.2) rearranges into the trapezoidal identity (34.4): the corrector H_P packages the same boundary jets that there summed to the Bernoulli polynomial part.

37.2 The exact all-mesh alias–Bernoulli identity

Before any threshold, the alias error has a *closed finite* probabilistic form. Writing $\widetilde{B}_r(t) = B_r(\{t\})$ for the periodized Bernoulli function and

$$C_{a,r}(M,\theta) := \mathbb{E}\big[Y^a\,\widetilde{B}_r(MY-\theta)\big] \tag{37.5}$$

for the Fabius variable Y, the layer-cake representation $F(x) = \mathbb{E}\,\mathbf{1}_{Y\leq x}$ and Faulhaber's formula for the random threshold sum $\sum_{a\geq\lceil MY-\theta\rceil}(a+\theta)^n$ give, for *every* integer $M \geq 1$, every $\theta \in [0,1]$, and every $n \geq 0$,

$$\boxed{\mathcal{A}^{\mathrm{raw}}_{M,\theta}[x^n] - I_n = \sum_{r=1}^{n+1} \frac{\binom{n}{r-1}}{r\,M^r}\Big[B_r(\theta) - (-1)^r C_{n+1-r,\,r}(M,\theta)\Big],} \tag{37.6}$$

where $\mathcal{A}^{\mathrm{raw}}$ is the uncorrected grid average; equivalently, the corrected rule's error (37.4) equals $\sum_r \frac{(-1)^{r+1}}{r!M^r} n^{\underline{r-1}}\, C_{n+1-r,r}(M,\theta)$. This is an exact identity, not an expansion: the Bernoulli addition and reflection formulas expand $B_{n+1}(M+\theta)$ and $B_{n+1}(MY+1-\{MY-\theta\})$, the $r=0$ term reproduces $I_n = (1-\mathbb{E}Y^{n+1})/(n+1)$, and no limit is taken. The Fourier series of $\widetilde{B}_r$ together with $\mathbb{E}[Y^a \mathrm{e}^{2\pi \mathrm{i}tY}] = \chi^{(a)}(-t)/(-2\pi \mathrm{i})^a$ shows that $C_{a,r}(M,\theta) = 0$ whenever $a < \nu_2(M)$ (the correlation annihilation form of theorem 37.3); below the threshold, (37.6) compresses the full infinite Fourier information into finitely many random periodic Bernoulli moments — the form best suited to exact arithmetic and to formalization.

37.3 Composite-mesh termination and the reflection reduction

Theorem 37.3 (2-adic termination on arbitrary meshes)**.** *If $s = \nu_2(M) \geq d+1$, then $\mathcal{A}_{M,\theta}[P] = I_P$ for* every *shift $\theta \in \mathbb{R}$.*

Proof. Each integral in (37.4) is a combination of derivatives of χ of order at most $d-j \leq d$ at $q = M\ell$, and χ vanishes there to order $\nu_2(M\ell) = s + \nu_2(\ell) \geq d+1$ (the order of Φ at $M\ell/2$, by (30.10)). □

The odd part of M is irrelevant: meshes $M = 2^s q$ with q odd are exactly as good as 2^s. (The condition is sufficient, not necessary — symmetry and phases can force extra cancellation, which is precisely what the next theorem exploits.)

Theorem 37.4 (Reflection reduction at the endpoint phase)**.** *Write $P^*(x) = P(1-x)$ and split $P = P_{\mathrm{s}} + P_{\mathrm{a}}$ into symmetric and antisymmetric parts about $x = \frac{1}{2}$. For the endpoint rule ($\theta = 0$),*

$$\mathcal{A}_{M,0}[P] + \mathcal{A}_{M,0}[P^*] = \int_0^1 P(x)\,\mathrm{d}x = I_P + I_{P^*}, \tag{37.7}$$

so the quadrature defect vanishes on P_{s} and is odd under reflection; if $P_{\mathrm{a}} = 0$ the endpoint rule is exact for every integer M, and in general $\nu_2(M) \geq \deg P_{\mathrm{a}} + 1$ suffices.

Proof. Pair $a \leftrightarrow M-a$ on the grid and $x \mapsto 1-x$ in the integral, using the reflection $F(1-x) = 1 - F(x)$: the paired Fabius samples add to plain polynomial samples, and the boundary corrections combine into the classical (exact) Euler–Maclaurin identity for the polynomial P itself. Then apply theorem 37.3 to P_{a}. □

For $P = x^p$ the antisymmetric degree is $p-1$ for even $p > 0$ and p for odd p, so the endpoint rule for I_p is exact once $\nu_2(M) \geq p$ (even p) or $\nu_2(M) \geq p+1$ (odd p): on dyadic meshes this recovers the threshold $\tau(p) = 2\lceil p/2 \rceil$ of (34.6) and extends it verbatim to every integer mesh with the same 2-adic valuation. Explicitly, whenever $\nu_2(M) \geq \tau(p)$,

$$I_p = \frac{1}{M}\sum_{a=1}^{M-1}\Big(\frac{a}{M}\Big)^p F\Big(\frac{a}{M}\Big) + \frac{1}{2M} - \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!}\, p^{\underline{2r-1}}\, M^{-2r}. \tag{37.8}$$

37.4 The shifted first defect and its forced phases

One level before the threshold the defect is again a single half-integer Dirichlet series, now phase-modulated. Extend the spectral series of section 35 by

$$D_{\cos}(s;\theta) = \sum_{\substack{q \geq 1\\ q \text{ odd}}} \frac{\Phi(q/2)}{q^s}\cos(2\pi q\theta), \qquad D_{\sin}(s;\theta) = \sum_{\substack{q \geq 1\\ q \text{ odd}}} \frac{\Phi(q/2)}{q^s}\sin(2\pi q\theta), \tag{37.9}$$

so $D(s) = D_{\cos}(s;0)$.

Theorem 37.5 (Shifted first defect on the Fabius side)**.** *Let P have degree $d \geq 1$ and leading coefficient a_d, and take the critical dyadic mesh $M = 2^d$. Then*

$$\mathcal{A}_{2^d,\theta}[P] - I_P = \frac{2\,a_d\,d!}{(2\pi 2^d)^{d+1}}\begin{cases}(-1)^{(d+1)/2}\,D_{\cos}(d+1;\theta), & d \text{ odd},\\ (-1)^{d/2+1}\,D_{\sin}(d+1;\theta), & d \text{ even}:\end{cases} \tag{37.10}$$

only the leading coefficient of P survives.

Proof. At aliases $q = 2^d\ell$, even ℓ meet zeros of order at least $d+1$; for odd ℓ only the $j = 0$ term of (37.4) with the leading monomial survives, and the leading jet (30.11) evaluates $\chi^{(d)}(2^d\ell) = -2^{-d}d!\,(2^{d-1}\ell)^{-d}\Phi(\ell/2)$. Pair $\pm\ell$; the parity of d decides cosine against sine. □

Corollary 37.6 (Parity-forced phases). *At dyadic level N the corrected shifted rule is exact through degree $N-1$ for every shift, and through degree N at the forced phases $\theta \equiv 0, \frac{1}{2} \pmod 1$ for even N and $\theta \equiv \frac{1}{4}, \frac{3}{4} \pmod 1$ for odd N.*

Proof. $\sin(2\pi q\theta) = 0$ at $\theta \in \{0, \frac{1}{2}\}$ for all integers q; $\cos(2\pi q\theta) = 0$ at $\theta \in \{\frac{1}{4}, \frac{3}{4}\}$ for all odd q. □

This is the Fabius-side companion of the Rvachev-side phase superconvergence corollary 36.7 — the same parity pattern by the same odd-harmonic mechanism, with different nodes and weights (the Fabius rule requires the Bernoulli corrector before the spectrum becomes visible; for up all endpoint jets vanish and no corrector is needed). Read backward at $\theta = 0$, theorem 37.5 is a Fabius–Euler–Maclaurin transference form of the spectral rationality theorem 35.2: the same values $D(2r)/\pi^{2r} \in \mathbb{Q}$ emerge from the corrected monomial rules one level below threshold.

The eleventh-wave twin extends the first defect from the dyadic mesh to every mesh at the critical valuation, in two equivalent closed forms. For $P = a_d x^d + \cdots$ and $M = 2^d q$ with q odd, the corrected error is the *single* periodic Bernoulli moment

$$\mathcal{A}_{M,\theta}[P] - I_P = a_d \frac{(-1)^{d+1}}{(d+1)\,M^{d+1}}\,\mathbb{E}\big[\widetilde{B}_{d+1}(qY+\theta)\big] = a_d \frac{(-1)^{d+1} d!}{(2\pi \mathrm{i} M)^{d+1}} \sum_{\substack{k\in\mathbb{Z}\setminus\{0\}\\ k \text{ odd}}} \frac{\Phi(kq/2)\,\mathrm{e}^{2\pi \mathrm{i} k\theta}}{k^{d+1}} \tag{37.11}$$

(only the $r = 1$ correlation of (37.6) survives, and the leading jet evaluates it); at $q = 1$ this is theorem 37.5. Two consequences deserve display. First, a parity superconvergence *one level below* the uniform threshold: for even d and $\theta \in \{0, \frac{1}{2}, 1\}$ the corrected rule is already exact at $\nu_2(M) = d$, because $Y \overset{d}{=} 1 - Y$ and odd q make the law of $\{qY + \theta\}$ symmetric under $u \mapsto 1 - u$ while $B_{d+1}(1-u) = -B_{d+1}(u)$ — extending the $\theta = 0$ statement of theorem 37.4 to the midpoint and upper rules. Second, at $q = 1$, $\theta = 0$ the defect is governed by the *Bernoulli moments* of the Fabius law, $\beta_m := \mathbb{E}B_m(Y)$:

$$\mathcal{A}_{2^d,0}[x^d] - I_d = \frac{(-1)^{d+1}\,\beta_{d+1}}{(d+1)\,2^{d(d+1)}}, \qquad \beta_2 = -\tfrac{1}{18}, \quad \beta_4 = \tfrac{23}{1350}, \quad \beta_6 = -\tfrac{1543}{119070}, \quad \beta_8 = \tfrac{1196113}{65063250}, \tag{37.12}$$

with $\beta_{2m+1} = 0$ by symmetry. Comparing the two forms of (37.11) at $q = 1$ yields the exact spectral identity

$$\boxed{\beta_{2m} \;=\; \mathbb{E}B_{2m}(Y) \;=\; (-1)^m \frac{2\,(2m)!}{(2\pi)^{2m}}\,D(2m),} \tag{37.13}$$

so the Bernoulli moments of the Fabius law are rational multiples of the spectral Dirichlet values — and theorem 37.11 settles their signs at once:

Corollary 37.7 (Alternating Bernoulli moments). $(-1)^m\beta_{2m} > 0$ *for every* $m \ge 1$. *Consequently the $q = 1$ endpoint defect (37.12) is nonzero for every odd d, with sign $(-1)^{(d+1)/2}$. (The eleventh-wave twin posed this sign law as a conjecture; the first-harmonic bound of theorem 37.11 proves it.)*

Proof. Equation (37.13) and theorem 37.11 give $\operatorname{sgn}(\beta_{2m}) = (-1)^m$. Substituting $2m = d + 1$ into (37.12) yields the asserted nonzero defect and its sign. □

Numerically the β_{2m} are also familiar from the companion polynomial volume: up to sign they are the canonical defect special values $\mathcal{E}_{2m-1}(0)$ of *Exact Polynomial Synthesis from Rvachev Up-Atoms* ($\mathcal{E}_1(0) = \frac{1}{18}$, $\mathcal{E}_3(0) = -\frac{23}{1350}$, $\mathcal{E}_5(0) = \frac{1543}{119070}$, $\mathcal{E}_7(0) = -\frac{1196113}{65063250}$), as both reduce to the same D-values through (37.13) and that volume's defect–quadrature duality.

37.5 The multiple-angle domination lemma and the complete phase zero sets

For general odd q the first harmonic $\Phi(q/2)$ is small and the absolute bound behind theorem 37.11 fails. The twelfth-wave report supplies the decisive refinement: a *relative* bound in which each harmonic is measured against the first one at the same odd scale.

Lemma 37.8 (Relative first-harmonic dominance). *For all positive odd integers q, ℓ,*

$$\boxed{|\Phi(q\ell/2)| \;\leq\; \frac{|\Phi(q/2)|}{\ell}.} \tag{37.14}$$

Consequently, for every $s \geq 2$,

$$\sum_{\substack{\ell \geq 3 \\ \ell \text{ odd}}} \frac{|\Phi(q\ell/2)|}{\ell^{s-1}} \;\leq\; |\Phi(q/2)| \sum_{\substack{\ell \geq 3 \\ \ell \text{ odd}}} \ell^{-s} \;\leq\; |\Phi(q/2)| \left(\frac{\pi^2}{8} - 1\right) \;<\; |\Phi(q/2)|\,. \tag{37.15}$$

Proof. $|\sin(\ell x)| \leq \ell\,|\sin x|$ gives $|\mathrm{sinc}_\pi(\ell y)| \leq |\mathrm{sinc}_\pi(y)|$ factorwise in (30.8) at $y = q/2^{j+1}$. At the leading factor $j = 0$ the numerators are exact: $q\ell$ and q are odd, so $|\sin(\pi q\ell/2)| = |\sin(\pi q/2)| = 1$ and the ratio of the leading factors is exactly $1/\ell$. Every later factor contributes a ratio at most one. □

Theorem 37.9 (Twisted spectral positivity and Thue–Morse signs). *For every odd $q \geq 1$ and every $s \geq 2$,*

$$D_q(s) := \sum_{\substack{\ell \geq 1 \\ \ell \text{ odd}}} \frac{\Phi(q\ell/2)}{\ell^s} \;\neq\; 0, \qquad \operatorname{sgn} D_q(s) = \operatorname{sgn} \Phi(q/2) = \varepsilon_{(q-1)/2}\,; \tag{37.16}$$

equivalently $\mathbb{E}\widetilde{B}_{2m}(qY) \neq 0$ for every odd q and $m \geq 1$, with sign $(-1)^m \varepsilon_{(q-1)/2}$. In particular the sufficient thresholds of theorems 37.3 and 37.4 are sharp at every *odd part, and the general-q first defect* (37.11) *never vanishes at a generic phase.*

Proof. The tail bound (37.15) with $s \geq 2$ leaves the first harmonic dominant; its sign is the Thue–Morse sign of $\Phi(q/2)$ by (35.2). The Bernoulli-moment form follows through (37.11) as in (37.13). (This settles what theorem 37.11 left open beyond $q = 1$.) □

Theorem 37.10 (Complete phase zero sets at the first failing level). *Let $d \geq 1$, q odd, $M = 2^d q$. The corrected Fabius rule $\mathcal{A}_{M,\theta}[x^d]$ equals I_d exactly for*

$$\theta \in \{0, \tfrac{1}{2}, 1\}\ (d\ \textit{even}), \qquad \theta \in \{\tfrac{1}{4}, \tfrac{3}{4}\}\ (d\ \textit{odd}), \tag{37.17}$$

and for no other *phase in $[0,1]$. The same holds for the Rvachev comb: at the first failing degree $p = m+1$ the only superconvergent shifts of $Q_{m,p}(\theta)$ are the parity-forced phases of corollary 36.7.*

Proof. With $x = 2\pi\theta$ the defect is, up to a nonzero constant, $S(x) = \Phi(q/2)\sin x + \sum_{\ell \geq 3 \text{ odd}} \Phi(q\ell/2)\,\ell^{-(d+1)}\sin(\ell x)$ for even d (cosines for odd d). By $|\sin \ell x| \leq \ell\,|\sin x|$ the tail has modulus at most $|\sin x| \sum_{\ell \geq 3} |\Phi(q\ell/2)|\,\ell^{-d} < |\Phi(q/2)|\,|\sin x|$ by lemma 37.8, so S has the sign of $\Phi(q/2)\sin x$ wherever $\sin x \neq 0$; its zeros are exactly those of $\sin x$. For odd d use $|\cos \ell x| \leq \ell\,|\cos x|$ (valid for odd ℓ, by shifting x by $\pi/2$). The Rvachev comb case is the $q = 1$ series (36.9) with the same parities. □

This upgrades both phase-classification conjectures — the Rvachev-side remark 36.8 and its Fabius-side companion stated below — from conjectures to theorems, and does so uniformly in the odd part of the mesh.

37.6 Spectral positivity: the conjecture becomes a theorem

Theorem 37.11 (First-harmonic dominance; $D(2r) > 0$ for all r)**.** *For every $r \geq 1$,*

$$D(2r) \;\geq\; \Phi(\tfrac{1}{2}) - \Big(\frac{\pi^2}{8} - 1\Big) \;>\; \frac{4}{9} - \frac{1}{4} \;>\; 0. \tag{37.18}$$

Proof. With $t_j = \pi/2^{j+1}$ one has $\Phi(\frac{1}{2}) = \prod_{j\geq 0} \sin(t_j)/t_j$. Since $\sin t \geq t - t^3/6$ gives $\sin(t)/t \geq 1 - t^2/6$ on $[0, \pi/2]$, and $\prod_j(1-u_j) \geq 1 - \sum_j u_j$ for $u_j \in [0,1]$,

$$\Phi(\tfrac{1}{2}) \;\geq\; 1 - \frac{1}{6}\sum_{j\geq 0} t_j^2 = 1 - \frac{\pi^2}{18} \;>\; \frac{4}{9}.$$

On the other hand $|\Phi(q/2)| \leq 1$ and $q^{2r} \geq q^2$ give

$$\sum_{\substack{q\geq 3\\ q\text{ odd}}} \frac{|\Phi(q/2)|}{q^{2r}} \leq \sum_{\substack{q\geq 3\\ q\text{ odd}}} \frac{1}{q^2} = \frac{\pi^2}{8} - 1 < \frac{1}{4},$$

so the first harmonic dominates the whole tail at every order. □

Corollary 37.12 (Odd-level sharpness)**.** *$\mathcal{R}_{m,m} \neq 0$ for every odd m, with $\operatorname{sgn} \mathcal{R}_{m,m} = (-1)^{(m+1)/2}$; the odd-$m$ half of conjecture 35.4 is therefore proved, and for every polynomial with nonzero antisymmetric part the sufficient threshold of theorem 37.4 cannot be lowered by one dyadic level. The even-m half of conjecture 35.4 (nonvanishing of $\mathcal{R}_{m,m+1}$, which mixes two jet series) remains open; remark 37.13 below reduces it to a single sawtooth moment.*

Proof. Combine theorem 37.11 with the odd-level defect identity (35.1) and, for the sharpness statement, with theorem 37.5 at $\theta = 0$ applied to the antisymmetric part (odd leading degree gives the cosine branch, whose value at $\theta = 0$ is $D(d+1) > 0$). □

Remark 37.13 (The even-m half is one sawtooth moment). The all-mesh identity localizes the open case completely. For $p = m+1$ at $\nu_2(M) = m$, correlation annihilation (section 37) kills every term of (37.6) except $r = 1, 2$, so the entire corrected error is the two-correlation expression

$$\mathcal{R}_{m,m+1} = \frac{C_{m+1,1}(M,\theta)}{M} - \frac{(m+1)\,C_{m,2}(M,\theta)}{2M^2}, \qquad C_{a,r} = \mathbb{E}\big[Y^a\,\widetilde{B}_r(MY-\theta)\big]. \tag{37.19}$$

The $r = 2$ correlation has exponent $a = m$ equal to the alias zero order, so only *leading* jets contribute: by the Fourier series of $\widetilde{B}_2$ and (30.11), the quantity $C_{m,2}(M,0)$ is an explicit rational-π-power multiple of $D(m+2)$, hence closed and nonzero by theorem 37.11. The even-m half of conjecture 35.4 is therefore equivalent to a single sawtooth-moment statement: that $\mathbb{E}\big[Y^{m+1}(\{MY\} - \frac{1}{2})\big]/M$ does not exactly cancel that closed $D(m+2)$ value — the first genuinely mixed row of the jet hierarchy research question 37.29, since $C_{m+1,1}$ carries the order-$(m+1)$ jets of χ at order-m zeros.

37.7 The exhaustion series and its closed tail

The "integration via summation" identity

$$\int_0^1 f = \sum_{j\geq 1} \Lambda_j[f], \qquad \Lambda_j[f] := 2^{-j} \sum_{a=1}^{2^j-1} (-1)^{a+1} f(a/2^j), \tag{37.20}$$

(Ruffa's generalized method of exhaustion) is the telescoping of nested endpoint-excluded Riemann sums: with $Q_j[f] = 2^{-j}\sum_{a=1}^{2^j-1} f(a/2^j)$ one has $\Lambda_j = Q_j - Q_{j-1}$, so the Jth partial sum of (37.20) is exactly $Q_J[f]$ and the identity holds for every Riemann-integrable f — the optimal natural hypothesis, with no bounded-variation assumption. In base b the level increment carries the weights 1 at nodes not divisible by b and $1-b$ at the others, and telescopes the same way.

For Fabius integrands the tail of (37.20) *closes*: past the threshold the corrected rule is exact, so the raw sums are an explicit exponential polynomial in the level.

Theorem 37.14 (Closed exhaustion tail). *Let P have endpoint threshold N_0 (so (37.8) holds at $M=2^N$, $N \ge N_0$), and write $c_r(P) = \frac{B_{2r}}{(2r)!}P^{(2r-1)}(1)$. Then, for $j \ge N_0+1$ and $J \ge N_0$,*

$$\Lambda_j[PF] = \frac{P(1)}{2^{j+1}} - \sum_{r\ge1} c_r(P)\,(2^{2r}-1)\,2^{-2rj}, \qquad I_P - Q_J[PF] = \frac{P(1)}{2^{J+1}} - \sum_{r\ge1} c_r(P)\,2^{-2rJ}, \quad (37.21)$$

both r-sums finite; in base b (both levels past threshold, which requires $\nu_2(b)>0$) the level increment is $(b-1)P(1)/(2b^j) - \sum_r c_r(P)(b^{2r}-1)b^{-2rj}$. For an odd base the zero multiplicity does not grow with the level and no finite termination arises from the present mechanism — but the corrected error is still supergeometric *in the level: for every $s>d+1$, $\left|\mathcal{A}_{b^N,\theta}[P] - I_P\right| \le 2\zeta(s)(2\pi b^N)^{-s}\,\|(PF)^{(s)}\|_{L^1}$, since the corrected integrand is C^∞-periodic; the dichotomy is exactness for even bases against faster-than-every-fixed-geometric-rate decay for odd ones. Variable mesh chains $M_N = \prod_{j\le N} b_j$ interpolate: exactness begins at the first N with $\sum_{j\le N}\nu_2(b_j) > d$.*

Proof. Rearrange (37.8) at consecutive levels and subtract. □

For example, $P=x^3$ has $N_0=4$, $c_1=\frac14$, $c_2=-\frac{1}{120}$, so for $j\ge5$ $\Lambda_j[x^3F] = 2^{-j-1} - \frac34 2^{-2j} + \frac18 2^{-4j}$ and $I_3 - Q_J = 2^{-J-1} - \frac14 2^{-2J} + \frac{1}{120}2^{-4J}$ for $J\ge4$: the nominally infinite exhaustion process becomes finite exact arithmetic.

37.8 Derivative-free rational extrapolation

Theorem 37.15 (Post-threshold annihilator). *Let $R = \lfloor (d+1)/2\rfloor$ and $\chi_R(z) = (z-\frac12)\prod_{r=1}^R (z-4^{-r})$, and let E shift the dyadic level, $(Ea)_j = a_{j+1}$. Then for $j\ge N_0$,*

$$\chi_R(E)\,\big(Q_j[PF] - I_P\big) = 0, \qquad \textit{hence} \qquad I_P = \frac{\sum_k a_k\, Q_{j+k}[PF]}{\chi_R(1)} \quad \textit{where } \chi_R(z) = \sum_k a_k z^k. \quad (37.22)$$

Proof. By (37.21), $Q_j - I_P$ is a finite combination of λ^j with $\lambda\in\{\frac12, 4^{-1},\dots,4^{-R}\}$, each killed by $E-\lambda$; applying $\chi_R(E)$ to $Q_j = I_P + (Q_j - I_P)$ leaves $\chi_R(1)I_P$. □

This is a derivative-free closure of the Euler–Maclaurin correction: the endpoint jets determine which geometric modes exist, and one eliminates them from consecutive raw sums alone. The even-mode part of the annihilator is naturally Gaussian-binomial at $q=\frac14$: $\prod_{r=1}^R(E-4^{-r}) = \sum_k (-1)^{R-k} 4^{-\binom{R-k+1}{2}} \begin{bmatrix} R \\ k\end{bmatrix}_{1/4} E^k$ — a second q-geometry beside the corpus's $q=\frac12$ dyadic-value formulas ($\frac12$ is the spatial refinement ratio, $\frac14$ the Bernoulli error ratio), connecting these filters to the q-Richardson calculus of the exponents volume. Explicitly,

$$I_n = \frac{Q_2 - 6Q_3 + 8Q_4}{3}\ (n=1,2), \qquad I_n = \frac{-Q_4 + 22Q_5 - 104Q_6 + 128Q_7}{45}\ (n=3,4), \quad (37.23)$$

with $Q_j = Q_j[x^nF]$; all samples are exact rationals by the dyadic evaluator of section 30.4, and the absorbed package verifies e.g. $(-Q_4+22Q_5-104Q_6+128Q_7)/45 = \frac{1207}{5400}$ for $n=3$ in exact arithmetic.

37.9 Composite-mesh Rvachev quadrature

The shifted quadrature theorem 36.1 extends verbatim from dyadic levels to arbitrary integer meshes:

Theorem 37.16 (Composite-mesh self-sampling)**.** *For every integer $M \geq 1$, every real θ, and every polynomial P with* $\deg P \leq \nu_2(M)$*,*

$$\frac{1}{M}\sum_{k\in\mathbb{Z}} P\Big(\frac{k+\theta}{M}\Big)\ \mathrm{up}\Big(\frac{k+\theta}{M}\Big) = \int_{-1}^{1} P(x)\ \mathrm{up}(x)\,\mathrm{d}x. \tag{37.24}$$

Proof. Shifted Poisson summation aliases at $M\ell$, where Φ vanishes to order $1+\nu_2(M\ell) \geq 1+\nu_2(M)$, killing all required derivatives. $\square$

Again the odd part of M is irrelevant — and here the 2-adic condition is *sharp*: the companion volume *Exact Polynomial Synthesis from Rvachev Up-Atoms* (`representations/Up_Polynomial_Synthesis/`) proves that the reproduction/quadrature order of the up-translate system at radius-to-spacing ratio M is exactly $\nu_2(M)$, with the general-ratio first defect carrying the values $\Phi(q\ell/2)$ for $M = 2^\nu q$. The sharpness is now kernel-verified at the transform level: `CombFirstDefect.lean` proves that at degree $p = \nu_2(M)+1$ the first Poisson alias survives — `Fabius.iteratedDeriv_rvachevFourierProduct_nat_mul_int_of_odd` evaluates the p-th derivative of Φ at an odd multiple $M\ell$ to its exact nonzero value $-p!\,\Phi(q'/2)/(M\ell)^p$ with q' the odd part of $M\ell$ (precisely the tabulated first-defect values), and `Fabius.fourier_monomialRvachevSchwartz_nat_int_ne_zero_of_odd` concludes that the transform of $x^{\nu_2(M)+1}\,\mathrm{up}(x/M)$ does *not* vanish at odd integer frequencies, so the quadrature order of the mesh-M comb is exactly $\nu_2(M)$. The defect itself is now a formal object: `CombDefectSeries.lean` proves the comb's spectral display absolutely summable at every degree and mesh (Schwartz decay of the transform), subtracts the zero mode, and identifies the quadrature defect with the convergent alias series (`Fabius.tsum_shifted_monomial_sub_integral`); at the threshold degree the even frequencies die by the sharp joint valuation count and the defect is carried by the odd frequencies alone (`Fabius.tsum_shifted_monomial_sub_integral_odd`) — the odd-level defect series of section 35, with the exact first-defect coefficients. The endpoint phenomenon of theorem 37.4 is formal on the up-side as well: coefficient parity (`Fabius.fourier_monomialRvachevSchwartz_neg`, $\widehat{f}_{2\pi}(-\ell) = (-1)^p \widehat{f}_{2\pi}(\ell)$) makes the odd-frequency defect series an odd function of the frequency whenever the threshold degree $\nu_2(M)+1$ is odd, so at the endpoint shift $\theta = 0$ the $\pm\ell$ aliases cancel exactly and the endpoint comb is exact *one degree beyond* the shifted threshold (`Fabius.tsum_monomial_eq_integral_of_odd_deg`) — every odd mesh, and every mesh of even two-adic valuation. At even threshold degree the same parity makes the aliases *reinforce*: the endpoint defect equals twice the one-sided odd-frequency series (`Fabius.tsum_shifted_monomial_sub_integral_even_deg`) — the up-side form of the one-sided D-series of section 35, with the exact first-defect coefficients as summands. The endpoint dichotomy at the threshold degree is thus fully formal: super-exactness when $\nu_2(M)+1$ is odd, the doubled Dirichlet fold when it is even. The odd-frequency support also gives the midpoint layer: a half-period shift flips every surviving alias, so the threshold defect is antiperiodic, $\mathrm{defect}(\theta+\frac{1}{2}) = -\,\mathrm{defect}(\theta)$ (`Fabius.tsum_shifted_monomial_sub_integral_add_half`), and the θ- and $(\theta+\frac{1}{2})$-combs average *exactly* to the integral at every mesh and every shift — the trapezoid–midpoint pairing, formal (`Fabius.tsum_shifted_monomial_add_half_add`). In particular the midpoint defect is the exact negative of the endpoint defect (`Fabius.tsum_half_shifted_monomial_sub_integral`), so the midpoint comb is super-exact at odd threshold degree (`Fabius.tsum_half_monomial_eq_integral_of_odd_deg`) and carries the mirrored Dirichlet fold at even threshold degree (`Fabius.tsum_half_shifted_monomial_sub_integral_even_deg`).

The quantitative input these defect series need — that the first harmonic is not swallowed by the tail — is now formal too. The sinc factor obeys the quadratic lower bound $1 - x^2/6 \leq \mathrm{sinc}_{\mathrm{rad}}\, x$ (`Fabius.one_sub_sq_div_six_le_sinc`), and the Weierstrass product inequality $1 - \sum u_i \leq \prod(1-u_i)$ (`Fabius.one_sub_sum_le_prod_one_sub`, `Fabius.one_sub_tsum_le_tpr` `od_one_sub`) — neither of which Mathlib carried — turns it into a bound on the whole dyadic product: $1 - 2t^2/9 \le \prod_n \mathrm{sinc}_{\mathrm{rad}}(t/2^n)$ for $t^2 \le 6$ (`Fabius.one_sub_le_prod_sinc_two_pow`, `Fabius.one_sub_le_tprod_sinc_two_pow`; the family is multipliable for *every* real t, `Fabius.multipliable_sinc_two_pow`). At the half-integer this gives the explicit constant the comb volume assumes, $\Phi(1/2) > 4/9$ (`Fabius.four_ninths_lt_tprod_sinc_pi_div_two`, and for the corpus transform itself `Fabius.four_ninths_lt_re_rvachevFourierProduct_half`). The companion *relative* bound is formal as well: every dilation contracts the sinc factor, $|\mathrm{sinc}_{\mathrm{rad}}(mx)| \le |\mathrm{sinc}_{\mathrm{rad}}\, x|$ for $m \ge 1$ (`Fabius.abs_sinc_nat_mul_le`), whence the dilated-product comparison at every odd ℓ (`Fabius.prod_abs_sinc_odd_dilate_le`), with sharpness witnesses showing the hypothesis $m \ge 1$ cannot be dropped.

The same bound gives an explicit *nonvanishing window*. At every real point the transform is the real dyadic sinc product, $\Phi(x) = \prod_{n\ge 0} \mathrm{sinc}_{\mathrm{rad}}(\pi x/2^n)$ (`Fabius.rvachevFourierProduct_ofReal_eq_tprod_sinc`), so $1 - 2(\pi x)^2/9 \le \Phi(x)$ for $(\pi x)^2 \le 6$ (`Fabius.one_sub_le_re_rvachevFourierProduct_ofReal`) and hence $\Phi(x) > 0$, with no zero at all, as soon as $2(\pi x)^2 < 9$ (`Fabius.re_rvachevFourierProduct_ofReal_pos`, `Fabius.rvachevFourierProduct_ofReal_ne_zero`); concretely on $|x| \le 2/3$ (`Fabius.re_rvachevFourierProduct_ofReal_pos_of_abs_le`), which needs only $\pi < 3.15$. The corpus already knew Φ is nonzero there; what is new is the explicit constant, which is what the alias estimates actually consume. Rescaling by $x = t/2\pi$ carries the window to the up-law's characteristic function itself: $\varphi_{\mathrm{up}}(t) = \Phi(t/2\pi)$ is real and strictly positive — in particular nonzero — for $t^2 < 18$, concretely on $|t| \le 4$ (`Fabius.charFun_rvachevMeasure_eq_ofReal`, `Fabius.re_charFun_rvachevMeasure_pos`, `Fabius.charFun_rvachevMeasure_ne_zero`, `Fabius.charFun_rvachevMeasure_ne_zero_of_abs_le_four`). Theorem 37.16 itself is now kernel-verified in full polynomial form: `Fabius.tsum_shifted_polynomial_eq_integral_nat` in `CompositeMeshExactness.lean` proves $\sum_{k\in\mathbb{Z}} P(\theta+k)\,\mathrm{up}((\theta+k)/M) = \int P(x)\,\mathrm{up}(x/M)\,\mathrm{d}x$ for every mesh $M \ge 1$, every real shift, and every $\deg P \le \nu_2(M)$ — the derivative vanishing at the alias frequencies $M\ell$ is exactly the valuation count $\nu_2(M\ell) + 1 \ge \nu_2(M) + 1$ (`Fabius.iteratedDeriv_rvachevFourier_nat_mul_int_eq_zero`), and the polynomial assembly runs at a general positive real scale (`Fabius.tsum_shifted_polynomial_eq_integral_of_forall_monomial`), of which the dyadic level (`Fabius.tsum_shifted_monomial_eq_integral_real` in `MonomialCombExactness.lean`) and the composite mesh are the two instances.

37.10 Phase cubature and the phase–resolution separation

The phase parameter is itself an exact integration variable; this fact is independent of smoothness, Fabius structure, and dyadic divisibility.

Theorem 37.17 (Exact phase average)**.** *For every integrable f on $[0,1]$ and every $M \ge 1$, put*

$$Q_{M,\theta}[f] := \frac{1}{M}\sum_{a=0}^{M-1} f\left(\frac{a+\theta}{M}\right).$$

Then

$$\boxed{\int_0^1 Q_{M,\theta}[f]\,\mathrm{d}\theta = \int_0^1 f(x)\,\mathrm{d}x.} \tag{37.25}$$

Proof. In the ath summand set $x = (a+\theta)/M$. Its factor $1/M$ is the Jacobian, and its phase interval maps to $[a/M, (a+1)/M]$. Summing those adjacent intervals partitions $[0,1]$ and proves (37.25). ☐

In the stable range $\nu_2(M) \ge d+1$ the shifted rule (37.2) makes $\theta \mapsto \mathcal{A}^{\mathrm{raw}}_{M,\theta}[P] - I_P$ an exact polynomial of degree at most $d+1$ whose nonconstant part is a zero-mean combination of Bernoulli polynomials (theorem 37.3). Combining that polynomial structure with theorem 37.17 yields a finite phase rule.

Theorem 37.18 (Finite phase cubature). *If $\nu_2(M) \geq d+1$ and $(\theta_j, w_j)_{j \leq s}$ is any quadrature rule on $[0,1]$ exact for polynomials of degree $\leq d+1$, then*

$$\boxed{I_P = \sum_{j=1}^{s} w_j \frac{1}{M} \sum_{a=0}^{M-1} P(x_a^{(j)}) F(x_a^{(j)}), \qquad x_a^{(j)} = \frac{a+\theta_j}{M},} \tag{37.26}$$

with no Bernoulli correction at all. *Simpson $(0, \frac{1}{2}, 1$; degrees $\leq 2)$, Boole $(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$; degrees $\leq 4)$, and shifted Gauss–Legendre rules (s points; degrees $\leq 2s-2$) are immediate instances; the dyadic-phase rules keep every Fabius sample at a dyadic rational, so they produce finite* rational certificates *for the moments I_p.*

Proof. For fixed P and M, theorem 37.3 writes the phase-dependent sample rule as $I_P + R(\theta)$, where R is a polynomial of degree at most $d+1$ and each of its nonconstant Bernoulli components has zero integral on $[0,1]$; equivalently, theorem 37.17 gives $\int_0^1 R(\theta)\, \mathrm{d}\theta = 0$. Exactness of the phase quadrature gives $\sum_j w_j R(\theta_j) = 0$ and $\sum_j w_j = 1$, which is precisely (37.26). Dyadic phases preserve dyadic sample arguments; exact dyadic Fabius values and rational quadrature weights therefore give rational certificates. ☐

The error at dyadic resolution $M = 2^m$ has the exact separated finite-rank form

$$\sum_{r \leq d+1} \frac{d^{\underline{r-1}}}{r!} B_r(\theta)\, 2^{-mr}.$$

Thus phase acts on the Bernoulli basis while each resolution step scales the rth mode by 2^{-r} (a Ruffa layer by $1-2^r$ at the fine mesh), so phase filters and resolution filters commute and can be designed independently. On the resolution side, general-phase rules carry *all* modes 2^{-r}, $1 \leq r \leq d+1$, and the $q = \frac{1}{2}$ Lagrange weights at geometric nodes, $w_j^{(R)} = (-1)^{R-j} 2^{-\binom{R-j+1}{2}} \begin{bmatrix} R \\ j \end{bmatrix}_{1/2} / (\frac{1}{2}; \frac{1}{2})_R$, delete them exactly — the general-θ companion of the parity-restricted annihilator (37.22), and the q-binomial machinery of the exponents volume specialized to this finite mode set.

37.11 Digital phase filters: Prouhet in the Bernoulli basis

On the phase side, signed Thue–Morse blocks annihilate initial Bernoulli modes with closed leading constants. For $s \geq 1$ let $t_j = (-1)^{\mathrm{wt}_2(j)} = \varepsilon_j$ and $\beta_{s,r} = \sum_{j=0}^{2^s-1} t_j\, B_r(j/2^s)$.

Theorem 37.19 (Thue–Morse–Bernoulli generating function).

$$\sum_{r \geq 0} \beta_{s,r} \frac{z^r}{r!} = \frac{z}{\mathrm{e}^z - 1} \prod_{\ell=1}^{s} \left(1 - \mathrm{e}^{z/2^\ell}\right), \qquad \textit{so} \qquad \beta_{s,r} = 0\ (r < s), \quad \beta_{s,s} = (-1)^s\, s!\, 2^{-\binom{s+1}{2}}. \tag{37.27}$$

Proof. $\sum_j t_j \mathrm{e}^{zj/2^s} = \prod_{\ell=1}^{s} (1 - \mathrm{e}^{z/2^\ell})$ by binary digit factorization; each factor is $-z/2^\ell + \mathcal{O}(z^2)$. ☐

This packages the Prouhet identities $\sum_j t_j j^r = 0$ $(r < s)$ directly in the shifted Euler–Maclaurin basis. Applied to stable shifted sums it gives exact finite identities among Fabius values: with $\Theta_{s;M,d} = \sum_{j<2^s} t_j\, \mathcal{A}^{\mathrm{raw}}_{M,j/2^s}[x^d]$,

$$\Theta_{s;M,d} = \sum_{r=s}^{d+1} \frac{d^{\underline{r-1}}}{r!} \beta_{s,r}\, M^{-r}, \qquad \Theta_{s;M,d} = 0\ (s \geq d+2),$$
$$\boxed{\Theta_{d+1;M,d} = (-1)^{d+1}\, d!\, 2^{-\binom{d+2}{2}}\, M^{-(d+1)}} \tag{37.28}$$

– e.g. $-1/(2M)$, $1/(8M^2)$, $-1/(32M^3)$, $3/(512M^4)$ for $d = 0, 1, 2, 3$; at dyadic M these are rational identities checkable in exact arithmetic, and descending s from $d+1$ recovers every Bernoulli mode by finite back substitution (phase spectroscopy). The binary signs are one instance of a complete radix and digit-mask calculus.

Theorem 37.20 (Root-of-unity Bernoulli filter). *Let $b \geq 2$, let $\omega^b = 1$ with $\omega \neq 1$, and let $s_b(j)$ be the sum of the base-b digits of j. Define*

$$\gamma^{(\omega)}_{b,s,r} := \sum_{j=0}^{b^s-1} \omega^{s_b(j)} B_r(j/b^s).$$

Then

$$\boxed{\sum_{r\geq 0} \gamma^{(\omega)}_{b,s,r} \frac{z^r}{r!} = \frac{z}{\mathrm{e}^z - 1} \prod_{\ell=1}^{s} \left(\sum_{a=0}^{b-1} \omega^a \mathrm{e}^{az/b^\ell} \right).} \tag{37.29}$$

Consequently

$$\gamma^{(\omega)}_{b,s,r} = 0 \quad (r < s), \qquad \gamma^{(\omega)}_{b,s,s} = (-1)^s s!\, b^{-s(s-1)/2} (1-\omega)^{-s}. \tag{37.30}$$

If $\nu_2(M) \geq d+1$, then the stable shifted Fabius rules obey

$$\sum_{j=0}^{b^s-1} \omega^{s_b(j)} \mathcal{A}^{\mathrm{raw}}_{M,j/b^s}[x^d] = \sum_{r=s}^{d+1} \frac{d^{\underline{r-1}}}{r!} \gamma^{(\omega)}_{b,s,r} M^{-r}. \tag{37.31}$$

For $s = d+1$ this is the single term $(-1)^{d+1} d!\, b^{-d(d+1)/2} (1-\omega)^{-(d+1)} M^{-(d+1)}$.

Proof. The Bernoulli generating function and the base-b digit expansion give

$$\begin{aligned} \sum_{r\geq 0} \gamma^{(\omega)}_{b,s,r} \frac{z^r}{r!} &= \frac{z}{\mathrm{e}^z - 1} \sum_{j<b^s} \omega^{s_b(j)} \mathrm{e}^{zj/b^s} \\ &= \frac{z}{\mathrm{e}^z - 1} \prod_{\ell=1}^{s} \sum_{a=0}^{b-1} \omega^a \mathrm{e}^{az/b^\ell}. \end{aligned}$$

Because $\sum_a \omega^a = 0$ and $\sum_a a\omega^a = -b/(1-\omega)$, the ℓth factor begins with $-b^{1-\ell} z/(1-\omega)$. Multiplying the leading terms proves (37.30). Finally substitute the exact stable Bernoulli expansion of theorem 37.3 and interchange the two finite sums. The constant term vanishes because the character weights sum to zero, giving (37.31) and its top-mode specialization. □

Theorem 37.21 (Digital masks of arbitrary cancellation order). *Let $c_0, \ldots, c_{b-1} \in \mathbb{C}$ and $C(z) := \sum_{a=0}^{b-1} c_a \mathrm{e}^{az}$. Suppose $C(z) = c^*_\rho z^\rho + \mathcal{O}(z^{\rho+1})$ at the origin, with $\rho \geq 1$. For $j = \sum_{\nu=0}^{s-1} a_\nu b^\nu$ put $w_s(j) := \prod_{\nu=0}^{s-1} c_{a_\nu}$ and*

$$\delta_{b,s,r} := \sum_{j=0}^{b^s-1} w_s(j) B_r(j/b^s).$$

Then

$$\sum_{r\geq 0} \delta_{b,s,r} \frac{z^r}{r!} = \frac{z}{\mathrm{e}^z - 1} \prod_{\ell=1}^{s} C(z/b^\ell), \tag{37.32}$$

and therefore

$$\delta_{b,s,r} = 0 \quad (r < \rho s), \qquad \delta_{b,s,\rho s} = (\rho s)! (c^*_\rho)^s b^{-\rho s(s+1)/2}. \tag{37.33}$$

Moreover, in the stable range $\nu_2(M) \geq d+1$,

$$\sum_{j=0}^{b^s-1} w_s(j) \mathcal{A}^{\mathrm{raw}}_{M,j/b^s}[x^d] = \sum_{r=\rho s}^{d+1} \frac{d^{\underline{r-1}}}{r!} \delta_{b,s,r} M^{-r}. \tag{37.34}$$

Thus $\rho s = d+1$ isolates the top Bernoulli mode, while $\rho s > d+1$ annihilates the entire filtered rule.

Proof. The same Bernoulli generating function turns the digit sum into $\prod_{\ell=1}^{s} C(z/b^\ell)$, proving (37.32). Its leading product is $(c_\rho^*)^s z^{\rho s} b^{-\rho(1+\cdots+s)}$, which proves (37.33). Since $C(0) = \sum_a c_a = 0$, the mask has zero total mass. Applying it to the exact finite Bernoulli expansion and interchanging finite sums proves (37.34). □

This gives a constructive design principle: choose a short digit symbol with several vanishing power moments and tensor it across digits. The resulting exponentially long phase mask has cancellation order linear in the digit depth; (37.28) is its binary one-state case.

37.12 Coherent radix exhaustion and tensor shells

Fix a radix $b \geq 2$ and $\alpha \in [0, 1)$. Put

$$\theta_N := b^N\alpha - \lfloor b^N\alpha \rfloor, \qquad \varepsilon_N := \lfloor b\theta_{N-1} \rfloor, \tag{37.35}$$

so $b\theta_{N-1} = \varepsilon_N + \theta_N$, and define the coherent tagged sum

$$Q_N^{(b,\alpha)}[f] := b^{-N} \sum_{j=0}^{b^N-1} f\left(\frac{j+\theta_N}{b^N}\right). \tag{37.36}$$

Theorem 37.22 (Coherent radix exhaustion). *Let $\zeta_b = \mathrm{e}^{2\pi\mathrm{i}/b}$. For every $N \geq 1$,*

$$\begin{aligned} \mathcal{E}_N^{(b,\alpha)}[f] &:= Q_N^{(b,\alpha)}[f] - Q_{N-1}^{(b,\alpha)}[f] \\ &= b^{-N} \sum_{j=0}^{b^N-1} \left(1 - b\,\mathbf{1}_{j\equiv\varepsilon_N \pmod b}\right) f\left(\frac{j+\theta_N}{b^N}\right) \qquad (37.37)\\ &= -b^{-N} \sum_{a=1}^{b-1} \zeta_b^{-a\varepsilon_N} \sum_{j=0}^{b^N-1} \zeta_b^{aj} f\left(\frac{j+\theta_N}{b^N}\right). \qquad (37.38) \end{aligned}$$

If f is Riemann integrable, then

$$\boxed{\int_0^1 f(x)\,\mathrm{d}x = f(\alpha) + \sum_{N=1}^{\infty} \mathcal{E}_N^{(b,\alpha)}[f].} \tag{37.39}$$

Proof. The orbit identity gives
$$\frac{k+\theta_{N-1}}{b^{N-1}} = \frac{bk+\varepsilon_N+\theta_N}{b^N}.$$
Thus the coarse grid is exactly the residue class $j \equiv \varepsilon_N \pmod b$ in the fine grid; subtracting it changes the fine weight from 1 to $1-b$ on precisely that class. This proves (37.37). Finite Fourier inversion of the residue indicator gives $1 - b\mathbf{1}_{j\equiv\varepsilon_N} = -\sum_{a=1}^{b-1} \zeta_b^{a(j-\varepsilon_N)}$, proving (37.38). The partial sum of (37.39) telescopes to $Q_N^{(b,\alpha)}[f]$ because $Q_0^{(b,\alpha)}[f] = f(\alpha)$; tagged Riemann-sum convergence finishes the proof. □

For $p \in \mathbb{N}_0$ write

$$Q_{M,\theta,p} := \frac{1}{M}\sum_{j=0}^{M-1} \left(\frac{j+\theta}{M}\right)^p F\left(\frac{j+\theta}{M}\right), \qquad h_N := b^{-N}.$$

Theorem 37.23 (Coherent Fabius–Bernoulli layer and tail). *If both b^N and b^{N-1} have 2-adic valuation at least $p+1$, then*

$$\mathcal{E}^{(b,\alpha)}_{N,p} := Q_{b^N,\theta_N,p} - Q_{b^{N-1},\theta_{N-1},p} = \sum_{r=1}^{p+1} \frac{p^{r-1}}{r!} h_N^r \left[B_r(\theta_N) - b^r B_r(\theta_{N-1})\right], \tag{37.40}$$

$$I_p - Q_{b^N,\theta_N,p} = -\sum_{r=1}^{p+1} \frac{p^{r-1}}{r!} B_r(\theta_N) h_N^r. \tag{37.41}$$

If b is even, both identities therefore hold for all N satisfying $(N-1)\nu_2(b) \geq p+1$.

Proof. At either stable mesh, theorem 37.3 and the definition (37.2) give

$$Q_{M,\theta,p} = I_p + \sum_{r=1}^{p+1} \frac{p^{r-1}}{r!} B_r(\theta) M^{-r}.$$

Subtract the formula at $M = b^{N-1}$ from the one at $M = b^N$ and use $b^{-(N-1)r} = b^r h_N^r$ to obtain (37.40). Moving the fine-grid correction to the other side gives (37.41). Finally $\nu_2(b^{N-1}) = (N-1)\nu_2(b)$, while the fine grid has at least as much valuation. □

Tensor products make the multidimensional closure equally explicit. Put

$$\mathcal{P}_{p,M}(\theta) := \sum_{r=1}^{p+1} \frac{p^{r-1}}{r!} B_r(\theta) M^{-r}. \tag{37.42}$$

Theorem 37.24 (Exact anisotropic tensor correction). *Let $\boldsymbol{p} = (p_1, \ldots, p_d)$, $\boldsymbol{M} = (M_1, \ldots, M_d)$, and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)$, with $\nu_2(M_j) \geq p_j + 1$ for every j. For the tensor-product rectangle rule and moment,*

$$Q^{(d)}_{\boldsymbol{M},\boldsymbol{\theta},\boldsymbol{p}} := \prod_{j=1}^{d} Q_{M_j,\theta_j,p_j}, \qquad I^{(d)}_{\boldsymbol{p}} := \prod_{j=1}^{d} I_{p_j},$$

one has

$$Q^{(d)}_{\boldsymbol{M},\boldsymbol{\theta},\boldsymbol{p}} = \prod_{j=1}^{d} \left(I_{p_j} + \mathcal{P}_{p_j,M_j}(\theta_j)\right), \tag{37.43}$$

$$\boxed{I^{(d)}_{\boldsymbol{p}} = \prod_{j=1}^{d} \left(Q_{M_j,\theta_j,p_j} - \mathcal{P}_{p_j,M_j}(\theta_j)\right)} \tag{37.44}$$

By linearity the same anisotropic rule integrates every polynomial times $\prod_j F(x_j)$ once each coordinate meets the largest exponent occurring in that coordinate.

Proof. The one-dimensional stable identity used in the preceding proof is $Q_{M_j,\theta_j,p_j} = I_{p_j} + \mathcal{P}_{p_j,M_j}(\theta_j)$. Multiplying these d identities proves the first formula. Solving each one for I_{p_j} and multiplying proves the second. Expansion into monomials and linearity give the polynomial statement. □

For the open dyadic rule define, after its stable threshold,

$$T_p(h) := I_p - Q_{1/h,0,p} = \frac{h}{2} - \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{2r-1} h^{2r}.$$

Corollary 37.25 (Exact tensor Ruffa shell and remaining tail). *Let $h = 2^{-N}$ and suppose both consecutive grids are stable in every coordinate. With $L^{(d)}_{N,\boldsymbol{p}} := \prod_j (I_{p_j} - T_{p_j}(h))$, the synchronous shell is*

$$\boxed{\mathcal{E}^{(d)}_{N,\boldsymbol{p}} = \prod_j \left(I_{p_j} - T_{p_j}(h)\right) - \prod_j \left(I_{p_j} - T_{p_j}(2h)\right),} \tag{37.45}$$

and its entire remaining tail is the finite polynomial

$$\boxed{I^{(d)}_{\boldsymbol{p}} - L^{(d)}_{N,\boldsymbol{p}} = \prod_j I_{p_j} - \prod_j \left(I_{p_j} - T_{p_j}(h)\right).} \tag{37.46}$$

Proof. The definition of T_p gives the displayed product for the level-N tensor rule. The level $N-1$ rule is the same product with $2h$; their difference is (37.45). Subtracting the level-N product from $\prod_j I_{p_j}$ proves (37.46). ☐

The raw tensor rule remains first order, with boundary term $\frac{h}{2}\sum_j \prod_{k\neq j} I_{p_k}$; phase averaging or coordinatewise Bernoulli subtraction removes it exactly. The Fabius moments also have a correction-free full-support realization: from $I_p = (1 - a_{p+1})/(p+1)$ and the binomial transfer to Rvachev moments, $\nu_2(M) \geq p+1$ gives $I_p = \frac{1}{p+1}[1 - 2^{-(p+1)} \sum_{j\leq p+1} \binom{p+1}{j} U_{M,\theta,j}]$ with U the full-support comb of theorem 37.16. Thus the same integral has a one-sided Bernoulli-corrected lattice rule and a full-support correction-free one.

37.13 Half-interval Rvachev rules and the boundary-topology trichotomy

Substituting $x \mapsto 1-x$ turns $x^n \operatorname{up}(x)$ on $[0,1]$ into $(1-x)^n F(x)$, whose endpoint jet at 1 is concentrated in the single order n. The whole Bernoulli corrector therefore collapses to one term: for $\nu_2(M) > n$ and every shift η,

$$\boxed{\int_0^1 x^n \operatorname{up}(x)\,\mathrm{d}x = \frac{1}{M}\sum_{a=0}^{M-1}\Big(\frac{a+\eta}{M}\Big)^n \operatorname{up}\Big(\frac{a+\eta}{M}\Big) + \frac{B_{n+1}(\eta)}{(n+1)\,M^{n+1}},} \tag{37.47}$$

and at the open endpoint grid ($\eta = 0$, $n \geq 1$) the correction is the single Bernoulli number $B_{n+1}/((n+1)M^{n+1})$. The exhaustion series of x^n up on $[0,1]$ accordingly either *terminates* — for even $n \geq 2$, $B_{n+1} = 0$, the open dyadic sum is exactly the integral from level $N \geq n$ on and every later detail vanishes — or, for odd n, has the single geometric mode $B_{n+1}\, 2^{-(n+1)N}/(n+1)$, giving the exact two-level recovery

$$\int_0^1 x^n \operatorname{up} = \frac{2^{n+1}\, Q_{N+1}[x^n \operatorname{up}] - Q_N[x^n \operatorname{up}]}{2^{n+1} - 1} \qquad (n \text{ odd},\ N > n). \tag{37.48}$$

The three support geometries thus align with the number of surviving Bernoulli modes: $x^n F$ on $[0,1]$ keeps the full endpoint jet ($\lfloor (n+1)/2 \rfloor + 1$ modes, theorem 37.14); the half-interval up keeps exactly one; and the full symmetric support keeps none — there theorem 37.16 terminates the exhaustion outright (level-N spacing 2^{1-N} integrates degree $\leq N-1$ exactly). Boundary topology alone dictates the exhaustion complexity.

37.14 Thue–Morse moment identities and the Bernoulli triangle

Substituting the exact dyadic evaluator (33.1)-style values into (37.8) at $M = 2^N$, $N \geq \tau(n)$, gives an infinite family of exact finite Thue–Morse identities for every Fabius moment,

$$\begin{aligned} I_n = \frac{2^{-N(n+1)-\binom{N+1}{2}}}{N!} \sum_{a=1}^{2^N-1} a^n \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor N/2 \rfloor} \binom{N}{2k} c_k \, (2a-2h-1)^{N-2k} \\ + 2^{-N-1} - \sum_{r=1}^{\lfloor (n+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} n^{\underline{2r-1}} 2^{-2rN}, \end{aligned} \tag{37.49}$$

whose left side is independent of N — the members are linked by the recurrence (37.22). Three Bernoulli mechanisms meet here: endpoint Bernoulli numbers encode the periodicity defect of PF; cumulant Bernoulli numbers encode the probability law behind F (through the moments c_k); and the sinc zeros force their difference to be a finite dyadic sample sum. This triangle can be used for symbolic elimination in either direction — solving for dyadic Fabius sums in Bell-polynomial terms, or extracting moment recurrences from families of exact grids.

37.15 Further directions from the absorbed report

Remark 37.26 (Fabius-side phase classification — resolved). The only zeros modulo one of $D_{\cos}(2r;\theta)$ are $\theta = \frac{1}{4}, \frac{3}{4}$, and the only zeros of $D_{\sin}(2r+1;\theta)$ are $\theta = 0, \frac{1}{2}$, uniformly in the odd part of the mesh: theorem 37.10, by the multiple-angle bounds $|\sin \ell x| \leq \ell\,|\sin x|$ and (odd ℓ) $|\cos \ell x| \leq \ell\,|\cos x|$ against lemma 37.8.

Conjecture 37.27 (Odd-base nontermination). *For odd b and any P with nonzero antisymmetric part, the Bernoulli-corrected base-b rules terminate at no finite level apart from explicitly classifiable phase cancellations; a route is the zero-set rigidity of χ on the alias sets $b^j\mathbb{Z} \setminus \{0\}$.*

Remark 37.28 (Twisted spectral positivity — resolved). The twisted series $\sum_{k \text{ odd}} \Phi(kq/2)k^{-2m}$ is nonzero for *every* odd q and $m \geq 1$, with Thue–Morse sign $\varepsilon_{(q-1)/2}$: this is theorem 37.9, via the relative dominance lemma 37.8 — precisely the “different mechanism” this question asked for.

Research question 37.29 (Higher-defect hierarchy at low valuation). *For $\nu_2(M) = v < d$, several jet layers survive; reduce all correlations $C_{a,r}(M,\theta)$ of (37.6) to periodic Bernoulli moments at the odd scale with coefficients in the sinc-product jets — the one-row case is (37.11), and the valuation-graded structure should parallel conjecture 35.6.*

Research question 37.30 (Complete mesh classification). *For fixed P and θ, are the exact integer meshes precisely the union of the 2-adically forced family (theorem 37.3), the reflection-symmetric family (theorem 37.4), and finitely many phase-alias cancellation families generated by the leading homogeneous part?*

Proposition 37.31 (All-orders complex-power Euler–Maclaurin expansion). *Fix $\alpha \in \mathbb{C}$ and use the real logarithm to define x^α for $0 < x \leq 1$. For every $0 < \theta < 1$ and $K \geq 1$, with $x_a = (a+\theta)/M$,*

$$\frac{1}{M} \sum_{a=0}^{M-1} x_a^\alpha F(x_a) = I(\alpha) + \sum_{r=1}^{K} \frac{\alpha^{\underline{r-1}}}{r!} B_r(\theta) M^{-r} + \mathcal{O}_{\alpha,K,\theta}\big(M^{-K-1}\big) \qquad (M \to \infty), \tag{37.50}$$

where $I(\alpha) = \int_0^1 x^\alpha F(x)\,\mathrm{d}x$ is entire in α. For $\alpha = p \in \mathbb{N}_0$ the endpoint series terminates after $p+1$ terms, and the remaining alias is the finite-order phenomenon studied above.

Proof. For every compact set of exponents and every integer m, flatness gives $F(x) = \mathcal{O}(x^N)$ near zero with N large enough that all parameter and x-derivatives of $x^\alpha F(x)$ through order m extend continuously and vanish at zero. At $x = 1$, the jump of the $(r-1)$st derivative of the one-periodic extension is $\alpha^{\underline{r-1}}$, because all positive derivatives of F vanish there and $F(1) = 1$. The shifted Euler–Maclaurin formula through order $K+1$ therefore gives the displayed Bernoulli terms and a remainder of order M^{-K-1}. Finally, on each compact set of α, the same flatness dominates $x^\alpha(\log x)^m F(x)$ for every m; differentiation under the integral is valid to all orders. Hence $I(\alpha)$ is entire. □

Research question 37.32 (Optimal truncation and Lambert-controlled aliases)**.** *For a general non-polynomial weight, quantify the alias remainder after subtracting a finite endpoint-jet corrector. For the proved complex-power expansion* (37.50)*, determine whether increasing K with the mesh yields a first-surviving-alias plus Lambert-controlled remainder split whose optimal truncation index follows the lower-Lambert scale of the endpoint asymptotics. Does the alias term dominate the exponentially small endpoint resurgence along suitable dyadic rays? In the companion regime $p \to \infty$ with $\nu_2(M) - p$ bounded, match the endpoint-localized Bernoulli terms to the order-p sinc-product alias jets through a Lambert–W saddle. Quantile transport of the exact PF rules through F^{-1} is a natural test case.*

Research question 37.33 (Minimal dyadic phase support)**.** *For each d, find the fewest dyadic phases with rational weights exact for all polynomials of degree $\le d+1$ on $[0,1]$ (Simpson and Boole are the first members), classify the minimizers under $\theta \leftrightarrow 1-\theta$, and minimize $\sum |w_j|$: inserted into theorem 37.18 this yields arithmetically optimal exact moment certificates. The digital masks provide signed rules of very high cancellation order; the positivity–denominator–node-count–conditioning tradeoff is the open part.*

Conjecture 37.34 (Automatic-filter Strang–Fix dictionary)**.** *For a compactly supported refinable density whose transform has a known cyclotomic zero divisor, every compatible finite automaton (mixed-radix, nonstationary digit symbols C_ℓ of orders ρ_ℓ) induces a digital phase filter of cancellation order $\sum \rho_\ell$ with leading constant the product of the local symbol jets; the Fabius–Thue–Morse identities* (37.28) *are the binary one-state case.*

The Lean-facing layer of this section is deliberately elementary: the jump lemma lemma 37.1, the finite-grid average (37.3), the coefficient recursion in (37.4), the reflection pairing, the finite geometric tail algebra of theorem 37.14, and the annihilator (37.22) are all finite algebra over the existing sinc zero-order and Poisson APIs; the exact tables of the absorbed package (`assets/companion-evidence/Fabius_Euler_Maclaurin_Report_Package/`) serve as regression data.

38 Complex, fractional, and negative powers

38.1 Entire dependence and the Hurwitz–Thue–Morse formula

At fixed level the comb $L_{m,\alpha}$ is a finite exponential sum, hence entire in α; and the continuous moment $I(\alpha) = (1 - d_{\alpha+1})/(\alpha+1)$ is entire by flatness (section 30.3). The finite comb has a closed form uniform in the exponent.

Theorem 38.1 (Hurwitz-zeta–Thue–Morse formula)**.** *For every $m \in \mathbb{N}_0$ and $\alpha \in \mathbb{C}$,*

$$\begin{aligned} L_{m,\alpha} = \frac{C_m}{M^{\alpha+1}} \sum_{a=0}^{M-2} \varepsilon_a \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j \sum_{i=0}^{m-2j} \binom{m-2j}{i} 2^i (-2a-1)^{m-2j-i} \\ \times \Big[\zeta(-\alpha - i,\, a+1) - \zeta(-\alpha - i,\, M)\Big], \end{aligned} \tag{38.1}$$

each bracket being an entire function of α (the poles of the two Hurwitz zetas cancel). At $\alpha = p \in \mathbb{N}_0$ this reduces to theorem 33.1 via $\zeta(-n, x) = -B_{n+1}(x)/(n+1)$; at negative integers it becomes a finite combination of generalized harmonic numbers.

Proof. Insert (30.29) into (31.1), interchange the finite sums, and expand $(2k - 2a - 1)^{m-2j} = \sum_i \binom{m-2j}{i}(2k)^i(-2a-1)^{m-2j-i}$; the remaining finite power sum is $\sum_{k=a+1}^{M-1} k^{\alpha+i} = \zeta(-\alpha - i, a+1) - \zeta(-\alpha - i, M)$. □

Corollary 38.2 (Negative integer powers are rational)**.** *For $q \in \mathbb{N}_0$,*

$$L_{m,-q} = M^{\,q-1} \sum_{k=1}^{M-1} \frac{F(k/M)}{k^q} \in \mathbb{Q}; \tag{38.2}$$

the case $q = 1$ is the harmonic Fabius comb $\sum_{k=1}^{M-1} F(k/M)/k$.

Proof. Every summand in (38.2) is rational: M^{q-1}, k^{-q}, and the exact dyadic value $F(k/M)$ are rational, and the sum is finite. The displayed identity is the definition of $L_{m,-q}$ with the vanishing endpoint omitted. □

A second entire interpolation, numerically preferable near negative integers where the Hurwitz representation contains large cancelling terms, is the Newton form

$$L_{m,\alpha} = M^{-\alpha-1} \sum_{r=0}^{M-1} (\Delta^r 0^\alpha) \sum_{k=r}^{M-1} \binom{k}{r} F(k/M), \tag{38.3}$$

from $k^\alpha = \sum_{r\le k} \binom{k}{r} \Delta^r 0^\alpha$. On the continuous side, expanding $F(x) = \mathrm{up}(1-x)$ around the flat point gives the rapidly convergent series

$$I(\alpha) = \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} a_r, \qquad a_r = \frac{d_{r+1}}{r+1} = \mathcal{O}_A(r^{-A})\ \forall A, \tag{38.4}$$

locally uniformly in α: flatness of up at 1 makes the half-moments a_r decay faster than every power, so the generalized binomial series converges normally. For $\alpha = p \in \mathbb{N}_0$ it terminates and recovers (30.25).

38.2 Complex-power Euler–Maclaurin and Richardson filtering

Theorem 38.3 (Complex-power expansion)**.** *For α in a compact $K \subset \mathbb{C}$ and every $R \ge 1$,*

$$L_{m,\alpha} = I(\alpha) - \frac{h}{2} + \sum_{r=1}^{R-1} \frac{B_{2r}}{(2r)!}\, \alpha^{\underline{2r-1}} h^{2r} + \mathcal{O}_{K,R}\big(h^{2R}\big) \qquad (m \to \infty). \tag{38.5}$$

Proof. The function $x^\alpha F(x)$, extended by 0 at $x = 0$, is C^∞ on $[0,1]$ uniformly for $\alpha \in K$: at 0 all derivatives vanish by flatness, and at 1 flatness of $1 - F$ gives $\big(x^\alpha F(x)\big)^{(n)}\big|_{x=1} = \alpha^{\underline{n}}$. The classical Euler–Maclaurin formula with R terms then has only the upper-endpoint corrections shown and a remainder $\mathcal{O}_{K,R}(h^{2R})$. □

When $\alpha = p \in \mathbb{N}_0$ the falling factorials terminate, and theorem 34.3 upgrades the truncated asymptotic to an exact identity above the threshold. For nonintegral α the series does not terminate, but it contains only even powers of h after the $h/2$ shift: the sequence $\widetilde{L}_m = L_{m,\alpha} + h/2$ Richardson-extrapolates with ratio 4, and for natural p the extrapolation *terminates in rational arithmetic*, returning I_p exactly once $m \ge \tau(p)$. Flatness also disarms the only apparent danger of negative exponents on the comb: the smallest node contributes

$$h^{1-s} F(h) = 2^{m(s-1)}\, C_m\, G_m(1), \tag{38.6}$$

which decays faster than every geometric sequence in m for fixed s, since $C_m = 2^{-m(m+1)/2}/m!$.

38.3 The Mellin transform of up

For $\Re\alpha > -1$ let $\mathcal{M}(\alpha) := \int_{-1}^{1} |x|^\alpha \operatorname{up}(x)\,\mathrm{d}x = 2\int_0^1 x^\alpha \operatorname{up}(x)\,\mathrm{d}x$. Since $\operatorname{up} - 1$ is flat at 0,

$$\mathcal{M}(\alpha) = \frac{2}{\alpha+1} + 2\int_0^1 x^\alpha\big(\operatorname{up}(x) - 1\big)\,\mathrm{d}x, \tag{38.7}$$

with an entire second term: $\mathcal{M}$ continues meromorphically to $\mathbb{C}$ with a *single* simple pole at $\alpha = -1$, of residue 2, and

$$\mathcal{M}(\alpha) = \frac{2d_{\alpha+1}}{\alpha+1} \tag{38.8}$$

identifies the continuation with the entire Fabius-law moment d_s. Thus d_s is simultaneously the analytic continuation of all absolute Rvachev moments — the meromorphic backbone of everything below.

38.4 The universal zeta correction

Lemma 38.4 (Cutoff-monomial summation)**.** *Let $\chi \in C_c^\infty([0,\infty))$ equal 1 near 0, let $0 < \theta \le 1$ and $\beta > -1$. Then for every $A > 0$*

$$h\sum_{k=0}^{\infty}\big(h(k+\theta)\big)^\beta \chi\big(h(k+\theta)\big) = \int_0^\infty x^\beta \chi(x)\,\mathrm{d}x + \zeta(-\beta,\theta)\,h^{\beta+1} + \mathcal{O}_{\beta,A,\chi}\big(h^A\big), \tag{38.9}$$

and the identity continues meromorphically in β with the integral read as a Mellin finite part; the two displayed residues collide only at $\beta = -1$.

Proof. Let $\chi^*(s) = \int_0^\infty \chi(x)x^{s-1}\,\mathrm{d}x$; integration by parts, $\chi^*(s) = -s^{-1}\int_0^\infty \chi'(x)x^s\,\mathrm{d}x$, shows χ^* is meromorphic with a single pole at $s = 0$ of residue 1 and rapid decay on vertical lines (as χ' has compact support in $(0,\infty)$). For $c > \beta+1$, Mellin inversion gives

$$h^{\beta+1}\sum_{k\ge 0}(k+\theta)^\beta \chi\big(h(k+\theta)\big) = \frac{1}{2\pi \mathrm{i}}\int_{(c)} \chi^*(s)\,\zeta(s-\beta,\theta)\,h^{\beta+1-s}\,\mathrm{d}s.$$

Shift the contour to $\Re s = -A$: the Hurwitz pole at $s = \beta+1$ contributes $\chi^*(\beta+1) = \int x^\beta\chi$, the pole of χ^* at $s = 0$ contributes $\zeta(-\beta,\theta)h^{\beta+1}$, and polynomial vertical bounds for $\zeta(\cdot,\theta)$ against the rapid decay of χ^* bound the shifted integral by $\mathcal{O}(h^{\beta+1+A})$. □

Theorem 38.5 (Fractional absolute powers)**.** *Fix $\alpha > -1$. For every $A > 0$,*

$$\boxed{U_m(\alpha) = \mathcal{M}(\alpha) + 2\zeta(-\alpha)\,h^{\alpha+1} + \mathcal{O}_{\alpha,A}\big(h^A\big).} \tag{38.10}$$

For $\alpha = p \in \mathbb{N}_0$ the remainder vanishes identically once $m \ge \tau(p)$, by theorem 36.10.

Proof. Split $x^\alpha \operatorname{up}(x) = x^\alpha\chi(x) + x^\alpha(\operatorname{up}(x) - \chi(x))$ on $[0,\infty)$ with χ as in the lemma, supported inside $[0,1)$. The second summand extends by zero to $C_c^\infty(0,\infty)$ — flatness of $\operatorname{up} - 1$ removes the origin, flatness of up the support endpoint — so Poisson summation makes its comb error $\mathcal{O}(h^A)$ for every A. Apply lemma 38.4 with $\theta = 1$ to the first summand and double by evenness. □

The striking feature is the *absence* of an infinite Euler–Maclaurin tail: flatness of $\operatorname{up} - 1$ kills every higher local coefficient, leaving one universal singular term. At integer exponents the trivial zeros $\zeta(-2r) = 0$ explain the even/odd parity of theorem 36.10. For a shifted comb the correction splits into its two one-sided lattice contributions:

Theorem 38.6 (Shifted Hurwitz corrections). *For fixed $\alpha > -1$, $0 < \theta < 1$, and every $A > 0$,*

$$h \sum_{k\in\mathbb{Z}} |h(k+\theta)|^{\alpha} \operatorname{up}\big(h(k+\theta)\big) = \mathcal{M}(\alpha) + h^{\alpha+1}\big[\zeta(-\alpha,\theta) + \zeta(-\alpha,1-\theta)\big] + \mathcal{O}(h^A), \tag{38.11}$$

$$h \sum_{k\in\mathbb{Z}} \operatorname{sgn}(k+\theta)\ |h(k+\theta)|^{\alpha} \operatorname{up}\big(h(k+\theta)\big) = h^{\alpha+1}\big[\zeta(-\alpha,\theta) - \zeta(-\alpha,1-\theta)\big] + \mathcal{O}(h^A). \tag{38.12}$$

Proof. The shifted lattice has positive distances $h(k+\theta)$ and negative distances $h(k+1-\theta)$, $k \geq 0$; apply lemma 38.4 with shifts θ and $1-\theta$ to the two singular halves, Poisson summation to the flat remainders, and add or subtract. The signed continuous integral vanishes by evenness. □

At natural exponents, the reflection identity for Bernoulli polynomials reconciles (38.11) with the exact shifted quadrature of theorem 36.1 and the phase zeros of corollary 36.7; a chosen complex branch on the negative axis is recovered from the absolute and signed combs via $x^{\alpha} = \frac{1+\mathrm{e}^{\pi \mathrm{i}\alpha}}{2}\,|x|^{\alpha} + \frac{1-\mathrm{e}^{\pi \mathrm{i}\alpha}}{2}\operatorname{sgn}(x)\,|x|^{\alpha}$.

38.5 Negative powers and the logarithmic transition

Proposition 38.7 (Renormalized negative powers). *For fixed $\alpha < -1$,*

$$U_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1} \longrightarrow \mathcal{M}(\alpha) \qquad (m\to\infty), \tag{38.13}$$

with superalgebraic convergence, $\mathcal{M}(\alpha)$ denoting the meromorphic value (38.7). At the pole,

$$\boxed{U_m(-1) = 2\log M + 2\gamma + 2\int_0^1 \frac{\operatorname{up}(x)-1}{x}\,\mathrm{d}x + \mathcal{O}_A\big(h^A\big) \quad \forall A>0.} \tag{38.14}$$

Proof. Away from $\alpha = -1$, lemma 38.4 in its meromorphically continued form gives (38.13). At $\alpha = -1+\varepsilon$ the two singular objects have cancelling poles:

$$\mathcal{M}(-1+\varepsilon) = \frac{2}{\varepsilon} + 2\int_0^1 \frac{\operatorname{up}(x)-1}{x}\,\mathrm{d}x + \mathcal{O}(\varepsilon), \qquad 2\zeta(1-\varepsilon)h^{\varepsilon} = -\frac{2}{\varepsilon} + 2\log M + 2\gamma + \mathcal{O}(\varepsilon),$$

and the finite part at $\varepsilon = 0$ is (38.14). □

Remark 38.8 (Elementary route and the Fabius constant). Through the bridge (31.8), the critical case is also elementary: $U_m(-1) = 2(H_{M-1} - L_{m,-1})$ with H_n the harmonic number, and the complete Euler–Maclaurin tails of H_{M-1} and of $L_{m,-1}$ (theorem 38.3 at $\alpha=-1$, where $(-1)^{\underline{2r-1}} = -(2r-1)!$ makes the corrections $-\sum B_{2r}h^{2r}/(2r)$) cancel *term by term*, leaving

$$U_m(-1) = 2\log M + 2\big(\gamma - I(-1)\big) + \mathcal{O}_A(h^A), \tag{38.15}$$

consistent with (38.14) via the elementary identity $\int_0^1 \frac{\operatorname{up}(x)-1}{x}\,\mathrm{d}x = -I(-1)$, i.e. the constant is $\gamma - \mathbb{E}[-\log Y]$. For integer $q>1$ the same cancellation gives the zeta-dominated law

$$U_m(-q) = 2\zeta(q)\,M^{q-1} - 2\left(\frac{1}{q-1} + I(-q)\right) + \mathcal{O}_A(h^A), \tag{38.16}$$

a discrete Mellin renormalization: the lattice singularity $2\zeta(q)M^{q-1}$ plus the finite part of the Rvachev integral. The only divergences of the whole family are the universal logarithm at $\alpha=-1$ and these zeta-driven powers below it; every Fabius comb $L_{m,\alpha}$, by contrast, converges for *every* complex α, since F is flat where the monomial is singular.

39 Iterated and fractional-order summation

39.1 The binomial kernel and two exact reductions

For a sequence $a = (a_k)$ let $(\Sigma a)_n = \sum_{k \le n} a_k$ and Σ^n its n-fold iterate. Induction with the hockey-stick identity gives the discrete Cauchy kernel

$$(\Sigma^n a)_K = \sum_{k \le K} \binom{K-k+n-1}{n-1} a_k. \tag{39.1}$$

Take $a_k = (k/M)^\alpha F(k/M)$ on $1 \le k \le M-1$ and define the normalized n-fold comb

$$L^{[n]}_{m,\alpha} := h^n \left(\Sigma^n a\right)_{M-1} = h^n \sum_{k=1}^{M-1} \binom{M-k+n-2}{n-1} \left(\frac{k}{M}\right)^\alpha F\left(\frac{k}{M}\right). \tag{39.2}$$

Two closed reductions to one-fold combs are available, one through Stirling numbers and one through elementary symmetric polynomials; they are equivalent, and each is more convenient in its own regime.

Theorem 39.1 (Exact n-fold reduction). *For $n \ge 1$ and every $\alpha \in \mathbb{C}$,*

$$\boxed{L^{[n]}_{m,\alpha} = \frac{M^{1-n}}{(n-1)!} \sum_{j=0}^{n-1} \begin{bmatrix} n-1 \\ j \end{bmatrix} M^j \sum_{i=0}^{j} (-1)^i \binom{j}{i} L_{m,\alpha+i}.} \tag{39.3}$$

Equivalently, with elementary symmetric polynomials e_r,

$$L^{[n]}_{m,\alpha} = \frac{1}{(n-1)!} \sum_{i=0}^{n-1} (-1)^i \, e_{n-1-i}\big(1,\, 1{+}h, \dots, 1{+}(n{-}2)h\big) \, L_{m,\alpha+i}. \tag{39.4}$$

An n-fold sum therefore requires only the n shifted base combs $L_{m,\alpha}, \dots, L_{m,\alpha+n-1}$.

Proof. The kernel is a polynomial of degree $n-1$ in the node: with $x = kh$,

$$h^{n-1} \binom{M-k+n-2}{n-1} = \frac{1}{(n-1)!} \prod_{s=0}^{n-2} (1 + sh - x),$$

since $(M-k-1+r)h = 1 + (r-1)h - x$ for $r = 1, \dots, n-1$. Expanding the product in powers of x through the elementary symmetric polynomials of $1, 1{+}h, \dots, 1{+}(n{-}2)h$, and recognizing $h \sum_k x^i (k/M)^\alpha F(k/M) = L_{m,\alpha+i}$, gives (39.4). Alternatively, expand the same product of consecutive terms as the rising factorial $\prod_{r=1}^{n-1}(M-k-1+r) = (M-k)^{\overline{n-1}} = \sum_j \begin{bmatrix} n-1 \\ j \end{bmatrix} (M-k)^j$ and then $(M-k)^j$ binomially; the identity $\sum_k k^i (k/M)^\alpha F(k/M) = M^{i+1} L_{m,\alpha+i}$ collects the powers of M into (39.3). □

Corollary 39.2 (Closed forms at natural degree). *For $p \in \mathbb{N}_0$ and $m \ge \tau(p+n-1)$, every shifted comb in (39.4) is given by the exact terminating polynomial (34.7); hence $L^{[n]}_{m,p}$ is an explicit polynomial in h with rational coefficients formed from $I_p, \dots, I_{p+n-1}$, Bernoulli numbers, and elementary symmetric polynomials. The corresponding full-support Rvachev sums close in the moment basis alone: for $m \ge p+n-1$,*

$$\begin{aligned} &h^n \sum_{k=-M}^{M} \binom{M-k+n-1}{n-1} (kh)^p \, \mathrm{up}(kh) \\ &\quad = \frac{1}{(n-1)!} \sum_{i=0}^{n-1} (-1)^i e_{n-1-i}\big(1{+}h, \dots, 1{+}(n{-}1)h\big) \, \mu_{p+i}. \end{aligned} \tag{39.5}$$

with $\mu_{2r} = c_r$, $\mu_{2r+1} = 0$, by theorem 36.1 applied to the degree-$(p{+}n{-}1)$ polynomial weight.

Proof. Insert the terminating formula (34.7) for each shifted comb in the finite linear combination (39.4); the largest degree is $p+n-1$, so the stated threshold applies to all terms simultaneously. For the full-support formula, expand the rising factor as in the proof of theorem 39.1 and apply theorem 36.1 to each monomial. Evenness of up then replaces its moments by $\mu_{2r} = c_r$ and $\mu_{2r+1} = 0$. ☐

The continuum limits are the classical Volterra kernels: for every $\alpha \in \mathbb{C}$,

$$\lim_{m\to\infty} L^{[n]}_{m,\alpha} = \frac{1}{(n-1)!}\int_0^1 (1-x)^{n-1}x^\alpha F(x)\,\mathrm{d}x = \frac{1}{(n-1)!}\sum_{i=0}^{n-1}(-1)^i\binom{n-1}{i}I(\alpha+i). \tag{39.6}$$

39.2 Fractional order

The generating-function form $(\Sigma^n a)_K = [z^K]\,(1-z)^{-n}A(z)$ continues canonically in the order: for $\Re\rho > 0$ define

$$L^{[\rho]}_{m,\alpha} := h^\rho \sum_{k=1}^{M-1}\frac{\Gamma(M-k+\rho-1)}{\Gamma(\rho)\,\Gamma(M-k)}\left(\frac{k}{M}\right)^\alpha F\left(\frac{k}{M}\right) = h^\rho\,[z^{M-1}]\,(1-z)^{-\rho}A_{m,\alpha}(z), \tag{39.7}$$

where $A_{m,\alpha}(z) = \sum_k (k/M)^\alpha F(k/M) z^k$. The discrete semigroup law $\Sigma^\rho\Sigma^\sigma = \Sigma^{\rho+\sigma}$ is immediate from $(1-z)^{-\rho}(1-z)^{-\sigma} = (1-z)^{-\rho-\sigma}$, and the fractional Volterra limit holds:

Theorem 39.3 (Fractional Volterra limit)**.** *For $\Re\rho > 0$, locally uniformly in $\alpha \in \mathbb{C}$,*

$$\lim_{m\to\infty} L^{[\rho]}_{m,\alpha} = \frac{1}{\Gamma(\rho)}\int_0^1 (1-x)^{\rho-1}x^\alpha F(x)\,\mathrm{d}x. \tag{39.8}$$

Proof. Uniform Stirling asymptotics give $h^{\rho-1}\Gamma(M-k+\rho-1)/[\Gamma(\rho)\Gamma(M-k)] \to (1-x)^{\rho-1}/\Gamma(\rho)$ at $x = k/M$; flatness of F at 0 and $\Re\rho > 0$ at $x = 1$ supply an integrable majorant after splitting off finitely many boundary nodes. ☐

For nonintegral ρ there is no finite Stirling reduction: the kernel is not polynomial. Two structured replacements are (i) a convergent Newton series in the shifted combs $L_{m,\alpha+i}$ with generalized binomial coefficients, and (ii) for the *singular* weights of section 38, a term-by-term application of lemma 38.4 to the Taylor expansion of the kernel at the singular point, which produces a finite string of corrections $\sum_{r\ge 0}\kappa_r\,\zeta(-\alpha-r,\theta)\,h^{\alpha+r+1}$ with κ_r the kernel's Taylor coefficients — more than one zeta term, precisely because the iterated kernel is not locally constant. Developing the two-variable object $(\rho,\alpha) \mapsto L^{[\rho]}_{m,\alpha}$ as a special function (Mellin–Barnes representation, joint asymptotics, arithmetic at rational ρ) is posed in section 42.

40 Arithmetic consequences

The exact formulas bound denominators. With the (nonsharp) level denominator

$$\Delta_m = m!\,2^{m(m+1)/2}\,(2\lfloor m/2\rfloor+1)!!\prod_{j=1}^{\lfloor m/2\rfloor}(2^{2j}-1), \qquad \Delta_m F(k/M) \in \mathbb{Z}\ \ \forall k, \tag{40.1}$$

one gets immediately $\Delta_m M^{p+1}L_{m,p} \in \mathbb{Z}$ and, with $\Lambda_M = \mathrm{lcm}(1,\dots,M-1)$, $\Delta_m\Lambda_M^q L_{m,-q} \in \mathbb{Z}$; the factorials and powers of M visible in (39.3) propagate the bounds to iterated sums. These are deliberately crude: the exact Euler–Maclaurin identity typically forces large cancellation between the moment denominators inside I_p and the Bernoulli powers of two, and above the threshold the true denominator of $L_{m,p}$ is governed by (34.8) alone.

Problem 40.1 (Sharp valuations). *Determine $\nu_2(\operatorname{den} L_{m,p})$ and the odd-prime valuations $v_\ell(\operatorname{den} L_{m,p})$ uniformly in m, p (likewise for iterated sums and negative powers). Above the threshold this is the arithmetic of d_{p+1}, Bernoulli denominators (von Staudt–Clausen), and powers of two; below it, the Bell–Prouhet formula (33.7) confines the signed part to $p+2$ terms and may expose a tractable valuation filtration through the Mersenne factors $2^{2j}-1$.*

41 Verification for Part I

Three fully independent exact routes to the same rational numbers were implemented in the absorbed sources and re-run for this consolidation:

1. direct summation of exact dyadic samples (algorithm 30.4 or the bit recursion (30.31));
2. the exchanged Bernoulli–Thue–Morse formula (33.1);
3. the Bell–Prouhet collapse (33.7).

The absorbed packages verify $L^{\text{direct}}_{m,p} = L^{\text{Bernoulli}}_{m,p} = L^{\text{Bell}}_{m,p}$ over representative ranges, the exactness matrix $\{\mathcal{R}_{m,p} = 0\} \Leftrightarrow \{m \geq \tau(p)\}$ through level 8 (table 5), the iterated reduction (39.3) against direct n-fold summation for $m \leq 6$, $p \leq 4$, $n \leq 4$, the shifted first-failure table 6, and the observed stabilization thresholds

p	0	1	2	3	4	5	6	7	8	9	10
first exact m	0	2	2	4	4	6	6	8	8	10	10
$\tau(p)$	0	2	2	4	4	6	6	8	8	10	10

in exact agreement with conjecture 35.4. For nonintegral exponents, high-precision scans confirm the universal zeta correction: the corrected sequences $U_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1}$ stabilize to 13–16 digits already by $m = 8$ (e.g. to 0.4913969866527990 at $\alpha = \frac{1}{2}$, 0.1707208835839732 at $\alpha = \frac{3}{2}$, 2.7852273785136552 at $\alpha = -\frac{1}{2}$), with no visible algebraic tail, exactly as theorem 38.5 predicts. For this volume the defect table and the exactness matrix were additionally reproduced by a fresh implementation (route 1 with the bit recursion), which is also the source of the values $D(6)$ and $D(8)$ in (35.6).

42 Frontier directions for Part I

Beyond the conjectures already stated (conjectures 35.4, 35.6 and 37.27 and problems 40.1 and 35.7; the former spectral-positivity, twisted-positivity, and both phase-classification conjectures are now theorems — theorems 37.9 to 37.11), the absorbed sources converge on the following programs.

Problem 42.1 (Row-polynomial geometry). *Write $\mathscr{F}_m(z) = Q_m(z)/(1-z)$. Determine the exact reciprocal symmetry of Q_m (cf. (32.9)), the multiplicities of its cyclotomic factors, and whether its noncyclotomic zeros obey a limiting curve or annulus law. The collision of the two descriptions of Q_m — Thue–Morse mask times moment kernel versus the functional equation — suggests a finite spectral factorization.*

- *Valuation-selective extrapolation.* Degree p activates only the shells $\nu_2(\ell) \leq p - m - 1$ (proposition 36.4); combining levels $m, \ldots, m + r$ with coefficients annihilating the first r shells should give *exact finite* filters rather than asymptotic Richardson extrapolation, with a natural q-binomial description at $q = 2^{-1}$.
- *Double scaling in negative powers.* For $q = q_m$ growing with m, the smallest node contributes $M^{q_m - 1}F(1/M)$ with $F(2^{-m})$ governed by the corpus's Lambert-W_{-1} endpoint asymptotics; balancing it against later nodes should yield a phase diagram in q_m/m — a direct meeting point of the endpoint asymptotics and the comb calculus.

- *Beyond-all-orders remainders.* After the zeta correction, the fractional remainder in (38.10) is smaller than every power of h; its logarithm should be controlled by the same saddle geometry and Lambert-W scales as the Fourier decay of Φ. An explicit transseries, uniform in α, is the natural target.
- *Inverse-Fabius combs.* For $h\sum_k x_k^p F^{-1}(x_k)$ the integrand has logarithmic/Lambert endpoint behavior rather than flatness; singular Euler–Maclaurin should produce non-power corrections, and the exact inverse-dyadic values of the corpus may permit arithmetic special cases.
- *Nonintegral scale ratios.* For $\prod_j \operatorname{sinc}_\pi(q^j\xi)$ with $1/q \notin \mathbb{N}_0$, exact lattice alignment fails; seek finite multiscale filters (Gaussian-binomial coefficients are the natural candidate) whose symbols vanish to prescribed order at the approximate aliases.
- *Base-b and digital masks.* The Bell–Prouhet mechanism needs only a digital mask with a high-order zero at $z = 1$; classify the masks that also create high-order zeros at the dual lattice, and hence an exact Euler–Maclaurin range, as in remark 36.3.

Formalization roadmap for Part I

The most modular Lean targets, in dependency order: the convolution identity (30.29) and row OGF with pole cancellation (finite algebra); the finite-product identity (33.5) and the Bell moments (33.6); the Bernoulli sum exchange (33.1); a periodic trapezoidal alias theorem for a function with one endpoint jump; the collapse theorem 34.3 from the already-formalized integer zero orders of Φ; the Stirling reduction (39.3); and only then the analytic Hurwitz/fractional layers. The collapse is especially suitable: after the zero-order lemma, every forbidden alias derivative is literally zero, so the essential spectral argument is finite.

Part VIII
Global polynomial interpolation on the dyadic comb

43 The interpolation problem and its classical context

Part I used the dyadic samples as *sums*; Part II asks what happens when a single global polynomial is required to pass *through* them. The Fabius function is an unusually clean test case for this question: the data are arithmetically ideal (exact rationals, by section 30.4), the target is C^∞ with exact symmetry, and it is infinitely flat at both endpoints — yet it is nowhere analytic, so every analyticity-based convergence mechanism is unavailable. The result is a particularly sharp laboratory for Runge instability, in which data accuracy, operator conditioning, and evaluation stability can be completely separated.

Three classical facts frame the study. For $M + 1$ equidistant nodes the Lebesgue constant of ordinary interpolation grows exponentially,

$$\Lambda_M^{\mathrm{eq}} \sim \frac{2^{M+1}}{\mathrm{e}\, M \log M} \tag{43.1}$$

[19, 20, 21]; barycentric evaluation [22, 23] stabilizes the *evaluation* of a given interpolant but cannot change this operator norm; and no method using equispaced samples can combine stability with unrestricted exponential convergence (Platte–Trefethen–Kuijlaars [25]; see also Coppersmith–Rivlin [24] for how a polynomial bounded on the grid can grow like 2^M between nodes). For the Fabius data the classical pointwise remainder is formally exact but useless: with $\left\|F^{(M+1)}\right\|_\infty = 2^{\binom{M+2}{2}}$ from (30.7), the factor $2^{(M+1)(M+2)/2}/(M+1)!$ overwhelms every nodal-product bound. The right diagnostics are instead the *exact cardinal geometry* of the comb (section 46) and the arithmetic of the exact data (section 49).

Within the corpus, two neighboring bodies of work are deliberately *not* duplicated here: the geometric-node (q-binomial) Lagrange formulas used as coefficient extractors, whose nodes are multiplicative rather than the additive comb j/M; and the stable local dyadic reconstructions (piecewise-linear/Faber–Schauder, repeated-integration splines). The audited corpus contained no global additive-comb Runge study and no endpoint-flat confluent construction before the six absorbed reports; their independent audits agree on this boundary (Appendix K).

44 Ordinary global Lagrange interpolation

Let $\mathcal{I}_M f$ denote the unique polynomial of degree at most M with $(\mathcal{I}_M f)(x_j) = f(x_j)$ at the comb nodes $x_j = j/M$, $0 \le j \le M$, and write $P_M := \mathcal{I}_M F$.

44.1 Barycentric and Newton forms

For equispaced nodes the barycentric weights are $\lambda_j = (-1)^j \binom{M}{j}$:

$$P_M(x) = \frac{\displaystyle\sum_{j=0}^{M} \frac{(-1)^j \binom{M}{j}}{x - x_j} F(x_j)}{\displaystyle\sum_{j=0}^{M} \frac{(-1)^j \binom{M}{j}}{x - x_j}} \qquad (x \notin \{x_j\}), \tag{44.1}$$

and the Newton-forward form is

$$P_M(x) = \sum_{k=0}^{M} \binom{Mx}{k} \Delta^k F(0), \qquad \Delta^k F(0) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} F(j/M). \tag{44.2}$$

The ordinary Pascal row here is a $q = 1$ object sitting on top of $q = \frac{1}{2}$ data — the mixed transform theme developed in section 49.

44.2 Symmetry, degree drop, and compressed forms

Proposition 44.1 (Symmetry and exact degree drop)**.** *For even M, P_M has rational coefficients,*

$$P_M(1-x) = 1 - P_M(x), \tag{44.3}$$

and $\deg P_M \le M - 1$*; equivalently the top forward difference vanishes, $\Delta^M F(0) = 0$. Consequently there is a rational polynomial R_M with*

$$P_M(x) = x + x(1-x)(2x-1)\, R_M\big(x(1-x)\big), \qquad \deg R_M \le \frac{M}{2} - 2, \tag{44.4}$$

and the interpolant of up *on the symmetric comb $t_j = -1 + 2j/M$ of $[-1,1]$ is even: $\mathcal{I}_M\,\mathrm{up}(t) = (1-t^2)\, T_M(t^2)$ with* $\deg T_M \le M/2 - 1$.

Proof. The polynomial $1 - P_M(1-x)$ matches the data by (30.5), so uniqueness gives (44.3). Centering at $x = \frac{1}{2}$, the function $P_M(\frac{1}{2} + y) - \frac{1}{2}$ is odd in y; its a priori degree bound M is even, so the leading coefficient vanishes. Then $P_M(x) - x$ is antisymmetric about $\frac{1}{2}$ and vanishes at $0, \frac{1}{2}, 1$; dividing by $x(1-x)(2x-1)$ leaves a reflection-symmetric polynomial, i.e. a polynomial in $x(1-x)$. The up statement is the same argument with evenness and the zeros at $t = \pm 1$. □

The compression halves the working dimension and provides exact symmetry checks for code; it does *not* improve conditioning — the compressed coefficients are as enormous as the original ones.

44.3 A 2-adic law for the barycentric weights

The real magnitudes of the weights drive the instability below, but on the dyadic comb the same weights are arithmetically stratified.

Theorem 44.2 (2-adic valuation of the comb weights). *For $M = 2^m$ and $1 \le j \le M-1$,*

$$\nu_2\binom{M}{j} = m - \nu_2(j). \tag{44.5}$$

Proof. $\binom{M}{j} = \frac{M}{j}\binom{M-1}{j-1}$, and every entry of row $M-1 = 2^m - 1$ is odd by Lucas's theorem, since all binary digits of $M-1$ are 1. □

The strict-interior formula (44.5) has the exact Lean counterpart `Fabius.twoPowChoose_padicValNat` in `FabiusFunction.PrimePowerBinomialValuation`. The same module proves a stronger arbitrary-prime identity for $0 < j \le p^m$, including the positive right endpoint $j = p^m$ that is excluded from the displayed dyadic theorem.

The related assertion for the interior weights $\binom{M-2}{j-1}$ of section 45 is separate from that exact Lean counterpart and remains unformalized: for $1 \le j \le M-1$, Lucas's theorem gives oddness exactly for odd j, with higher valuations determined by the binary carries of $j-1$ against $M-2$.

Thus the weights that are catastrophic in the real norm are perfectly regular 2-adically — a hint (made precise in conjecture 51.5) that a renormalized 2-adic interpolation problem on the same data may be far better behaved than the archimedean one.

44.4 Endpoint-jet defects: the Stirling transform

Differentiating (44.2) at 0 with $(Mx)^{\underline{k}} = \sum_r s(k,r) M^r x^r$ ($s(k,r)$ the signed Stirling numbers):

Proposition 44.3 (Exact endpoint jets). *For $0 \le r \le M$,*

$$(P_M)^{(r)}(0) = r!\, M^r \sum_{k=r}^{M} \frac{s(k,r)}{k!}\, \Delta^k F(0); \tag{44.6}$$

in particular the endpoint slope is the mixed binomial transform

$$(P_M)'(0) = M \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} \frac{F(j/M)}{j}. \tag{44.7}$$

Proof. Differentiate the Newton formula (44.2). Since $Mx^{\underline{k}} = \sum_{j=0}^{k} s(k,j) M^j x^j$, its rth derivative at zero is $r! M^r s(k,r)$, which proves (44.6). For $r = 1$ use $s(k,1) = (-1)^{k-1}(k-1)!$, expand $\Delta^k F(0) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} F(j/M)$, and note $F(0) = 0$. The coefficient of $F(j/M)$ becomes

$$M(-1)^{j-1} \sum_{k=j}^{M} \frac{1}{k}\binom{k}{j} = M(-1)^{j-1}\frac{1}{j}\sum_{k=j}^{M}\binom{k-1}{j-1} = M(-1)^{j-1}\frac{1}{j}\binom{M}{j},$$

the hockey-stick identity, giving (44.7). □

Since $F^{(r)}(0) = 0$ for $r \ge 1$, every value (44.6) is a *purely interpolatory* boundary defect, computable as an exact rational number. Constraining the first r endpoint derivatives (section 45) removes exactly the first r defects; this explains why low-order constraints help at finite M, and says nothing about the remaining global modes — the subject of section 46.

44.5 The observed Runge transition

High-precision scans with exact rational data (protocol in section 48) show the classical Runge geometry with a sharp onset:

M	$\|F-P_M\|_\infty$ (sampled)	leftmost maximizer	sampled Λ_M	$\log_2\|F-P_M\|_\infty$
4	3.997×10^{-2}	0.0923	2.21	-4.6
8	9.487×10^{-3}	0.0406	1.10×10^{1}	-6.7
16	3.797×10^{-3}	0.0154	9.35×10^{2}	-8.0
32	4.989	0.0071	2.43×10^{7}	2.3
64	3.589×10^{7}	0.0031	4.41×10^{16}	25.1
128	3.837×10^{22}	0.0013	3.54×10^{35}	75.0

Table 7: Ordinary interpolation of F on the comb j/M. Errors are sampled maxima on dyadic grids of mesh 2^{-12}–2^{-13}; the Lebesgue constants are sampled on the same grids. The maximizer migrates toward the endpoint on the scale $1/M$.

The error *decreases* through $M=16$ — a misleading impression of convergence — then explodes in narrow boundary layers. Because samples and comparison values are exact rationals and the precision was raised until every displayed digit stabilized, this is a property of the interpolant, not of the arithmetic.

A still cleaner observable is the exact endpoint derivative (44.7), whose true value should be 0:

M	sign	$\log_2\lvert(P_M)'(0)\rvert$	$M-\log_2\lvert(P_M)'(0)\rvert$
16	+	-0.681	16.681
32	−	10.816	21.184
64	+	34.823	29.177
128	−	85.942	42.058
256	−	206.287	49.713
512	−	450.491	61.509
1024	+	949.989	74.011
2048	+	1957.083	90.917
4096	−	3992.945	103.055
8192	−	8070.574	121.426

Table 8: The exact rational endpoint derivative defect of the ordinary interpolant, computed from (44.7) before taking logarithms. The binary deficit in the last column is consistent with a quadratic function of $\log_2 M$ plus lower-order oscillation; the sign pattern is irregular.

Numerical observation 44.4. The interpolants P_{2^m} exhibit the classical endpoint oscillation of Runge's phenomenon, with $\log_2\|F-P_M\|_\infty \approx M-o(M)$ at the tested sizes and an exact endpoint derivative growing almost like 2^M.

Conjecture 44.5 (Fabius Runge divergence)**.**

$$\|F-P_{2^m}\|_\infty \longrightarrow \infty; \qquad \frac{1}{M}\log_2\|F-P_M\|_\infty \longrightarrow 1.$$

The stronger limit is asserted along dyadic M, possibly with a subexponential, dyadically oscillatory correction.

Conjecture 44.6 (Endpoint derivative law)**.** *There are constants $c_1, c_2>0$ with*

$$M-c_2(\log_2 M)^2 \le \log_2\left|(P_M)'(0)\right| \le M-c_1(\log_2 M)^2 \tag{44.8}$$

for all large dyadic M outside a possible sparse zero set, the remainder after the quadratic-log term containing a bounded or slowly growing log-periodic component tied to the endpoint phase of F.

The operator theory of section 46 makes divergence plausible but does not prove it for this specific data vector: that requires controlling the cancellation between exponentially large cardinal terms and the special Thue–Morse structure of the exact values — the central open problem of Part II (section 51). A proof of conjecture 44.6 would already imply nonuniform convergence, and with a quantitative Markov inequality, a large supremum norm in a boundary interval.

45 The endpoint-flat Hermite family

The most natural use of the flatness (30.6) is to impose the *true* endpoint derivatives — all zero — on the interpolant. The resulting confluent problem has an exact factorization that converts it into ordinary interior interpolation of a deflated quotient, exposes the entire conditioning geometry, and contains ordinary Lagrange interpolation as its zeroth member.

45.1 The smoothstep and the factorization

For $r \geq 0$ define the symmetric beta-polynomial smoothstep and the endpoint weight

$$S_r(x) := I_x(r+1, r+1) = \frac{(2r+1)!}{(r!)^2} \int_0^x t^r (1-t)^r \, \mathrm{d}t = \frac{(2r+1)!}{(r!)^2} \sum_{k=0}^{r} (-1)^k \binom{r}{k} \frac{x^{r+k+1}}{r+k+1}, \tag{45.1}$$

$$W_r(x) := \big[x(1-x)\big]^{r+1}. \tag{45.2}$$

S_r is the unique polynomial of degree $2r+1$ with $S_r(0) = 0$, $S_r(1) = 1$, vanishing derivatives through order r at both endpoints, and $S_r(1-x) = 1 - S_r(x)$; the first members are $S_0 = x$, $S_1 = 3x^2 - 2x^3$, $S_2 = 10x^3 - 15x^4 + 6x^5$.

Because $F - S_r$ has zeros of order at least $r+1$ at both endpoints, the deflated quotient

$$g_r(x) := \frac{F(x) - S_r(x)}{W_r(x)} \tag{45.3}$$

extends to a C^∞ function on $[0, 1]$ with $g_r(1-x) = -g_r(x)$. Let $\mathcal{I}^\circ_{M-2}$ denote ordinary interpolation of degree at most $M-2$ on the *interior* comb $x_j = j/M$, $1 \leq j \leq M-1$.

Theorem 45.1 (Endpoint-flat factorization)**.** *For $M = 2^m \geq 4$ and $r \geq 0$, define*

$$H_{M,r} := S_r + W_r \cdot \mathcal{I}^\circ_{M-2}\, g_r. \tag{45.4}$$

Then $H_{M,r}$ is the unique polynomial of degree at most $M + 2r$ satisfying

$$H_{M,r}(j/M) = F(j/M) \quad (0 \leq j \leq M), \qquad H^{(q)}_{M,r}(0) = H^{(q)}_{M,r}(1) = 0 \quad (1 \leq q \leq r), \tag{45.5}$$

and it satisfies the exact error identity and symmetry

$$F - H_{M,r} = W_r \cdot \big(g_r - \mathcal{I}^\circ_{M-2} g_r\big), \qquad H_{M,r}(1-x) = 1 - H_{M,r}(x). \tag{45.6}$$

All coefficients are rational; the actual degree is at most $M + 2r - 1$; and $H_{M,0} = P_M$. Explicitly, the interior data are

$$\mathcal{I}^\circ_{M-2} g_r \text{ interpolates } \quad g_r(j/M) = \frac{F(j/M) - S_r(j/M)}{\big[\frac{j}{M}(1 - \frac{j}{M})\big]^{r+1}}, \qquad 1 \leq j \leq M-1, \tag{45.7}$$

and the compressed form $H_{M,r}(x) = S_r(x) + W_r(x)(2x-1)\, R_{M,r}\big(x(1-x)\big)$ holds with $\deg R_{M,r} \leq$ *$M/2 - 2$.*

Proof. The second term of (45.4) has degree at most $(M-2)+(2r+2) = M+2r$. At an interior node the definition of g_r gives the value conditions; at 0 and 1 the factor W_r vanishes to order $r+1$, so values and derivatives through order r are those of S_r, which are correct. For uniqueness, the homogeneous problem asks for a polynomial of degree at most $M+2r$ divisible by $x^{r+1}(1-x)^{r+1}\prod_{j=1}^{M-1}(x-j/M)$, of degree $M+2r+1$; only the zero polynomial qualifies. Subtracting (45.4) from $F = S_r + W_r g_r$ gives the error identity. Symmetry follows by applying uniqueness to $1 - H_{M,r}(1-x)$; the odd central part then kills the top even coefficient, and dividing the antisymmetric interior interpolant by $2x-1$ produces the compressed form as in proposition 44.1. For $r = 0$: $S_0 = x$, $W_0 = x(1-x)$, and $H_{M,0}$ matches all $M+1$ Lagrange conditions with degree $\leq M$. □

Remark 45.2 (Rvachev versions). Two distinct up problems reduce to the same machinery. *(i) Half comb.* On $[-1,0]$, $\mathrm{up}(t) = F(1+t)$, so interpolation on $t_j = -1+j/M$ is the Fabius problem verbatim. For the symmetric comb $t_j = -1+2j/M$ of $[-1,1]$, both endpoint values vanish, no smoothstep is needed, and

$$H^{\mathrm{up}}_{M,r}(t) = (1-t^2)^{r+1}\, T_{M,r}(t^2), \qquad \deg T_{M,r} \leq \tfrac{M}{2} - 1, \tag{45.8}$$

with $T_{M,r}$ interpolating $\mathrm{up}(t_j)/(1-t_j^2)^{r+1}$; evenness is forced by uniqueness. *(ii) Full integer comb.* For the comb $y_j = j/M$, $-M \leq j \leq M$, of $[-1,1]$, the degree-$2M$ interpolant R_M of up is even, so $R_M(y) = Q_M(y^2)$ where Q_M interpolates the data $\mathrm{up}(j/M)$ at the *quadratically spaced* nodes j^2/M^2 — an exact reduction of the symmetric equispaced problem to a one-sided problem on a quadratic mesh, useful both computationally and conceptually (the quadratic mesh clusters near 0, not near the dangerous support endpoints). The endpoint-flat versions carry the factor $(1-y^2)^{r+1}$ exactly as in (i).

45.2 Barycentric implementation

On the interior comb the scaled barycentric weights are

$$w_j = (-1)^{j-1}\binom{M-2}{j-1}, \qquad 1 \leq j \leq M-1, \tag{45.9}$$

and

$$\mathcal{I}^{\circ}_{M-2} g_r(x) = \frac{\displaystyle\sum_{j=1}^{M-1} \frac{w_j\, g_r(x_j)}{x - x_j}}{\displaystyle\sum_{j=1}^{M-1} \frac{w_j}{x - x_j}}. \tag{45.10}$$

No confluent Vandermonde system is ever formed: the exact rational data (45.7) feed a standard second-form barycentric evaluation [22]. High precision remains essential — the formula avoids avoidable roundoff but cannot repair the exponentially ill-conditioned dependence on the $g_r(x_j)$ quantified next.

45.3 Why the textbook remainder explains nothing

The confluent Hermite remainder is formally exact: with $K = M+2r+1$, for each x there is ξ_x with

$$F(x) - H_{M,r}(x) = \frac{F^{(K)}(\xi_x)}{K!}\, x^{r+1}(x-1)^{r+1} \prod_{j=1}^{M-1}\Big(x - \frac{j}{M}\Big). \tag{45.11}$$

Inserting the sharp norms (30.7) produces the factor $2^{K(K+1)/2}/K!$ — astronomically large, and equally so in the regimes where the actual error is 10^{-6}. The failure of (45.11) to be predictive is another signature of nowhere analyticity: one extreme global derivative norm ignores all cancellation. The informative decomposition is (45.6) together with the cardinal geometry of the next section, and, for the actual Fabius data, the arithmetic cancellation phenomena of section 49.

46 Weighted Lebesgue geometry and the instability theorems

46.1 The value-data operator and its cardinal functions

Hold the endpoint values and jets exact and perturb the interior values by δ_j. By (45.4) the interpolant moves by

$$\delta H(x) = W_r(x) \sum_{j=1}^{M-1} \frac{\ell_j^\circ(x)}{W_r(x_j)}\, \delta_j, \tag{46.1}$$

where ℓ_j° are the interior cardinal polynomials. The relevant operator norm from ℓ^∞ data is therefore the *weighted Lebesgue function and constant*

$$\mathscr{L}_{M,r}(x) = \sum_{j=1}^{M-1} |L_{j,r}(x)|\,, \qquad L_{j,r}(x) := \left[\frac{x(1-x)}{x_j(1-x_j)}\right]^{r+1} \ell_j^\circ(x), \qquad \mathscr{L}_{M,r} = \max_{[0,1]} \mathscr{L}_{M,r}(x). \tag{46.2}$$

This is a lower bound on the conditioning of *any* implementation of the same polynomial operator, independent of the evaluation strategy; and it is invariant under the affine change to $[-1,1]$, so all statements below hold verbatim for the Rvachev versions of remark 45.2. Note $r = 0$ recovers (the interior part of) the ordinary Lebesgue constant, and the weight ratio in $L_{j,r}$ is the two-sided competition in miniature: $W_r(x)$ *suppresses* output near the endpoints while $W_r(x_j)^{-1}$ *amplifies* data from nodes near them.

Everything is computable through one gamma-product formula.

Lemma 46.1 (Gamma form of the interior cardinals)**.** *Let $y = Mx \in (0, M)$ be a non-node. Then, for $1 \le j \le M-1$,*

$$\left|\ell_j^\circ(x)\right| = \frac{\Gamma(y)\,\Gamma(M-y)\,\,|\sin \pi y|}{\pi\,\,|y-j|\,\,\Gamma(j)\,\Gamma(M-j)}. \tag{46.3}$$

Proof. With $\omega(y) = \prod_{k=1}^{M-1}(y-k)$, reflection gives $|\omega(y)| = \Gamma(y)\Gamma(M-y)\,|\sin \pi y|\,/\pi$, and $|\omega'(j)| = (j-1)!\,(M-1-j)! = \Gamma(j)\Gamma(M-j)$; divide. ☐

46.2 Exact modes

Three explicit evaluations of single terms of (46.2) organize the entire theory. Write them as *modes*, each an exact lower bound for $\mathscr{L}_{M,r}$.

Proposition 46.2 (Exact cardinal modes)**.** *Let M be even.*

1. Boundary-to-center mode *($x = \frac{1}{2M}$, $j = \frac{M}{2}$):*

$$\mathscr{L}_{M,r} \ge A_{M,r} := A_M\, a_M^{r+1}, \qquad A_M = \frac{\Gamma(M-\frac{1}{2})}{\sqrt{\pi}\,(\frac{M}{2}-\frac{1}{2})\,\Gamma(M/2)^2}, \qquad a_M = \frac{2M-1}{M^2}, \tag{46.4}$$

with $A_M \sim 2^M/(\pi\sqrt{2}\,M)$; hence for every fixed r

$$A_{M,r} \sim \frac{2^{\,M+r+1}}{\pi\sqrt{2}\,M^{\,r+2}}. \tag{46.5}$$

2. Center-to-boundary mode *($x = \frac{1}{2} + \frac{1}{2M}$, $j = 1$):*

$$\mathscr{L}_{M,r} \ge B_{M,r} := B_M\, b_M^{r+1}, \qquad B_M = \frac{\Gamma(\frac{M}{2}-\frac{1}{2})^2}{\pi\,\Gamma(M-1)} \sim 2^{\,2-M}\sqrt{\frac{2}{\pi M}}, \qquad b_M = \frac{M+1}{4}. \tag{46.6}$$

3. Interior entropy modes*: for any fixed* $a \in (0, \frac{1}{2})$*, with* $\mathsf{H}(a) = -a\log a - (1-a)\log(1-a)$*,*

$$\frac{1}{M}\log\left|\ell^{\circ}_{M/2}(a)\right| \longrightarrow \log 2 - \mathsf{H}(a) > 0, \qquad \frac{W_r(a)}{W_r(1/2)} = \big[4a(1-a)\big]^{r+1}. \tag{46.7}$$

At arithmetically clean points the entropy modes are exact gamma products; e.g. at $x = \frac{1}{3}$ *(never a node, since* $3 \nmid M$*),*

$$\big|L_{M/2,r}(\tfrac{1}{3})\big| = \Big(\frac{8}{9}\Big)^{r+1}\frac{3\sqrt{3}}{\pi M}\,\frac{\Gamma(M/3)\,\Gamma(2M/3)}{\Gamma(M/2)^2} = \Theta\Big(M^{-1}\mathrm{e}^{\delta M}(8/9)^r\Big), \tag{46.8}$$

where $\delta = \log 2 - \mathsf{H}(\frac{1}{3}) = \frac{5}{3}\log 2 - \log 3 = 0.0566330\ldots$*.*

Proof. All parts are lemma 46.1 at the stated points plus the elementary weight ratios; the gamma-product manipulations (duplication for B_M, reflection at $y = M/3$ for (46.8), Stirling for the asymptotics) are collected in Appendix F. □

The three modes have transparent meanings. Mode (i) measures how a perturbation of the *central* value contaminates the first half-cell: exponentially at $r = 0$, damped by only one factor $\sim 2/M$ per extra jet order. Mode (ii) measures how a perturbation at the *first interior node* contaminates the center: negligible at $r = 0$ but amplified by $(M+1)/4$ per jet order, because the weight $W_r(x_1)^{-1}$ explodes. The entropy modes fill in the whole interior profile. The instability theory is simply the statement that these mechanisms cannot all be small at once.

46.3 Fixed order and all orders

Theorem 46.3 (Every fixed jet order is exponentially unstable)**.** *For every fixed* $r \ge 0$*,* $\mathscr{L}_{M,r} \ge A_{M,r} = (1+o(1))\,2^{M+r+1}\big(\pi\sqrt{2}\,M^{r+2}\big)^{-1}$ *grows exponentially in* M*. Each additional endpoint derivative buys only one algebraic factor* $\asymp M^{-1}$ *against* 2^M*.*

Proof. The half-cell mode in proposition 46.2 is one of the cardinal terms whose absolute sum defines $\mathscr{L}_{M,r}$, so $\mathscr{L}_{M,r} \ge A_{M,r}$. For fixed r, the exact gamma product in (46.4) and Stirling's formula give the displayed asymptotic; its exponential factor is 2^M, while increasing r contributes only the fixed power M^{-r}. □

Theorem 46.4 (No jet schedule stabilizes the family)**.** *There exist absolute constants* $c, C > 0$ *and* M_0 *such that for every dyadic* $M \ge M_0$ *and* every *integer* $r \ge 0$*,*

$$\boxed{\mathscr{L}_{M,r} \ge C\,\mathrm{e}^{cM}.} \tag{46.9}$$

Consequently no sequence $r = r(M)$ *of endpoint orders makes the endpoint-flat interpolation operators subexponentially conditioned on the dyadic comb.*

Proof. Fix $a \in (0, \frac{1}{2})$, say $a = \frac{1}{4}$, and put $\alpha = \log 2 - \mathsf{H}(a) > 0$ and $\eta = \alpha/\big(2\log\frac{1}{4a(1-a)}\big) > 0$ (for $a = \frac{1}{4}$, $4a(1-a) = \frac{3}{4}$).

Moderate orders, $r \le \eta M$. Keep the single entropy-mode term $j = M/2$ at $x = a$ (perturbing a by $\mathcal{O}(1/M)$ to avoid nodes changes nothing at exponential scale). Its logarithm is

$$M\big(\alpha + o(1)\big) + (r+1)\log\big[4a(1-a)\big] \ge M\alpha - \eta M\log\tfrac{1}{4a(1-a)} + o(M) = \tfrac{\alpha}{2}M + o(M),$$

so the mode is at least $\mathrm{e}^{c_1 M}$ for large M.

Large orders, $r > \eta M$. Use mode (ii):

$$\log B_{M,r} = (r+1)\log\frac{M+1}{4} - M\log 2 + \mathcal{O}(\log M) \ge \eta M\log\frac{M+1}{4} - M\log 2 + \mathcal{O}(\log M),$$

which is eventually of order $M\log M$, in particular $\ge c_2 M$. Take $c = \min(c_1, c_2)$. □

Remark 46.5 (Interpretation, and the two regimes). Raising r first damps the boundary-to-center modes (each entropy mode carries $[4a(1-a)]^{r+1} < 1$), but the division by $W_r(x_j)$ in (45.7) eventually makes data at near-endpoint nodes explosively influential in the interior — mode (ii). Endpoint flatness is a *preconditioner with two opposing regimes*, not a stability cure. The proof deliberately uses only single cardinal terms; the exact identities (46.4)–(46.8) make each ingredient a finite gamma-product statement, which is also what makes the theorem a realistic formalization target (section 52).

46.4 Matching derivatives at every node is worse

The other reading of "use derivative data" — prescribe $F^{(s)}(x_j)$ for $0 \le s \le q$ at *every* node — is exactly computable too, since the derivative tower (30.7) makes every datum $F^{(s)}(j/M) = 2^{\binom{s+1}{2}} F_{\mathrm{gl}}(2^s j/M)$ an explicit rational number. It fails faster.

Theorem 46.6 (First-order global Hermite amplification)**.** *Let $\mathcal{H}_M$ interpolate F and F' at all $M+1$ comb nodes (degree $\le 2M+1$), and let ℓ_j be the full-comb cardinals. The cardinal multiplying a value perturbation at x_j is $h_j = [1 - 2\ell_j'(x_j)(x - x_j)]\,\ell_j^2$; at the central node $\ell'_{M/2}(x_{M/2}) = 0$ by symmetry, and at $x = \frac{1}{2M}$*

$$\left|\ell_{M/2}(\tfrac{1}{2M})\right| = \frac{\Gamma(M+\frac12)}{2\sqrt{\pi}\,(\frac{M}{2}-\frac12)\,\Gamma(\frac{M}{2}+1)^2} \sim \frac{\sqrt{2}}{\pi}\,\frac{2^M}{M^2}, \tag{46.10}$$

so the value-data Lebesgue constant of the scheme obeys

$$\Lambda_M^{C^1} \ge \left|h_{M/2}(\tfrac{1}{2M})\right| \sim \frac{2}{\pi^2}\,\frac{4^M}{M^4}. \tag{46.11}$$

Proof. $\ell'_{M/2}(x_{M/2}) = M\sum_{k\ne M/2}(M/2-k)^{-1} = 0$ by symmetry of the full comb; the gamma product is lemma 46.1 adapted to the full node set (Appendix F); and $h_{M/2} = \ell^2_{M/2}$ at this point. ☐

Squaring the cardinals squares the exponential. The numerical confirmation is immediate and extends to higher confluence:

M	q (derivatives matched)	degree	sampled max error
4	1	9	1.148×10^{-2}
4	2	14	6.555×10^{-3}
4	3	19	6.887×10^{-3}
8	1	17	3.736×10^{-3}
8	2	26	4.435×10^{-3}
8	3	35	2.735×10^{-1}
16	1	33	2.272
16	2	50	5.089×10^{2}
32	1	65	1.538×10^{7}
64	1	129	1.352×10^{23}

Table 9: Full-comb confluent Hermite interpolation with exact rational values *and* derivatives. The explosion arrives at half the size of the ordinary case (table 7), consistent with the squared cardinal geometry.

Conjecture 46.7 (Higher confluent amplification)**.** *For fixed $s \ge 1$, matching derivatives through order s at all nodes gives $\Lambda^{\mathrm{confl}}_{M,s} \ge C_s\, 2^{(s+1)M} M^{-\alpha_s}$ for an explicit α_s; theorem 46.6 is $s = 1$ with $\alpha_1 = 4$.*

The moral is not that derivative information is useless — section 50 shows local Hermite data succeeding completely — but that *where* confluent conditions are imposed matters more than their exactness: piling them onto one global equispaced polynomial inherits, and can square, the same cardinal geometry.

47 The balance law and the logarithmically thin window

The operator is exponentially unstable for every jet order — and yet the exact Fabius data exhibit a narrow range of r in which the actual error drops by many orders of magnitude (section 48). The location of that window is predicted almost perfectly by balancing the two exact modes of proposition 46.2.

47.1 Exact and asymptotic balance orders

Definition 47.1 (Two-mode balance order). For even $M > 4$ define $r_{\mathrm{bal}}(M)$ by

$$r_{\mathrm{bal}}(M) + 1 := \frac{\log(A_M/B_M)}{\log(b_M/a_M)}, \tag{47.1}$$

the unique real order at which the two exact mode bounds $A_{M,r} = A_M a_M^{r+1}$ and $B_{M,r} = B_M b_M^{r+1}$ of (46.4)–(46.6) are equal.

Proposition 47.2 (Asymptotic balance law and window). *As $M \to \infty$ through powers of two,*

$$r_{\mathrm{bal}}(M) = \frac{M \log 2}{\log M}\Big(1 + \mathcal{O}\big(\tfrac{1}{\log M}\big)\Big); \tag{47.2}$$

the orders at which the two modes separately cross 1 are $r_\pm + 1 \sim M \log 2/\log(M/2)$ and $\sim M \log 2/\log(M/4)$, so the window in which both exponential mechanisms are simultaneously tamed has width only

$$r_+(M) - r_-(M) \sim \frac{M(\log 2)^2}{(\log M)^2}. \tag{47.3}$$

Retaining only the leading logarithms of the two modes (equivalently, balancing $M \log 2 + r(\log 2 - \log M)$ against $-M \log 2 + r(\log M - \log 4)$) gives the closed dyadic form

$$r_{\mathrm{bal}}^{\log}(M) = \frac{2M \log 2}{2 \log M - 3 \log 2} = \frac{M}{\log_2 M - \frac{3}{2}} \overset{M=2^m}{=} \frac{2^{m+1}}{2m-3}, \tag{47.4}$$

with the same asymptotics.

Proof. Insert the Stirling expansions $\log A_M = M \log 2 - \log M - \log(\pi\sqrt{2}) + \mathcal{O}(M^{-1})$, $\log B_M = -(M-2)\log 2 + \frac{1}{2}\log\frac{2}{\pi M} + \mathcal{O}(M^{-1})$, $\log a_M = -\log(M/2) + \log(1 - \frac{1}{2M})$, $\log b_M = \log(M/4) + \log(1 + \frac{1}{M})$ (Appendix G) into (47.1): the numerator is $2M \log 2 + \mathcal{O}(\log M)$ and the denominator $2\log M + \mathcal{O}(1)$. The thresholds solve $A_{M,r} = 1$ and $B_{M,r} = 1$ respectively, and the window width is an elementary expansion of the difference of reciprocal logarithms. □

At finite M the exact order (47.1) and the leading-log order (47.4) differ noticeably — by $\mathcal{O}(M/(\log M)^2)$, the same size as the window — and the observed optima sit between them:

M	$r_{\mathrm{bal}}(M)$ exact	$r_{\mathrm{bal}}^{\log}(M)$	error-optimal r (scanned)	$\mathscr{L}$-optimal r	minimum sampled error
16	4.11	6.40	5	5	2.73×10^{-5}
32	7.16	9.14	8	8	7.62×10^{-7}
64	12.42	14.22	14	14	1.78×10^{-6}
128	21.57	23.27	23	23	3.98×10^{-3}
256	37.76	39.38	40	41	9.33×10^{8}

Table 10: Balance predictions against the scanned optima for F. The error-optimal and stability-optimal orders coincide or nearly coincide, and both track the balance laws to within one or two units — while the attained minimum error itself eventually *diverges* (see section 48). The Rvachev scans give the same optimal orders at $M = 16, 32, 64$, although the $[-1, 1]$ problem uses no smoothstep: the window location is governed by node geometry and endpoint multiplicity, the target only by how completely the large modes cancel.

Warning 47.3. The balance order minimizes the *worst-case* amplification envelope, not the error. At $r \approx r_{\text{bal}}$ the sampled weighted Lebesgue constant at $M = 64$ is still $\mathscr{L}_{64,14} \approx 2.8 \times 10^3$ — and the entropy modes (46.8) remain exponentially large. The observed 10^{-6} errors therefore rest on highly structured cancellation in the exact Thue–Morse data, and tiny relative perturbations of the samples can destroy them. Function-specific accuracy and operator stability must never be conflated in this problem.

47.2 Lambert-W appearances

Two independent Lambert structures attach to the window.

Budget inversion. Solving the leading law $r \sim M \log 2 / \log M$ for the largest comb a fixed jet budget r can serve gives

$$M_{\max}(r) \sim -\frac{r}{\log 2}\, W_{-1}\Big(-\frac{\log 2}{r}\Big), \tag{47.5}$$

with W_{-1} the lower real branch — the same branch that governs the endpoint asymptotics of F itself in the corpus.

Peak migration. For jet ratio $\rho = r/M$, the exponential rate of the weighted central cardinal at position x is

$$\Phi_\rho(x) = \log 2 - \mathsf{H}(x) + \rho \log\big[4x(1-x)\big], \qquad \Phi'_\rho(x) = \log\frac{x}{1-x} + \rho\,\frac{1-2x}{x(1-x)}, \tag{47.6}$$

and in the regime $\rho \downarrow 0$, $r \to \infty$ the boundary saddle solves $x \log(1/x) \sim \rho$, i.e.

$$x_\rho \sim -\frac{\rho}{W_{-1}(-\rho)}. \tag{47.7}$$

Conjecture 47.4 (Lambert peak law)**.** *In that regime the dominant weighted-cardinal peak associated with central nodes is located at* $x_{M,r} = -\dfrac{r/M}{W_{-1}(-r/M)}\big(1 + o(1)\big)$*, up to a bounded log-periodic modulation induced by the dyadic structure of the exact data. (The observed migration of the* error *maximizer at* $r = 0$*, table 7, scales like* $1/M$*, the* $\rho \to 0$ *endpoint of the same family.)*

47.3 The all-orders correction program

Both balance laws use only first-order Stirling expansions. Every correction to $\log A_M$, $\log B_M$, $\log a_M$, $\log b_M$ is an explicit Bernoulli-polynomial series, and reversion of (47.1) can be organized with complete Bell polynomials, exactly as in the corpus's all-orders Lambert expansions. The result is a computable hierarchy

$$r_{\text{opt}}(M) \sim \frac{M \log 2}{\log M}\Big(1 + \frac{c_1}{\log M} + \frac{c_2 \log\log M}{(\log M)^2} + \cdots\Big), \tag{47.8}$$

whose coefficients for the *actual* optimum should acquire function-specific Thue–Morse corrections. Since the exact order (47.1) is trivial to evaluate numerically, the value of (47.8) is conceptual: it places the interpolation window in the same asymptotic calculus (Stirling, Bernoulli, Bell, Lambert) as the Fabius endpoint itself. Testing the corrected law against exact minimizers at $M = 512, 1024$ is a concrete computational project (section 51).

48 High-precision experiments

48.1 Protocol

All three absorbed interpolation studies follow the same discipline, and their overlapping measurements agree to all displayed digits:

1. every interpolation datum (value or derivative) is an exact rational, produced by the evaluators of section 30.4 and the derivative tower (30.7);
2. interpolants are evaluated by scaled barycentric formulas ((44.1), (45.10)) in `mpmath`, at 70–300 decimal digits, raised until every displayed digit stabilizes;
3. error maxima are *sampled* on complete dyadic grids of stated mesh (2^{-11}–2^{-13}), not certified continuous suprema; symmetry restricts the search to $[0, \frac{1}{2}]$;
4. independent formulations cross-check: ordinary interpolation is computed both directly and as $H_{M,0}$; full-grid Hermite uses confluent divided differences; symmetry, nodal reproduction, endpoint jets, and known small values are verified before each scan.

The scripts, raw CSV data, and figures live under `assets/` in the three interpolation packages.

48.2 Endpoint jets: suppression, window, reversal

Target	M	$r = 0$	$r = 1$	$r = 2$	$r = 4$	$r = 8$
F	8	9.49×10^{-3}	1.75×10^{-3}	1.50×10^{-3}	1.64×10^{-3}	2.78×10^{-2}
F	16	3.80×10^{-3}	2.68×10^{-4}	1.37×10^{-4}	4.93×10^{-5}	1.44×10^{-4}
F	32	4.99	1.82×10^{-1}	9.75×10^{-3}	1.11×10^{-4}	7.62×10^{-7}
F	64	3.59×10^{7}	6.26×10^{5}	1.68×10^{4}	3.18×10^{1}	4.76×10^{-3}
up	8	2.02×10^{-1}	3.07×10^{-2}	1.96×10^{-2}	1.34×10^{-2}	2.55×10^{-1}
up	16	1.97×10^{-1}	2.67×10^{-2}	4.17×10^{-3}	9.85×10^{-4}	1.06×10^{-2}
up	32	4.50×10^{2}	1.46×10^{1}	6.76×10^{-1}	4.45×10^{-3}	4.08×10^{-5}
up	64	1.05×10^{10}	1.75×10^{8}	4.52×10^{6}	7.72×10^{3}	9.49×10^{-1}

Table 11: Sampled maximum errors at fixed jet orders (up on the symmetric comb of $[-1, 1]$; level-11/-12 check grids). Low-order constraints already buy several orders of magnitude at moderate M.

Scanning all orders at $M = 64$ exposes the sharp U-shape:

r	error for F	error for up	sampled $\mathscr{L}_{64,r}$
0	3.59×10^{7}	1.05×10^{10}	4.40×10^{16}
8	4.76×10^{-3}	9.49×10^{-1}	6.40×10^{6}
10	1.55×10^{-4}	2.73×10^{-2}	2.20×10^{5}
12	8.30×10^{-6}	1.26×10^{-3}	1.29×10^{4}
13	1.94×10^{-6}	2.12×10^{-4}	–
14	1.78×10^{-6}	6.19×10^{-4}	2.83×10^{3}
16	7.79×10^{-5}	1.95×10^{-2}	4.37×10^{4}
18	3.25×10^{-3}	7.31×10^{-1}	–
20	1.72×10^{-1}	3.65×10^{1}	8.46×10^{7}
24	1.13×10^{3}	–	5.49×10^{11}

Table 12: The $M = 64$ jet scan. Both targets are U-shaped with minima at $r = 13$–14, exactly where the sampled weighted Lebesgue constant is also smallest – but that smallest value is still $\approx 2.8 \times 10^{3}$.

At larger sizes the optimized error stops improving and then diverges:

M	ordinary error	best scanned error (at r)	$\mathscr{L}_{M,r}$ at that r
32	4.99	7.62×10^{-7} (8)	1.30×10^{1}
64	3.59×10^{7}	1.78×10^{-6} (14)	2.83×10^{3}
128	3.84×10^{22}	3.98×10^{-3} (23)	6.19×10^{9}
256	–	9.33×10^{8} (40)	4.28×10^{24}

Table 13: Best endpoint-flat errors across sizes. At $M = 128$ the optimum has already deteriorated to 10^{-3}; at $M = 256$ optimization over all jet orders no longer produces a usable approximation — the first unambiguous sign of the second divergence regime conjectured in conjecture 51.2.

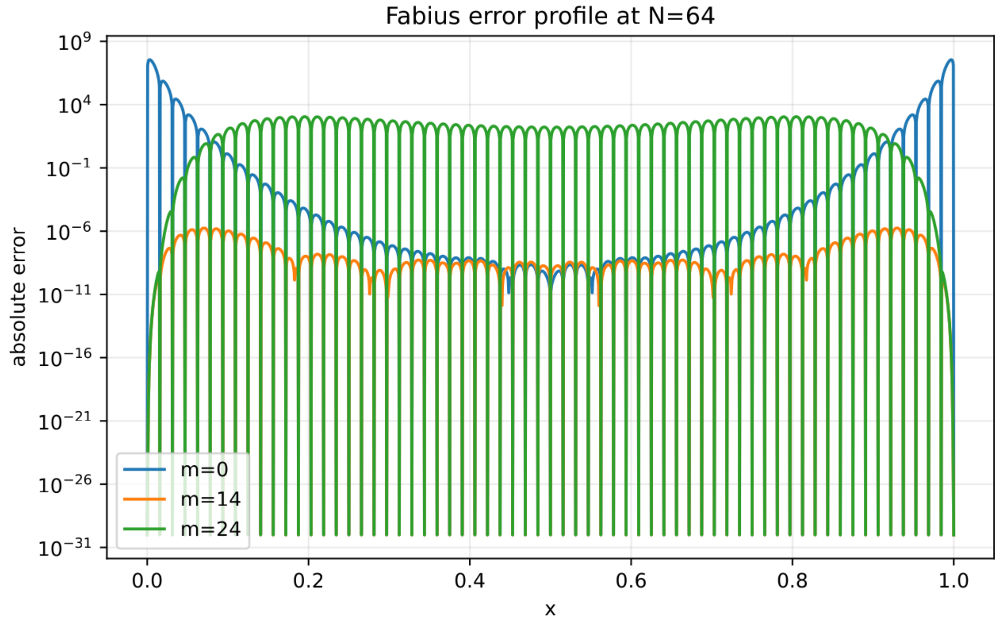

Figure 8: Absolute error profiles at $M = 64$: the ordinary interpolant ($r = 0$) with its giant boundary spike, the tuned order $r = 14$, and the overconstrained $r = 24$ with the reverse instability emerging from the center. (Figure from the absorbed source with N denoting the comb size M.)

48.3 Interpretation

1. *Runge behavior occurs for the specific functions*, not merely for the operator: exact data, high precision, and refined check grids leave the explosions unchanged.
2. *Correct endpoint derivatives matter at finite M*: they remove the first spurious jets (44.6) and can improve errors by factors up to $\sim 10^{13}$.
3. *The benefit has a finite window* of width $\asymp M/(\log M)^2$, centered on the balance order; past it, the $W_r(x_j)^{-1}$ amplification takes over, exactly as the mode geometry predicts.
4. *Function-specific cancellation is not stability*: at the optimum the operator amplification is still 10^3-scale at $M = 64$ and grows without bound; by $M = 256$ no jet order helps. The cancellation that produces the 10^{-6} errors is a property of the Thue–Morse structure of the exact samples — understanding it is the arithmetic frontier of section 49.

49 The dyadic hierarchy and arithmetic connections

49.1 Nested corrections and the midpoint surplus

Dyadic combs are nested, so successive interpolants differ by exactly factorable corrections:

$$P_{2M}(x) - P_M(x) = \omega_M(x)\, R_{M-1}(x), \qquad \omega_M(x) = \prod_{j=0}^{M}\Big(x - \frac{j}{M}\Big), \tag{49.1}$$

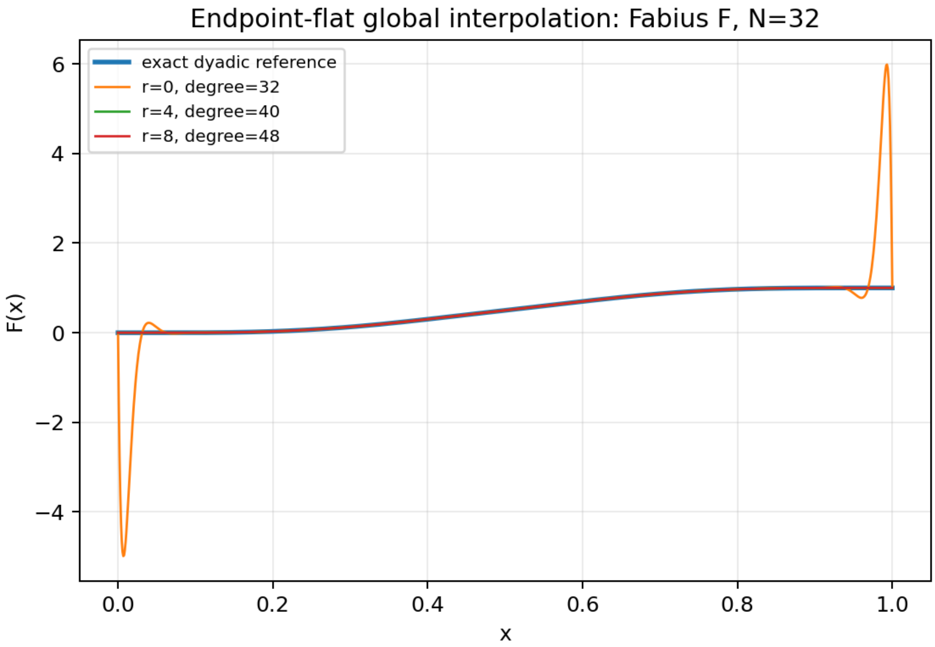

Figure 9: Fabius interpolation on the $M = 32$ comb: ordinary interpolation has entered the boundary-layer regime while moderate endpoint jets recover excellent accuracy — temporarily.

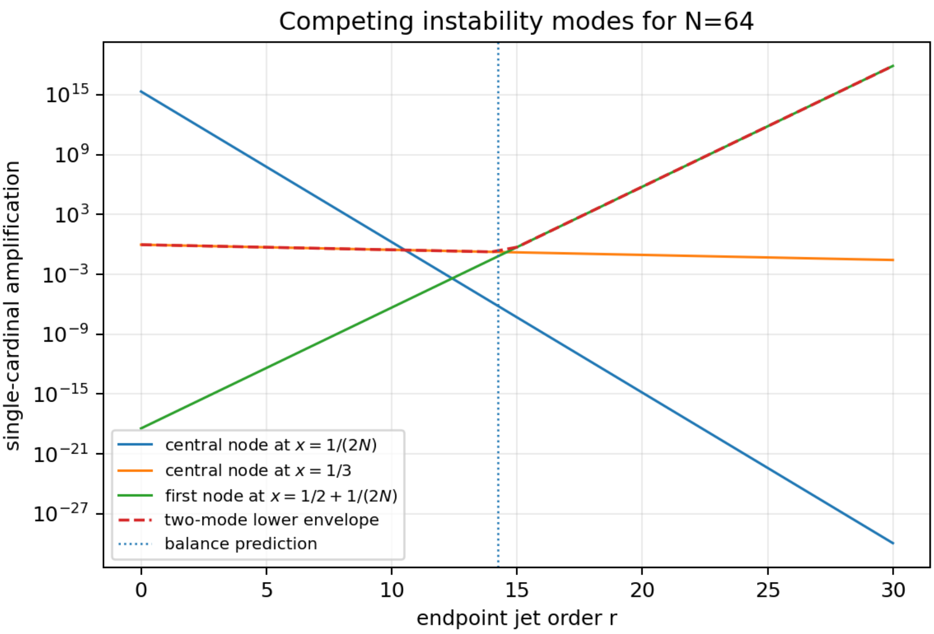

Figure 10: The explicit value-cardinal modes at $M = 64$ as functions of the jet order: raising r suppresses the boundary-to-center and entropy modes but amplifies the first-interior-node mode; their envelope — the mechanism of theorem 46.4 — never becomes small.

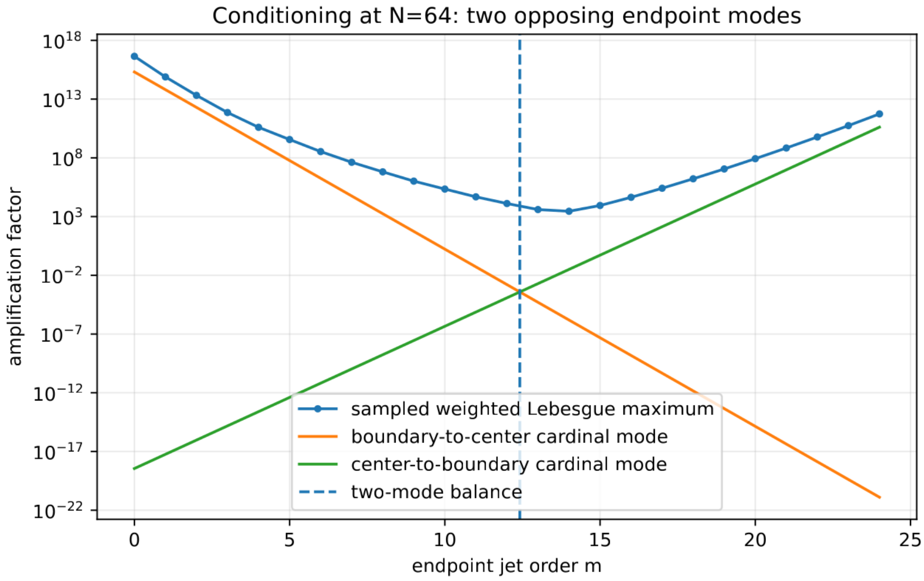

Figure 11: Sampled weighted Lebesgue maximum at $M = 64$ against the two exact mode lower bounds; their crossing predicts the useful finite-M window.

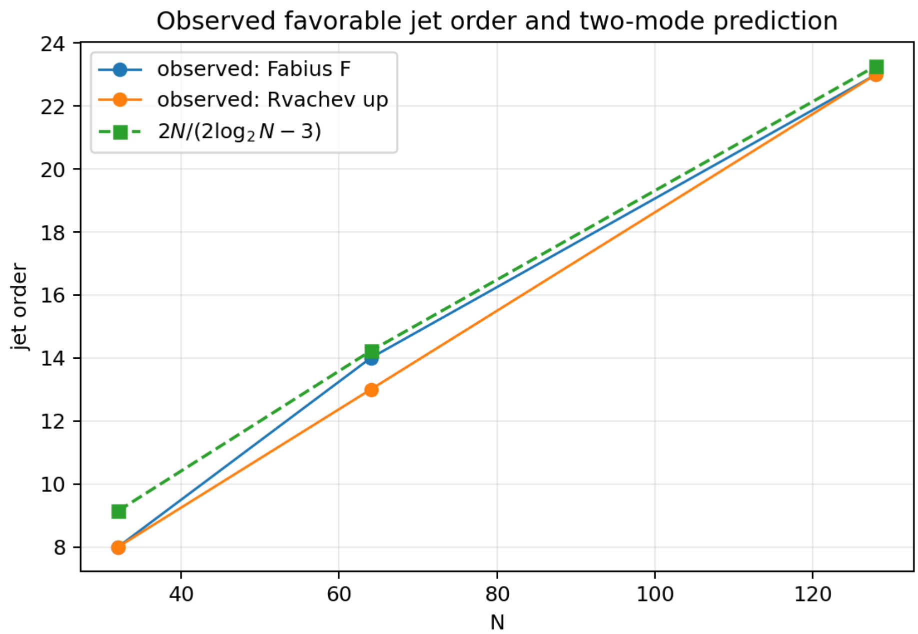

Figure 12: Observed optimal jet orders against the balance law (47.4).

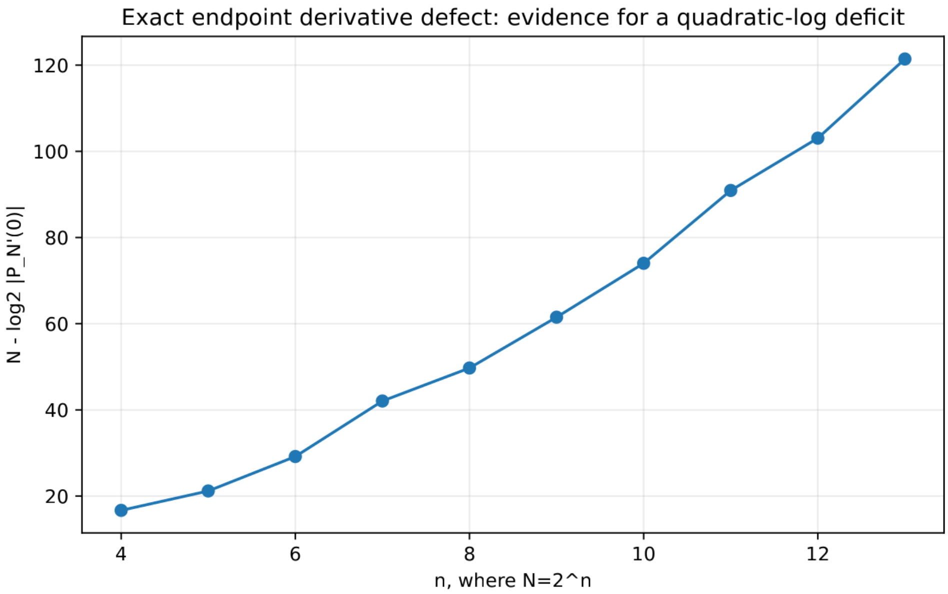

Figure 13: The binary deficit $M - \log_2 |(P_M)'(0)|$ of table 8, consistent with a quadratic in $\log_2 M$ plus oscillation (conjecture 44.6).

$$H_{2M,r}(x) - H_{M,r}(x) = \big[x(1-x)\big]^{r+1} \prod_{j=1}^{M-1} \Big(x - \frac{j}{M}\Big) R_{M-1,r}(x), \tag{49.2}$$

where in each case the quotient $R_{M-1,\cdot} \in \Pi_{M-1}$ is determined by the M *midpoint surpluses*

$$\sigma_{M,j} = F\bigg(\frac{2j+1}{2M}\bigg) - P_M\bigg(\frac{2j+1}{2M}\bigg) \qquad (\text{resp. } H_{M,r}), \tag{49.3}$$

by degree counting: both differences vanish at every coarse node (and carry the endpoint multiplicities in the flat case), and the midpoint values fix the quotient. The surpluses are exact rationals, so the hierarchy is a roundoff-free Runge diagnostic: a boundary surplus whose normalized size grows signals the instability of the next level before it is computed, and cells with small surpluses can stop refining — a mathematically transparent adaptive scheme specialized to nested dyadic data.

49.2 Thue–Morse cancellation and the mixed transform

Every Newton coefficient $\Delta^k F(0)$ and every cardinal evaluation is a finite signed sum of exact dyadic values, hence (by theorem 30.2) a finite Thue–Morse expression. Prouhet cancellation annihilates low-degree polynomial trends of complete blocks — the same mechanism as theorem 33.2 — which is the structural reason the function-specific error near r_{bal} can be so much smaller than the operator norm: large cardinals meet blockwise signed moment identities in the data. The cleanest single observable is the endpoint defect (44.7),

$$D_M := (P_M)'(0) = M \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} \frac{F(j/M)}{j} = M \int_0^1 \sum_{j=1}^{M} (-1)^{j-1} \binom{M}{j} F\Big(\frac{j}{M}\Big)\, t^{j-1}\, \mathrm{d}t, \tag{49.4}$$

a *mixed transform*: an ordinary (q=1) alternating Pascal row applied to q=$\frac{1}{2}$-structured data. The integral form invites a saddle analysis after inserting the exact evaluator; the entropy concentration of the Pascal row near $j = M/2$, the Prouhet block cancellation, the lower-Lambert endpoint scale of $F(j/M)$ for small j, and a residual log-periodic phase all compete in table 8. A rigorous asymptotic for (49.4) would connect several currently separate parts of the corpus and would settle conjecture 44.6.

The contrast between the two node geometries in the corpus is now sharp:

Geometry	Nodes	Natural weights
additive dyadic comb	$j/2^m$	$(-1)^{j-1}\binom{M-2}{j-1}$ (2-adic law (44.5))
geometric/q comb	q^j or -2^j	Gaussian-binomial products

Both are Vandermonde inverses, over different node groups. A mixed transform interpolating across additive location while filtering across dyadic scale — for instance through the nested surplus polynomials (49.2) — is a natural place to look for an additive–multiplicative bridge; the valuation-selective filters of section 42 are the Part-I face of the same idea.

49.3 Fourier, Lambert, and inverse-function connections

Spectral view. A global polynomial has no localized Fourier response: its boundary oscillation is broad spectral leakage. A cellwise operator (below) acts as a compactly supported quasi-interpolant whose multiplier can be compared with the sinc product (30.8) near the origin — a route to sharpened local error constants that respects the compact-support structure of up.

Lambert phases. The corpus's endpoint asymptotics carry a nonconstant periodic correction in a lower-Lambert phase. Three observables here plausibly lock to it: the boundary-layer maximizer (scale $1/M$ at $r=0$, conjecture 47.4 in general), the fluctuation $r_{\text{opt}}-r_{\text{bal}}$, and the log-periodic component of the derivative-defect deficit (conjecture 44.6).

Inverse function. F^{-1} has infinite endpoint *slopes*, the opposite pathology to flatness. Direct equispaced interpolation of F^{-1} must fail at least as badly, and the endpoint-flat machinery does not transfer; the natural schemes are parametric interpolation of the exact pairs $(F(j/M), j/M)$, a Lambert-coordinate transformation that removes the known singular asymptotic before interpolating the slowly varying correction, or monotone local rational/Hermite methods. The Part-I comb calculus faces the same issue for $\sum x_k^p F^{-1}(x_k)$ (section 42).

50 Stable reconstruction on the same comb

The negative results concern one global polynomial whose degree grows with the sample count. The same exact samples support provably stable schemes; the global/local contrast is the practical conclusion of Part II.

50.1 Local Hermite interpolation: the cure

For $s\ge 0$ let $\mathcal{H}^{\text{loc}}_{M,s}f$ be the piecewise polynomial of degree $2s+1$ matching $f^{(k)}$, $0\le k\le s$, at both ends of every cell $[j/M,(j+1)/M]$ — all data exact rationals via the derivative tower.

Theorem 50.1 (Local Hermite error, with exact Fabius constants)**.** *For $f\in C^{2s+2}$,*

$$\left\|f-\mathcal{H}^{\text{loc}}_{M,s}f\right\|_\infty \le \frac{\left\|f^{(2s+2)}\right\|_\infty}{(2s+2)!}\left(\frac{1}{4M^2}\right)^{s+1}; \tag{50.1}$$

in particular, inserting (30.7),

$$\boxed{\left\|F-\mathcal{H}^{\text{loc}}_{M,s}F\right\|_\infty \le \frac{2^{(s+1)(2s+1)}}{(2s+2)!}\,M^{-2s-2}, \qquad \left\|\mathrm{up}-\mathcal{H}^{\text{loc}}_{M,s}\,\mathrm{up}\right\|_\infty \le \frac{2^{\binom{2s+3}{2}}}{(2s+2)!}\,M^{-2s-2},} \tag{50.2}$$

the second bound for M equal cells on $[-1,1]$. The local value and derivative cardinals have norms bounded independently of M: the scheme is uniformly stable.

Proof. On a cell of length h_0 the two-point Hermite remainder is $f^{(2s+2)}(\xi)\,(x-a)^{s+1}(x-b)^{s+1}/(2s+2)!$, bounded by $(h_0^2/4)^{s+1}$ times the derivative norm. For F, $h_0 = 1/M$ and the factor $4^{-(s+1)}$ lowers the derivative exponent $\binom{2s+3}{2}$ to $(s+1)(2s+1)$; for up on $[-1,1]$, $h_0 = 2/M$ makes $h_0^2/4 = M^{-2}$ with no extra power of two. Rescaling any cell to $[0,1]$ shows the local cardinal basis depends only on s. ☐

With exact values and exact derivatives ($s = 1$, cubic case) the predicted M^{-4} convergence is observed cleanly:

M	4	8	16	32	64
sampled error	9.67×10^{-4}	5.03×10^{-4}	2.06×10^{-5}	2.29×10^{-6}	1.58×10^{-7}

Table 14: Piecewise-cubic Hermite errors for F (level-12 check grid): steady algebraic convergence on the very samples where the global polynomial explodes.

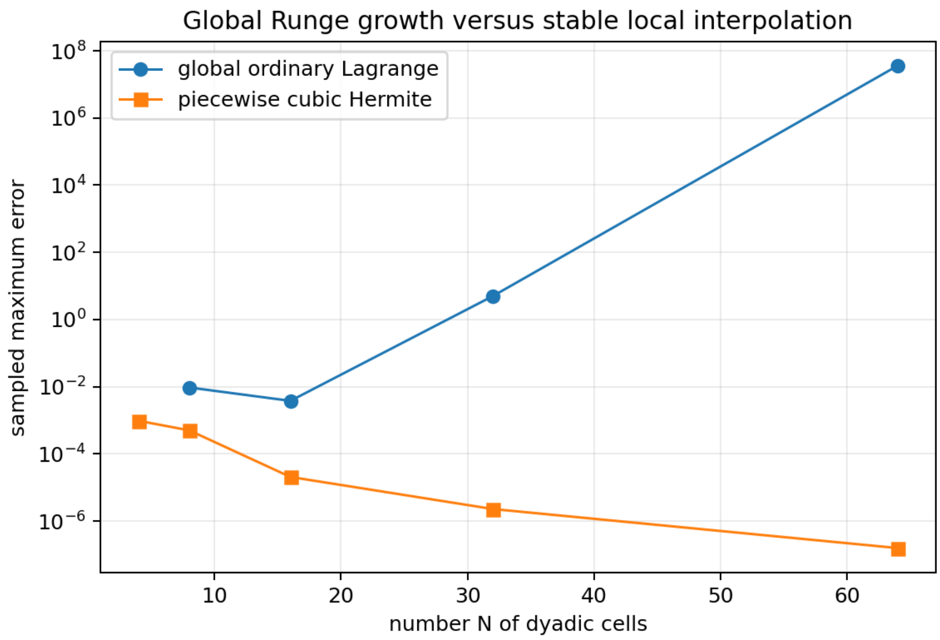

Figure 14: Global ordinary interpolation versus the cellwise cubic method on identical dyadic samples.

50.2 The wider menu

Between the failed global extreme and the local cure lies a hierarchy of methods that keep the exact dyadic data:

- *Oversampled least squares.* Fit degree $d \asymp \sqrt{M}$ to all $M+1$ samples, optionally imposing exact endpoint flatness through the factorization (45.4) with Q fitted rather than interpolated: stable root-exponential accuracy is the best compatible with equispaced data [25, 30].
- *Mock-Chebyshev subset interpolation.* Interpolate only on the comb points nearest to Chebyshev nodes, or constrain a least-squares fit to them; constrained mock-Chebyshev methods extend to Hermite data [31], for which the dyadic derivative tower supplies exact samples.
- *Rational barycentric families.* Floater–Hormann interpolants have logarithmic Lebesgue growth on equispaced nodes [28, 29], and rational Hermite variants accept the exact derivative data; applying them to the deflated quotient g_r preserves endpoint flatness exactly.

- *Mapped and Lambert coordinates.* Interpolate in a coordinate that respects $x \leftrightarrow 1-x$ and opens the endpoint scale — a Lambert coordinate fitted to the known endpoint asymptotics being the natural non-generic choice (cf. Kosloff–Tal-Ezer maps).
- *Atomic/multiresolution bases.* Expand in scaled translates of up itself (the refinement equation makes the dyadic recursion exact), or in Faber–Schauder/spline systems — the representation-theoretic route developed elsewhere in the corpus, with certified local error control.
- *Constrained minimax.* Impose $p(0) = 0$, $p(1) = 1$, $p^{(q)}(0) = p^{(q)}(1) = 0$ and minimize the residual on the comb in a stable basis for g_r.

Practical hierarchy

(1) exact dyadic evaluation for data; (2) local piecewise/atomic reconstruction for stability; (3) global approximation only with degree well below the sample count or after a change of basis/geometry; (4) full-degree global interpolation only for small M or as an exact symbolic identity — never as an asymptotic approximation scheme.

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51 Conjectures and frontier problems for Part II

The proved landscape is: operator instability at every jet order (theorems 46.3, 46.4 and 46.6), exact structural identities (theorem 45.1, proposition 46.2, theorem 44.2), and a stable local alternative (theorem 50.1). The open frontier concerns the *specific* Fabius data. Besides conjectures 47.4 to 46.7:

Conjecture 51.1 (Optimal jet order)**.** *Let r_M^F and r_M^{up} minimize the true uniform errors of the endpoint-flat families, and let $r_M^{\mathscr{L}}$ minimize $\mathscr{L}_{M,r}$. All three satisfy*

$$r_M = \frac{M\log 2}{\log M}\big(1+o(1)\big), \qquad \textit{indeed} \qquad r_M = r_{\mathrm{bal}}(M) + \mathcal{O}\left(\frac{\log M}{\log\log M}\right), \tag{51.1}$$

with a stronger $\mathcal{O}(1)$ version along phase-locked dyadic subsequences. (Scanned data: table 10.)

Conjecture 51.2 (Optimized divergence)**.** *$E_M^* := \min_{r\ge 0}\|F - H_{M,r}\|_\infty$ does not tend to zero; more strongly $\log E_M^*$ is eventually linear in M up to $o(M)$, and for the Rvachev version $\limsup E_M^{*,\mathrm{up}} = +\infty$. The measured sequence $7.6\times 10^{-7} \to 1.8\times 10^{-6} \to 4.0\times 10^{-3} \to 9.3\times 10^{8}$ at $M = 32, 64, 128, 256$ (table 13) is the evidence; a proof must show the Thue–Morse cancellation cannot keep pace with the exponential envelope of theorem 46.4.*

Conjecture 51.3 (Boundary-layer profile)**.** *Along a dyadic subsequence there are normalizations A_M with $A_M\big(P_M(\xi/M) - F(\xi/M)\big) \to \mathcal{E}(\xi)$ locally for $\xi > 0$, the profile governed jointly by the entropy saddle of the uniform cardinals and the lower-Lambert endpoint asymptotic of F.*

Conjecture 51.4 (Surplus arithmetic)**.** *After clearing the natural common denominator (40.1), the midpoint surplus vector $(\sigma_{M,j})_j$ of (49.3) factors as a finite binomial transform of a Thue–Morse block; its sign changes and its largest boundary entries obey a depth-m recursion. (This would make the nested hierarchy of section 49 an arithmetic object and is the most concrete route to conjecture 44.5: showing a boundary surplus cannot stay small at all dyadic levels.)*

Conjecture 51.5 (2-adic stability counterpart)**.** *The valuation law (44.5) admits a renormalized 2-adic interpolation operator on the dyadic data with polynomially bounded norm growth — the archimedean catastrophe and the 2-adic regularity of the same weights being two valuations of one object. Identify the correct Mahler-type basis and compare with the Newton form (44.2).*

Problem 51.6 (Noise threshold). *For tuned $r = r_{\mathrm{bal}}(M)$, quantify the largest relative perturbation of the exact samples that preserves the observed accuracy. The crude bound $\mathscr{L}_{M,r}^{-1}$ is pessimistic; the structured condition number should project noise onto the self-similar residual subspace.*

Research question 51.7 (Sign structure of the differences). *Do the signs or 2-adic valuations of $\Delta^k F(0)$ follow a Thue–Morse or q-binomial law along $M = 2^m$? Such a law would control the cancellations in* (44.2) *and could turn conjecture 44.5 into a lower-bound proof.*

Three research routes toward the divergence conjectures, in increasing ambition: (i) insert the exact evaluator into one selected cardinal evaluation and exploit blockwise Thue–Morse cancellation until a dominant term survives; (ii) run a two-variable saddle analysis of $\Delta^k F(0)$ in (k, m), Bernoulli–Bell corrections included; (iii) prove the surplus recursion of conjecture 51.4. On the constructive side, the corrected balance expansion (47.8) should be computed and tested against exact minimizers at $M = 512$ and 1024, separately for F and up.

52 Lean formalization program for Part II

The new results separate cleanly into finite algebra, exact gamma identities, and elementary asymptotics — a good match for the repository's formalization style. In dependency order:

1. *Comb combinatorics*: nodes, barycentric weights, the 2-adic law (44.5), symmetry and degree drop (proposition 44.1), Newton form and the Stirling jet transform (44.6).
2. *Endpoint-flat algebra*: the smoothstep (45.1) as a finite polynomial with its jet and symmetry properties; divisibility by W_r; existence, uniqueness, and the error identity of theorem 45.1.
3. *Cardinal identities*: the finite-product forms of lemma 46.1 and the exact modes (46.4)–(46.8) as factorial/binomial identities (gamma notation as corollaries).
4. *Instability*: explicit Stirling bounds for factorials; the two-regime dichotomy of theorem 46.4; the Hermite squares of theorem 46.6.
5. *Local stability*: the two-point Hermite remainder and the derivative-norm specialization (50.2).
6. *Exact certificates*: kernel-checked rational error values at small M (as statements about finite grids, not continuum suprema), and the exact endpoint defects of table 8.

A natural module decomposition:

Module	Contents
`DyadicCombLagrange`	nodes, weights, symmetry, Newton form, 2-adic law
`EndpointFlatHermite`	smoothstep, factorization, uniqueness, error identity
`WeightedLebesgue`	perturbation operator, cardinal functions
`DyadicCombModes`	exact $A/B/$entropy-mode identities
`DyadicCombInstability`	fixed-r and all-r exponential lower bounds
`FabiusInterpolationDefect`	exact endpoint derivative transform
`LocalHermiteStability`	cellwise remainder and Fabius constants

This would extend the corpus's geometric-node Lagrange formalization to the additive comb without duplicating it; the Part-I roadmap (section 42) shares the first module.

A Formula dictionary

This appendix collects the principal transforms in one place. It is meant as a reference sheet rather than a substitute for the proofs.

A.1 General geometric comb

For $x_j = cq^j$, $0 < q < 1$, define

$$N_k(x) = \prod_{r=0}^{k-1}(x - cq^r), \qquad \omega_{n+1}(x) = N_{n+1}(x).$$

Then

Table 15: Core geometric-comb formulas.

Object	Formula
Newton/q-Pochhammer basis	$N_k(x) = (-c)^k q^{\binom{k}{2}}(x/c; q^{-1})_k$
Divided difference	$[f; c, \ldots, cq^k] = D_q^k f(c)/[k]_q!$
Explicit data transform	$c^{-k}\sum_{j=0}^{k}(-1)^j q^{-jk+\binom{j+1}{2}} f(cq^j)/((q;q)_j(q;q)_{k-j})$
Monomial to Newton	$x^m = \sum_{k=0}^{m} c^{m-k}\begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x)$
Newton to monomial	$N_k(x) = \sum_{m=0}^{k}(-1)^{k-m}c^{k-m}q^{\binom{k-m}{2}}\begin{bmatrix} k \\ m \end{bmatrix}_q x^m$
Barycentric weight	$\lambda_{j,n} = c^{-n}(-1)^j q^{\binom{j+1}{2}-nj}/((q;q)_j(q;q)_{n-j})$
Accumulation weight	$\mu_{j,n} = (-1)^{n-j}q^{\binom{n-j+1}{2}}/((q;q)_j(q;q)_{n-j})$
Endpoint Lebesgue constant	$\mathscr{L}_n(0) = (-q;q)_n/(q;q)_n$
Global Lebesgue asymptotic	$\mathscr{L}_n^* \sim 2(-q;q)_\infty q^{-\binom{n}{2}}/(\mathrm{e}n)$
Monomial residual	$x^{n+1+r} - \mathcal{I}_n x^{n+1+r} = N_{n+1}(x)h_r(x, c, cq, \ldots, cq^n)$
Geometric specialization	$h_r(c, cq, \ldots, cq^n) = c^r\begin{bmatrix} n+r \\ n \end{bmatrix}_q$
Jackson moment	$\int_0^c N_k(x)\ \mathrm{d}_q x = (-1)^k c^{k+1} q^{\binom{k+1}{2}}/[k+1]_q$

A.2 Half-base Fabius formulas

For $q = 1/2$, $d_n = \mathbb{E}X^n$, and $a_n = d_n/n!$,

Table 16: Fabius and Rvachev specialization.

Object	Formula
Moment recurrence	$(2^n - 1)a_n = \sum_{j<n} a_j/(n-j+1)!$
Rvachev half-moment	$d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\ \mathrm{d}t$
Inverse-dyadic value	$F(2^{-n}) = 2^{-\binom{n}{2}}a_n$
Newton coefficient	$\nu_k = (q;q)_k^{-1}\sum_{j=0}^{k}(-1)^j\begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} a_j$
Finite interpolant	$\mathcal{I}_n F(x) = \omega_{n+1}(x)(q;q)_n^{-1}\sum_{j=0}^{n}(-1)^j\begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} a_j/(x - q^j)$
Boundary residue generating function	$\sum_{n\ge 0} R_n z^n = (qz;q)_\infty \sum_{j\ge 0} q^{\binom{j}{2}} a_j z^j/(q;q)_j$
Boundary bound	$\lvert R_n\rvert \le \mathrm{e}(q;q)_\infty^{-2} q^{\lfloor n^2/4\rfloor}$
Fixed-jet bound	$\lvert(\mathcal{I}_n F)^{(m)}(0)\rvert \le \mathrm{e}m!\binom{n}{m}(q;q)_\infty^{-2} q^{\lfloor n^2/4\rfloor - mn + \binom{m}{2}}$

Object	Formula
Rvachev relation	$\mathrm{up}(t) = F(1-\lvert t\rvert)$ and $\mathcal{I}_n\,\mathrm{up} = 1 - \mathcal{I}_n F$ on the center-directed positive comb

A.3 Atomic synthesis formulas

For a degree bound n, $h = 2^{-n}$ and $K_n = \{-2^{n+1}+1, \dots, 2^{n+1}-1\}$,

$$\mathcal{A}p = \mathcal{M}_u(D)^{-1}p = \sum_{r\ge 0} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}, \tag{A.1}$$

$$p(x) = h \sum_{k\in K_n} (\mathcal{A}p)(kh)u(x-kh), \qquad \deg p \le n, \tag{A.2}$$

$$\ell_{j,n}(x) = h \sum_{k\in K_n} (\mathcal{A}\ell_{j,n})(kh)u(x-kh), \tag{A.3}$$

$$(hB)D = I_{n+1}. \tag{A.4}$$

B Selected exact dyadic data

B.1 Moments and inverse-dyadic values

Table 17: Regularized moments $a_n = d_n/n!$ and the corresponding values $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$.

n	a_n	$F(2^{-n})$
0	1	1
1	$1/2$	$1/2$
2	$5/36$	$5/72$
3	$1/36$	$1/288$
4	$143/32400$	$143/2073600$
5	$19/32400$	$19/33177600$
6	$1153/17146080$	$1153/561842749440$
7	$583/85730400$	$583/179789679820800$
8	$1616353/2623350240000$	$1616353/704200217922109440000$

B.2 First geometric Newton coefficients

For

$$\mathcal{I}_n F(x) = \sum_{k=0}^{n} \nu_k \prod_{r=0}^{k-1} (x - 2^{-r}),$$

the exact coefficients begin

Table 18: First exact half-base Newton coefficients.

k	ν_k
0	1
1	1
2	$-26/27$
3	$-128/27$

k	ν_k
4	$-4253248/637875$
5	$189673472/19774125$
6	$134859284873216/1648153544625$
7	$21867230161534976/209315500167375$
8	$-34928636063215783195 4432/204161105975753390625$

Large numerator and denominator sizes are not numerical artifacts; they are exact consequences of composing the Fabius moment recurrence with the reciprocal Gaussian transform.

C Additional derivations

C.1 Partial fractions for the q-exponential perpetuity

For completeness, consider

$$L_q(s) = \prod_{j=0}^{\infty}(1+sq^j)^{-1}.$$

At the simple pole $s = -q^{-k}$, the coefficient of $(1+sq^k)^{-1}$ is

$$\begin{aligned} c_k &= \left[\prod_{j\neq k}(1-q^{j-k})\right]^{-1} \\ &= \frac{(-1)^k q^{k(k+1)/2}}{(q;q)_k (q;q)_\infty}. \end{aligned}$$

Indeed, the factors with $j < k$ give $(-1)^k q^{-k(k+1)/2}(q;q)_k$, and the factors with $j > k$ give $(q;q)_\infty$. Since

$$\mathcal{L}^{-1}\left[\frac{1}{1+sq^k}\right](y) = q^{-k}e^{-y/q^k},$$

we recover (17.11).

C.2 Equivalence of the power formulas

For an integer $d = n+1+r$, (14.7) and (14.8) agree. Indeed,

$$\begin{aligned} q^{dn}\frac{(q^{1-d};q)_n}{(q;q)_n} &= q^{(n+1+r)n}\frac{\prod_{j=0}^{n-1}(1-q^{-n-r+j})}{(q;q)_n} \\ &= (-1)^n q^{\binom{n+1}{2}}\frac{(q;q)_{n+r}}{(q;q)_n(q;q)_r} \\ &= (-1)^n q^{\binom{n+1}{2}}\begin{bmatrix} n+r \\ n \end{bmatrix}_q. \end{aligned}$$

This calculation is often useful when moving between Mellin exponents and integer Taylor moments.

C.3 A generating function for the smooth remainder on exponentials

Take $f(x) = e^{tx}$ in (17.2). Using (17.6),

$$(\mathcal{G}_{n,q}e^{t\cdot})(A) - 1 = \frac{(-1)^n A^{n+1} q^{\binom{n+1}{2}} t^{n+1}}{(n+1)!} \mathbb{E}e^{tY_{n,A,q}}$$
$$= (-1)^n A^{n+1} q^{\binom{n+1}{2}} t^{n+1} \sum_{r=0}^{\infty} \frac{A^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q}{(n+r+1)!} t^r. \tag{C.1}$$

The same series follows from (14.9). The equality is a useful consistency check between the analytic-tail and B-spline pictures.

D Selected finite identities

For quick reference, the principal finite formulas are collected here.

$$\Omega_{n+1}(x) = x^{n+1}(a/x; q)_{n+1},$$
$$\Phi_k(x) = (-a)^k q^{\binom{k}{2}} (x/a; q^{-1})_k,$$
$$\Phi_k(aq^j) = (-a)^k q^{\binom{k}{2}} (q; q)_k \begin{bmatrix} j \\ k \end{bmatrix}_q,$$
$$\beta_{n,j} = \frac{(-1)^j q^{-jn+\binom{j+1}{2}}}{a^n (q; q)_j (q; q)_{n-j}},$$
$$\lambda_{n,j}^{(q)} = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}},$$
$$\sum_{j=0}^{n} \lambda_{n,j}^{(q)} z^j = \frac{\prod_{r=1}^{n} (z - q^r)}{(q; q)_n},$$
$$f[a, aq, \ldots, aq^k] = \frac{D_q^k f(a)}{[k]_q!},$$
$$\sum_{j=0}^{n} |\lambda_{n,j}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n},$$
$$M_{n,q}(s) = q^{ns} \frac{(q^{1-s}; q)_n}{(q; q)_n},$$
$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mathbb{E}X_q^n \quad (0 < q \le 1/2),$$
$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{\mu_n}{n!},$$
$$A_n = \frac{1}{(1/2; 1/2)_n} \sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 \frac{\mu_j}{j!}.$$

E Pseudocode for exact dyadic computation

The following high-level algorithm uses rational arithmetic throughout.

```
mu[0] = 1
M[0]  = 1
for n = 1..N:
    mu[n] = (sum_{k=0}^{n-1} binom(n,k)*mu[k]/(n-k+1)) / (2^n-1)
    M[n]  = M[n-1]*(2^n-1)

for n = 0..N:
    Fdyadic[n] = 2^(-n*(n-1)/2) * mu[n] / n!
    A[n] = 2^(n*(n+1)/2) *
           sum_{j=0}^n (-1)^j * mu[j] / (j!*M[j]*M[n-j])
    E[n] = sum_{j=0}^n (-1)^(n-j) * 2^j * mu[j]
           / (j!*M[j]*M[n-j])
```

The attached Python implementation additionally verifies all transforms by independent formulas, evaluates Newton partial sums at rational targets, and computes the high-precision Fourier-product experiment.

F Gamma-product calculations

F.1 Half-cell central cardinal (mode A)

In the scaled variable $y = Mx$ with interior nodes $1, \dots, M-1$ and $j = M/2$, at $y = \frac{1}{2}$: $\prod_{k=1}^{M-1} \left|k - \frac{1}{2}\right| = \Gamma(M - \frac{1}{2})/\Gamma(\frac{1}{2})$; deleting the j factor divides by $\left|j - \frac{1}{2}\right| = \frac{M}{2} - \frac{1}{2}$, and the denominator of the cardinal is $(j-1)!\,(M-1-j)! = \Gamma(M/2)^2$, giving A_M in (46.4). The weight ratio is $\left[\frac{1}{2M}(1 - \frac{1}{2M})\right]/\frac{1}{4} = (2M-1)/M^2 = a_M$. Stirling gives $A_M \sim 2^M/(\pi\sqrt{2}\,M)$.

F.2 Near-central boundary-node cardinal (mode B)

For $j = 1$ at $y = (M+1)/2$ with $M = 2K$: $\left|\prod_{k=2}^{M-1}(y-k)\right| = \left[\Gamma(K - \frac{1}{2})/\Gamma(\frac{1}{2})\right]^2$ (the products on the two sides of the half-integer are equal), and the denominator is $(M-2)! = \Gamma(M-1)$, giving B_M. The weight ratio is $\left[(\frac{1}{2} + \frac{1}{2M})(\frac{1}{2} - \frac{1}{2M})\right]/\left[\frac{1}{M}(1 - \frac{1}{M})\right] = (M+1)/4 = b_M$. The duplication formula turns B_M into $\dfrac{2^{2-M}}{\sqrt{\pi}}\,\dfrac{\Gamma((M-1)/2)}{\Gamma(M/2)} \sim 2^{2-M}\sqrt{2/(\pi M)}$; equivalently $B_M = \binom{M-2}{M/2-1}/16^{\,M/2-1}$.

F.3 The one-third mode

At $y = M/3$ (dyadic M is never divisible by 3): $\prod_{k=1}^{M-1}(M/3 - k) = \Gamma(M/3)/\Gamma(1 - 2M/3)$, and reflection gives $1/\left|\Gamma(1 - 2M/3)\right| = \Gamma(2M/3)\left|\sin(2\pi M/3)\right|/\pi = \frac{\sqrt{3}}{2\pi}\Gamma(2M/3)$. Removing the central factor $|M/3 - M/2| = M/6$ and dividing by $\Gamma(M/2)^2$ gives (46.8); the exponential part of the gamma ratio is $\exp\{M[\frac{1}{3}\log\frac{1}{3} + \frac{2}{3}\log\frac{2}{3} - \log\frac{1}{2}]\} = \mathrm{e}^{M(\log 2 - \mathsf{H}(1/3))}$.

F.4 Full-comb central cardinal

For the full node set $0, \dots, M$ at $y = \frac{1}{2}$, $j = M/2$: $\prod_{k=0}^{M}\left|\frac{1}{2} - k\right| = \frac{1}{2}\,\Gamma(M + \frac{1}{2})/\Gamma(\frac{1}{2})$; deleting the central factor divides by $\left|\frac{1}{2} - \frac{M}{2}\right| = \frac{M-1}{2}$, and the denominator is $(M/2)!\,(M/2)! = \Gamma(\frac{M}{2}+1)^2$, giving (46.10); Stirling's formula then yields $\left|\ell_{M/2}(\frac{1}{2M})\right| \sim \sqrt{2}\,2^M/(\pi M^2)$, whose square is (46.11).

G Mode asymptotics

Stirling's expansion in logarithmic form,

$$\log\Gamma(z) = \left(z - \tfrac{1}{2}\right)\log z - z + \tfrac{1}{2}\log(2\pi) + \sum_{r=1}^{R-1}\frac{B_{2r}}{2r(2r-1)\,z^{2r-1}} + \mathcal{O}\big(z^{-2R+1}\big), \tag{G.1}$$

applied to (46.4)–(46.6) yields

$$\log A_M = M\log 2 - \log M - \log(\pi\sqrt{2}) + \mathcal{O}(M^{-1}), \tag{G.2}$$

$$\log B_M = -(M-2)\log 2 + \tfrac{1}{2}\log\frac{2}{\pi M} + \mathcal{O}(M^{-1}), \tag{G.3}$$

$$\log a_M = -\log(M/2) + \log\big(1 - \tfrac{1}{2M}\big), \qquad \log b_M = \log(M/4) + \log\big(1 + \tfrac{1}{M}\big), \tag{G.4}$$

which prove proposition 47.2. Retaining further Bernoulli terms and reverting with complete Bell polynomials yields the all-orders program (47.8).

H Row reciprocity

With $A_m(z) = \sum_{k=0}^{M-1} F(k/M)z^k$ and $F((M-k)/M) = 1 - F(k/M)$,

$$z^M A_m(1/z) = \sum_{k=1}^{M} F\big((M-k)/M\big)z^k = \sum_{k=1}^{M}\big(1 - F(k/M)\big)z^k = \frac{z(1-z^M)}{1-z} - A_m(z) - z^M, \tag{H.1}$$

which is (32.9). This exact reciprocity forces the palindromic structure of the low-level row numerators (32.6)–(32.8).

I Core algorithms

Listing 1: Streaming the exact dyadic Fabius row (algorithm 30.4); d holds the exact Fabius-law moments of (30.20).

```
def fabius_row_exact(m):
    M = 2**m
    d = fabius_law_moments(m)          # Fractions d_0..d_m
    S = [0]*(m+1)                      # signed prefix power sums
    scale = Fraction(1, 2**(m*(m-1)//2) * factorial(m))
    row = []
    for a in range(M+1):
        row.append(scale * sum(comb(m,j)*d[j]*S[m-j]
                               for j in range(m+1)))
        if a == M: break
        eps = -1 if bin(a).count('1') % 2 else 1
        S = [S[0]+eps] + [sum(comb(k,q)*S[q] for q in range(k+1))
                           for k in range(1, m+1)]
    return row
```

Listing 2: Endpoint-flat interpolation in barycentric form (theorem 45.1).

```
S = lambda r,x: (factorial(2*r+1)//(factorial(r)**2)
                 * sum((-1)**k*comb(r,k)*x**(r+k+1)/(r+k+1)
                       for k in range(r+1)))
W = lambda r,x: (x*(1-x))**(r+1)
g = [(F(x_j)-S(r,x_j))/W(r,x_j) for x_j in interior_nodes]
w = [(-1)**(j-1)*comb(M-2,j-1) for j in range(1,M)]
H = S(r,x) + W(r,x)*barycentric(x, interior_nodes, g, w)
```

The absorbed packages add the sparse-difference row generation ($\mathcal{O}(mM)$), confluent divided differences for full-grid Hermite, weighted-Lebesgue scans, and all figure/table generation; the canonical entry points, dependencies, and replay commands are recorded in `assets/README.md` and `assets/VALIDATION.md`.

J Status ledger

Table 19: Logical and formal status of the principal claims. “Paper proof” means a manuscript proof in this report, not a Lean declaration.

Claim	Manuscript	Lean boundary	Location
Newton/q-Pochhammer identification	Paper proof	Finite q-Pochhammer API exists; no report-specific N_k theorem found	Section 2
Iterated D_q and divided differences	Paper proof	No one-to-one declaration found	Theorem 2.2
Gaussian Pascal basis pair	Paper proof	Gaussian reciprocity/inversion ingredients formalized; basis pair not packaged	Theorem 3.1
Complete-homogeneous monomial residual	Paper proof	Geometric residual moments formalized; arbitrary evaluation remainder not packaged	Theorem 5.2
Analytic infinite Newton convergence	Paper proof	No one-to-one declaration found	Theorem 6.1
Endpoint Lebesgue formula and limit	Paper proof	Exact finite rational ℓ^1 identity formalized; real infinite-product limit not	Theorem 7.1
Boundary profile and global Lebesgue asymptotic	Paper proof	No one-to-one declaration found	Theorems 7.2 and 7.3
Newton–Jackson formula	Paper proof	No one-to-one declaration found	Theorem 8.1
Fabius half-base Gaussian transform	Paper proof	Inverse-dyadic moment and Gaussian ingredients formalized; transform not packaged	Theorem 9.3
Fabius residue and fixed-jet bounds	Paper proof	No one-to-one declarations found	Theorems 10.1 and 10.3
Fabius gap upper bound	Paper proof	No one-to-one declaration found	Theorem 10.5
Two-sided Fabius gap asymptotic	Conjecture	Unproved	Conjecture 10.6
Second-order Lebesgue expansion	Conjecture	Unproved	Conjecture 13.1
Hermite saddle displacement	Conjecture	Unproved	Conjecture 13.5
Dyadic fixed-radius up synthesis	Paper proof	Directly formalized by `normalized_sum_Ioo_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`	Theorem 11.1
Generic scalar cardinal-decoder synthesis	Paper proof	Generic scalar decoder synthesis formalized by `normalized_sum_Ioo_lagrangeRvachevDecoder_mul_shifted_rvachevUp`	Theorem 11.3
Geometric closed decoder formula	Paper proof	Elementary-symmetric expression and Gaussian prefactor not formalized	Theorem 11.3
Atomic coefficient factorization and full interpolation loop	Paper proof	Directly formalized by `lagrangeRvachevAtomCoefficient_eq_deconvolved_interpolate` and `sum_Ioo_lagrangeRvachevAtomCoefficient_mul_shifted_rvachevUp`	Corollary 11.4

Claim	Manuscript	Lean boundary	Location
Nodewise biorthogonality and decoder row sum	Paper proof	Directly formalized componentwise by `normalized_sum_Ioo_lagrangeRvachevDecoder_eval_node` and `sum_lagrangeRvachevDecoder_eq_one`; no Matrix wrapper	Theorem 11.5
Stochastic encoder and Matrix right inverse	Paper proof	No typed Matrix or row-stochastic encoder package found	Theorem 11.5
Minimum-variation atomic decoder	Open problem	Unproved; no optimization or conditioning theorem found	Problem 13.7
Growing-order Hermite anchoring	Open problem	Unproved	Problem 8.6

K Source and asset provenance

This appendix is intentionally revision-backed. The four live source manuscripts and all 180 files in their parent subtree are pinned at commit `73f0b373126ef22a3b5dccadfa7b99d61d445345`. The executable extractor in `audit/` reconstructs their result inventory directly from that commit, while `source_disposition.csv` records the SHA-256 digest and final disposition of every path. Thus deleting an absorbed report does not erase either its bytes or its theorem-level identity.

K.1 The four peer manuscripts

Historical package	Canonical contribution
`Dyadic_Comb_Frontiers`	Additive dyadic foundations, exact comb sums and quadrature, spectral and phase calculus, and endpoint-flat global interpolation. That manuscript was itself an editorial merge of nine earlier packages, itemized below.
`geometric_comb_q_fabius_report`	Canonical finite geometric-comb spine: affine covariance, Jackson–Newton calculus, Gaussian-Pascal changes of basis, complete-homogeneous residuals, sharp Lebesgue asymptotics, Fabius boundary recovery, and the Gaussian–Appell Rvachev decoder.
`geometric_comb_interpolation_report`	Confluent modal filters, regular variation, geometric-knot B-splines, finite geometric-uniform splines, cumulants, and even-entire Fourier-scale reconstruction.
`geometric_comb_interpolation_report-3`	Arbitrary-target cardinals, Mellin symbols and logarithmic modes, Taylor/Richardson limits, regular summability, superflat divergence, the general geometric-uniform moment transform, and the reciprocal Rvachev sinc-product filter.

The four manuscripts are peers, not successive editions. Similar statements therefore receive multiple source rows in the concordance but one strongest canonical destination. In particular, the q-Pochhammer nodal polynomial, geometric Vandermonde determinant, Gaussian-Pascal factorization, Jackson divided differences, accumulation-point barycentric row, universal monomial residual, and analytic infinite Newton series are not repeated in the independent-extension part.

K.2 The nine absorbed additive-dyadic sources

The additive-dyadic manuscript already absorbed the following packages. The first six main TeX sources are directly recoverable from the parent of commit `d022736a`; the later three are recoverable inside the deleted incoming ZIP blobs named below. Their exact source hashes and theorem-level mapping are retained in Git history and in the concordance notes.

Historical package	Material retained in the additive-dyadic parts
`Fabius_Dyadic_Comb_Sums_Report_Package`	Row generating function, Bernoulli–Thue–Morse and Bell–Prouhet formulae, exact collapse, spectral defects, Hurwitz continuation, and iterated/fractional sums.
`fabius-dyadic-comb-sums-report`	Moment EGF, probabilistic dyadic convolution, sparse row algorithm, early spectral values, Richardson filtering, and tensor cubature.

Historical package	Material retained in the additive-dyadic parts
`fabius_dyadic_comb_report_final`	Arbitrarily shifted exactness, 2-adic alias hierarchy, Bell–Bernoulli jets, absolute/negative powers, universal zeta correction, and phase superconvergence.
`fabius_dyadic_interpolation_report`	Symmetry compression, smoothstep factorization, exact cardinal modes, fixed-order instability, Lambert balance, full-grid Hermite amplification, and local stability.
`fabius_interpolation_report`	Exact balance window, entropy-mode all-order instability, endpoint derivative defects, mixed transforms, and stable alternatives.
`Fabius_Rvachev_Dyadic_Interpolation_Report`	Valuation law, error-identity form of endpoint-flat factorization, even Rvachev reduction, optimized-divergence data, and peak-migration asymptotics.
`Fabius_Euler_Maclaurin_Report_Package`	Bernoulli periodization, arbitrary-mesh termination, reflection, phase defects, positivity, closed exhaustion tails, and derivative-free extrapolation. Source archive: `a348a5b1^:drafts/incoming/Fabius_Euler_Maclaurin_Report.zip`.
`fabius_rvachev_exhaustion_euler_maclaurin_bundle`	All-mesh alias–Bernoulli identity, general first defect, half-interval Rvachev rule, two-level recovery, and Gaussian phase filters. Source archive: `9d4f6771^:drafts/incoming/fabius_rvachev_exhaustion_euler_maclaurin_bundle.zip`.
`fabius_ruffa_phase_calculus`	Multiple-angle domination, twisted spectral positivity, complete phase zero sets, finite phase cubature, digital Bernoulli filters, coherent radix exhaustion, and tensor shells. Source archive: `563e4d35^:drafts/incoming/fabius_ruffa_phase_calculus.zip`.

The old nested READMEs, audit prose, report PDFs, stale checksum ledgers, and three files explicitly announcing that their content was “not shipped” were not mathematical evidence. Their nonduplicative provenance and reproduction instructions have been distilled into this appendix and the canonical asset documentation. Scripts, exact data, generated tables, textual outputs, and valid diagnostic PNGs survive under `assets/companion-evidence/<source-slug>/`. The only byte-identical payload pair was a two-line requirements file, now represented once with two provenance rows.

K.3 What the audit establishes

The source-result extractor and publication validator are intentionally separate. The former proves that the immutable source columns of the theorem concordance still agree with the pinned manuscripts. The latter checks chapter inclusion, environment balance, labels and references, explicit proof coverage, conjecture labelling, source-disposition completeness, and live asset hashes. Three strict serial pdfLaTeX passes and rendered-page inspection are additional publication evidence; neither static checking nor numerical replay is presented as a substitute for a mathematical proof.

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